

Protocol

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This trial protocol has been provided by the authors to give readers additional information about the work.

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Title Page**Protocol Title:**

A randomized, double-blind, placebo-controlled, parallel-group, multicenter Phase 3 study to investigate the efficacy and safety of **FInerenone**, in addition to standard of care, on the progression of kidney disease in patients with **Non-Diabetic Chronic Kidney Disease**

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1. Protocol Summary

1.1 Synopsis

Protocol Title: A randomized, double-blind, placebo-controlled, parallel-group, multicenter Phase 3 study to investigate the efficacy and safety of finerenone, in addition to standard of care, on the progression of kidney disease in patients with non-diabetic chronic kidney disease

Brief Title: Efficacy and safety of finerenone in patients with non-diabetic CKD

Rationale:

This is a randomized, double-blind, parallel-group study to investigate the efficacy and safety of finerenone, i.e., 10 mg and 20 mg once daily, compared with placebo, on the progression of non-diabetic chronic kidney disease in male and female patients over 18 years of age, over minimum of 32-months of study intervention per participant, in addition to standard of care.

Population:

This study will include patients with non-diabetic CKD of different disease etiology at risk of progressive renal function decline.

Objectives and Endpoints

Objectives	Endpoints
Primary	
To demonstrate that finerenone in addition to SoC, is superior to placebo in delaying the progression of kidney disease.	Mean rate of change as measured by the total slope of eGFR from baseline to Month-32.
Secondary	
- To demonstrate the beneficial effect of finerenone in addition to SoC as compared to placebo. - To assess the safety of finerenone in addition to SoC as compared to placebo	- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF or CV death - Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$ - Time to the composite of hospitalization for HF or CV death - Number of participants with TEAEs, TESAEs and AESI

AESI = Adverse events of special interest, CV = Cardiovascular, eGFR = Estimated glomerular filtration rate, HF = Heart failure, SoC = Standard of care, TEAEs = Treatment-emergent adverse events, TESAEs = Treatment-emergent serious adverse events

Overall Design:

This is a double blind parallel-group intervention study in non-diabetic chronic kidney disease patients with 2 treatment groups, including a placebo treatment group.

Participants who fulfill all eligibility criteria will be randomized in a 1:1 ratio to receive finerenone or placebo, in addition to standard of care:

- Finerenone 10 mg/OD or 20mg/OD
- Placebo

All participants will receive blinded study intervention (i.e. finerenone 10mg/20mg or Placebo) OD on top of individually tolerated maximum labeled doses of either an ACEI or an ARB. The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/ 1.73m^2 (CKD-EPI), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/ 1.73m^2 (CKD-EPI). Subjects who start on the lower dose of 10 mg will be uptitrated to the target dose of 20 mg from Month 1 on treatment onwards provided potassium is ≤ 4.8 mmol/L, and eGFR decrease is less than 30% of the last measured value. Down-titration or interruption of study intervention is permitted at any time during the study for safety reasons and study intervention has to be interrupted if potassium is > 5.5 mmol/L. Visits will take place at screening, baseline, Month 1, Month 3 and from then every 3 months up to Month 12, and thereafter every 4 months until the planned end of treatment. The minimum per participant treatment duration will be 32 months. Additional visits will take place 4 weeks after each up-titration, interruption for > 7 days or treatment re-initiation, and 1 month after the end of treatment. eGFR as the primary endpoint measure and potassium as the key safety parameter as described will be assessed at each visit. UACR and UPCR will be collected at screening, baseline, at Month 1, Month 3, Month 6, Month 12, Month 24, at the end of treatment and 1 month after treatment discontinuation.

In order to offer participants and sites increased flexibility in trial conduct over the duration of this multi-year study, a "hybrid" DCT model may be implemented where local regulations and legislations allow. This model integrates telemedicine technology into certain parts of the clinical research process and supports management of research activities, including remote data collection. For the selected visits identified in the schedule of activities, remote visit activities such as Home Health Nurse visits, completed in conjunctions with a telehealth call and DtP study intervention shipment will be available for use. The sponsor will select qualified vendors to offer these services or, in some cases, with prior written sponsor approval, sites may conduct some of these remote visit activities themselves (e.g. home health nurse visits or DtP from the site to the participant).

Brief Summary:

The purpose of this study is to assess efficacy and safety of finerenone (10 mg and 20 mg, OD) compared to placebo in participants with non-diabetic chronic kidney disease. Study details include:

- Study duration per participant: from approximately 33 months to approximately 50 months (with an expectation of approximately 17 months recruitment duration).
- Treatment duration per participant: from 32 to up to approximately 49 months (with an expectation of approximately 17 months recruitment duration)
- Visit frequency: approximately once every 1-4 months

Number of Participants:

Approximately 3160 participants will be screened to achieve 1580 randomly assigned to study intervention and 1500 evaluable participants for an estimated total of 750 evaluable participants per intervention group.

Intervention Groups and Duration:

Study Periods	Duration
1. Screening	1. 1 to 14 days
2. Intervention	2. 32 to approximately 49 months
3. Follow-up	3. approximately 1 month
Total number of participants	Total study duration
1580 randomized	approximately 33 to 50 months

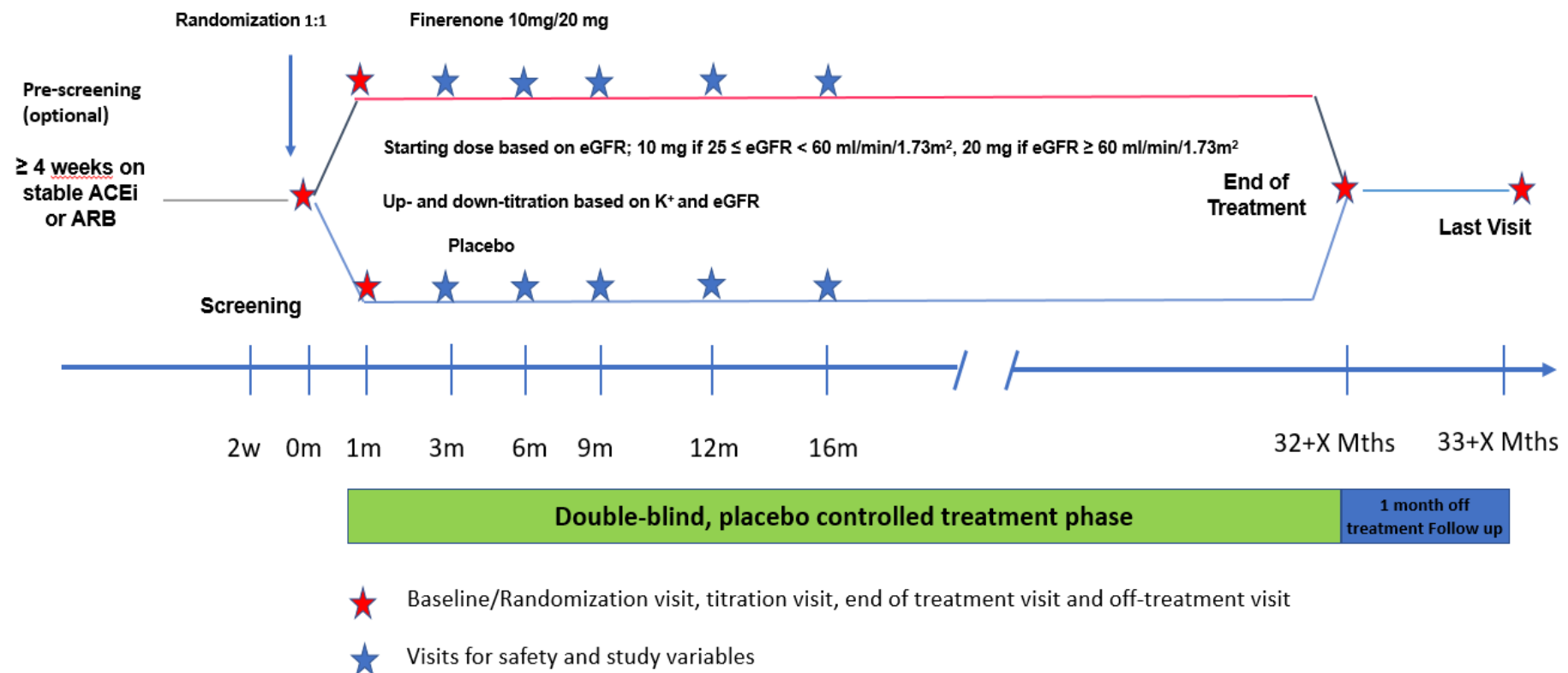
Intervention groups:

- Group 1: 10 mg to 20 mg of finerenone OD
- Group 2: placebo

Data Monitoring/Other Committee: Yes

1.2 Schema

Figure 1–1: Study Schema



Study duration and the number of study Visits will depend on the time of enrollment of the participant

ACEi = Angiotensin-converting enzyme inhibitor, ARB = Angiotensin receptor blocker, eGFR = Estimated glomerular filtration rate, K = potassium, m = Month, mths = Months, W = Week

1.3 Schedule of Activities (SoA)**Table 1–1: Schedule of Activities**

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up titration or reinitiation		at planned end of treatment	1 month after last dose	
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff)
Visit Window (days)				±5	±5	±7	±7	±7	±7	±7	±7			±7	
Informed consent and eligibility check															
Pre-screening consent	•														
Informed consent		•													
Informed consent for genetic sampling (optional)		•													Not relevant for China. Can be done at Visits 0 or 1
I/E criteria		•	•												
Baseline characteristics															
Demography		•													
Full physical examination with height and weight			•												
weight only		•		•	•	•	•	•	•	•		•		•	
Medical history		•													Including kidney biopsy results as available
Randomization			•												
Questionnaires															
EQ-5D-5L			•					•	•*			•			EQ-5D-5L Should be completed before any other evaluations. *End of treatment only

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up-titration or reinitiation		at planned end of treatment	1 month after last dose	
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff)
Visit Window (days)				±5	±5	±7	±7	±7	±7	±7	±7			±7	
Laboratory evaluations															
Serum or urine pregnancy test		●						●							WOCBP only. Urine or serum, as required by local regulations
Full clinical chemistry (incl eGFR and K ⁺); central lab		●													
Creatinine/eGFR, K ⁺ ; Na ⁺ central lab			● ^{**}	● ^{**}	● ^{**}	●	●	● ^{**}	●	●		● [*]		● [*]	* If central laboratory sample cannot be taken, a local laboratory results may be used instead. **Na ⁺ will be measured only at these visits.
Creatine/eGFR, potassium; local lab		● [*]	●	●	●	● ^{**}	●	●	●	● ^{**}	●				*Only potassium sample needed ** At DCT visits local lab samples will not be collected
UACR/UPCR; central lab		●	●	●	●	● [*]		● [*]	● [*]			●		●	* At Visits 4, 6, 9 (months 6, 12, 24) and EoT only
UACR local screening	●														
Biomarker (serum, plasma and urine)			●		●			●	● [*]			●		●	*At end of treatment only
Biomarker (blood for			●												Not relevant for China.

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up titration or reinitiation		at planned end of treatment	1 month after last dose	
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff)
Visit Window (days)				±5	±5	±7	±7	±7	±7	±7	±7			±7	
pharmacogenetics)															Optional, only participants who consented. Can be taken also on Visit 3.
PK blood sample				•	•	•*	•*	•*							See Table 1–2 and Section 8.4 for information on study intervention intake on these visits. *at Visits 4-6 only. PK sample will be taken at Visit 5 only if it has not been taken at Visits 2-4 (i.e. for a DCT visit).
Further safety evaluations and Study intervention															
12-lead ECG			•						•			•			
Vital signs		•	•	•	•	•	•	•	•	•	•	•		•	
Study intervention dispense/administration, or return			•	•	•	•	•	•	•*	•		•			*Study intervention will not be dispensed at EoT
DCT package provision				•								•			for participants enrolled in the DCT model
DCT package return														•	
End of study phone call													•		Only for participants who have discontinued

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up-titration or reinitiation		at planned end of treatment	1 month after last dose	
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff)
Visit Window (days)				±5	±5	±7	±7	±7	±7	±7	±7			±7	
															treatment prior to planned end of treatment. At the end of study AE/SAE/OEs will be collected by phone.
AE/ SAE/ OE and concomitant medication review		-----→													

One month is counted as 30 days

AE = Adverse event, ECG = Electrocardiogram, eGFR = estimated glomerular filtration rate, EoS = End of Study Visit, EQ-5D-5L = EuroQoL 5-dimension 5-level questionnaire, FU = Follow-up, I/E = Inclusion/exclusion, K+ = potassium, OE = Outcome event, PD = premature discontinuation, PK = Pharmacokinetics, SAE = Serious adverse event, UACR = Urine albumin/creatinine ratio, UPCR = Urine protein/creatinine ratio, UT = Up-titration

- Pre-screening visit is optional
- Premature discontinuation (PD) visit is to be carried out if the participant permanently discontinues the study intervention or the study prior to declaration of End of Study. PD Visit should take place within 7 days from the last study drug intake. If more than 7 days has passed from the last study intervention intake, the participant may move directly to the FU Visit.
- 32 Months equals end of treatment for the last participant randomized globally.

Table 1–2: PK sampling schedule

Visit	V2	V3	V4	V5	V6	Notes
Month	1	3	4	9	12	
Study intervention intake on visit days	Take at home	Taken at site	Taken at home	Taken at site	Taken at home	Document timing of last study drug intake before PK sampling
PK sample (trough)		•		•*		Document exact time of study intervention intake on the day prior to the visit. *PK sample will be taken at Visit 5 only if it has not been taken at Visits 2-4 (i.e. for a DCT visit).
PK sample (peak, 1.5-10 hours after intake of study intervention)	•		•		•	Document exact time of intake on visit day.

PK = Pharmacokinetics

2. Introduction

Finerenone (BAY 94-8862) is a next-generation, oral, selective, non-steroidal MRA that is being developed for once daily treatment of non-diabetic chronic kidney disease.

2.1 Study Rationale

Diverse pathologies induce injury and inflammation resulting in a common pathophysiological mechanism in CKD where glomerular injury leads to proteinuria, interstitial inflammation, and, ultimately, fibrosis. Finerenone is a mineralocorticoid receptor antagonist (MRA) acting by targeting inflammatory and fibrotic pathways mediated by mineralocorticoid receptor overactivation in the heart and kidney ([Kolkhof and Barfacker 2017](#)). Finerenone protected from kidney function decline and reduced the risk of CV events in the phase 3 FIDELIO study in participants with CKD and T2D. Efficacy was observed independent of the hyperglycemic state ([Bakris et al. 2020](#)). It also reduced albuminuria in diabetic and non-diabetic participants with CKD in the ARTS ph2a study ([Pitt et al. 2014](#)). Based on the available pre-clinical and clinical data finerenone is believed to show efficacy in delaying the progression of kidney disease in diabetic and non-diabetic patients.

2.2 Background

In 2017, globally 1.2 million people died from CKD and the global all-age mortality rate from CKD increased 41.5% between 1990 and 2017. In 2017, 697.5 million cases of all-stage CKD were recorded accounting for a global prevalence of 9.1%. Globally, 4.6% of total mortality were caused by deaths due to CKD and cardiovascular disease attributable to impaired kidney function. High blood pressure accounted for the largest proportion of CKD burden in east Asia, eastern Europe, tropical Latin America, and western sub-Saharan Africa, whereas diabetes was the leading risk factor for CKD in all other regions. A higher prevalence of CKD stages 1–3 in females and higher mortality in males suggest that males progress to ESKD more rapidly ([GBD Chronic Kidney Disease Collaboration 2020](#)).

In populations with a high prevalence of diabetes and hypertension, it can be difficult to distinguish CKD due to hypertension and diabetes from CKD due to other disorders. In other countries, other causes of CKD may be as frequent as hypertension and diabetes or coexist with them ([KDIGO 2012](#)). The most common glomerulonephritis globally is IgAN with varying prevalence by geographic region showing a higher incidence in Pacific Asian regions ([Rodrigues et al. 2017](#)). After diabetes, hypertension is the second most common cause for newly requiring dialysis followed by glomerulonephritis ([KDIGO 2012](#), [Kramer et al. 2020](#), [Saran et al. 2019](#)). ACEI and ARB are considered standard of care for patients with non-diabetic CKD and albuminuria ([Kidney Disease Improving Global Outcomes \(KDIGO\) 2013](#)), however, existing evidence is mainly based on efficacy seen in diabetic patients ([Brenner et al. 2001](#), [Lewis et al. 2001](#)) and comparable small numbers of non-diabetic patients were studied. Recently completed studies ([Heerspink et al. 2020](#), [Perkovic et al. 2019](#)) showed a benefit of SGLT2is in patients CKD with and without diabetes.

Despite its relevance for global health, CKD often progresses slowly and mostly asymptotically until kidney failure occurs, limiting the development of new medications due to long study durations and high patient numbers ([Levey et al. 2020](#)). Addressing this problem, a scientific workshop was held by the National Kidney Foundation in collaboration with FDA and EMA to evaluate changes in UACR and GFR as surrogate endpoints for kidney

failure in clinical trials. Meta-analyses of clinical trials which were conducted around this workshop revealed that treatment effects on GFR slope from baseline (total) and from 3-month follow-up (chronic) of 0.5–1.0 mL/min/1.73 m² per year strongly predict benefits on clinical endpoints such as doubling of serum creatinine, GFR < 15 mL/min/1.73 m², or ESKD ([Inker et al. 2019](#)). Based on these analyses which were consistent across subgroups of different disease etiology and treatment it was concluded that total and chronic eGFR slope can play a role as an endpoint for clinical trials of kidney disease progression in CKD. Obtaining before-treatment eGFR and final eGFR after withdrawing treatment (off-treatment effect) for the reversibility of acute effects was considered supportive. Recently, surrogates for eGFR have been emerging as a more commonly accepted endpoint in phase 3 clinical trials ([Komers et al. 2020](#), [Torres et al. 2017](#)).

Finerenone is already studied in more than 13,000 patients with CKD in T2D in the phase 3 FIDELIO-DKD and FIGARO-DKD outcome trials. In the completed positive FIDELIO-DKD trial, 5674 patients were treated for a median of 2.67 years to achieve an 18% lower hazard for the primary renal composite endpoint of kidney failure, a sustained eGFR decrease of ≥ 40% from baseline for at least 4 weeks or renal death (7.59 and 9.08 per 100 patient years, respectively). The secondary renal endpoint that included a sustained eGFR decline of 57% or more was also achieved (HR 0.76 [95% CI, 0.65 to 0.90], p=0.0012; i.e. RRR of 24%) further supporting the renal benefit ([Bakris et al. 2020](#)). At 3 years a difference of 0.65 mL/min/1.73m² per year was seen between finerenone and placebo for the difference in total eGFR slope (from baseline to end of study) in the overall population, the chronic eGFR slope difference (calculated from Month 4 onwards) was 1.37 mL/min/1.73m² per year. This difference was more pronounced in patients with higher albuminuria (UACR ≥ 500mg/g) independent of their baseline eGFR. In these patients, a total eGFR slope difference of > 0.7 mL/min/1.73m² per year was observed earlier, after 32 months (following the 4 monthly visit schedule).

These data show that the effect observed on the eGFR slope translated into a significant benefit on the primary endpoint and thus, confirms the predictive value of eGFR slope difference in a large finerenone outcome trial in patients with CKD and T2D.

Based on these findings and while the beneficial renal effect of finerenone has already been proven in a related population, this study will investigate the effect of finerenone in patients with non-diabetic CKD etiologies including hypertension and chronic glomerulonephritis such as IgAN and FSGS; disease etiologies such as polycystic kidney disease or active lupus nephritis requiring a specific treatments will be excluded. Polycystic kidney disease has a primary disease mechanism which is not directly affected by MRA action. Kidney diseases with vasculitic etiology are likely to require immunosuppressive therapy and the study results might be confounded.

To also account for the initial decline, total eGFR slope will be used as the primary endpoint. Since the use of eGFR slope as the primary endpoint is most applicable to populations which show a certain degree of progression ([Greene et al. 2019](#)), only patients with albuminuria (UACR ≥ 200mg/g), a key predictor of progression independent of disease etiology will be included in this study ([Anderson et al. 2020](#), [Sarnak and Astor 2011](#)) and the majority of patients will have an UACR value ≥ 500mg/g. Chronic eGFR slope (from Month 3 until end of treatment) will be assessed as well describing the chronic benefit of finerenone translating into long-term preservation of kidney function. In addition, to show the persistent benefit of finerenone after treatment discontinuation eGFR change from baseline to 1 month after end of treatment will be assessed. The composite of kidney failure, confirmed eGFR decrease ≥ 57%

from baseline, hospitalization for heart failure or CV death will be a secondary endpoint in the trial providing supportive clinical evidence.

2.3 Benefit/Risk Assessment

2.3.1 Risk Assessment

The main risk observed with finerenone in previous studies was hyperkalemia. The population in this study has an inherent risk of hyperkalemia due to their underlying disease (as serum potassium tends to increase with decreasing eGFR) and due to their standard of care therapy (ACEI/ARB) (Bianchi and Regolisti 2019). In FIDELIO-DKD, hyperkalemia increased by approximately two-fold with finerenone treatment (18.3% finerenone vs 9.0% placebo). However, most treatment-emergent hyperkalemia events were non-serious, mild or moderate in intensity, and resolved. There were no fatal hyperkalemia TEAEs. Treatment-emergent hyperkalemia AEs of relevant clinical consequence (i.e. that led to permanent study intervention discontinuation [2.3% vs 0.9% for finerenone vs placebo, respectively] or hospitalization [1.4 vs 0.3%]) constituted a small proportion. Further, there was no evidence for an increased incidence of any severe clinical cardiac manifestations of hyperkalemia (e.g. bradycardia, ventricular arrhythmia or sudden cardiac death). Overall, the risk of hyperkalemia was manageable and expected based on finerenone's mode of action.

An increase from baseline in mean serum potassium in the first month of treatment of approximately 0.2 mmol/L was observed in the finerenone arm compared to placebo, with stable mean measurements from Month 4 onwards in the finerenone arm. The between-group difference of 0.2 mmol/L was consistently observed across different subgroups of baseline serum potassium and eGFR at screening, including participants with eGFR 25 to <45 mL/min/1.73 m².

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Study Intervention finerenone		
Hyperkalemia	- Incidence of TEAEs hyperkalemia: finerenone 18.3% vs placebo 9.0% - Incidence of TEAEs hyperkalemia leading to hospitalization: finerenone 1.4% vs. placebo 0.3% - Incidence of TEAEs hyperkalemia leading to permanent discontinuation: finerenone 2.3% vs placebo 0.9% - Incidence of TEAEs hyperkalemia with fatal outcome: finerenone 0% vs placebo 0% - Treatment emergent serum potassium (central lab) >5.5 mmol/L: finerenone 21.4% vs placebo 9.2% - TE serum potassium (central lab) >6 mmol/L: finerenone	- Dose initiation if K⁺ is ≤ 4.8 mmol/L at screening - Up-titration if K⁺ is ≤ 4.8 mmol/L and eGFR has not declined > 30% compared to the previous visit - Dose interruption if K⁺ is > 5.5 mmol/L, restart if ≤ 5.0 mmol/L - Exclusion of concomitant use of potassium sparing diuretics, MRAs, renin inhibitors, ARNIs - Exclusion of potent CYP3A4 inhibitors - Dosing will be based on eGFR and participants with a screening eGFR between 25 and < 60 mL/min/1.73m² will start with the lower dose of 10mg.

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
	4.5% vs placebo 1.4%	
Study Procedures		
Participation in this study presents a minimal risk as the documented AEs observed in prior studies were mild and completely resolved.	Frequent blood sampling (by single vein puncture and/or indwelling cannula) may be accompanied by mild pain, hematoma, feeling faint or dizzy, and in rare cases inflammation of the vessel wall or injury of a nerve.	All study assessments will be carried out by trained clinical staff.

2.3.2 Benefit Assessment

In the recently completed FIDELIO-DKD study, Finerenone was superior to placebo in delaying the progression of kidney disease (HR 0.825 [95% CI, 0.732 to 0.928], $p=0.0014$; i.e. RRR of 18%). Primary endpoint in FIDELIO-DKD was a composite of kidney failure, a sustained decrease in eGFR of 40% or more from baseline, or renal death. The treatment effect was reflected both in reductions in kidney failure and in sustained eGFR decrease from baseline, and the benefit of treatment was sustained throughout the duration of the study. The absolute risk reduction was 3.4% at Month 36 for the primary renal endpoint, corresponding to an NNT with finerenone to prevent a primary endpoint event of 29 participants. The secondary renal endpoint that included a sustained eGFR decline of 57% or more in the FAS was also achieved (HR 0.76 [95% CI, 0.65 to 0.90], $p=0.0012$; i.e. RRR of 24%), adding to the robustness and internal validity of the study with respect to the primary outcome. The results for the primary and secondary renal composite endpoints were generally consistent across evaluated subgroups, including region, eGFR, UACR, HbA1c, systolic blood pressure and serum potassium at baseline ([Bakris et al. 2020](#)). Based on these findings and while the beneficial renal effect of finerenone has already been proven in a related population and is independent of the hyperglycemic state, this study 21177 will investigate the effect of finerenone in participants with non-diabetic CKD etiologies including hypertension and chronic glomerulonephritis. Evidence of efficacy is supported by data on albuminuria reduction in diabetic and non-diabetic patients from the phase 2a ARTS study as well as by preclinical data ([Barrera-Chimal et al. 2018](#), [Kolkhof et al. 2017](#), [Lattenist et al. 2017](#)).

2.3.3 Overall Benefit Risk Conclusion

Finerenone successfully delayed the onset of the composite of kidney failure, a sustained eGFR decrease of $\geq 40\%$ from baseline for at least 4 weeks or renal death in the phase 3 FIDELIO-DKD study. At 3 years a difference of 0.65 mL/min/1.73m² per year was seen between finerenone and placebo for the difference in total eGFR slope (from baseline to end of study) in the overall population, the chronic eGFR slope difference (calculated from Month 4 onwards) was 1.37 mL/min/1.73m² per year reflecting the long-term benefit. An increased risk of hyperkalemia was observed with finerenone, however, the majority of these events were non-serious and the number of events leading to clinical consequences was low. The analysis of data collected in the context of dose-titration and interruption indicates that hyperkalemia with finerenone treatment is manageable using a serum potassium-guided dose titration regimen. This is supported by simulations of the exposure-serum potassium model, which

demonstrated that even at higher exposures, no increased risk for hyperkalemia is observed when dose-titration is applied.

To reduce the risk of hyperkalemia in this study, only participants with a potassium value of ≤ 4.8 mmol/L at the screening visit will be included and the starting dose will be reduced to 10mg if eGFR is between 25 and < 60 mL/min/1.73m² at screening. Potassium and eGFR will be monitored at each visit and guide up-and down-titration in line with the dose titration scheme used in FIDELIO-DKD.

To date, ACEI and ARB constitute the only available treatment option to delay kidney disease progression for patients with non-diabetic CKD, in the near future, SGLT2i may become a new treatment option based on recently presented data. Based on its mode of action and the consistent benefit seen in a related population, finerenone is expected to show efficacy in patients with non-diabetic CKD. The main risk of finerenone treatment, hyperkalemia, has been proven to be manageable even in the advanced CKD population of FIDELIO-DKD; efficacy was also consistent in participants at higher risk (i.e. with lower eGFR, higher baseline potassium). Based on the available data and the high medical need in the studied population, inclusion of patients into this study is considered favorable.

3. Objectives, Endpoints and Estimands

Objectives	Endpoints
Primary	
To demonstrate that finerenone in addition to SoC, is superior to placebo in delaying the progression of kidney disease.	Mean rate of change as measured by the total slope of eGFR from baseline to Month-32.
Secondary	
- To demonstrate the beneficial effect of finerenone in addition to SoC as compared to placebo. - To assess the safety of finerenone in addition to SoC as compared to placebo	- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF or CV death - Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$ - Time to the composite of hospitalization for HF or CV death - Number of participants with TEAEs, TESAEs and AESI
Exploratory	
- To be supportive of the primary efficacy endpoint by further demonstrating the effect of finerenone in addition to SoC as compared to placebo. - To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (supportive of the key secondary endpoints) - To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (other exploratory endpoints)	- Chronic eGFR slope from Month-3 to end of planned treatment-period - eGFR change from baseline to 1 month after end of treatment (off-treatment) - Time to onset of kidney failure (ESRD or sustained eGFR decline below 15 ml/min/1.73m²) - Time to onset of ESRD (chronic dialysis or transplant) - Time to onset of sustained eGFR decrease below 15 mL/min/1.73m² - Time to onset of sustained eGFR decrease of $\geq 57\%$ from baseline - Time to CV death - Time to first hospitalization for HF or urgent heart failure visit - Number of participants with UACR reduction $\geq 30\%$ from baseline at Month 6 - Change in UACR from baseline - Time to the composite of sustained eGFR decrease of $\geq 40\%$ or kidney failure - Time to first CV event (CV death, non-fatal MI, non-fatal stroke or transient ischemic attack, hospitalization for HF or urgent HF visit) - Time to CV hospitalization - Time to all-cause mortality - Time to all-cause hospitalization - Number of participants with new diagnosis of atrial fibrillation or atrial flutter - Number of participants with new diagnosis of HF

Objectives	Endpoints
Exploratory: Biomarkers	
To further investigate the study intervention and similar drugs (e.g., mode-of-action-related effects, safety) and to further investigate pathomechanisms deemed relevant to CKD and associated health problems	Various biomarkers (e.g., diagnostic, safety, pharmacodynamic, monitoring, or potentially predictive biomarkers)
Exploratory: Quality of Life	
To further investigate the quality of life parameters	Change from baseline in EQ-5D-5L

AESI = Adverse events of special interest, CV = Cardiovascular, eGFR = Estimated glomerular filtration rate, ESRD = End stage renal disease, EQ-5D-5L = EuroQol Group 5-dimension 5-level questionnaire, HF = Heart failure, MI = Myocardial infarction, SoC = Standard of care, TEAEs = Treatment-emergent adverse events, TESAes = Treatment-emergent serious adverse events, UACR = Urinary albumin-to-creatinine ratio

Estimand

The attributes constituting the primary estimand are the following:

- Population: The population consists of non-diabetic CKD patients, further defined by the in- and exclusion criteria.
- Treatment: All participants will receive blinded study intervention (i.e. finerenone 10mg/20mg or placebo) OD on top of their individually tolerated maximum labeled doses of either an ACEI or an ARB.
- Variable: eGFR values over time collected every 1 to 4 months during the planned treatment period
- Intercurrent events: The three most important intercurrent events to consider are
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant, and
 - Premature treatment discontinuation/extended interruption of treatment.

For premature treatment discontinuation/extended interruption of treatment, a treatment policy strategy will be applied. In the treatment policy strategy, participants will be followed up for their planned treatment visits after discontinuing treatment, and eGFR values after discontinuation of treatment will be included in the analysis. Details on the incorporation of intercurrent events (ICEs) considered as death, initiation of chronic dialysis, or kidney transplant and as such addressed by the appropriate estimand strategy will be provided in the SAP.

- Population-level summary: Total eGFR slope from baseline to 32-months comparing the finerenone treatment group with the placebo group as estimated from a random coefficient linear spline mixed-effect model.

4. Study Design

4.1 Overall Design

This is a parallel-group intervention study in non-diabetic chronic kidney disease patients with 2 treatment groups, including a double-blinded placebo treatment group.

Pre-screening (Optional)

An assessment of UACR is permitted under the Pre-screening visit and may be measured locally. This is optional and may be performed if the investigator wishes to obtain a recent UACR value to help assess if a participant is likely to be eligible. Pre-screening UACR values cannot be used in lieu of the centrally assessed urinalysis at Screening. In the exceptional case where no previous evidence of albuminuria or proteinuria is documented in the participant's medical records, the pre-screening value can be used instead. In such situations, the screening visit can only take place 3 months after pre-screening.

Screening

Approximately 3160 participants will be screened to achieve approximately 1580 participants randomly assigned to study intervention.

Randomization

This study is a randomized, double-blind, placebo-controlled, parallel-group, multicenter study with a variable treatment duration of a minimum of 32 months, up to a maximum of approximately 49 months (final treatment duration will depend on recruitment time). Participants will be randomized 1:1 to either finerenone or placebo OD in a blinded fashion on top of their SoC therapy including an ACEI or an ARB.

Randomization will be stratified by screening UACR (≤ 1000 mg/g vs. > 1000 mg/g) and baseline SGLT2 inhibitor use (yes/no) to allow equal distribution of those characteristics between the treatment arms.

Treatment

All participants will receive blinded study intervention (i.e. finerenone 10mg/20mg or placebo) OD on top of an individually tolerated maximum labeled dose of either an ACEI or an ARB. The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/1.73m² (CKD-EPI), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/1.73m² (CKD-EPI). Participants who start on the lower dose of 10 mg will be up-titrated to the target dose of 20 mg from Month 1 on treatment onwards provided potassium is ≤ 4.8 mmol/L, and eGFR decrease is less than 30% of the last measured value. Down-titration or interruption of study intervention is permitted at any time during the study for safety reasons and study intervention has to be interrupted if potassium is > 5.5 mmol/L. Visits will take place at screening, baseline, Month 1, Month 3 and from then every 3 months up to Month 12. Thereafter, if End of Study has not been declared, participants will continue on 4-monthly visits until the End of Study.

Additional safety visits will take place 4 weeks after each up-titration or after restart of study intervention following treatment interruption for > 7 days. eGFR as the primary endpoint measure and potassium as the key safety parameter will be assessed at each visit, UACR and UPCR will be collected at timepoints shown in SoA. Up- and down-titration of study intervention will be based on local potassium and eGFR values.

End of treatment

The minimum planned treatment duration will be 32 months. All participants will stay in the study until the last randomized participant has reached 32 months of treatment. At this point, all participants remaining in the study and still on treatment will end treatment phase, and move to the follow-up phase. Thus, treatment duration will be from 32 to approximately 49 months depending on the recruitment time. Participants who have already discontinued treatment at that time, will be contacted via phone to collect outcome events and adverse events.

Follow-up

The follow-up visit will take place approximately 1 month after last administration of study intervention.

4.1.1 Decentralized Clinical Trial Model

In order to offer participants and sites increased flexibility in trial conduct over the duration of this multi-year study, a “hybrid” DCT model may be implemented where feasible and local laws/regulations allow.

4.1.1.1 Remote Visits

For selected visits as outlined in Section 1.3, remote visit activities are available for use in place of on-site visits. Where utilized, remote visits will be divided into two parts. For the first part, home nurses will visit the participant at home to complete the designated study procedures, such as the collection of lab samples and measurement of vital signs (see Section 6.4), or the option to visit a Patient Service Centre may be possible for collection of lab samples and measurement of vital signs in the US. For the second part of the visit (once the lab results are available from the samples collected by the home nurse), the Investigator, or designee, will conduct the specified study procedures with the participant via telehealth video call. The investigator will also review the investigational product dose. The relevant IxRS transaction will be made and resupply of study intervention requested. Study intervention will be shipped DtP in appropriate shipping containers (see Section 6 for more information). Due to the timelines associated with shipping new study intervention to the participants' home for remote visits, it is possible that study intervention resupply doesn't happen within the scheduled visit window. This is acceptable and not considered a protocol deviation.

4.1.1.2 Technology

At sites where local regulations permit remote visit activities, participants may have the option to participate in the remote visits using their own mobile device ('Bring Your Own Device'; BYOD) *or* be provided with a study-specific provisioned mobile device. If the participant wishes to use their own device, but their own smartphone model and/or operating system does not meet the minimum technology requirements, they will be provisioned with a mobile device.

- If the participant decides to receive or requires a provisioned mobile device, the device will be distributed to him/her by the Investigator and/or authorized Site representative at Visit 2 to facilitate their participation in the study.

- Participants who prematurely discontinue from treatment prior to this timepoint will instead have the provisioned device, if needed, provided to them at the Premature Discontinuation visit to facilitate telehealth video calls during the follow-up period.

If the participant decides to use his/her own mobile device, he/she will be provided with a study-specific mobile app to support the DCT elements of the study. This mobile app will be downloaded by the participant on his/her own smartphone. The study-specific mobile app will support the participant to complete study-specific activities such as:

- Receive reminders/notifications for timely activity completion
- Surveys (i.e. participant feedback)
- Scheduling and conducting Telehealth Video Calls with the Investigator and/or delegate

The investigator and authorized site representatives will have access to a website portal via unique access credentials to access the study data from the completed activities noted above.

4.2 Scientific Rationale for Study Design

Based on the effect seen in the recently completed phase 3 FIDELIO-DKD study, eGFR slope has been chosen to be the primary outcome measure in this label-expanding study in non-diabetic CKD. In the FIDELIO-DKD study, mean eGFR values at baseline were similar for finerenone and placebo. Similar to other compounds affecting the RAAS system, an initial acute eGFR decline of 2-3 mL/min/1.73m² was observed for patients on finerenone. From Month 4 onwards until the end of the study, the decline in eGFR was slower in the finerenone group than the placebo group indicating long-term protection of treatment with finerenone. The acute decline was most pronounced at month 1 and may even ameliorate earlier than month 4 which was the visit time point in FIDELIO-DKD. Thus, to describe the eGFR slope behavior over the first months precisely and adequately reflect the point of change (from acute to chronic slope), a 3 monthly visit schedule is planned during the first year. Thereafter, a 4-monthly visit schedule will continue until the planned end of treatment to enable continuous data collection and limit variability. After the 32 months fixed treatment period, treatment continuation for those randomized early will allow the collection of additional data on the chronic eGFR slope. A post-treatment visit will enable the assessment of a persistent effect on eGFR and UACR after discontinuation.

The approximate power (> 90%) calculation was based on a total slope difference of at least 0.7 mL/min/1.73m² per year at month 32 of treatment. The assumption and initial parameter estimate of total eGFR slope are based on the effect seen in FIDELIO-DKD subpopulations at the corresponding time point where eGFR values were collected following a 4 monthly visit schedule.

Upper thresholds of potassium (4.8 mmol/L) and lower thresholds for eGFR (25 mL/min/1.73m²) for study intervention initiation and dosing have been chosen as confirmed effective and safe in the recently completed FIDELIO-DKD 16244 study.

4.3 Justification for Dose

Doses of 10mg and 20mg have been confirmed to be effective and safe in the recently completed FIDELIO-DKD study in participants with CKD and T2D. 10mg will be the starting dose for participants with a lower eGFR of 25 to < 60mL/min/1.73m², and participants will be

up-titrated to the target dose of 20mg after one month on treatment. Throughout the study, up- and down-titration will be based on potassium and eGFR levels. The analysis of data collected in the context of dose-titration and interruption from the FIDELIO-DKD study indicates that hyperkalemia with finerenone treatment is manageable using a serum potassium-guided dose titration regimen.

4.4 End of Study Definition

The end of the study is defined as the date of the last visit of the last participant in the study globally.

The primary completion is defined as the date of the last on-treatment visit of the last participant in the study globally.

A participant is considered to have completed the study if he/she has completed all phases of the study, including the last visit.

5. Study Population

This study will investigate the effect of finerenone in patients with non-diabetic CKD including different disease etiologies like hypertensive nephropathy and chronic glomerulonephritis such as IgA nephropathy (IgAN) and focal segmental glomerulosclerosis (FSGS). Patients with autosomal dominant or autosomal recessive polycystic kidney disease requiring specific treatment will be excluded from the study as well as patients with lupus nephritis or ANCA vasculitis and patients receiving immunotherapy for other primary or secondary renal diseases within 6 months prior to enrollment. This will ensure the exclusion of patients at high risk of flares and likely to require immunosuppressive therapy. To ensure a pre-specified ratio for a population at risk of progressive renal function decline, the number of participants with eGFR ≥ 25 but < 60 mL/min/1.73 m² (CKD-EPI) and UACR ≥ 200 but < 500 mg/g is planned to be capped at approximately 10% of the total population. The upper limit of UACR will be 3500 mg/g excluding those participants with proteinuria in the nephrotic range.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1 Inclusion Criteria

Participants are eligible to be included in the study only if all of the following criteria apply:

Age

1. Participant must be 18 years of age inclusive at the time of signing the informed consent.

Type of Participant and Disease Characteristics

2. A clinical diagnosis of chronic kidney disease and:
 - eGFR ≥ 25 but < 60 mL/min/1.73m² and UACR of ≥ 200 but < 500 mg/g at screening¹ or eGFR ≥ 25 but < 90 mL/min/1.73m² and UACR of ≥ 500 but ≤ 3500 mg/g at screening, and
 - Documentation of albuminuria/proteinuria² in the participant's medical records at least 3 months prior to screening. If no value is available prior to pre-screening, the pre-screening could be used to identify participants.
3. Stable and max. tolerated labeled dose of an ACEI or ARB for at least 4 weeks prior to screening as documented in the participant's medical records
4. K⁺ ≤ 4.8 mmol/L at screening

Re-assessment is allowed once for individual lab values (UACR; eGFR, potassium). If one of the 3 UACR measurements is missing but the other 2 are valid, these values can be used to assess participant's eligibility for this study. Local laboratory value for potassium should be used for screening reference.

Sex and Contraceptive/Barrier Requirements

5. Male and/or female

Women of childbearing potential can only be included in the study if a pregnancy test is negative at the Screening Visit and if they agree to use adequate contraception as listed in Section 10.4. Contraceptive use by women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies. Women are considered not of childbearing potential if they fulfill the criteria displayed in Section 10.4.

Informed Consent

6. Capable of giving signed informed consent as described in Section 10.1.3 which includes compliance with the requirements and restrictions listed in the ICF and in this protocol

5.2 Exclusion Criteria

Participants are excluded from the study if any of the following criteria apply:

Medical Conditions

1. Established diagnosis of Type 1 or 2 Diabetes mellitus, or HbA1c $\geq 6.5\%$ (48 mmol/mol)
2. Autosomal dominant or autosomal recessive polycystic kidney disease
3. Lupus nephritis or ANCA-associated vasculitis or any other primary or secondary kidney disease requiring immunosuppressive therapy within 6 months prior to screening

¹ capped at approximately 10% of the total population

² quantitative or semiquantitative measurement documented in the medical records

4. History of organ transplantation
5. Acute kidney injury requiring dialysis within 6 months prior to screening
6. Uncontrolled arterial hypertension with mean sitting systolic blood pressure ≥ 160 mmHg, mean sitting diastolic blood pressure ≥ 100 mmHg at screening
7. Symptomatic heart failure with reduced ejection fraction with class 1A indication for MRAs
8. UACR > 3500 mg/g at the screening visit
Note: One re-assessment is allowed in case UACR is > 3500 mg/g in one of the three urine samples collected at the Screening Visit.
9. CV event within 3 months prior to screening (heart failure decompensation, myocardial infarction, stroke, transient ischemic attack, pulmonary embolism, elective coronary artery bypass grafting) or elective percutaneous coronary intervention within 1 months prior to screening
10. Known hypersensitivity to the study treatment (active substance or excipients)
11. Addison's disease
12. Hepatic insufficiency classified as Child-Pugh C

Prior/Concomitant Therapy

13. Concomitant therapy with eplerenone, spironolactone, esaxerenone, any renin inhibitor, sacubitril/valsartan combination, or potassium-sparing diuretic which cannot be discontinued at least 4 weeks prior to the Screening Visit
14. Concomitant therapy with both ACEI and ARBs in case one of those cannot be discontinued at least 4 weeks prior to the Screening Visit.
15. Concomitant therapy with potent CYP3A4 inhibitors or inducers and/or the moderate CYP3A4 inhibitor erythromycin (to be stopped at least 7 days before randomization) (see Section 10.6)

Prior/Concurrent Clinical Study Experience

16. Previous assignment to treatment during this study
17. Simultaneous participation in another interventional clinical study (e.g. Phase 1 to 3 clinical studies) or treatment with any investigational medicinal product within 30 days prior to randomization

Other Exclusions

18. Any other condition or therapy, which would make the participant unsuitable for this study and will not allow participation for the full planned study period (e.g. active malignancy or other condition limiting life expectancy to less than 12 months)
19. Any other history, condition, therapy, or uncontrolled intercurrent illness which could in the opinion of the investigator affect compliance with study requirements.
20. Pregnant or breast-feeding or intention to become pregnant during the study

21. Close affiliation with the investigational site; e.g. a close relative of the investigator, dependent person (e.g. employee or student of the investigational site).

5.3 Lifestyle Considerations

No restrictions to the participant's customary lifestyle are required.

5.4 Screen Failures

Screen failures are defined as participants who consent to participate in the clinical study but are not subsequently randomly assigned to study intervention. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the Consolidated Standards of Reporting Trials (CONSORT) publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, and eligibility criteria.

Sites will also have the option of a pre-screening visit to examine UACR. This UACR value is not valid to determine eligibility, and is only to be used to assess whether the participant may progress to a formal screening visit. Participants will be required to provide informed consent for this UACR test, and very limited data will be reported. Such pre-screening failures will not be considered as part of the study population. See Section 1.3 for assessments at the pre-screening visit.

Individuals who do not meet the criteria for participation in this study (screen failure) may be re-screened once. The following conditions are pre-requisites of re-screening:

- A minimum of 1 month between the initial Screening Visit and re-screening.
- Before re-screening, new documented informed consent must be obtained
- Allocation of a new participant identification number
- All assessments for the study must be repeated

5.5 Criteria for Temporarily Delaying Randomization/ Study Intervention Administration

The following conditions may allow a participant to be randomized or started on study intervention once the conditions have resolved and the participant is otherwise eligible:

- Current febrile illness (temperature 38.0°C [100.4° F]) or other acute illness within 48 hours before study intervention administration.
- Pre-screening showing albuminuria, if not documented previously in the participant's health records. In this case, the screening visit may be delayed by at least 3 months for the inclusion criterion 2 to be fulfilled.

6. Study Intervention(s) and Concomitant Therapy

Study intervention is defined as any investigational intervention(s), marketed product(s), placebo, or medical device(s) intended to be administered to a study participant according to the study protocol.

6.1 Study Intervention(s) Administered

Table 6–1: Study intervention

Arm Name	Finerenone	Placebo
Intervention Name	Finerenone/ BAY 94-8862	Placebo for Finerenone
Type	drug	drug
Dose	tablet	tablet
Formulation		
Unit Dose	10mg and 20mg	N/A
Strength(s)		
Dosage Level(s)	10mg once daily or 20mg once daily	once daily
Route of Administration	oral	oral
Use	experimental	placebo
Packaging and Labeling	Study intervention will be provided in a plastic bottle. Each bottle will be labeled as required per country requirement	Study intervention will be provided in a plastic bottle. Each bottle will be labeled as required per country requirement

Study intervention will be dispensed at the study visits summarized in SoA. Returned study intervention should not be re-dispensed to the participants.

As shown in [Table 6–2](#), the starting dose of finerenone will depend on the eGFR value at the Screening Visit.

Dosage of study intervention

At randomization, participants with an eGFR between 25 to < 60 mL/min/1.73m² at the Screening Visit (central laboratory results) will be assigned to the lower dose of finerenone (10 mg) or placebo in addition to SoC, whilst participants with an eGFR ≥ 60 mL/min/1.73m² will be assigned to the higher dose of finerenone (20 mg) or placebo in addition to SoC.

Table 6–2 Dosage of study intervention for administration

eGFR value at the Screening Visit, based on central laboratory results:	25 to < 60 mL/min/1.73m ²		≥ 60 mL/min/1.73m ²	
Participant randomized to group:	<u>Finerenone</u>	<u>Placebo</u>	<u>Finerenone</u>	<u>Placebo</u>
Receives	10 mg finerenone OD + SoC	Placebo OD + SoC	20 mg finerenone OD + SoC	Placebo OD + SoC

Table 6–2 Dosage of study intervention for administration

eGFR value at the Screening Visit, based on central laboratory results:	25 to < 60 mL/min/1.73m ²		≥ 60 mL/min/1.73m ²	
Participant randomized to group:	<u>Finerenone</u>	<u>Placebo</u>	<u>Finerenone</u>	<u>Placebo</u>
Study intervention intake	One tablet of study intervention once daily, preferably in the morning at approximately the same time each day. The study intervention can be taken with a glass of water, with or without food. Note: Study intervention will be administered at home, except on the days of Visit 2 and Visit 4 when the tablet will be taken at the study site.			
Missed intake	• If >8 hours before the next scheduled dose, the participant should take one tablet of study intervention as soon as possible. • If ≤8 hours of the next scheduled dose, the participant should wait and take the next tablet of study intervention at the usual time.			
Up-titration of dose Allowed from Visit 2 (Month 1) onwards provided that: • Potassium is ≤4.8 mmol/L ^a • eGFR decrease is less than 30% below the value measured at the last scheduled visit ^a • Must be documented in eCRF	20 mg finerenone OD, maintain SoC	Sham-titrate, maintain SoC	Not applicable	Not applicable
Down-titration of dose • Only for safety reasons • Allowed any time during the study (e.g. between scheduled visits) • Must be documented in eCRF • An unscheduled safety visit is performed within an adequate timeframe proposed by the investigator	• If at higher dose of study intervention, down-titrate to lower dose of study intervention, maintain SoC • If at lower dose of study intervention, interrupt study intervention, maintain SoC			

Abbreviations: eCRF = electronic case report form; eGFR = estimated glomerular filtration rate; OD = once daily; SoC = standard of care

^a Potassium and eGFR according to local laboratory values

Up- and down-titration of study intervention

The investigator is encouraged to up-titrate the dose of study intervention at any time once the participant has been on a stable dose for 4 weeks (± 7 days) (e.g. from Month 1 onwards for participant starting study intervention on the lower dose) either at a regular visit or an 4-week safety Visit. Up-titration can be performed provided:

- the potassium concentration (local laboratory value) is ≤ 4.8 mmol/L,

AND

- eGFR decrease (local laboratory value) is less than 30% of the value measured at the last regular visit.

Participants who started with or were up-titrated to the target dose (20 mg OD) of the study intervention but who do not tolerate this dose may be down-titrated at any point during the study, including between-scheduled visits if required for safety reasons. These participants may be up-titrated again based on the rules provided above. If the participant is already at the lower dose, the study intervention can be interrupted at the investigator's discretion (refer to Section 6.5 for guidance). If the study intervention is interrupted for more than 7 days, the study intervention should be re-started at the lower dose (10 mg OD).

Subsequent to an up-titration or re-start of study intervention after interruption of study intervention intake for more than 7 days, the investigator should perform a 4-week safety Visit (± 7 days) after titration or restart, in order to monitor potassium levels and renal function (see Section 1.3). If a regular study visit will be scheduled to take place 4 weeks (± 7 days) after up-titration, the monitoring of potassium and renal function is assured and no 4-week safety Visit has to be performed in addition. With down-titrations, the unscheduled safety visit may need a smaller time window and should be performed at the investigator's discretion.

At any point during study intervention treatment, if blood potassium is found to be elevated, study intervention treatment should be adjusted according to guidance provided in Section 6.5.

Both up- and down-titrations will be at the investigator's discretion and will depend on the general clinical condition of the individual participant. The investigator should attempt to reach the maximum (target) dose of study intervention as long as safety is not compromised and the guidance for the up- and down titration is followed (See Section 6.5).

If up-titration will take place subsequent to a remote (+ Telehealth Video Call) visit, the participant must be instructed to continue taking the current dose until the new study intervention supply is received via DtP shipment. Therefore, actual up-titration of dose will take place several days after the completion of the Telehealth Video Call part of the visit. The start date of the new dose must be documented in the eCRF.

If laboratory results subsequent to a remote (+Telehealth Video Call) indicate the need for a down-titration, the participant must visit the site to return their current study intervention and be supplied with new intervention. Down-titration must always be performed at an on-site visit to allow participants to be supplied with the new lower dose of study intervention treatment as soon as possible.

All titrations, including the reasons for down-titration or for not dispensing the 20 mg dose, must be documented in the eCRF.

Medical Devices

No medical devices will be provided by the Sponsor in this study.

6.2 Preparation/Handling/Storage/Accountability

1. The investigator or designee must confirm study interventions have been received and, if a temperature data logger has been included, appropriate temperature conditions have been maintained during transit. Any discrepancies are to be reported and resolved before use of the study interventions
2. Only participants randomized in the study may receive study intervention and only authorized site staff (or remote pharmacy designee if applicable, where local regulations permit, for any sites utilizing remote visits) may supply or administer study intervention. All study intervention must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions with access limited to the investigator and authorized site staff.
3. The investigator or the head of the institution (where applicable) is responsible for study intervention accountability, reconciliation, and record maintenance (i.e., receipt, reconciliation, and final disposition records). Drug returns, reconciliation and destruction return information will be captured in IxRS. For sites where DtP is completed by site staff (including site pharmacy) shipping study intervention directly to individual study participants, the investigator retains responsibility for these activities. The site must receive written approval from the sponsor before engaging in direct to participant shipments that originate from the site.

For sites utilizing remote visits, where DtP distribution of study intervention occurs via a remote pharmacy designee, the investigator delegates distribution responsibility to the sponsor. In such circumstances, the investigator maintains responsibility and control for dispensing via IxRS by confirming whether or not DtP distribution of study intervention can be triggered. Where a remote pharmacy designee is used, the sponsor contracts a distribution depot with pharmacy licensing and delegates responsibility for storage and handling via DtP distribution of study intervention until handover to the participant. Participants who have a remote visit and receive study intervention via DtP will return all unused investigational product and empty bottles to the study site at their next scheduled on-site visit.

4. Further guidance and information for the final disposition of unused study interventions are provided in investigator site file.

6.3 Measures to Minimize Bias: Randomization and Blinding

All participants will be centrally assigned to randomized study intervention using IxRS. Before the study is initiated, the telephone number and call-in directions for the IxRS and/or the log in information and directions for the IxRS will be provided to each site.

Treatment allocation will be done according to a computer generated randomization list specified by the sponsor or delegate.

IxRS will be programmed with blind-breaking instructions. In case of an emergency, the investigator has the responsibility for determining if unblinding of a participant's intervention assignment is warranted. If the investigator is unavailable, and a treating physician not

associated with the study requests emergency unblinding, the emergency unblinding requests are forwarded to the emergency medical advice 24 hours/7 day service. Participant safety must always be the first consideration in making such a determination. If the investigator decides that unblinding is warranted, the investigator should make every effort to contact the sponsor prior to unblinding a participant's intervention assignment unless this could delay emergency treatment of the participant. If a participant's intervention assignment is unblinded, the sponsor must be notified within 24 hours after breaking the blind. The date and reason that the blind was broken must be recorded in the source documentation and case report form, as applicable.

Sponsor pharmacovigilance staff may unblind the intervention assignment for any participant with an SAE. If the SAE requires that an expedited regulatory report be sent to one or more regulatory agencies, a copy of the report, identifying the participant's intervention assignment, may be sent to investigators in accordance with local regulations and/or sponsor policy.

6.4 Study Intervention Compliance

When participants self-administer study intervention(s) at home, compliance with study intervention will be assessed at each visit. Compliance will be assessed by direct questioning and counting of returned tablets during the site visits.

An adequate record of receipt, distribution, and return/destruction of all study treatment must be captured on the dispensing log and in the IxRS. Intervention start and stop dates, including dates for intervention delays and/or dose reductions will also be recorded.

Overall compliance with study intervention intake should be between 80% and 120% of the scheduled dose at the end of study intervention treatment.

Deviation(s) from the prescribed dosage regimen should be recorded. Study intervention tablets not returned will be considered to have been taken unless otherwise specified. At the end of the study, any remaining drugs will be collected and returned to the sponsor, destruction depot or destroyed at the site. Any discrepancies between the returned and expected returned study interventions should be explained.

6.5 Dose Modification

The dose of study intervention should be adjusted (up- or down-titration) on the basis of potassium and eGFR levels obtained from the local laboratory, except for remote visits where central lab values are used.

Monitoring of Blood Potassium and Dose Adjustment

Hyperkalemia is a frequent event in participants with CKD with decreasing eGFR, and one of the main risks identified from previous studies with MRAs including finerenone. In FIDELIO-DKD a 2-fold increase in treatment-emergent hyperkalemia was observed for participants treated with finerenone compared to those treated with placebo which was consistent throughout subgroups. However, numbers of events leading to discontinuation and hospitalization were low and no fatal events were reported. Potassium will be monitored

closely during this study, with assessments of potassium concentration scheduled at visits specified in Section 1.3 SoA.

During treatment with study intervention, at on-site visits all assessments of potassium, including re-tests, will be performed at both local and/or central laboratories. Due to sample handling and extended transportation time to the central laboratory, falsely elevated serum potassium values in centrally (but not locally) analyzed samples may be observed. Since the value of potassium in the local sample is usually not affected by this and results from the local laboratory are available earlier, local samples will be taken not only for safety purposes but also for dose adjustments.

Re-checking of potassium values must be done at the latest within 72 hours of the investigator being aware of results that need confirmation (> 5.5 mmol/L). The investigator can perform a re-test at any time if they consider it necessary to confirm the potassium concentration of the local or the central sample. For remote visits if the central laboratory serum potassium result is > 5.5 mmol/L following blood sampling during a remote visit, as soon as the investigator is aware of the result, the participant should be contacted promptly and instructed to withhold further study intervention until the second sample can be collected locally. Potassium should be rechecked within 72 hours of initial potassium result awareness.

Adjustment (lowering) of participant's maximum tolerated dose of baseline ACEI/ARB medication should not be made solely for the purpose of continuing with the same study intervention dosing.

Adjustment of dose after start of study intervention intake based on blood sample potassium levels is provided in Table 6–3.

Table 6–3 Potassium from blood sample: guidance for dose adjustment at Visit 2 and subsequent visits

Blood potassium (mmol/L)	Action
First sample:	
≤ 4.8	If on lower dose of study intervention, up-titrate to higher dose. If on higher dose of study intervention, continue on the same dose.
4.9 to 5.5	Continue on the same dose.
> 5.5	Withhold study intervention and re-check potassium within 72 hours.
Second and subsequent samples:	
≤ 5.0	Re-start study intervention at lower dose.
> 5.0	Continue to withhold study intervention; continue to monitor potassium and restart study intervention at the lower dose only if potassium is ≤ 5.0

NOTE: lower dose = 10 mg once daily; higher dose = 20 mg once daily.

The following aspects have also to be taken into consideration:

- Potassium should be measured 4 weeks (± 7 days) after re-starting treatment or dose adjustment, especially after up-titration.
- If central laboratory results for potassium differ from local laboratory results to such an extent that the investigator is uncomfortable with the dose adjustment guidance shown in Table 6–3, blood sampling may be repeated at the investigator's discretion.

- If the participant is already on the lower dose of study intervention but hyperkalemia recurs soon after a previous event of hyperkalemia with interruption of study intervention, and there is no explanation for the recurring hyperkalemia event other than intake of study intervention, permanent discontinuation of study intervention is recommended.
- If blood potassium is > 6.5 mmol/L, an ECG should be obtained.

6.6 Continued Access to Study Intervention after the End of the Study

No intervention is planned following the end of the study.

6.7 Treatment of Overdose

In this trial, an overdose is defined as any occasion when the participant has taken (accidentally or intentionally) any dose higher than the maximal target dose prescribed in the protocol. Overdose of study interventions should be treated as clinically indicated based on symptoms and signs. There is no specific reversal agent for finerenone and the Sponsor does not recommend specific treatment for an overdose.

In the event of an overdose, the investigator should:

- Contact the Medical Monitor.
- Evaluate the participant to determine, whether study intervention should be interrupted or whether the dose should be reduced. This may happen in consultation with the Medical Monitor.
- Closely monitor the participant for any AE/SAE and laboratory abnormalities as clinically indicated (at least 1 day).
- Document the quantity of the excess dose as well as the duration of the overdose in the eCRF.

Decisions regarding dose interruptions or modifications will be made by the investigator in consultation with the Medical Monitor based on the clinical evaluation of the participant.

Overdose per se will not be reported as an AE/SAE unless it is an intentional overdose taken with possible suicidal/self-harming intent (see Section [10.3.1](#)).

6.8 Concomitant Therapy

Any medication or vaccine (including over-the-counter or prescription medicines, recreational drugs, vitamins, and/or herbal supplements) or other specific categories of interest that the participant is receiving at the time of enrollment or receives during the study must be recorded along with:

- Reason for use
- Dates of administration including start and end dates
- Dosage information including dose and frequency
- Route of administration

The Medical Monitor should be contacted if there are any questions regarding concomitant or prior therapy.

All participants should be treated according to regional standards of care for CV risk factors (e.g., blood pressure, lipids, and antithrombotic treatment), and CKD complications (hyperphosphatemia, hyperparathyroidism, hyperkalaemia, acidosis and renal anaemia).

6.8.1 Standard of Care (SoC) medications

To be eligible, the participant needs to be on stable and for the participant maximum tolerated labelled daily dose of ACEI or ARB for at least 4 weeks before the screening visit

The maximum tolerated labeled dose for ACEIs or ARBs should be the highest dose within the ranges shown in [Table 6–4](#) or according to local labels, as applicable, which the participant can safely tolerate. This dose should not be below the minimum labeled dose in order to maximize therapeutic benefit of background SoC.

Table 6–4 Dose ranges of ACEIs and ARBs for adults

Angiotensin-converting enzyme inhibitors		Angiotensin receptor blockers	
Generic name	Dose Range (mg/day)*	Generic name	Dose Range (mg/day)*
Benazepril	20 – 40	Candesartan	16 – 32
Captopril	50 – 450	Eprosartan	400 – 800
Cilazapril	1 – 5	Irbesartan	150 – 300
Enalapril	10 – 40	Losartan	50 – 100
Fosinopril	20 – 80	Olmesartan	20 – 40
Lisinopril	20 – 40	Telmisartan	40 – 80
Moexipril	7.5 – 30	Valsartan	80 – 320
Perindopril	4 – 16		
Quinapril	20 – 80		
Ramipril	2.5 – 20		
Trandolapril	2 – 4		

Source: Adapted from Table 28 of reference ([Diabetes_&_CKD_Work_Group 2007](#))

If the maximum labeled dose cannot be reached, the reason(s) should be documented in the eCRF.

The dosage of SoC therapies should not be reduced in order to solely facilitate maintenance of study intervention.

6.8.2 SGLT2 inhibitors

If the participant is assessed by the investigator as having an indication for an SGLT2 inhibitor, it should be started preferably prior to inclusion into the study, and the participant should be on a stable dose for at least 1 month prior to screening.

In the exceptional case where the participant is started on an SGLT2 inhibitor after randomization, then participant must be on a stable dose of study intervention for at least 1 month before SGLT2 inhibitor initiation.

6.8.3 Potassium supplements and lowering agents

Potassium supplementation can be prescribed during the course of the study. To assure the potassium value is within the normal range, investigators are advised to monitor potassium carefully at each visit and to adapt the dose of potassium supplementation accordingly.

Potassium-lowering agents (e.g. sodium polystyrene sulfonate, calcium polystyrene sulfonate) are allowed to be started during treatment with study intervention.

Any use of potassium supplementation and potassium-lowering agents must be documented in the eCRF.

6.8.4 Low dose steroids

Immunosuppressive therapy, other than topical, inhaled or low dose steroids, is not allowed during the study. Oral low dose steroids (dose equivalent of daily 5-10 mg of prednisolone) is allowed if used for another non-renal indication.

The following therapies are NOT permitted:

- Concomitant therapy with eplerenone, spironolactone, esaxerenone, any renin inhibitor, or potassium-sparing diuretic which cannot be discontinued at least 4 weeks prior to the Screening Visit and during study intervention treatment
- Concomitant therapy with both an ACEI **and** an ARB which cannot be discontinued at least 4 weeks prior to the Screening Visit and during study intervention treatment
- Concomitant therapy with an ARNI (valsartan/sacubitril) which cannot be discontinued at least 4 weeks prior to the screening visit and during study intervention treatment
- Concomitant therapy with potent cytochrome P450 isoenzyme 3A4 (CYP3A4) inhibitors or inducers and/or the moderate CYP3A4 inhibitor erythromycin must be stopped at least 7 days before randomization and is not allowed to be started during study intervention treatment.

A list of the most common medication regarded as potent CYP3A4 inhibitors or inducers is provided in Section 10.6.

Prior medication that was discontinued to comply with the eligibility criteria of the study should also be recorded in the eCRF.

Caution: Increases in finerenone exposure in combination with the following moderate CYP3A4 inhibitors cannot be excluded: amiodarone, aprepitant, bicalutamide, chloramphenicol, imatinib, mifepristone, norfloxacin, tacrolimus, verapamil, lapatinib, dasatinib, and nilotinib.

Caution should also be exercised with non-steroidal anti-inflammatory agents or trimethoprim or trimethoprim / sulfamethoxazole. Finerenone treatment may be interrupted before trimethoprim or trimethoprim / sulfamethoxazole is started.

6.8.5 Rescue Medicine

Not applicable

7. Discontinuation of Study Intervention and Participant Discontinuation/Withdrawal**Table 7–1: Discontinuation of study intervention and participant discontinuation**

Study intervention discontinuation	Participant discontinuation
Must be permanently discontinued	
Pregnancy of the study participant	At their own request or at the request of their legally acceptable representative. At any time during the study and without giving reasons, a participant may decline to participate further. The participant will not suffer any disadvantage as a result
The investigator is of the opinion that continuation of treatment with study intervention is harmful to the participant's well-being	
The randomization code is broken by the investigator, or other responsible person, when knowledge of the participant's treatment is required	
Any investigational drug other than the study intervention is used.	
May be permanently discontinued	
Any suspected drug-related AE or SAE	If, in the investigator's opinion, continuation of the study would be harmful to the participant's well-being
If any exclusion criterion applies during treatment	At the specific request of the sponsor and in liaison with the investigator.
If a significant violation of the protocol occurs, as defined by the sponsor and the coordinating investigator.	

7.1 Discontinuation of Study Intervention

In rare instances, it may be necessary for a participant to permanently discontinue study intervention. If study intervention is permanently discontinued, the participant will remain in the study and continue with the regular study visit schedule until the end of the study. Additionally, post treatment eGFR, UACR and UPCR assessments will be performed after treatment discontinuation. See the SoA for data to be collected at the time of discontinuation of study intervention and follow-up and for any further evaluations that need to be completed.

7.1.1 Temporary Discontinuation of Study Intervention

Interruption of study intervention is permitted at any time during the study for safety reasons. See Section 6.5 for guidance.

Upon temporary interruption of the study intervention due to hyperkalemia, eGFR decrease, (S)AE, OE, intolerability or any other reason, intake should be resumed as soon as medically acceptable at the discretion of the investigator. There is no defined maximum time limit for temporary interruption. In all cases, the reason for study intervention interruption must be recorded in the eCRF and the participant's medical records.

If the study intervention is interrupted for more than 7 days, the re-start should be performed at the 10mg dose and the investigator should schedule a 4-week safety visit (± 7 days) in order to monitor potassium levels and renal function (see Table 6–2). If a regular visit will be scheduled to take place 4 weeks ± 7 days after up-titration or re-start, the monitoring of potassium and renal function is assured and no 4-week safety visit has to be performed in addition.

7.1.2 Permanent Discontinuation of Study Intervention

- In general, participants who permanently discontinue study intervention are expected to continue in the study and to attend all protocol specified study visits, and should be encouraged to perform the required assessments described in the visit schedule. PK sampling and pregnancy testings as well as local laboratory sampling will not be continued.
- If a participant no longer on study intervention is unable to attend the clinic for a study visit, and a remote visit is not possible, a telephone consultation may be performed to determine if relevant health events / endpoints (e.g. development of CV or renal complications) have occurred.
- Ideally, a face to face visit should be performed at least once a year. Expected frequency of telephone contacts should be in line with the standard visit schedule.
 - Ad hoc additional telephone contacts may also be requested (e.g. close to the end of the study). These may be made to the participant themselves or to other contact as provided by the participant, e.g. a next of kin, primary physician (or local equivalent).

7.2 Participant Discontinuation/Withdrawal from the Study

Withdrawal from treatment, and withdrawal from the study visits do not constitute withdrawal of consent. The definition of withdrawal of consent is deemed to be when the participant does not agree to any kind of direct/in direct follow up and actively refuses any further collection of data. Therefore this should only occur in rare and exceptional circumstances.

In this study, eGFR values are crucial to the analysis and must be collected until the planned end of the study, as participants will still be part of the study even if participants are no longer taking study intervention. Therefore, all efforts will be taken to motivate participants to comply with all study procedures and to continue to be followed until the end of the study. The same applies to outcome events and vital status data.

- A participant may withdraw from the study at any time at his/her own request, or may be withdrawn at any time at the discretion of the investigator for safety, behavioral, or compliance reasons. This is expected to be uncommon.
- At the time of discontinuing from the study, if possible, a PD visit should be conducted, as shown in the SoA. See SoA for data to be collected at the time of study discontinuation and follow-up and for any further evaluations that need to be completed.
- The participant will be permanently discontinued both from the study intervention and from the study at that time.
- If the participant withdraws consent for disclosure of future information, the sponsor may retain and continue to use any data collected before such a withdrawal of consent.

7.3 Lost to Follow Up

A participant will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and is unable to be contacted by the study site.

The following actions must be taken if a participant fails to return to the clinic for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible and counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether or not the participant wishes to and/or should continue in the study.
- Before a participant is deemed lost to follow up, the investigator or designee must make every effort to regain contact with the participant (where possible, telephone calls and, if necessary, a certified letter to the participant's last known mailing address or local equivalent methods). These contact attempts should be documented in the participant's medical record.
- Should the participant continue to be unreachable, he/she will be considered to have withdrawn from the study.
- Site personnel, or an independent third party, will attempt to collect the vital status of the participant within legal and ethical boundaries for all participants randomized, including those who did not get study intervention. Public sources may be searched for vital status information. If vital status is determined as deceased, this will be documented and the participant will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

Discontinuation of specific sites or of the study as a whole are handled as part of Section [10.1](#).

8. Study Assessments and Procedures

In the event of a significant trial-continuity issue (e.g. caused by a pandemic), alternate strategies for participant visits, assessments, medication distribution and monitoring may be implemented by the sponsor or the investigator, as per local health authority / ethics

requirements. For example, use of remote visit activities may be expanded to include other visits than those outlined in Section 1.3 with prior written agreement of the sponsor.

- Study procedures and their timing are summarized in the SoA. Protocol waivers or exemptions are not allowed.
- Immediate safety concerns should be discussed with the sponsor immediately upon occurrence or awareness to determine if the participant should continue or discontinue study intervention.
- Adherence to the study design requirements, including those specified in the SoA, is essential and required for study conduct.
- All screening evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria. The investigator will maintain a screening log to record details of all participants screened and to confirm eligibility or record reasons for screening failure, as applicable.
- Procedures conducted as part of the participant's routine clinical management (e.g., blood count) and obtained before signing of the ICF may be utilized for screening or baseline purposes provided the procedures met the protocol-specified criteria and were performed within the time frame defined in the SoA.
- Laboratory results that could unblind the study will not be reported to investigative sites or other blinded personnel until the study has been unblinded.

8.1 Efficacy Assessments

Planned time points for all efficacy assessments are provided in the SoA.

8.1.1 eGFR

The primary efficacy variable of interest is eGFR values over time, which will be collected every 3 months for 12 months and afterwards every 4 months until the end-of planned treatment period and an off-treatment measurement after 4-weeks post-treatment.

The laboratory parameter eGFR will be calculated by the central laboratory based on the CKD-EPI formula. The formula used for calculating eGFR (in mL/min/1.73 m²) will be the CKD-EPI equation (Levey et al. 2009).

In addition, for safety reasons and titration purposes, GFR will also be assessed locally (as specified in Section 1.3). In case of unavailability of the central eGFR measurement, computed eGFR (CKD-EPI) from local measurements may be used for the primary and secondary endpoint analyses.

8.1.2 UACR and UPCR

Early morning void urine samples will be collected on 2 consecutive days before each visit, at screening and baseline visits on 3 consecutive days before the visit. All samples can be collected at home.

UACR and UPCR will be assessed by Central Lab.

At screening, UACR levels for inclusion have to be confirmed by 3 samples. If one of the 3 UACR measurements is invalid but the other 2 are valid, these values can be used to assess participant's eligibility for this study.

UPCR is analyzed to support extrapolation of efficacy from the adult to future pediatric development for finerenone.

8.1.3 EuroQoL 5-dimension 5-level (EQ-5D-5L)

The EQ-5D-5L is a self-administered preference-based generic measure of health status which includes five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression ([EuroQol-GROUP 2013](#)). Participants provide a rating for each question on a five-level Likert scale: having no problems, having slight problems, having moderate problems, having severe problems and being unable to do/having extreme problems. In addition, participants are asked to self-rate their own health today on a vertical 0-100 unit visual analogue scale, with 0 corresponding to "the worst health you can imagine", and 100 corresponding to "the best health you can imagine".

EQ-5D-5L will be collected on eDiary at timepoints indicated in the SoA.

The reported data will be analyzed descriptively. The utility analyses based on collected EQ-5D-5L data are planned to be conducted in separate post-hoc analyses.

8.2 Safety Assessments

Planned time points for all safety assessments are provided in the SoA.

8.2.1 Physical Examinations

A complete physical examination will include, at a minimum, assessments of the cardiovascular and respiratory systems. Height and weight will also be measured and recorded. Investigators should pay special attention to clinical signs related to previous serious illnesses.

A complete clinical examination will be conducted at timepoints indicated in the SoA.

8.2.2 Vital Signs

- Vital signs (to be taken before blood collection for laboratory tests) will consist of 1 pulse and 3 blood pressure measurements (3 consecutive blood pressure readings will be recorded at intervals of at least 1 minute). The average of the 3 blood pressure readings will be recorded.
- Blood pressure and pulse measurements will be assessed in sitting position with a completely automated device. Manual techniques will be used only if an automated device is not available.
- Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the participant in a quiet setting without distractions (e.g., television, cell phones).

8.2.3 Electrocardiograms

- 12-lead ECG will be obtained as outlined in the SoA (see Section 1.3), in a supine position, using an ECG machine, with a preferred speed of 50 mm/s, that automatically calculates the heart rate and measures PR, QRS, QT, and QTc intervals. Evaluation for safety will be performed and signed by an investigator. If there is any clinical indication, unscheduled 12-lead ECGs can be performed at any point during the study. An additional 12-lead ECG should be obtained if the potassium > 6.5 mmol/l.
- The interpretation of the tracing must be made locally by a qualified physician.
- Clinically significant abnormalities should be reported as AEs or outcome event as appropriate (e.g. new onset atrial fibrillation) when not already reported in the medical history or in one of the previous recordings and in case of worsening of previous findings.
- The date of the recording must be documented on the ECG section of the eCRF. Each ECG, paper or electronic tracing should be labeled with the study number, participant number, and date and kept in the source documents at the study site.

8.2.4 Clinical Safety Laboratory Assessments

Potassium as the key safety parameter will be collected at each visit in central and/or local laboratory. Local laboratory values will be used to guide dose titration at on site visits and central laboratory values will be used for remote visits, where applicable (See Section 6.5 for further information).

- See Section 10.2 for the list of clinical laboratory tests to be performed and the SoA for the timing and frequency.
- The investigator must review the laboratory report, document this review, and record any clinically significant changes occurring during the study as an AE. The laboratory reports must be filed with the source documents.
- Abnormal laboratory findings associated with the underlying disease are not considered clinically significant unless judged by the investigator to be more severe than expected for the participant's condition.
- All laboratory tests with values considered clinically significantly abnormal during participation in the study or within 3 days after the last dose of study intervention should be repeated until the values return to normal or baseline or are no longer considered clinically significant by the investigator or the medical monitor.
 - If clinically significant values do not return to normal/baseline within a period of time judged reasonable by the investigator, the etiology should be identified, and the sponsor notified.
 - All protocol-required laboratory tests, as defined in Section 10.2, must be conducted in accordance with the laboratory manual and the SoA.

- If laboratory values from non-protocol specified laboratory tests performed at the institution's local laboratory require a change in participant management or are considered clinically significant by the investigator (e.g., SAE or AE or dose modification), then the results must be recorded.

8.2.5 Pregnancy Testing

Urine or serum pregnancy test will be conducted at timepoints indicated in the SoA (Section 1.3).

8.3 Adverse Events (AEs), Serious Adverse Events (SAEs) and Other Safety Reporting

The definitions of AEs and SAEs can be found in Section 10.3.

AEs will be reported by the participant (or, when appropriate, by a caregiver, surrogate, or the participant's legally authorized representative).

The investigator and any qualified designees are responsible for detecting, documenting, and recording events that meet the definition of an AE or SAE either via on-site visit, or telephone or video-conferencing contact.

Sites utilizing remote visit activities: home nurses will not actively enquire about or collect AE/SAE information. However, if an AE/SAE is self-evident or spontaneously reported by the participant, home nurses will capture relevant details which will be made available to the investigator as soon as possible upon conclusion of the home nurse visit. In addition, in the case of suspected SAEs, the home nurse will telephone to immediately inform the investigator. If the nurse is not able to speak to the investigator or qualified designee directly, he/she is responsible for continuing to attempt this until direct contact is made and should ensure this is done on the same day that the home nurse visit took place. In all cases, the investigator is responsible for evaluating potential AEs/SAEs, ensuring their timely entry into the eCRF as well as for SAE reporting to the sponsor within the required timeframe (see Section 8.3.1). The investigator/designee is then responsible for obtaining further information if needed e.g. contacting the participant, obtaining records from other sources or deciding if an on-site follow-up visit is required.

The investigator and qualified designees remain responsible for following up AEs that are serious, considered related to the study intervention or study procedures, or that caused the participant to discontinue the study intervention (see Section 7).

The method of recording, evaluating, and assessing causality of AEs and SAEs and the procedures for completing and transmitting SAE reports are provided in Section 10.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

(S)AEs will be collected from the start of randomization until the last follow-up visit at the time points specified in the SoA (Section 1.3). (S)AEs which are related to protocol-required study procedures (e.g., (S)AE related to invasive study procedures) will be recorded as (S)AEs from the signing of the ICF.

Any medical occurrences/conditions that begin in the period between signing ICF and randomization, and which are not related to a protocol-required study procedure, will be recorded on the Medical History/Current Medical Conditions, not as AEs.

With the exception of hyperkalemia, which is collected as an AESI, non-serious AEs are not mandatory to be collected in the eCRF. Instead, only AEs leading to permanent discontinuation of study intervention and AEs considered related to study intervention by the investigator, or considered related to protocol required procedures will be collected. Any other AEs should only be collected if deemed relevant by the investigator.

All SAEs will be recorded and reported to the sponsor or designee immediately and under no circumstance should this exceed 24 hours of learning of the event, as indicated in Section 10.3. The investigator will submit any updated SAE data to the sponsor within 24 hours of it being available.

Investigators are not obligated to actively seek information on AEs or SAEs after conclusion of the study participation. However, if the investigator learns of any SAE, including a death, at any time after a participant has been discharged from the study, and he/she considers the event to be reasonably related to the study intervention or study participation, the investigator must promptly notify the sponsor.

8.3.2 Method of Detecting AEs and SAEs

Care will be taken not to introduce bias when detecting AEs and/or SAEs. Open-ended and non-leading verbal questioning of the participant is the preferred method to inquire about AE occurrences.

8.3.3 Follow-up of AEs and SAEs

After the initial AE/SAE report, the investigator is required to proactively follow each participant at subsequent visits/contacts. All SAEs will be followed until resolution, stabilization, the event is otherwise explained, or the participant is lost to follow-up (as defined in Section 7.3). Further information on follow-up procedures is provided in Section 10.3.

8.3.4 Regulatory Reporting Requirements for SAEs

Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of participants and the safety of a study intervention under clinical investigation are met.

The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRB/ IEC, and investigators.

An investigator who receives an investigator safety report describing an SAE or other specific safety information (e.g., summary or listing of SAEs) from the sponsor will review and then file it along with the Investigator's Brochure and will notify the IRB/IEC, if appropriate according to local requirements.

Investigator safety reports must be prepared for SUSAR according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.

8.3.5 Pregnancy

- Details of all pregnancies in female participants and, if indicated, female partners of male participants will be collected after the start of study intervention and until Follow-up visit.
- If a pregnancy is reported, the investigator will record pregnancy information on the appropriate form and submit it to the sponsor within 24 hours of learning of the female participant or female partner of male participant (after obtaining the necessary signed informed consent from the female partner) pregnancy.
- For contraceptive guidance and collection of pregnancy information see Section 10.4.

8.3.6 Disease-Related Outcome Events Not Qualifying as AEs or SAEs

The following disease-related events (DREs) are common in participants with CKD and will not be reported as (S)AEs, but entered into the clinical trial database as OEs:

- Kidney failure defined as:
 - ESRD: Initiation of chronic dialysis (haemo- or peritoneal dialysis) for at least 30 days or renal transplantation.
 - Decrease of eGFR to less than 15 mL/min/1.73m², confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 57\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 40\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- CV death
- Non-fatal myocardial infarction
- Non-fatal stroke or transient ischemic attack (TIA)
- Hospitalization for heart failure or urgent heart failure visit
- Other CV hospitalization
- New diagnosis of atrial fibrillation or atrial flutter
- New diagnosis of heart failure

Sustained eGFR decreases of less than 15 mL/min/1.73m², of $\geq 57\%$ from baseline, and of $\geq 40\%$ from baseline will be programmatically derived, and not entered into the eCRF by the investigator.

Because these events are typically associated with the disease under study, they will not be reported according to the standard process for expedited reporting of SAEs even though the

event may meet the definition of an SAE. These events will be recorded in the eCRF. These DREs will be monitored by an unblinded independent Data Monitoring Committee, on a routine basis.

8.3.7 Adverse Events of Special Interest (AESI)

Hyperkalemia will be an AESI in this study and collected in the eCRF. Only hyperkalemia that qualifies as an SAE will be reported to PV within 24 hours of awareness.

8.4 Pharmacokinetics

For the investigation of systemic exposure to finerenone and its relationship with treatment effects, the plasma concentrations of finerenone will be determined at different time points using a sparse sampling approach in all participants. The plasma concentration versus time data collected on the visits as outlined in SoA (see Section 1.3) will be evaluated descriptively, separated by dose and visit. Plots will be prepared of all individual plasma concentrations vs. actual relative study times (time of sample collection after time of study intervention administration).

At Visits 2 and 4, a trough sample for the determination of finerenone plasma concentrations will be drawn before intake of study intervention. At this visit, study intervention will be administered at the study site by study personnel and the exact time of study intervention intake on the day before the visit and on the day of the visit and the exact sampling time will be recorded in the eCRF. The study personnel should contact the participant prior to Visits 2 and 4 to remind them not to take the study intervention as usual in the morning at home and to remind him to properly document time of intake. In case the trough sample cannot be taken at visit 2 or 4 (e.g. in case of a remote visit, or if study intervention was taken at home), this may be postponed to Visit 5.

At Visits 3 and 6 one blood sample for the determination of finerenone plasma concentrations will be drawn during the visit 1.5-10 hours after study intervention intake at home. In case the sample cannot be taken at visit 3, this may be postponed to Visit 5. The participants should be advised to take their drug as usual in the morning at home and recall the time of drug intake or note the time of drug intake on the contact card. The exact time of study intervention intake and the exact sampling times will be recorded in the eCRF.

Pharmacokinetic and exposure-response analysis may be performed using population approaches (popPK and popPK/PD, e.g., by non-linear mixed effect modeling). Such evaluations will be described in a separate analysis plan and will be reported separately. Such evaluations may be started prior to database lock. If this is applicable, appropriate measures will be taken to maintain blinding of the study team, e.g., data will be stored separately, and members of the study team will neither have access to the randomization list nor to individual data.

Details about the collection, processing, storage and shipment of samples will be provided separately (e.g., sample handling sheets or laboratory manual).

Pharmacodynamic parameters are not evaluated in this study.

8.5 Genetics and/or Pharmacogenomics

Genetic as well as non-genetic analyses may be part of the biomarker investigations in this study if approved by local IECs / IRBs and competent authorities. Genetic investigations may be of any kind, including whole genome sequencing.

Participation is optional. Participants who do not wish to participate in the genetic research may still participate in the study. Details on sample handling will be provided separately. See Section 8.6. for details.

8.6 Biomarkers

In this study, genetic as well as non-genetic biomarkers will be investigated.

Timing of collection

The planned time points of sample collection are provided in the SoA in Section 1.3. If deemed necessary, the sampling time points or frequency according to the SoA may be adjusted.

Specimen Types

The following sample types will be collected for biomarker analysis:

- Plasma
- Serum
- Whole blood
- Urine

PK or safety lab samples (leftovers) may be used for biomarker investigations if deemed necessary during the study conduct. The exact specimen type for a particular biomarker (e.g. serum or EDTA plasma) will be provided in separate documents (e.g. sample handling sheets or lab manual).

Sample handling and storage

Details on the collection, processing, shipment and storage of samples will also be provided in separate documents (e.g. sample handling sheets or lab manual). Samples may be stored for a maximum of 15 years (or according to local regulations) following the end of the study at a facility selected by the Sponsor to enable further analyses.

Reporting

Biomarker investigations may be reported separately (e.g. in a biomarker evaluation report) except for biomarkers described in Section 9.4 or the SAP.

8.6.1 Genetic investigations

Genetic analyses are primarily done to address first, altered drug metabolism, in particular genetic variants that may influence finerenone exposure and second, factors predisposing to human diseases.

Genotyping solutions such as Axiom PMRA includes selected, clinically significant pathogenic variants including those implicated for actionable genetic risk. The Axiom PMRA helps to better understand the interplay between human genetics and the susceptibility to complex diseases (e.g. CKD) and to translate that understanding to potential clinical use. The Axiom PMRA includes novel risk variants that have been implicated in human diseases and conditions, including cardiovascular disease variants.

Pharmacogenetic analyses relate to gene variants such as single nucleotide polymorphisms of drug transporters and/or drug metabolizing enzymes including enzymes of the cytochrome P450 system, the Solute Carrier Family or uridine 5'-diphosphate-glucuronosyltransferase. Different variants predispose to altered metabolism of drugs and may result in differences in finerenone exposure. Specific genetic variants may arise during the clinical program.

Genetic analyses may be pivotal to understanding the pharmacology of finerenone and support its safe and efficacious use resulting in an improved benefit/risk ratio.

8.6.2 Laboratory biomarkers

Biomarkers will be evaluated in samples collected before, during and after treatment in order to determine the impact of study intervention. These markers fall into the following categories:

- A. **Biomarkers related to the mode-of-action of the study intervention and/or functional markers of the cardiovascular system:** vasoactive agents (aldosterone etc.), NT-proBNP/BNP
- B. **OMICS / Multiplex Analysis:** A sample will be collected for semi-targeted/untargeted omics analysis (e.g. metabolomics by mass spectrometry, OLINK Proximity Extension Assays (Explore 1536®) in order to identify '*de novo*' pharmacodynamic biomarkers, biomarkers which support the understanding of the mode-of-action and/or biomarkers which may indicate disease progression.
- C. **Further biomarkers related to the mode of action or the safety of** study intervention and similar drugs may be examined. The same applies to further biomarkers deemed relevant to renal and (cardio)vascular diseases and associated health problems. These investigations may include e.g. diagnostic, safety, PD, monitoring, or potentially predictive biomarkers.

8.7 Immunogenicity Assessments

Not applicable.

8.8 Health Economics

Health Economics parameters are not evaluated in this study.

9. Statistical Considerations

9.1 Statistical Hypotheses

Testing procedure of primary and key secondary efficacy endpoints

The primary and the key secondary endpoints will be tested under a confirmatory hierarchical testing procedure. If the primary efficacy endpoint achieves the significance level of 0.05, the corresponding null hypothesis can be rejected, and the remaining secondary endpoints will be tested at a two-sided level of 0.05 according to the following hierarchy. The three hypotheses listed will be tested in the pre-specified hierarchical sequence according to their order specified below. Consequently, confirmatory testing of the first, second and third hypotheses requires rejection of each preceding null hypothesis. In this setting, each hypothesis will be assessed at a two-sided significance level of 0.05 while keeping the global type I error rate of 0.05.

1. Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or CV death
2. Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$
3. Time to the composite of HF hospitalization or CV death

9.2 Sample Size Determination

Approximately 3160 participants will be screened to achieve 1580 randomly assigned to study intervention.

The approximate power of t-tests for comparing eGFR slope parameters (mL/min/1.73m² per year) between treatment groups is based on a two-slope linear spline mixed-effects model with specified model parameters, approximately 17 months recruitment period, at least 32 months per participant follow-up, and designated sample sizes. The power estimate from the simulation is based on initial parameter values obtained from FIDELIO-DKD data, consistent with the parameters of the linear spline mixed effect model described by (Vonesh et al. 2019). A total sample size of 1500 participants (750 per treatment arm) will have > 90% power to demonstrate the superiority of finerenone to placebo with respect to total eGFR slope from baseline to month 32 at a two-sided significance level of 5%. The sample size calculation includes assuming a 5% annual event rate resulting from intercurrent events (due to kidney failure or all-cause mortality). The >90% power calculation is based on an assumption of a 0.7 mL/min/1.73m² per year total eGFR slope difference. A total sample size of 1500 will have an 80% power if we assume a 0.527 mL/min/1.73m² per year total eGFR slope difference of finerenone to placebo.

An estimated total of 1580 participants (790 per treatment arm) will be randomized, considering an approximately 5% overall discontinuation rate in the study. All participants will be randomized in a 1:1 ratio to receive treatment with either finerenone OD or placebo.

Further participants may be enrolled in case of an exceptionally high drop-out rate due to a trial-continuity issue.

9.3 Analysis Set

Analysis sets are defined in [Table 9–1](#).

Table 9–1: Analysis sets

Participant Analysis Set	Description
Enrolled	All participants who sign the ICF
Safety analysis set (SAF)	All randomized participants who have taken at least one dose of the study intervention. Participants will be analyzed according to the intervention they received.
Full analysis set (FAS)	All randomized participants. Participants will be analyzed according to the intervention they were randomized to.
Pharmacokinetic analysis set (PKS)	All finerenone-treated participants with at least 1 valid finerenone plasma concentration and without validity findings which would interfere with the evaluation of the PK data.

9.4 Statistical Analyses

The statistical analysis plan will be finalized prior to database lock and it will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints, including primary and key secondary endpoints.

9.4.1 General Considerations

Statistical analysis will be performed using SAS; the version used will be specified in the SAP.

Randomization will be stratified for the use of SGLT2 inhibitor at baseline (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g).

The most important subgroups besides stratification factors are given below (with further details and subgroups provided in the SAP):

- Primary cause of CKD
- eGFR category at baseline (eGFR < 45 mL/min/1.73 m², 45 to < 60 mL/min/1.73 m², ≥ 60 mL/min/1.73 m²)
- UACR category at baseline: UACR < 300 mg/g, UACR 300 to 1000mg/g, UACR > 1000 mg/g
- Baseline serum potassium (≤ 4.5 , > 4.5 mmol/L)
- Region
- Race
- Sex
- Age group
- CVD history

It is anticipated that in these proposed subgroups for analysis, differences in treatment effects may be observed according to the screening or baseline characteristics defined, due in part to the differences in the risk of clinical events expected in the different subgroups.

9.4.1.1 Planned Methods of Analysis

A log-normal distribution is assumed for serum creatinine and UACR. For all other continuous variables, a normal distribution is assumed. The distributional assumptions will be investigated. If necessary, nonparametric methods or transformation of the data will be considered.

All variables will be analyzed by treatment arm using descriptive statistical methods. The number of data available, mean, SD, minimum, median, and maximum, will be calculated for continuous data. The geometric mean and SD will be provided instead of the arithmetic mean and SD for the variables where lognormal distributions are assumed. Frequency tables will be generated for categorical data.

Baseline values will be defined as the mean central lab value from screening and randomization visit (Visits 0 and 1). If the last observation available prior to randomization is the measurement from the Screening Visit, this would be used as the baseline value. This also includes assessments from a local laboratory, in case that prior to randomization, no assessment from the central laboratory is available. Otherwise baseline values will be missing. If more than one measurement was planned for a scheduled time point, for example blood pressure measurements and heart rate, the mean value of the last set of measurements per time point prior to randomization will be used for the calculation of the baseline value.

In the event of repeated measurements for Pre-treatment (Visit 0) and Visit 1 (baseline), the closest measurement prior to randomization will be used for analysis instead of the scheduled measurements. At all visits post-randomization and unless stated otherwise, only the values at scheduled measurements will be used for analysis.

Further details on the statistical analyses will be provided in the SAP that will be approved before database release.

9.4.1.2 Demographic and Baseline characteristics

Demographic and baseline characteristics including disease characteristics obtained before the first study intervention administration, e.g.;

- age, gender, ethnicity, race, region (North America, Europe, Asia, Latin America and others), body weight, body height, body mass index, smoking history (never, former, current smoker), alcohol consumption
- primary Cause of CKD, baseline eGFR, categories of baseline eGFR (<30 mL/min/1.73m², 30 to <45 mL/min/1.73m², 45 to <60 mL/min/1.73m², 60 to <75 mL/min/1.73m², 75 to <90 mL/min/1.73m², ≥ 90 mL/min/1.73m²) baseline UACR, categories of baseline UACR 30 to <300 mg/g, ≥ 300 mg/g), serum creatinine, serum potassium, categories of baseline serum potassium (≤ 4.5 mmol/L and > 4.5 mmol/L), vital signs parameters (i.e. systolic blood pressure, diastolic blood pressure, heart rate)

Demographic and baseline characteristics will be summarized using descriptive statistics by treatment group and all participants for full analysis set population and safety population. Age for each participant will be defined as the number of years between informed consent date and birth date. For summarizing race, if multiple races are recorded in the database, then 'multiple race' will be used in the summary table. The study sample sizes will be summarized by the trial unit (site) for enrolled participants. All participants with signed ICF will be included in the enrolled participants.

9.4.1.3 Incomplete or Missing Data

It is important to minimize the amount of missing data for the analysis of the primary estimand. Certain measures will be taken to guarantee the availability of as many eGFR values as possible (e.g. DCT implementation including the option to have remote visits and access to walk-in labs, participant engagement activities including participant education on the disease and the trial, early site training on the importance of retention, option to use local lab samples in case central lab shipment is not possible).

Data handling rules for incomplete data will be described in the specified section in the SAP if needed.

9.4.2 Primary Endpoint(s)

9.4.2.1 Mean rate of change as measured by the total slope of eGFR from baseline to Month-32

The primary efficacy variable of interest is eGFR values over time collected every 3 months for 12 months and afterwards every 4 months until end-of planned treatment period. The population-level summary is the mean rate of change of eGFR as measured by total slope from baseline to month-32.

The primary analysis of the primary efficacy variable will be performed according to participants' randomized intervention and will be based on the FAS.

A descriptive summary of eGFR will be presented by treatment group and visit, including absolute and relative changes to baseline.

The rate of change of eGFR will be compared between the finerenone and the placebo group by estimating the total eGFR slope using available assessment from baseline to planned end of treatment period. We assume that changes in the mean response can be expressed in terms of a linear trend, and the treatment effect can be expressed in terms of the difference in slope between finerenone and placebo.

For analysis of the total slope, the serial change in eGFR will be modeled using a two-slope linear spline mixed-effects model in which a fixed change point will define acute and chronic eGFR slope at month-3. The two-slope linear spline mixed-effects model is based on the method described by Vonesh et al. in section 2.1 “A two-slope linear spline mixed-effects models for eGFR” (Vonesh et al. 2019). The acute and chronic phases are differentiated because finerenone showed acute effects on eGFR that differed from its long-term effects on disease progression in FIDELIO-DKD. In addition to fixed effects for the treatment (finerenone vs. placebo), time (continuous) and a treatment by time interaction, the model will

include fixed effects for stratification factors used in the stratified randomization (i.e., baseline SGLT2i use (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g)) and for baseline eGFR (continuous). A random effect for the intercept, acute slope (baseline to month-3), and chronic slope (month-3 to planned end of treatment period) will be included to account for variation in progression rates of participants over time. An unstructured covariance will be used to model the within-participant errors. The time variable represents the duration from baseline to each eGFR assessment during acute and chronic phases. This model will include all observed eGFR measurements except those obtained after the onset date of ESRD. The analysis will be carried out using maximum likelihood procedures. Linear contrasts will be constructed to estimate the acute, chronic, and total slope in eGFR. LS means of the acute, chronic and total eGFR slope for finerenone and placebo group with 95% confidence interval and corresponding two-sided p-value will be presented.

The time-dependent total eGFR slope until month-32 will be calculated for the two treatment groups. The difference of total eGFR slope from baseline to month-32 between the finerenone and the placebo treatment group will be calculated, and corresponding two-sided 95% confidence interval and p-value will be presented. The total slope difference of finerenone and placebo will be estimated at 32 months after baseline, considering the anticipated data availability and duration of treatment. The study will be considered successful if the p-value is statistically significant (i.e. below 0.05).

Model Specification

For reasons of clarity, only the fixed effects for treatment and time and their interaction will be included in the model specification outlined in the following. In contrast to that, the fixed effects for baseline eGFR and for the stratification factors used in the stratified randomization (i.e., baseline SGLT2i use [yes/no] and screening UACR [≤ 1000 mg/g vs. > 1000 mg/g]) will not be shown in the model formula as their inclusion is straightforward.

As described in (Vonesh et al. 2019), we will formulate the two-slope linear spline mixed-effects model as follows:

$$Y_{ij} = (\beta_{0c} + \beta_{1c}t_{ij} + \beta_{2c} \max\{t_{ij} - t^*, 0\}) (1 - X_i) + (\beta_{0f} + \beta_{1f}t_{ij} + \beta_{2f} \max\{t_{ij} - t^*, 0\}) X_i + b_{0i} + b_{1i} t_{ij} + b_{2i} \max\{t_{ij} - t^*, 0\} + \epsilon_{ij}$$

where,

Y_{ij} denotes the response variable, eGFR, for the i-th participant on the j-th occasion,
 X_i be a treatment group indicator variable with $X_i = 0$ if the i-th participant is randomized to the placebo group and $X_i = 1$ if randomized to finerenone group,
 $\beta_k = (\beta_{0k} \beta_{1k} \beta_{2k})'$ denote the regression parameters for placebo ($k=c$) and finerenone ($k=f$),

ϵ_{ij} denote within-participant residuals,

$b_i = (b_{0i} \ b_{1i} \ b_{2i})'$ denote between-participant random effects,

t^* denotes the change point (knot at month 3),

$\max\{t_{ij} - t^*, 0\}$ be the truncated line function

where t_{ij} are the time points at which eGFR is measured for the i-th participant across v_i occasions ($j = 1, 2, \dots, v_i$).

Model assumptions:

- $(\epsilon_{ij} | b_i)$ are conditionally independent normally distributed random variables with mean 0 and within-participant variance σ_{ij}^2
- Participant-specific random effects b_i that are independent multivariate normally distributed with mean vector 0 and 3x3 covariance matrix that reflects a common unstructured covariance matrix

The interpretation of the regression parameter $\beta_k = (\beta_{0k} \beta_{1k} \beta_{2k})'$ in terms of chronic and acute slope for placebo ($k=c$) or finerenone ($k=f$), respectively, is as follows:

β_{0k} denotes the intercept for treatment group k ,

β_{1k} is the acute slope up through but excluding time t^* (knot at month 3) for treatment group k ,

β_{2k} is the directional change in slope starting at time t^* (knot at month 3) for treatment group k , and

$\beta_{1k} + \beta_{2k}$ corresponds to the chronic slope for treatment group k .

Finally, the total-slope up to time t describing the rate of change in eGFR from baseline to a specific time $t > t^*$ (t = total study time, here: 32 months) is defined as

$$\{[\beta_{0k} + \beta_{1k} \cdot t + \beta_{2k} \max(t - t^*, 0)] - \beta_{0k}\} / t$$

9.4.2.2 Supportive analysis of the primary efficacy endpoint

Chronic eGFR slope, defined as the rate of eGFR change from Month 3 after randomization to the end of the planned treatment period

The eGFR slope will be determined separately, from randomization to month 3 (acute slope) and after month 3 to end of planned study treatment (chronic slope). The chronic eGFR slope will be analyzed using the same model as described for the primary analysis ([Vonesh et al. 2019](#)), with month 3 defined as the 'change point' in the model and the time variable representing the duration from baseline to each eGFR assessment during acute and chronic phases.

LS means of the chronic eGFR slope of finerenone and placebo group, as well as the treatment difference (finerenone minus placebo) with a 95% confidence interval and corresponding two-sided p-value will be presented.

eGFR change from baseline to 1 month after the end of treatment period (off-treatment)

A post-treatment eGFR measurement will be conducted for all participants 1 month after end of the planned treatment period.

As a supportive analysis an ANCOVA model will be fitted to the above change from baseline eGFR value from baseline to the off-treatment (1 month), including the factors treatment group, stratification factors, total time on treatment, and the baseline eGFR as a covariate. Corresponding two-sided 95% confidence intervals and p-value will be computed.

9.4.2.3 Sensitivity analysis of the primary efficacy endpoint

Sensitivity analysis related to MAR assumption

It is considered that intermittent missing eGFR values (e.g. due to one missed visit) will not lead to a biased estimation of the slope; however, for monotone missing data patterns (e.g.

missingness due to lost to follow-up), several sensitivity analyses under a ‘missing not at random (MNAR)’ assumption will be considered. For this ‘control-based pattern imputation methods’ or a ‘tipping point analysis’ could be employed.

Sensitivity analysis with alternative estimands

For the ICEs of death or start of renal replacement therapy (chronic dialysis or transplant) during the trial, additional sensitivity analyses for the primary efficacy endpoint will be performed. Further details will be included in the statistical analysis plan, where the above-mentioned ICEs will be handled with different strategies.

For the ICE permanent treatment discontinuation, it is planned to investigate an estimand using a “while-on-treatment strategy” to assess the slope difference while the participants were taking study medication.

9.4.3 Secondary Efficacy Endpoints

Secondary efficacy endpoints will be assessed by the investigator based on their clinical judgement. For each endpoint the investigator will record information specific to that type of endpoint in the eCRF. Event criteria will be provided to facilitate event evaluation and edit checks will be in place to detect potential inconsistencies.

To further explore the chronic and persistent benefit of finerenone on kidney disease progression and CV morbidity and mortality, the following secondary variables will be evaluated:

Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or CV death

The secondary efficacy variable is the time to the first occurrence of the composite endpoint of onset of kidney failure, a sustained decrease of $\text{eGFR} \geq 57\%$ from baseline, HF hospitalization or CV death.

Randomized participants without an event of the composite endpoint at the time of analysis will be censored at the date of their last visit or date of non-CV death. Participants without information about the composite endpoint after baseline will be censored at Day 1.

In order to evaluate whether finerenone is superior to placebo in prolonging the time to the first event of the composite endpoint, the corresponding null hypothesis will be tested using the log-rank test at the two-sided 5% significance level.

A Cox proportional regression model will be used in order to provide a point estimate of the hazard ratio with a corresponding two-sided 95% confidence interval. In addition, Kaplan-Meier curves will provide the cumulative proportions of events by treatment groups.

Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$

Time to the composite kidney endpoint will be analyzed analogously to the composite kidney and CV endpoint, with modifications to the censoring rules as the events will differ.

Time to the composite of HF hospitalization or CV death

Time to first event of HF hospitalization or CV death will be conducted analogously to the above secondary composite endpoints, with modifications to the censoring rules as the events will differ.

HF hospitalization events will be counted up to the day of withdrawal from the study or death. Randomized participants without a HF hospitalization event or CV death at the time of analysis will be censored at the date of their last contact up to and including the end of study visit or date.

9.4.4 Secondary Safety endpoint

Number of participants with TEAEs, TESAEs and AESI:

The number and incidence of participants with TEAEs and TESAEs will be summarized by treatment group and overall using MedDRA terms grouped by Primary System Organ Class and Preferred Term. Tables and/or narratives of treatment-emergent deaths, serious and significant AEs, including early discontinuation due to AEs, will also be provided. For details on other analysis of TEAEs and TESAEs see Section 9.4.6.

AEs that started or worsened after the first dose of study intervention up to 3 days after any temporary or permanent interruption of study intervention will be considered as TEAEs.

9.4.5 Exploratory Endpoints

The analysis of other exploratory endpoints, including those listed below, are described in the SAP.

- Time to onset of kidney failure (ESRD or sustained eGFR decline below 15 ml/min/1.73m²)
- Time to onset of ESRD (chronic dialysis or transplant)
- Time to onset of eGFR decrease to less than 15 mL/min/1.73 m² sustained over at least 4 weeks
- Time to onset of eGFR decrease of ≥ 57 % from baseline sustained over at least 4 weeks
- Time to CV death
- Time to first hospitalization for HF or urgent HF visit
- Number of participants with UACR decrease of at least 30% from baseline at Month 6
- Change in UACR from baseline
- Time to occurrence of the composite endpoint of kidney failure or eGFR decrease of ≥ 40 % from baseline sustained over at least 4 weeks
- Time to first CV event (CV death, non-fatal MI, non-fatal stroke or transient ischemic attack, hospitalization for HF or urgent HF visit)
- Time to CV hospitalization

- Time to death from any cause (all-cause mortality)
- Time to first all-cause hospitalization
- Number of participants with new diagnosis of atrial fibrillation or atrial flutter
- Number of participants with new diagnosis of HF
- Change in Quality of Life (QoL) summary scores measured by the following health-related quality of life (HRQoL) questionnaire EuroQol Group 5 dimension, 5 level questionnaire (EQ-5D-5L)

9.4.6 Other Safety Analysis

All analyses on safety and tolerability data will be performed in the SAF. Details on safety analyses will be described in the SAP.

The following safety variables will be assessed during the study:

- SAEs and AEs leading to discontinuation of treatment with study intervention
- Change in serum potassium from baseline
- Number of participants with serum potassium > 5.5 mmol/L
- Number of participants with serum potassium > 6.0 mmol/L
- Number of participants with hospitalization for hyperkalemia
- Number of participants discontinuing study intervention permanently due to hyperkalemia
- Change in vital signs from baseline
- Number of participants with hospitalization for worsening of renal function
- Number of participants discontinuing study intervention permanently due to worsening of renal function
- Changes in laboratory values

9.5 Interim Analysis

No formal interim analysis is planned for this study.

10. Supporting Documentation and Operational Considerations

10.1 Appendix 1: Regulatory, Ethical, and Study Oversight Considerations

10.1.1 Regulatory and Ethical Considerations

- This study will be conducted in accordance with the protocol and with the following:
 - Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines
 - Applicable ICH Good Clinical Practice (GCP) Guidelines
 - Applicable laws and regulations
- The protocol, protocol amendments, ICF, Investigator Brochure, and other relevant documents (e.g., advertisements) must be submitted to an IRB/IEC by the investigator and reviewed and approved by the IRB/IEC before the study is initiated.
- Any amendments to the protocol will require IRB/IEC approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants. Any substantial modification of the protocol will be submitted to the competent authorities as substantial amendments for approval, in accordance with ICH Good Clinical Practice and national and international regulations.
- Protocols and any substantial amendments to the protocol will require health authority approval prior to initiation except for changes necessary to eliminate an immediate hazard to study participants.
- The investigator will be responsible for the following:
 - Providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC
 - Notifying the IRB/IEC of SAEs or other significant safety findings as required by IRB/IEC procedures
 - Providing oversight of the conduct of the study at the site and adherence to requirements of ICH guidelines, the IRB/IEC, and all other applicable local regulations

10.1.2 Financial Disclosure

Investigators and sub-investigators will provide the sponsor with sufficient, accurate financial information as requested to allow the sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

10.1.3 Informed Consent Process

- The investigator or his/her representative will explain the nature of the study to the participants or their legally authorized representative and answer all questions regarding the study.
- Participants must be informed that their participation is voluntary. Participants or their legally authorized representative will be required to sign a statement of

informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, Health Insurance Portability and Accountability Act (HIPAA) requirements, where applicable, and the IRB/IEC or study center.

- The medical record must include a statement that documented informed consent was obtained before the participant was enrolled in the study and the date the written consent was obtained. The authorized person obtaining the informed consent must also sign the ICF.
- Any study specific procedure can be performed only if written informed consent form is available.
- Participants must be re-consented to the most current version of the ICF(s) during their participation in the study.
- A copy of the ICF(s) must be provided to the participants or their legally authorized representative.
- A pre-screening consent form will be signed prior to any optional pre-screening visits, which may be used at some sites to assess UACR prior to full screening.

Participants who are rescreened are required to sign a new ICF.

10.1.4 Data Protection

- Participants will be assigned a unique identifier by the sponsor. Any participant records, datasets or biological samples that are transferred to the sponsor will contain the identifier only; participant names or any information which would make the participant identifiable will not be transferred.
- The participant must be informed that his/her personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the participant who will be required to give consent for their data to be used as described in the informed consent
- The participant must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.

10.1.5 Committees Structure

10.1.5.1 Steering Committee

The Steering Committee, which consists of external experts in the area of nephrology, and cardiology, will ensure the overarching integrity of the study.

- Review feedback from the independent DMC
- Input into endpoint event assessment
- Provision of recommendations regarding amendments to the protocol
- Contribution to and oversight of publications and communication of study results
- Overseeing overall blinded event rates to ensure that they meet protocol projections.

Decisions will be made by consensus; in the absence of consensus, the co-Chairs of the steering Committee will provide their recommendation to the Sponsor for final determination.

10.1.5.2 Independent data monitoring committee

Ongoing safety monitoring during the conduct of the study will be performed by an external and independent DMC. An independent statistical analysis center will be involved in processing unblinded safety data for the DMC. Analysis periods and procedures will be defined in an operational charter (DMC charter) filed in the study file. The independent DMC will periodically review and assess all outcome events as well as efficacy and safety data from the study for imbalances in an unblinded manner. It is believed that in this way, participant safety can continue to be monitored throughout the duration of the trial, and the integrity of the study maintained. If unexpected safety issues are identified, specific amendments will be implemented based on the recommendation of the DMC. Following data review, the DMC will provide written recommendations that will be transferred to the chairmen of the Steering Committee and Bayer. DMC opinions and recommendations will be notified by Bayer as soon as possible to the competent authorities and the IECs where they qualify for expedited reporting.

10.1.6 Dissemination of Clinical Study Data

Results of clinical drug trials will be provided on the publicly funded website www.ClinicalTrials.gov and EU Clinical Trials Register in line with the applicable regulations.

In addition, Result Summaries of Bayer's sponsored clinical trials in drug development phases II, III and IV and Phase I studies in patients are provided in the Bayer Trial Finder application after marketing authorization approval in line with the position of the global pharmaceutical industry associations laid down in the "Joint Position on the Disclosure of Clinical Trial Information via Clinical Trial Registries and Databases".

Bayer commits to sharing upon request from qualified scientific and medical researchers patient-level clinical trial data, study-level clinical trial data, and protocols from clinical trials in patients for medicines and indications approved in the United States (US) and European Union (EU) on or after January 01, 2014 as necessary for conducting legitimate research.

All Bayer-sponsored clinical trials are considered for publication in the scientific literature irrespective of whether the results of the clinical trials are positive or negative.

10.1.7 Data Quality Assurance

- All participant data relating to the study will be recorded on printed or electronic CRF unless transmitted to the sponsor or designee electronically (e.g., laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the CRF.
- Guidance on completion of CRFs will be provided in the CRF Completion Guidelines.
 - Home nurses will record source data directly into a DCT study app. The clinical data recorded by home nurses is readily visible in the study app to the

investigator/authorized site personnel and will be referenced when making entries into the eCRF. Source data collected in the study app can also be viewed by study monitors for ongoing source data verification purposes.

- The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.
- QTLs will be pre-defined in the integrated data review plan to identify systematic issues that can impact participant safety and/or reliability of study results. These pre-defined parameters will be monitored during the study and important deviations from the QTLs and remedial actions taken will be summarized in the clinical study report.
- Monitoring details describing strategy (e.g., risk-based initiatives in operations and quality such as Risk Management and Mitigation Strategies and Analytical Risk-Based Monitoring), methods, responsibilities and requirements, including handling of noncompliance issues and monitoring techniques (central, remote, or on-site monitoring) are provided in the Monitoring Plan.
- The sponsor or designee is responsible for the data management of this study including quality checking of the data.
- The sponsor assumes accountability for actions delegated to other individuals (e.g., Contract Research Organizations).
- Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for 15 years after study completion unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

10.1.8 Source Documents

- Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.
- Data reported on the eCRF or entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.
- Definition of what constitutes source data can be found in the Source Data Location List or equivalent.
- The investigator must maintain accurate documentation (source data) that supports the information entered in the CRF.
- Study monitors will perform ongoing source data verification to confirm that data entered into the CRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

10.1.9 Study and Site Start and Closure

First Act of Recruitment

The study start date is the date on which the clinical study will be open for recruitment of participants.

The first act of recruitment is the first participant screened in the study and will be the study start date.

Study/Site Termination

The sponsor or designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the sponsor. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to:

For study termination:

- Discontinuation of further study intervention development

For site termination:

- Failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines
- Inadequate or no recruitment (evaluated after a reasonable amount of time) of participants by the investigator
- Total number of participants included earlier than expected.

If the study is prematurely terminated or suspended, the sponsor shall promptly inform the investigators, the IECs/IRBs, the regulatory authorities, and any contract research organization(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

10.1.10 Publication Policy

- The results of this study may be published or presented at scientific meetings. If this is foreseen, the investigator agrees to submit all manuscripts or abstracts to the sponsor before submission. This allows the sponsor to protect proprietary information and to provide comments.
- The sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the sponsor will generally support publication of multicenter studies only in their entirety and not as individual site data. In this case, a coordinating investigator will be designated by mutual agreement.
- Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

10.2 Appendix 2: Clinical Laboratory Tests

- The tests detailed in [Table 10–1](#) will be performed by the central laboratory.
- Protocol-specific requirements for inclusion or exclusion of participants are detailed in [Section 5](#) of the protocol.
- Additional tests may be performed at any time during the study as determined necessary by the investigator or required by local regulations.

Investigators must document their review of each laboratory safety report.

Laboratory evaluations (hematology, HbA_{1c}, clinical chemistry and urinalysis parameters) are shown in [Table 10–1](#). Hematology, HbA_{1c} and (full) clinical chemistry are considered as ‘Full central laboratory’ assessments.

Laboratory evaluations will be performed according to the visit schedule (see [1.3](#)).

Table 10–1 Central laboratory parameters

Parameter	Component
Hematology	White blood cell count (WBC), red blood cell count (RBC), hemoglobin, hematocrit, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), MCH concentration (MCHC), platelets
Biomarker sample	For biomarkers blood, serum, plasma and urine samples are needed (See Section 8.6 for further information)
Genetic testing	Genetics (See Section 8.6 for further information)
Glycated hemoglobin HbA_{1c} (HbA_{1c})	
Clinical chemistry (full)	Aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (AP), gamma glutamyl transpeptidase (GGT), lactate dehydrogenase (LDH), creatine kinase (CK), cholesterol (high density lipoprotein [HDL], low density lipoprotein [LDL], total), triglycerides, creatinine, eGFR (CKD-EPI), blood urea nitrogen, uric acid, bilirubin, sodium, serum potassium, magnesium, total protein, albumin, high-sensitivity C-reactive protein (hs-CRP)
Clinical chemistry (limited)	Serum creatinine, eGFR (CKD-EPI), serum potassium, serum sodium
Urinalysis	Urine albumin-to-creatinine ratio (UACR), Urine protein/creatinine ratio (UPCR)

10.2.1.1 Local laboratory

In addition to samples for the central laboratory, other blood samples will be taken for screening purposes, and also for safety from Visit 1 (Day 1; baseline) onwards (see [Section 1.3](#)) for analysis at the local laboratory. These samples may be obtained up to 72 hours before a

scheduled visit. For sites utilizing remote visit activities, blood safety samples collected at remote visits will be analyzed at the central laboratory only.

The following clinical chemistry parameters must be measured for as long as participant is not prematurely discontinued from the study intervention:

- potassium from blood sample.
- creatinine (eGFR will be calculated and recorded in the eCRF by the investigator using the CKD-EPI formula, or the JP equation for sites located in Japan) from blood sample.

Potassium values should be recorded using a single decimal point (e.g. 4.5 mmol/L or mEq/L). In the event of hyperkalemia, see Section 6.5 for guidance on treatment.

Up-titration of the study intervention will be based on local potassium and eGFR values for on site visits. For remote visits it will be based on central laboratory potassium results and eGFR. In all cases, this must be documented in the eCRF. Down-titration of the study intervention will occur for safety reasons only (for guidance, see Section 6.5 and Table 6–3).

In women of childbearing potential, a **pregnancy test** will be performed locally, at the Screening Visit and following randomization at yearly visits as long as the participant is not permanently discontinued from the study intervention (since more than 7 days). However, pregnancy tests in participant of childbearing potential should be performed more frequently (e.g. at every visit) if this is required by national / institutional regulations. Both serological and urine tests are acceptable.

10.3 Appendix 3: AEs and SAEs: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

10.3.1 Definition of AE

AE Definition
• An AE is any untoward medical occurrence in a clinical study participant, associated with the use of study intervention, whether or not considered related to the study intervention. • NOTE: An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) associated with the use of study intervention.

Events Meeting the AE Definition
• Any abnormal laboratory test results (hematology, clinical chemistry, or urinalysis) or other safety assessments (e.g., ECG, radiological scans, vital signs measurements), including those that worsen from baseline, considered clinically significant in the medical and scientific judgment of the investigator (i.e., not related to progression of underlying disease). • Exacerbation of a chronic or intermittent pre-existing condition including

either an increase in frequency and/or intensity of the condition.

- New conditions detected or diagnosed after study intervention administration even though it may have been present before the start of the study.
 - Signs, symptoms, or the clinical sequelae of a suspected intervention-intervention interaction.
 - Signs, symptoms, or the clinical sequelae of a suspected overdose of either study intervention or a concomitant medication. Overdose per se will not be reported as an AE/SAE unless it is an intentional overdose taken with possible suicidal/self-harming intent. Such overdoses should be reported regardless of sequelae.
 - “Lack of efficacy” or “failure of expected pharmacological action” per se will not be reported as an AE or SAE. Such instances will be captured in the efficacy assessments. However, the signs, symptoms, and/or clinical sequelae resulting from lack of efficacy will be reported as AE or SAE if they fulfill the definition of an AE or SAE.
-

Events **NOT** Meeting the AE Definition

- Any clinically significant abnormal laboratory findings or other abnormal safety assessments which are associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant's condition.
 - The disease/disorder being studied or expected progression, signs, or symptoms of the disease/disorder being studied, unless more severe than expected for the participant's condition.
 - Medical or surgical procedure (e.g., endoscopy, appendectomy): the condition that leads to the procedure is the AE.
 - Situations in which an untoward medical occurrence did not occur (social and/or convenience admission to a hospital).
 - Anticipated day-to-day fluctuations of pre-existing disease(s) or condition(s) present or detected at the start of the study that do not worsen.
-

10.3.2 Definition of SAE

An SAE is defined as any AE that, at any dose:

a. Results in death

b. Is life-threatening

- The term 'life-threatening' in the definition of 'serious' refers to an event in which the participant was at risk of death at the time of the event. It does not refer to an event, which hypothetically might have caused death, if it were more severe.
-

c. Requires inpatient hospitalization or prolongation of existing hospitalization

- In general, hospitalization signifies that the participant has been admitted (usually involving at least an overnight stay) at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AEs. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether "hospitalization" occurred or was necessary, the AE should be considered serious.
 - Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.
-

d. Results in persistent or significant disability/incapacity

- The term disability means a substantial disruption of a person's ability to conduct normal life functions.
 - This definition is not intended to include experiences of relatively minor medical significance such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma (e.g., sprained ankle) which may interfere with or prevent everyday life functions but do not constitute a substantial disruption.
-

e. Is a congenital anomaly/birth defect

f. Is a suspected transmission of any infectious agent via an authorized medicinal product

g. Other situations:

- Medical or scientific judgment should be exercised by the investigator in deciding whether SAE reporting is appropriate in other situations such as significant medical events that may jeopardize the participant or may require medical or surgical intervention to prevent one of the other outcomes listed in the above definition. These events should usually be considered serious.
 - Examples of such events include invasive or malignant cancers, intensive treatment for allergic bronchospasm, blood dyscrasias, convulsions, or development of intervention dependency or intervention abuse.
-

10.3.3 Recording and Follow-Up of AE and/or SAE

AE and SAE Recording

- When an AE/SAE occurs, it is the responsibility of the investigator to review all documentation (e.g., hospital progress notes, laboratory reports, and diagnostics reports) related to the event.
 - The investigator will then record all relevant AE/SAE information.
 - It is not acceptable for the investigator to send photocopies of the participant's
-

medical records to Sponsor in lieu of completion of the required form.

- There may be instances when copies of medical records for certain cases are requested by Sponsor. In this case, all participant identifiers, with the exception of the participant number, will be redacted on the copies of the medical records before submission to Sponsor.
- The investigator will attempt to establish a diagnosis of the event based on signs, symptoms, and/or other clinical information. Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the AE/SAE.

Assessment of Intensity

- The investigator will make an assessment of intensity for each AE and SAE reported during the study and assign it to 1 of the following categories:
- Mild: An event that is easily tolerated by the participant, causing minimal discomfort and not interfering with everyday activities.
- Moderate: An event that causes sufficient discomfort to interfere with normal everyday activities.
- Severe: An event that prevents normal everyday activities. An AE that is assessed as severe should not be confused with an SAE. Severe is a category utilized for rating the intensity of an event; and both AEs and SAEs can be assessed as severe.

An event is defined as 'serious' when it meets at least 1 of the predefined outcomes as described in the definition of an SAE, NOT when it is rated as severe.

Assessment of Causality

- The investigator is obligated to assess the relationship between study intervention and each occurrence of each AE/SAE.
- A “reasonable possibility” of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out.
- The investigator will use clinical judgment to determine the relationship.
- Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study intervention administration will be considered and investigated.
- The investigator will also consult the Investigator’s Brochure (IB) and/or Product Information, for marketed products, in his/her assessment.
- For each AE/SAE, the investigator **must** document in the medical notes that he/she has reviewed the AE/SAE and has provided an assessment of causality.
- There may be situations in which an SAE has occurred and the investigator has minimal information to include in the initial report to Sponsor. However, **it is very important that the investigator always make an assessment of causality for every event before the initial transmission** of the SAE data to Sponsor.
- The investigator may change his/her opinion of causality in light of follow-up information and send an SAE follow-up report with the updated causality assessment.
- The causality assessment is one of the criteria used when determining regulatory reporting requirements.

Follow-up of AEs and SAEs

- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by Sponsor to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.
- If a participant dies during participation in the study or during a recognized follow-up period, the investigator will provide Sponsor with a copy of any post-mortem findings including histopathology.
- New or updated information will be recorded in the originally submitted documents.
- The investigator will submit any updated SAE data to Sponsor within 24 hours of receipt of the information.

10.3.4 Reporting of SAEs

SAE Reporting to Sponsor via an Electronic Data Collection Tool

- The primary mechanism for reporting an SAE to Sponsor will be the electronic data collection tool.
 - If the electronic system is unavailable, then the site will use the paper SAE data transmission (see next section) to report the event within 24 hours.
 - The site will enter the SAE data into the electronic system as soon as it becomes available.
 - After the study is completed at a given site, the electronic data collection tool will be taken off-line to prevent the entry of new data or changes to existing data.
 - If a site receives a report of a new SAE from a study participant or receives updated data on a previously reported SAE after the electronic data collection tool has been taken off-line, then the site can report this information on a paper SAE form (see next section) or to the medical monitor by telephone.
 - Contacts for SAE reporting can be found in the investigator site file.
-

SAE Reporting to Sponsor via Paper Data Collection Tool

- Email transmission of the SAE paper data collection tool is the preferred method to transmit this information to the Sponsor.
 - In rare circumstances and if email transmission is not feasible, notification by telephone is acceptable with a copy of the SAE data collection tool sent by overnight mail or courier service.
 - Initial notification via telephone does not replace the need for the investigator to complete and sign the SAE data collection tool within the designated reporting time frames.
 - Contacts for SAE reporting can be found in the investigator site file.
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10.4 Appendix 4: Contraceptive Guidance and Collection of Pregnancy Information

Definitions:

Woman of Childbearing Potential (WOCBP)

A woman is considered fertile following menarche and until becoming post-menopausal unless permanently sterile (see below).

If fertility is unclear (e.g., amenorrhea in adolescents or athletes) and a menstrual cycle cannot be confirmed before first dose of study intervention, additional evaluation should be considered.

Women in the following categories are not considered WOCPB

1. Premenarchal
2. Premenopausal female with 1 of the following:

- Documented hysterectomy
- Documented bilateral salpingectomy
- Documented bilateral oophorectomy

For individuals with permanent infertility due to an alternate medical cause other than the above, (e.g., mullerian agenesis, androgen insensitivity), investigator discretion should be applied to determining study entry.

Note: Documentation can come from the site personnel's: review of the participant's medical records, medical examination, or medical history interview.

3. Postmenopausal female

- A postmenopausal state is defined as no menses for 12 months without an alternative medical cause.
A high FSH level in the postmenopausal range may be used to confirm a postmenopausal state in women not using hormonal contraception or HRT. However, in the absence of 12 months of amenorrhea, confirmation with more than one FSH measurement > 40 mIU/mL (for US only: FSH levels > 40 mIU/mL and estradiol < 20 pg/mL) is required.
- Females on HRT and whose menopausal status is in doubt will be required to use one of the non-estrogen hormonal highly effective contraception methods if they wish to continue their HRT during the study. Otherwise, they must discontinue HRT to allow confirmation of postmenopausal status before study enrollment.

Contraception Guidance for women:

According to pre-clinical and clinical data, Finerenone does not indicate teratogenicity/fetotoxicity in early pregnancy (please refer to the current version of the Investigator's Brochure for details). Based on these data, women of child-bearing potential can be included into the trial if reliable contraception is used.

Adequate contraception is defined as any combination of at least 2 effective methods of birth control. Highly effective birth control methods are hormonal contraception (progesterone only), combined hormonal contraception (estrogen and progesterone), intrauterine devices and intrauterine hormone-releasing systems. One of these methods should be combined with a supplementary barrier method (preferably male condom) due to possible drug-to drug interactions which might reduce efficacy of the hormonal contraception. Contraception should be used until 30 days after last intake of study intervention.

Male participants do not have to use condoms in the study, because there is no indication of male-mediated developmental toxicity. Therefore, female partners of male participants are also not required to use contraception.

Collection of Pregnancy Information:**Male participants with partners who become pregnant**

- The investigator will attempt to collect pregnancy information on any male participant's female partner who becomes pregnant while the male participant is in this study. This applies only to male participants who receive finerenone.
- After obtaining the necessary signed informed consent from the pregnant female partner directly, the investigator will record pregnancy information on the appropriate form and submit it to the sponsor within 24 hours of learning of the partner's pregnancy. The female partner will also be followed to determine the outcome of the pregnancy. Information on the status of the mother and child will be forwarded to the sponsor. Generally, the follow-up will be no longer than 6 to 8 weeks following the estimated delivery date. Any termination of the pregnancy will be reported regardless of fetal status (presence or absence of anomalies) or indication for the procedure.

Female participants who become pregnant

- The investigator will collect pregnancy information on any female participant who becomes pregnant while participating in this study. The initial information will be recorded on the appropriate form and submitted to the sponsor within 24 hours of learning of a participant's pregnancy.
- The participant will be followed to determine the outcome of the pregnancy. The investigator will collect follow-up information on the participant and the neonate and the information will be forwarded to the sponsor. Generally, follow-up will not be required for longer than 6 to 8 weeks beyond the estimated delivery date. Any termination of pregnancy will be reported, regardless of fetal status (presence or absence of anomalies) or indication for the procedure.
 - While pregnancy itself is not considered to be an AE or SAE, any pregnancy complication or elective termination of a pregnancy for medical reasons will be reported as an AE or SAE.
 - A spontaneous abortion (occurring at <22 weeks gestational age) or still birth (occurring at >22 weeks gestational age) is always considered to be an SAE and will be reported as such.
- Any post-study pregnancy related SAE considered reasonably related to the study intervention by the investigator will be reported to the sponsor as described in Section 8.3.4. While the investigator is not obligated to actively seek this information in former study participants, he or she may learn of an SAE through spontaneous reporting.
- Any female participant who becomes pregnant while participating in the study will discontinue study intervention or be withdrawn from the study.

10.5 Appendix 5: Calculating the Child Pugh score

The severity of liver disease (Table 10–2) will determine the Child Pugh score (Table 10–3).

Table 10–2 Grading of severity of liver disease, adapted from (Pugh et al. 1973)

Factor	+1	+2	+3
Bilirubin (mg/dL)	<2	2 – 3	>3
Albumin (g/dL)	>3.5	2.8 – 3.5	<2.8
International Normalized Ratio	<1.7	1.7 – 2.3	>2.3
Ascites	None	Mild	Moderate / Severe
Encephalopathy	None	Grade I - II	Grade III – IV

Table 10–3 Classification using the added score from Table 10–2, adapted from (Pugh et al. 1973)

Child-Pugh Class	A	B	C
Points	5 – 6	7 – 9	10 – 15

10.6 Appendix 6: Guidance on use of common CYP3A4 inhibitors or inducers

Table 10–4 lists the most common medication regarded as potent CYP3A4 inhibitors or inducers.

Table 10–4 Cytochrome P450: list of concomitant medication

Excluded CYP3A4 inducers	Excluded CYP3A4 inhibitors	Allowed CYP3A4 inhibitors
• avasimibe • bosentan • carbamazepin • efavirenz	• amprenavir • atazanavir and other inhibitors of human HIV protease • boceprevir • clarithromycin	• amiodarone • aprepitant • bicalutamide • chloramphenicol
• enzalutamide • etravirine • fosphenytoin • hypericum perforatum / St John's wort	• cobicistat • conivaptan > 20 mg • crizotinib • danoprevir	• conivaptan ≤ 20 mg • fluconazole ≤ 200 mg • imatinib • mifepristone
• lersivirine • mephénytoin • methylphenobarbital • mitotane	• darunavir • delavirdine • diltiazem • dronedarone	• norfloxacin • tacrolimus • verapamil • lapatinib
• modafinil • nafcillin • nevirapine • oxcarbazepine	• elvitegravir • erythromycin • fluconazole > 200 mg • fosamprenavir	• dasatinib • nilotinib
• phenobarbital • phenytoin • primidone • rifabutin	• grapefruit • idelalisib • indinavir • itraconazole	
• rifampicin • semagacestat • talviraline • thioridazine	• ketoconazole • lopinavir • miconazole • nefazodone	
• troglitazone • vemurafenib	• nelfinavir • posaconazole • ritonavir • saquinavir	
	• telaprevir • telithromycin • tipranavir • troleandomycin • voriconazole	

HIV = Human immunodeficiency virus

10.7 Appendix 7: Country-specific Requirements**China:**

- Genetic and non-genetic biomarker samples will not be performed in China.
- For China, sample leftovers must not be repurposed after measurement.
- For China, with regard to Section 8.6.2 Laboratory biomarkers, NT-proBNP as a functional marker of the cardiovascular system is the only additional biomarker that will be assessed in central laboratory.

Japan:

- The investigator must report immediately all non-approved medical device failures which could cause health damage, as well as any health damage that may be causally associated with a non-approved medical device failure. For this reporting, the forms provided are to be used and sent to the designated recipient.

10.8 Appendix 8: Abbreviations

ACEI(s)	Angiotensin-converting enzyme inhibitor(s)
AE(s)	Adverse event(s)
AESI(s)	Adverse event(s) of special interest
ANCA	Anti-neutrophilic cytoplasmic autoantibody
ANCOVA	Analysis of covariance
ARB(s)	Angiotensin receptor blocker(s)
ARNI(s)	Angiotensin receptor neprilysin inhibitor(s)
ARTS	Mineralocorticoid Receptor Antagonist Tolerability Study
ARTS-DN	Mineralocorticoid Receptor Antagonist Tolerability Study–Diabetic Nephropathy
ARTS-HF	Mineralocorticoid Receptor Antagonist Tolerability Study–Heart Failure
BNP	B-type natriuretic peptide
BYOD	Bring your own device
CI	Confidence interval
CKD	Chronic kidney disease
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CONSORT	Consolidated Standards of Reporting Trials
CV	Cardiovascular
CYP3A4	Cytochrome P450 isoenzyme 3A4
DCT	Decentralized clinical trial
DMC	Data Monitoring Committee
DRE(s)	Disease-related event(s)
DtP	Direct-to-patient
ECG	Electrocardiogram
eCRF	Electronic case report form
EDTA	Ethylene diamine tetraacetic acid
eGFR	Estimated glomerular filtration rate
EMA	European Medicines Agency

ESKD	End stage kidney disease
ESRD	End stage renal disease
EQ-5D-5L	EuroQoL Group 5-dimension, 5-level questionnaire
EU	European Union
EudraCT	European Union Drug Regulating Authorities Clinical Trials
FAS	Full analysis set
FDA	United States Food and Drug Administration
FSGS	Focal segmental glomerulosclerosis
FSH	Follicle-stimulating hormone
FU	Follow-up
GCP	Good Clinical Practice
GFR	Glomerular filtration rate
HF	Heart failure
HR	Hazard ratio
HRT	Hormone replacement therapy
IB	Investigator's Brochure
ICE(s)	Intercurrent event(s)
ICF	Informed consent form
ICH	International Council on Harmonisation
IgAN	Immunoglobulin A nephropathy
IEC	Independent Ethics Committee
IRB	Institutional Review Board
IxRS	Interactive voice / web response system
MAR	Missing at random
MedDRA	Medical Dictionary for Regulatory Activities
MI	Myocardial infarction
MR	Mineralocorticoid receptor
MRA	Mineralocorticoid receptor antagonist
NNT	Number needed to treat
NT-proBNP	N-terminal prohormone B-type natriuretic peptide
NYHA	New York Heart Association
OD	Once daily
OE	Outcome event
PD	Premature discontinuation
PK	Pharmacokinetics
PKS	Pharmacokinetic analysis set
PMRA	Precision Medicine Array
popPK	Population pharmacokinetics
QTLs	Quality tolerance limits
RRR	Relative risk reduction
SAE(s)	Serious adverse event(s)
SAF	Safety analysis set
SAP	Statistical analysis plan

SD	Standard deviation
SGLT2	Sodium-glucose co-transporter-2
SGLT2i	Sodium-glucose co-transporter-2 inhibitor
SoA	Schedule of activities
SoC	Standard of care
SOP	Standard operating procedure
SUSAR(s)	Suspected unexpected serious adverse reaction(s)
TEAE(s)	Treatment-emergent adverse event(s)
TESAE(s)	Treatment-emergent serious adverse event(s)
UACR	Urinary albumin-to-creatinine ratio
UPCR	Urinary protein-to-creatinine ratio
US(A)	United States (of America)
WOCBP	Women of child-bearing potential

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Title Page**Protocol Title:**

A randomized, double-blind, placebo-controlled, parallel-group, multicenter Phase 3 study to investigate the efficacy and safety of **FInerenone**, in addition to standard of care, on the progression of kidney disease in patients with **Non-Diabetic Chronic Kidney Disease**

Protocol Number: 21177

Protocol Version: 5.1

Amendment Number: 4

Compound Number: BAY 94-8862 / Finerenone

Brief Title: Efficacy and safety of finerenone in patients with non-diabetic CKD

Study Phase: 3

Acronym: FIND-CKD

Sponsor Name and Legal Registered Address:

Non-US: Bayer AG, 51368 Leverkusen, Germany

US territory: Bayer HealthCare Pharmaceuticals Inc., 100 Bayer Boulevard, P.O. Box 915, Whippany, NJ 07981-0915, USA

Regulatory Agency Identifier Number(s):

IND: 117847

EudraCT: 2021-000421-27

EU-CT: 2023-506897-11-00

Protocol Date: 06 AUG 2025

Medical Monitor name and contact information will be provided separately.

Name: PPD

Role: PPD

This protocol was approved via electronic/digital signature(s) as indicated on the signature page appended to this document.

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Document History Table

DOCUMENT HISTORY			
Document	Version	Date	Comments (if applicable)
Amendment 4	5.1	06 AUG 2025	EU-1 Local requirements specific to the participating EU countries that follow the EU CTR added in Section 10.7
Amendment 3	5.0	10 APR 2025	Global amendment that includes changes applicable to all countries
Amendment 2	4.0	16 AUG 2023	Global amendment that includes changes applicable to all countries
Amendment 1	3.0	01 DEC 2021	
Clinical Study Protocol	2.0	05 JUL 2021	Version to correct inconsistencies in the protocol
Clinical Study Protocol	1.0	09 JUN 2021	Version submitted to the IEC/IRB in Denmark and Malaysia

Protocol Amendment Summary of Changes Table**Amendment 4 (06 AUG 2025)**

This amendment implements region-specific modifications of the global protocol amendment 3, dated 10 APR 2025, to meet the regional requirements in the participating EU countries. A description of the region-specific changes and a brief rationale are outlined in Section [10.7.3](#).

Overall Rationale for the Amendment:

Section # and Name	Description of Change	Brief Rationale
Appendix 10.7 Country/region-specific requirements	EU-specific changes are listed in Section 10.7.3 .	To follow the request from the competent authorities of the participating EU countries.

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1. Protocol Summary

1.1 Synopsis

Protocol Title: A randomized, double-blind, placebo-controlled, parallel-group, multicenter Phase 3 study to investigate the efficacy and safety of finerenone, in addition to standard of care, on the progression of kidney disease in patients with non-diabetic chronic kidney disease

Brief Title: Efficacy and safety of finerenone in patients with non-diabetic CKD

Rationale:

This is a randomized, double-blind, parallel-group study to investigate the efficacy and safety of finerenone, i.e., 10 mg and 20 mg once daily, compared with placebo, on the progression of non-diabetic chronic kidney disease in male and female patients over 18 years of age, over minimum of 32-months of study intervention per participant, in addition to standard of care.

Population:

This study will include patients with non-diabetic CKD of different disease etiology at risk of progressive renal function decline.

Objectives and Endpoints

Objectives	Endpoints
Primary	
To demonstrate that finerenone in addition to SoC, is superior to placebo in delaying the progression of kidney disease.	Mean rate of change as measured by the total slope of eGFR from baseline to Month-32.
Secondary	
- To demonstrate the beneficial effect of finerenone in addition to SoC as compared to placebo. - To assess the safety of finerenone in addition to SoC as compared to placebo	- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF or CV death - Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$ - Time to the composite of hospitalization for HF or CV death - Number of participants with TEAEs, TESAEs and AESI

AESI = Adverse events of special interest, CV = Cardiovascular, eGFR = Estimated glomerular filtration rate, HF = Heart failure, SoC = Standard of care, TEAEs = Treatment-emergent adverse events, TESAEs = Treatment-emergent serious adverse events

Overall Design:

This is a double blind parallel-group intervention study in non-diabetic chronic kidney disease patients with 2 treatment groups, including a placebo treatment group.

Participants who fulfill all eligibility criteria will be randomized in a 1:1 ratio to receive finerenone or placebo, in addition to standard of care:

- Finerenone 10 mg/OD or 20 mg/OD
- Placebo

All participants will receive blinded study intervention (i.e. finerenone 10 mg/20 mg or Placebo) OD on top of individually tolerated maximum labeled doses of either an ACEI or an ARB. The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/ 1.73m^2 (CKD-EPI), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/ 1.73m^2 (CKD-EPI). Subjects who start on the lower dose of 10 mg will be uptitrated to the target dose of 20 mg from Month 1 on treatment onwards provided potassium is ≤ 4.8 mmol/L, and eGFR decrease is less than 30% of the last measured value. Down-titration or interruption of study intervention is permitted at any time during the study for safety reasons and study intervention has to be interrupted if potassium is > 5.5 mmol/L. Visits will take place at screening, baseline, Month 1, Month 3 and from then every 3 months up to Month 12, and thereafter every 4 months until the planned end of treatment. The minimum per participant treatment duration will be 32 months. Additional visits will take place 4 weeks after each up-titration, interruption for > 7 days or treatment re-initiation, and 1 month after the end of treatment. eGFR as the primary endpoint measure and potassium as the key safety parameter as described will be assessed at each visit. UACR and UPCR will be collected at screening, baseline, at Month 1, Month 3, Month 6, Month 12, Month 24, at the end of treatment and 1 month after treatment discontinuation.

In order to offer participants and sites increased flexibility in trial conduct over the duration of this multi-year study, a "hybrid" DCT model may be implemented where local regulations and legislations allow. This model integrates telemedicine technology into certain parts of the clinical research process and supports management of research activities, including remote data collection. For the selected visits identified in the schedule of activities, remote visit activities such as Home Health Nurse visits, completed in conjunctions with a telehealth call and DtP study intervention shipment will be available for use. The sponsor will select qualified vendors to offer these services or, in some cases, with prior written sponsor approval, sites may conduct some of these remote visit activities themselves (e.g. home health nurse visits or DtP from the site to the participant).

Brief Summary:

The purpose of this study is to assess efficacy and safety of finerenone (10 mg and 20 mg, OD) compared to placebo in participants with non-diabetic chronic kidney disease. Study details include:

- Study duration per participant: from approximately 33 months to approximately 50 months (with an expectation of approximately 17 months recruitment duration).
- Treatment duration per participant: from 32 to up to approximately 49 months (with an expectation of approximately 17 months recruitment duration)
- Visit frequency: approximately once every 1-4 months

Number of Participants:

Approximately 3160 participants will be screened to achieve 1580 randomly assigned to study intervention and 1500 evaluable participants for an estimated total of 750 evaluable participants per intervention group.

Intervention Groups and Duration:

Study Periods	Duration
1. Screening	1. 1 to 14 days
2. Intervention	2. 32 to approximately 49 months
3. Follow-up	3. approximately 1 month
Total number of participants	Total study duration
1580 randomized	approximately 33 to 50 months

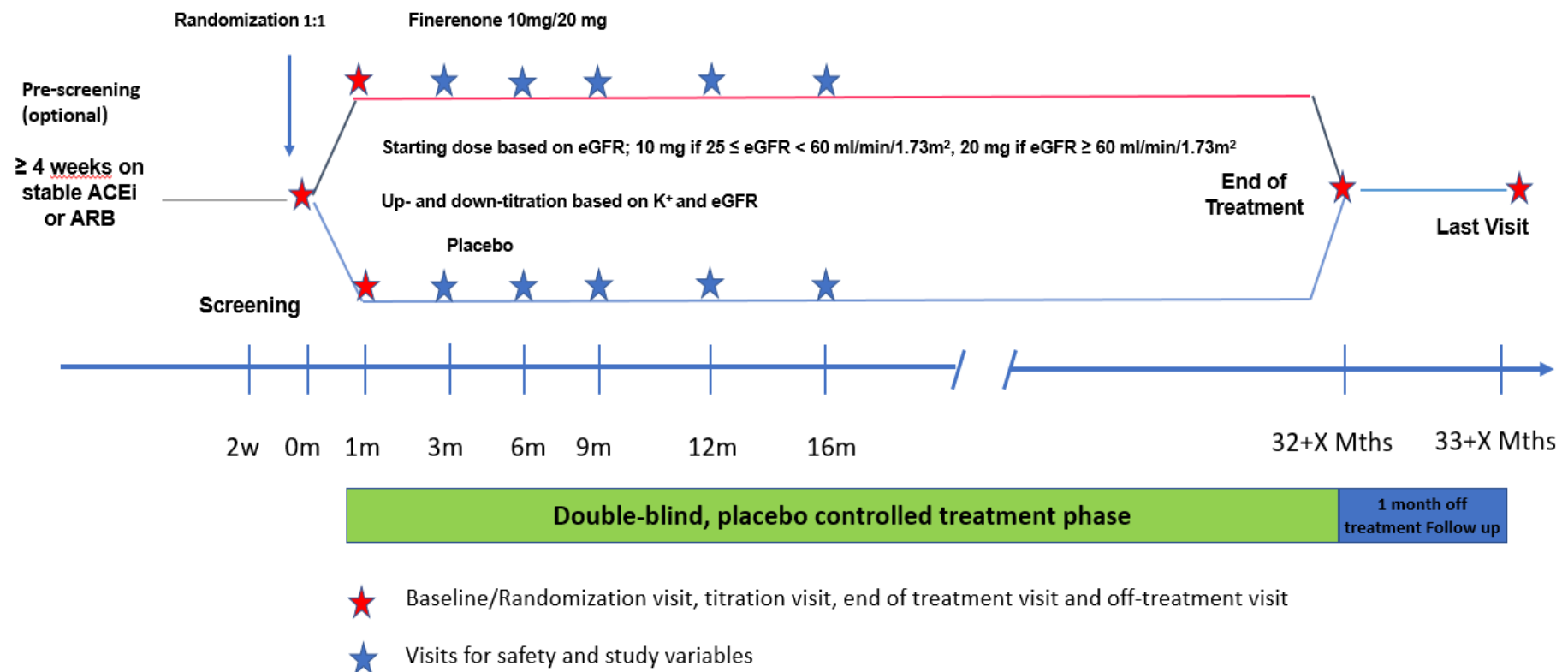
Intervention groups:

- Group 1: 10 mg to 20 mg of finerenone OD
- Group 2: placebo

Data Monitoring/Other Committee: Yes

1.2 Schema

Figure 1-1: Study Schema



Study duration and the number of study Visits will depend on the time of enrollment of the participant

ACEi = Angiotensin-converting enzyme inhibitor, ARB = Angiotensin receptor blocker, eGFR = Estimated glomerular filtration rate, K = potassium, m = Month, mths = Months, W = Week

1.3 Schedule of Activities (SoA)**Table 1-1: Schedule of Activities**

Procedure	pre- scree- ning ^a	Scree- -ning	Intervention Period (Months)								4-week safety visit	PD _b	EoS	Follow- up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post uptitration or reinitiation		at planned end of treatment	1 month after last dose	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff). At selected sites, UT visit can be performed at home if necessary. Samples will be sent to central lab.
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	
Visit Window (days)				±5	±5	±7	±7	±7	±7 ^e	±7	±7			±7	
Informed consent and eligibility check															
Pre-screening consent	●														
Informed consent		●													May be performed up to 30 days in advance of the screening visit
Informed consent for genetic sampling (optional)		●													Not relevant for China. Can be done at Visits 0 or 1
I/E criteria		●	●												
Baseline characteristics															
Demography		●													
Full physical examination with height and weight			●												
weight only		●		●	●	●	●	●	●	●		●		●	
Medical history		●													Including kidney biopsy results as available
Randomization			●												
Questionnaires															
EQ-5D-5L			●					●	● [*]			●			EQ-5D-5L Should be completed before any other evaluations. *End of treatment only

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up titration or reinitiation		at planned end of treatment	1 month after last dose	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff). At selected sites, UT visit can be performed at home if necessary. Samples will be sent to central lab.
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	
Visit Window (days)				±5	±5	±7	±7	±7	±7 ^e	±7	±7			±7	
Laboratory evaluations															
Serum or urine pregnancy test		•						•							WOCBP only. Urine or serum, as required by local regulations
Full clinical chemistry (incl eGFR and K ⁺); central lab		•													
Creatinine/eGFR ^d , K ⁺ ; Na ⁺ central lab			•**	•**	•**	•	•	•**	•	•		•*		•*	* If central laboratory sample cannot be taken, a local laboratory results may be used instead. **Na+ will be measured only at these visits.
Creatine/eGFR, potassium; local lab ^f		•*	•	•	•	•**	•	•	•	•**	•				*Only potassium sample needed ** At DCT visits local lab samples will not be collected
UACR/UPCR; central lab		•	•	•	•	•*		•*	•*			•		•	* At Visits 4, 6, 9 (months 6, 12, 24) and EoT only
UACR local screening	•														
Cystatin C ^d (from serum), central lab		•		•		•	•		•*	•					*Month 32 only
Biomarker ^d (serum [inc cystatin C], plasma and urine)			•		•			•	•*			•		•**	*At end of treatment only ** At DCT visits sample collection is not mandatory if swift sample

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up titration or reinitiation		at planned end of treatment	1 month after last dose	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff). At selected sites, UT visit can be performed at home if necessary. Samples will be sent to central lab.
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	
Visit Window (days)				±5	±5	±7	±7	±7	±7 ^e	±7	±7			±7	
															processing - according to sample handling documents - is not feasible.
Biomarker (blood for pharmacogenetics)			•												Not relevant for China. Optional, only participants who consented. Can be taken also on Visit 3.
PK blood sample				•	•	•*	•*	•*							See Table 1-2 and Section 8.4 for information on study intervention intake on these visits. *at Visits 4-6 only. PK sample will be taken at Visit 5 only if it has not been taken at Visits 2-4 (i.e. for a DCT visit).
Further safety evaluations and Study intervention															
12-lead ECG			•						•			•			
Vital signs		•	•	•	•	•	•	•	•	•	•	•		•	
Study intervention dispense/administration, or return			•	•	•	•	•	•	•*	•		•			*Study intervention will not be dispensed at EoT
DCT package provision				•								•			

Procedure	pre-screening ^a	Screening	Intervention Period (Months)								4-week safety visit	PD ^b	EoS	Follow-up	Notes
Month	Up to 30 days prior to screening	14 to 3 days before Day 1	Day 1	1	3	6, 16 and 28	9 and 20	yearly	32/ EoT ^c	40 and 44	28 days post up-titration or reinitiation		at planned end of treatment	1 month after last dose	At selected sites, Visits 4, 7, 10, 13 and FU can be carried out as 2-step remote visits (home nurse visit followed by telehealth call with site staff). At selected sites, UT visit can be performed at home if necessary. Samples will be sent to central lab.
Visit number		0	1	2	3	4, 7 and 10	5 and 8	6, 9 12 and 15	11, EoT	13 and 14	UT		EoS	FU	
Visit Window (days)				±5	±5	±7	±7	±7	±7 ^e	±7	±7			±7	
DCT package return														•	for participants enrolled in the DCT model
End of study phone call													•		Only for participants who have discontinued treatment prior to planned end of treatment who have not had a physical Month 32/EoT Visit after the declared end of study period. At the end of study AE/SAE/OEs will be collected by phone.
AE/ SAE/ OE and concomitant medication review		----->													

One month is counted as 30 days

AE = Adverse event, DCT = Decentralized clinical trial, ECG = Electrocardiogram, eGFR = estimated glomerular filtration rate, EoS = End of Study Visit, EoT = End of Treatment, EQ-5D-5L = EuroQoL 5-dimension 5-level questionnaire, FU = Follow-up, I/E = Inclusion/exclusion, K+ = potassium, Na+ = Sodium, OE = Outcome event, PD = premature discontinuation, PK = Pharmacokinetics, SAE = Serious adverse event, UACR = Urine albumin/creatinine ratio, UPCR = Urine protein/creatinine ratio, UT = Up-titration, WOCBP = Women of child-bearing potential

- Pre-screening visit is optional
- Premature discontinuation (PD) visit is to be carried out if the participant permanently discontinues the study intervention or the study prior to start of the End of Study Period. PD Visit should take place within 7 days from the last study drug intake. If more than 7 days has passed from the last study intervention intake, the participant may move directly to the FU Visit.
- The start of the end of study period will be declared when the last randomized participants reach their 32nd Month in the study. All participants already past month 32 prior to the date of the start of end of study period will attend EoT visits after this date. For any participant due their month 32 visit after the declared start of end of study period, these visits will be performed as EoT visits within the month 32 visit window.
- If central eGFR results or cystatin C/biomarker samples are missed at a scheduled visit, the patient should be retested within 30 days of the scheduled visit.
- Visit window applies to Visit 11 only.
- Local lab Potassium is not required if participant has prematurely discontinued from study intervention or is on a temporary interruption of study intervention.

Table 1-2: PK sampling schedule

Visit	V2	V3	V4	V5	V6	Notes
Month	1	3	6	9	12	
Study intervention intake on visit days	Taken at site	Taken at home	Taken at site	Taken at home or at site ^a	Taken at home	Document timing of last study drug intake before PK sampling. (i.e. for trough samples, the relevant intake time is from the previous day as intake at site is only after PK sampling)
PK sample predose (trough) ^c	•		•	•*		Document exact time of study intervention intake on the day prior to the visit and on the day of the visit. *PK sample will be taken at Visit 5 only if it has not been taken at Visits 2 or 4 (e.g. for a DCT visit). ^b
PK sample (peak, 1.5-10 hours after intake of study intervention) ^c		•		•*	•	Document exact time of intake on visit day. *PK sample will be taken at Visit 5 only if it has not been taken at Visit 3. ^b

DCT = Decentralized clinical trial, PK = Pharmacokinetics, V= Visit

a) If trough sample is to be taken, the study intervention should be taken at site after sampling. If a peak sample is to be taken, study intervention should be taken at home.

b) samples taken in error at visit 5, will be analyzed.

c) samples that were planned as peak but taken as trough, or vice versa, will be analyzed. It is important to document the actual time of last study intervention before the sample.

2. Introduction

Finerenone (BAY 94-8862) is a next-generation, oral, selective, non-steroidal MRA that is being developed for once daily treatment of non-diabetic chronic kidney disease.

2.1 Study Rationale

Diverse pathologies induce injury and inflammation resulting in a common pathophysiological mechanism in CKD where glomerular injury leads to proteinuria, interstitial inflammation, and, ultimately, fibrosis. Finerenone is a mineralocorticoid receptor antagonist (MRA) acting by targeting inflammatory and fibrotic pathways mediated by mineralocorticoid receptor overactivation in the heart and kidney ([Kolkhof and Barfacker 2017](#)). Finerenone protected from kidney function decline and reduced the risk of CV events in the phase 3 FIDELIO study in participants with CKD and T2D. Efficacy was observed independent of the hyperglycemic state ([Bakris et al. 2020](#)). It also reduced albuminuria in diabetic and non-diabetic participants with CKD in the ARTS ph2a study ([Pitt et al. 2014](#)). Based on the available pre-clinical and clinical data finerenone is believed to show efficacy in delaying the progression of kidney disease in diabetic and non-diabetic patients.

2.2 Background

In 2017, globally 1.2 million people died from CKD and the global all-age mortality rate from CKD increased 41.5% between 1990 and 2017. In 2017, 697.5 million cases of all-stage CKD were recorded accounting for a global prevalence of 9.1%. Globally, 4.6% of total mortality were caused by deaths due to CKD and cardiovascular disease attributable to impaired kidney function. High blood pressure accounted for the largest proportion of CKD burden in east Asia, eastern Europe, tropical Latin America, and western sub-Saharan Africa, whereas diabetes was the leading risk factor for CKD in all other regions. A higher prevalence of CKD stages 1–3 in females and higher mortality in males suggest that males progress to ESKD more rapidly ([GBD Chronic Kidney Disease Collaboration 2020](#)).

In populations with a high prevalence of diabetes and hypertension, it can be difficult to distinguish CKD due to hypertension and diabetes from CKD due to other disorders. In other countries, other causes of CKD may be as frequent as hypertension and diabetes or coexist with them ([KDIGO 2012](#)). The most common glomerulonephritis globally is IgAN with varying prevalence by geographic region showing a higher incidence in Pacific Asian regions ([Rodrigues et al. 2017](#)). After diabetes, hypertension is the second most common cause for newly requiring dialysis followed by glomerulonephritis ([KDIGO 2012](#), [Kramer et al. 2020](#), [Saran et al. 2019](#)). ACEI and ARB are considered standard of care for patients with non-diabetic CKD and albuminuria ([Kidney Disease Improving Global Outcomes \(KDIGO\) 2013](#)), however, existing evidence is mainly based on efficacy seen in diabetic patients ([Brenner et al. 2001](#), [Lewis et al. 2001](#)) and comparable small numbers of non-diabetic patients were studied. Recently completed studies ([Heerspink et al. 2020](#), [Perkovic et al. 2019](#)) showed a benefit of SGLT2is in patients CKD with and without diabetes.

Despite its relevance for global health, CKD often progresses slowly and mostly asymptotically until kidney failure occurs, limiting the development of new medications due to long study durations and high patient numbers ([Levey et al. 2020](#)). Addressing this problem, a scientific workshop was held by the National Kidney Foundation in collaboration with FDA and EMA to evaluate changes in UACR and GFR as surrogate endpoints for

kidney failure in clinical trials. Meta-analyses of clinical trials which were conducted around this workshop revealed that treatment effects on GFR slope from baseline (total) and from 3-month follow-up (chronic) of 0.5–1.0 mL/min/1.73 m² per year strongly predict benefits on clinical endpoints such as doubling of serum creatinine, GFR < 15 mL/min/1.73 m², or ESKD (Inker et al. 2019). Based on these analyses which were consistent across subgroups of different disease etiology and treatment it was concluded that total and chronic eGFR slope can play a role as an endpoint for clinical trials of kidney disease progression in CKD. Obtaining before-treatment eGFR and final eGFR after withdrawing treatment (off-treatment effect) for the reversibility of acute effects was considered supportive. Recently, surrogates for eGFR have been emerging as a more commonly accepted endpoint in phase 3 clinical trials (Komers et al. 2020, Torres et al. 2017).

Finerenone is already studied in more than 13,000 patients with CKD in T2D in the phase 3 FIDELIO-DKD and FIGARO-DKD outcome trials. In the completed positive FIDELIO-DKD trial, 5674 patients were treated for a median of 2.67 years to achieve an 18% lower hazard for the primary renal composite endpoint of kidney failure, a sustained eGFR decrease of $\geq 40\%$ from baseline for at least 4 weeks or renal death (7.59 and 9.08 per 100 patient years, respectively). The secondary renal endpoint that included a sustained eGFR decline of 57% or more was also achieved (HR 0.76 [95% CI, 0.65 to 0.90], $p=0.0012$; i.e. RRR of 24%) further supporting the renal benefit (Bakris et al. 2020). At 3 years a difference of 0.65 mL/min/1.73m² per year was seen between finerenone and placebo for the difference in total eGFR slope (from baseline to end of study) in the overall population, the chronic eGFR slope difference (calculated from Month 4 onwards) was 1.37 mL/min/1.73m² per year. This difference was more pronounced in patients with higher albuminuria (UACR ≥ 500 mg/g) independent of their baseline eGFR. In these patients, a total eGFR slope difference of > 0.7 mL/min/1.73m² per year was observed earlier, after 32 months (following the 4 monthly visit schedule).

These data show that the effect observed on the eGFR slope translated into a significant benefit on the primary endpoint and thus, confirms the predictive value of eGFR slope difference in a large finerenone outcome trial in patients with CKD and T2D.

Based on these findings and while the beneficial renal effect of finerenone has already been proven in a related population, this study will investigate the effect of finerenone in patients with non-diabetic CKD etiologies including hypertension and chronic glomerulonephritis such as IgAN and FSGS; disease etiologies such as polycystic kidney disease or active lupus nephritis requiring specific treatments will be excluded. Polycystic kidney disease has a primary disease mechanism which is not directly affected by MRA action. Kidney diseases with vasculitic etiology are likely to require immunosuppressive therapy and the study results might be confounded.

To also account for the initial decline, total eGFR slope will be used as the primary endpoint. Since the use of eGFR slope as the primary endpoint is most applicable to populations which show a certain degree of progression (Greene et al. 2019), only patients with albuminuria (UACR ≥ 200 mg/g), a key predictor of progression independent of disease etiology will be included in this study (Anderson et al. 2020, Sarnak and Astor 2011) and the majority of patients will have an UACR value ≥ 500 mg/g. Chronic eGFR slope (from Month 3 until end of treatment) will be assessed as well describing the chronic benefit of finerenone translating into long-term preservation of kidney function. In addition, to show the persistent benefit of finerenone after treatment discontinuation eGFR change from baseline to 1 month after end of treatment will be assessed. The composite of kidney failure, confirmed eGFR decrease $\geq 57\%$

from baseline, hospitalization for heart failure or CV death will be a secondary endpoint in the trial providing supportive clinical evidence.

2.3 Benefit/Risk Assessment

2.3.1 Risk Assessment

The main risk observed with finerenone in previous studies was hyperkalemia. The population in this study has an inherent risk of hyperkalemia due to their underlying disease (as serum potassium tends to increase with decreasing eGFR) and due to their standard of care therapy (ACEI/ARB) ([Bianchi and Regolisti 2019](#)). In FIDELIO-DKD, hyperkalemia increased by approximately two-fold with finerenone treatment (18.3% finerenone vs 9.0% placebo). However, most treatment-emergent hyperkalemia events were non-serious, mild or moderate in intensity, and resolved. There were no fatal hyperkalemia TEAEs. Treatment-emergent hyperkalemia AEs of relevant clinical consequence (i.e. that led to permanent study intervention discontinuation [2.3% vs 0.9% for finerenone vs placebo, respectively] or hospitalization [1.4 vs 0.3%]) constituted a small proportion. Further, there was no evidence for an increased incidence of any severe clinical cardiac manifestations of hyperkalemia (e.g. bradycardia, ventricular arrhythmia or sudden cardiac death). Overall, the risk of hyperkalemia was manageable and expected based on finerenone's mode of action.

An increase from baseline in mean serum potassium in the first month of treatment of approximately 0.2 mmol/L was observed in the finerenone arm compared to placebo, with stable mean measurements from Month 4 onwards in the finerenone arm. The between-group difference of 0.2 mmol/L was consistently observed across different subgroups of baseline serum potassium and eGFR at screening, including participants with eGFR 25 to <45 mL/min/1.73 m².

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
Study Intervention finerenone		
Hyperkalemia	- Incidence of TEAEs hyperkalemia: finerenone 18.3% vs placebo 9.0% - Incidence of TEAEs hyperkalemia leading to hospitalization: finerenone 1.4% vs. placebo 0.3% - Incidence of TEAEs hyperkalemia leading to permanent discontinuation: finerenone 2.3% vs placebo 0.9% - Incidence of TEAEs hyperkalemia with fatal outcome: finerenone 0% vs placebo 0% - Treatment emergent serum potassium (central lab) >5.5 mmol/L: finerenone 21.4% vs placebo 9.2%	- Dose initiation if K⁺ is ≤ 4.8 mmol/L at screening - Up-titration if K⁺ is ≤ 4.8 mmol/L and eGFR has not declined > 30% compared to the previous visit - Dose interruption if K⁺ is > 5.5 mmol/L, restart if ≤ 5.0 mmol/L - Exclusion of concomitant use of potassium sparing diuretics, MRAs, renin inhibitors - Exclusion of potent CYP3A4 inhibitors - Dosing will be based on eGFR and participants with a screening eGFR between 25 and < 60 mL/min/1.73m² will start with the lower dose of 10mg.

Potential Risk of Clinical Significance	Summary of Data/Rationale for Risk	Mitigation Strategy
	TE serum potassium (central lab) >6 mmol/L: finerenone 4.5% vs placebo 1.4%	
Study Procedures		
Participation in this study presents a minimal risk as the documented AEs observed in prior studies were mild and completely resolved.	Frequent blood sampling (by single vein puncture and/or indwelling cannula) may be accompanied by mild pain, hematoma, feeling faint or dizzy, and in rare cases inflammation of the vessel wall or injury of a nerve.	All study assessments will be carried out by trained clinical staff.

2.3.2 Benefit Assessment

In the recently completed FIDELIO-DKD study, Finerenone was superior to placebo in delaying the progression of kidney disease (HR 0.825 [95% CI, 0.732 to 0.928], $p=0.0014$; i.e. RRR of 18%). Primary endpoint in FIDELIO-DKD was a composite of kidney failure, a sustained decrease in eGFR of 40% or more from baseline, or renal death. The treatment effect was reflected both in reductions in kidney failure and in sustained eGFR decrease from baseline, and the benefit of treatment was sustained throughout the duration of the study. The absolute risk reduction was 3.4% at Month 36 for the primary renal endpoint, corresponding to an NNT with finerenone to prevent a primary endpoint event of 29 participants. The secondary renal endpoint that included a sustained eGFR decline of 57% or more in the FAS was also achieved (HR 0.76 [95% CI, 0.65 to 0.90], $p=0.0012$; i.e. RRR of 24%), adding to the robustness and internal validity of the study with respect to the primary outcome. The results for the primary and secondary renal composite endpoints were generally consistent across evaluated subgroups, including region, eGFR, UACR, HbA1c, systolic blood pressure and serum potassium at baseline ([Bakris et al. 2020](#)). Based on these findings and while the beneficial renal effect of finerenone has already been proven in a related population and is independent of the hyperglycemic state, this study 21177 will investigate the effect of finerenone in participants with non-diabetic CKD etiologies including hypertension and chronic glomerulonephritis. Evidence of efficacy is supported by data on albuminuria reduction in diabetic and non-diabetic patients from the phase 2a ARTS study as well as by preclinical data ([Barrera-Chimal et al. 2018](#), [Kolkhof et al. 2017](#), [Lattenist et al. 2017](#)).

2.3.3 Overall Benefit Risk Conclusion

Finerenone successfully delayed the onset of the composite of kidney failure, a sustained eGFR decrease of $\geq 40\%$ from baseline for at least 4 weeks or renal death in the phase 3 FIDELIO-DKD study. At 3 years a difference of 0.65 mL/min/1.73m² per year was seen between finerenone and placebo for the difference in total eGFR slope (from baseline to end of study) in the overall population, the chronic eGFR slope difference (calculated from Month 4 onwards) was 1.37 mL/min/1.73m² per year reflecting the long-term benefit. An increased risk of hyperkalemia was observed with finerenone, however, the majority of these events were non-serious and the number of events leading to clinical consequences was low. The analysis of data collected in the context of dose-titration and interruption indicates that

hyperkalemia with finerenone treatment is manageable using a serum potassium-guided dose titration regimen. This is supported by simulations of the exposure-serum potassium model, which demonstrated that even at higher exposures, no increased risk for hyperkalemia is observed when dose-titration is applied.

To reduce the risk of hyperkalemia in this study, only participants with a potassium value of ≤ 4.8 mmol/L at the screening visit will be included and the starting dose will be reduced to 10mg if eGFR is between 25 and < 60 mL/min/1.73m² at screening. Potassium and eGFR will be monitored at each visit and guide up-and down-titration in line with the dose titration scheme used in FIDELIO-DKD.

To date, ACEI and ARB constitute the only available treatment option to delay kidney disease progression for patients with non-diabetic CKD, in the near future, SGLT2i may become a new treatment option based on recently presented data. Based on its mode of action and the consistent benefit seen in a related population, finerenone is expected to show efficacy in patients with non-diabetic CKD. The main risk of finerenone treatment, hyperkalemia, has been proven to be manageable even in the advanced CKD population of FIDELIO-DKD; efficacy was also consistent in participants at higher risk (i.e. with lower eGFR, higher baseline potassium). Based on the available data and the high medical need in the studied population, inclusion of patients into this study is considered favorable.

3. Objectives, Endpoints and Estimands

Objectives	Endpoints
Primary	
To demonstrate that finerenone in addition to SoC, is superior to placebo in delaying the progression of kidney disease.	Mean rate of change as measured by the total slope of eGFR from baseline to Month-32.
Secondary	
- To demonstrate the beneficial effect of finerenone in addition to SoC as compared to placebo. - To assess the safety of finerenone in addition to SoC as compared to placebo	- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF or CV death - Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$ - Time to the composite of hospitalization for HF or CV death - Number of participants with TEAEs, TESAEs and AESI
Exploratory	
- To be supportive of the primary efficacy endpoint by further demonstrating the effect of finerenone in addition to SoC as compared to placebo. - To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (supportive of the key secondary endpoints) - To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (other exploratory endpoints)	- Chronic eGFR slope from Month-3 to end of planned treatment-period - eGFR change from baseline to 1 month after end of treatment (off-treatment) - Time to onset of kidney failure (ESRD or sustained eGFR decline below 15 mL/min/1.73m²) - Time to onset of ESRD (chronic dialysis or transplant) - Time to onset of sustained eGFR decrease below 15 mL/min/1.73m² - Time to onset of sustained eGFR decrease of $\geq 57\%$ from baseline - Time to CV death - Time to first hospitalization for HF or urgent heart failure visit - Number of participants with UACR reduction $\geq 30\%$ from baseline at Month 6 - Change in UACR from baseline - Time to the composite of sustained eGFR decrease of $\geq 40\%$ or kidney failure - Time to first CV event (CV death, non-fatal MI, non-fatal stroke or transient ischemic attack, hospitalization for HF or urgent HF visit) - Time to CV hospitalization - Time to all-cause mortality - Time to all-cause hospitalization - Number of participants with new diagnosis of atrial fibrillation or atrial flutter - Number of participants with new diagnosis of HF - Outpatient worsening of HF

Objectives	Endpoints
Exploratory: Biomarkers	
To further investigate the study intervention and similar drugs (e.g., mode-of-action-related effects, safety) and to further investigate pathomechanisms deemed relevant to CV/CKD and associated health problems	Various biomarkers (e.g., diagnostic, safety, pharmacodynamic, monitoring, or potentially predictive biomarkers)
Exploratory: Quality of Life	
To further investigate the quality of life parameters	Change from baseline in EQ-5D-5L

AESI = Adverse events of special interest, CKD = Chronic kidney disease, CV = Cardiovascular, eGFR = Estimated glomerular filtration rate, ESRD = End stage renal disease, EQ-5D-5L = EuroQol Group 5-dimension 5-level questionnaire, HF = Heart failure, MI = Myocardial infarction, SoC = Standard of care, TEAEs = Treatment-emergent adverse events, TESAEs = Treatment-emergent serious adverse events, UACR = Urinary albumin-to-creatinine ratio

Estimand

The attributes constituting the primary estimand are the following:

- Population: The population consists of non-diabetic CKD patients, further defined by the in- and exclusion criteria.
- Treatment: All participants will receive blinded study intervention (i.e. finerenone 10mg/20mg or placebo) OD on top of their individually tolerated maximum labeled doses of either an ACEI or an ARB.
- Variable: eGFR values over time collected every 1 to 4 months during the planned treatment period
- Intercurrent events: The three most important intercurrent events to consider are
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant, and
 - Premature treatment discontinuation/extended interruption of treatment.

For premature treatment discontinuation/extended interruption of treatment, a treatment policy strategy will be applied. In the treatment policy strategy, participants will be followed up for their planned treatment visits after discontinuing treatment, and eGFR values after discontinuation of treatment will be included in the analysis. Details on the incorporation of intercurrent events (ICEs) considered as death, initiation of chronic dialysis, or kidney transplant and as such addressed by the appropriate estimand strategy will be provided in the SAP.

- Population-level summary: Total eGFR slope from baseline to 32-months comparing the finerenone treatment group with the placebo group as estimated from a random coefficient linear spline mixed-effect model.

4. Study Design

4.1 Overall Design

This is a parallel-group intervention study in non-diabetic chronic kidney disease patients with 2 treatment groups, including a double-blinded placebo treatment group.

Pre-screening (Optional)

An assessment of UACR is permitted under the Pre-screening visit and may be measured locally. This is optional and may be performed if the investigator wishes to obtain a recent UACR value to help assess if a participant is likely to be eligible. Pre-screening UACR values cannot be used in lieu of the centrally assessed urinalysis at Screening. In the exceptional case where no previous evidence of albuminuria or proteinuria is documented in the participant's medical records, the pre-screening value can be used instead. In such situations, the screening visit can only take place 3 months after pre-screening.

Screening

Approximately 3160 participants will be screened to achieve approximately 1580 participants randomly assigned to study intervention.

Randomization

This study is a randomized, double-blind, placebo-controlled, parallel-group, multicenter study with a variable treatment duration of a minimum of 32 months, up to a maximum of approximately 49 months (final treatment duration will depend on recruitment time). Participants will be randomized 1:1 to either finerenone or placebo OD in a blinded fashion on top of their SoC therapy including an ACEI or an ARB.

Randomization will be stratified by screening UACR (≤ 1000 mg/g vs. > 1000 mg/g) and baseline SGLT2 inhibitor use (yes/no) to allow equal distribution of those characteristics between the treatment arms.

Treatment

All participants will receive blinded study intervention (i.e. finerenone 10mg/20mg or placebo) OD on top of an individually tolerated maximum labeled dose of either an ACEI or an ARB. The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/1.73m² (CKD-EPI), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/1.73m² (CKD-EPI). Participants who start on the lower dose of 10 mg will be up-titrated to the target dose of 20 mg from Month 1 on treatment onwards provided potassium is ≤ 4.8 mmol/L, and eGFR decrease is less than 30% of the last measured value. Down-titration or interruption of study intervention is permitted at any time during the study for safety reasons and study intervention has to be interrupted if potassium is > 5.5 mmol/L. Visits will take place at screening, baseline, Month 1, Month 3 and from then every 3 months up to Month 12. Thereafter, if End of Study has not been declared, participants will continue on 4-monthly visits until the End of Study.

Additional safety visits will take place 4 weeks after each up-titration or after restart of study intervention following treatment interruption for > 7 days. eGFR as the primary endpoint measure and potassium as the key safety parameter will be assessed at each visit, UACR and UPCR will be collected at timepoints shown in SoA. Up- and down-titration of study intervention will be based on local potassium and eGFR values.

End of treatment

The minimum planned treatment duration will be 32 months. All participants will stay in the study until the last randomized participant has reached their 32nd month of treatment. At this point, all participants remaining in the study and still on treatment will end treatment phase, and move to the follow-up phase. Thus, treatment duration will be from 32 to approximately 49 months depending on the recruitment time. Participants who have already discontinued treatment at that time, will be contacted via phone to collect outcome events and adverse events.

Follow-up

The follow-up visit will take place approximately 1 month after last administration of study intervention.

4.1.1 Decentralized Clinical Trial Model

In order to offer participants and sites increased flexibility in trial conduct over the duration of this multi-year study, a “hybrid” DCT model may be implemented where feasible and local laws/regulations allow.

4.1.1.1 Remote Visits

For selected visits as outlined in Section 1.3, remote visit activities are available for use in place of on-site visits. Where utilized, remote visits will be divided into two parts. For the first part, home nurses will visit the participant at home to complete the designated study procedures, such as the collection of lab samples and measurement of vital signs (see Section 6.4), or the option to visit a Patient Service Centre may be possible for collection of lab samples and measurement of vital signs in the US. For the second part of the visit (once the lab results are available from the samples collected by the home nurse), the Investigator, or designee, will conduct the specified study procedures with the participant via telehealth video call. The investigator will also review the investigational product dose. The relevant IxRS transaction will be made and resupply of study intervention requested. Study intervention will be shipped DtP in appropriate shipping containers (see Section 6 for more information). Due to the timelines associated with shipping new study intervention to the participants' home for remote visits, it is possible that study intervention resupply doesn't happen within the scheduled visit window. This is acceptable and not considered a protocol deviation.

4.1.1.2 Technology

At sites where local regulations permit remote visit activities, participants may have the option to participate in the remote visits using their own mobile device ('Bring Your Own Device'; BYOD) **or** be provided with a study-specific provisioned mobile device. If the participant wishes to use their own device, but their own smartphone model and/or operating system does not meet the minimum technology requirements, they will be provisioned with a mobile device.

- If the participant decides to receive or requires a provisioned mobile device, the device will be distributed to him/her by the Investigator and/or authorized Site representative at Visit 2 to facilitate their participation in the study.

- Participants who prematurely discontinue from treatment prior to this timepoint will instead have the provisioned device, if needed, provided to them at the Premature Discontinuation visit to facilitate telehealth video calls during the follow-up period.

If the participant decides to use his/her own mobile device, he/she will be provided with a study-specific mobile app to support the DCT elements of the study. This mobile app will be downloaded by the participant on his/her own smartphone. The study-specific mobile app will support the participant to complete study-specific activities such as:

- Receive reminders/notifications for timely activity completion
- Surveys (i.e. participant feedback)
- Scheduling and conducting Telehealth Video Calls with the Investigator and/or delegate

The investigator and authorized site representatives will have access to a website portal via unique access credentials to access the study data from the completed activities noted above.

4.2 Scientific Rationale for Study Design

Based on the effect seen in the recently completed phase 3 FIDELIO-DKD study, eGFR slope has been chosen to be the primary outcome measure in this label-expanding study in non-diabetic CKD. In the FIDELIO-DKD study, mean eGFR values at baseline were similar for finerenone and placebo. Similar to other compounds affecting the RAAS system, an initial acute eGFR decline of 2-3 mL/min/1.73m² was observed for patients on finerenone. From Month 4 onwards until the end of the study, the decline in eGFR was slower in the finerenone group than the placebo group indicating long-term protection of treatment with finerenone. The acute decline was most pronounced at month 1 and may even ameliorate earlier than month 4 which was the visit time point in FIDELIO-DKD. Thus, to describe the eGFR slope behavior over the first months precisely and adequately reflect the point of change (from acute to chronic slope), a 3 monthly visit schedule is planned during the first year. Thereafter, a 4-monthly visit schedule will continue until the planned end of treatment to enable continuous data collection and limit variability. After the 32 months fixed treatment period, treatment continuation for those randomized early will allow the collection of additional data on the chronic eGFR slope. A post-treatment visit will enable the assessment of a persistent effect on eGFR and UACR after discontinuation.

The approximate power (> 90%) calculation was based on a total slope difference of at least 0.7 mL/min/1.73m² per year at month 32 of treatment. The assumption and initial parameter estimate of total eGFR slope are based on the effect seen in FIDELIO-DKD subpopulations at the corresponding time point where eGFR values were collected following a 4 monthly visit schedule.

Upper thresholds of potassium (4.8 mmol/L) and lower thresholds for eGFR (25 mL/min/1.73m²) for study intervention initiation and dosing have been chosen as confirmed effective and safe in the recently completed FIDELIO-DKD 16244 study.

4.3 Justification for Dose

Doses of 10mg and 20mg have been confirmed to be effective and safe in the recently completed FIDELIO-DKD study in participants with CKD and T2D. 10mg will be the starting dose for participants with a lower eGFR of 25 to < 60mL/min/1.73m², and

participants will be up-titrated to the target dose of 20mg after one month on treatment. Throughout the study, up- and down-titration will be based on potassium and eGFR levels. The analysis of data collected in the context of dose-titration and interruption from the FIDELIO-DKD study indicates that hyperkalemia with finerenone treatment is manageable using a serum potassium-guided dose titration regimen.

4.4 End of Study Definition

The end of the study is defined as the date of the last visit of the last participant in the study globally.

The primary completion is defined as the date of the last on-treatment visit of the last participant in the study globally.

A participant is considered to have completed the study if he/she has completed all phases of the study, including the last visit.

5. Study Population

This study will investigate the effect of finerenone in patients with non-diabetic CKD including different disease etiologies like hypertensive nephropathy and chronic glomerulonephritis such as IgA nephropathy (IgAN) and focal segmental glomerulosclerosis (FSGS). Patients with autosomal dominant or autosomal recessive polycystic kidney disease requiring specific treatment will be excluded from the study as well as patients with lupus nephritis or ANCA vasculitis and patients receiving immunotherapy for other primary or secondary renal diseases within 6 months prior to enrollment. This will ensure the exclusion of patients at high risk of flares and likely to require immunosuppressive therapy. To ensure a pre-specified ratio for a population at risk of progressive renal function decline, the number of participants with eGFR ≥ 25 but < 60 mL/min/1.73 m² (CKD-EPI) and UACR ≥ 200 but < 500 mg/g is planned to be capped at approximately 10% of the total population. The upper limit of UACR will be 3500 mg/g excluding those participants with proteinuria in the nephrotic range.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

5.1 Inclusion Criteria

Participants are eligible to be included in the study only if all of the following criteria apply:

Age

1. Participant must be 18 years of age inclusive at the time of signing the informed consent.

Type of Participant and Disease Characteristics

2. A clinical diagnosis of chronic kidney disease and:
 - $\text{eGFR} \geq 25$ but $< 60 \text{ mL/min/1.73m}^2$ and UACR of ≥ 200 but $< 500 \text{ mg/g}$ at screening¹ or $\text{eGFR} \geq 25$ but $< 90 \text{ mL/min/1.73m}^2$ and UACR of ≥ 500 but $\leq 3500 \text{ mg/g}$ at screening, and
 - Documentation of albuminuria/proteinuria² in the participant's medical records at least 3 months prior to screening. If no value is available prior to pre-screening, the pre-screening could be used to identify participants.
3. Stable and max. tolerated labeled dose of an ACEI or ARB for at least 4 weeks prior to screening as documented in the participant's medical records
4. $\text{K}^+ \leq 4.8 \text{ mmol/L}$ at screening

Re-assessment is allowed once for individual lab values (UACR; eGFR, potassium). If one of the 3 UACR measurements is missing but the other 2 are valid, these values can be used to assess participant's eligibility for this study. Local laboratory value for potassium should be used for screening reference.

Sex and Contraceptive/Barrier Requirements

5. Male and/or female

Women of childbearing potential can only be included in the study if a pregnancy test is negative at the Screening Visit and if they agree to use adequate contraception as listed in Section 10.4. Contraceptive use by women should be consistent with local regulations regarding the methods of contraception for those participating in clinical studies. Women are considered not of childbearing potential if they fulfill the criteria displayed in Section 10.4.

Informed Consent

6. Capable of giving signed informed consent as described in Section 10.1.3 which includes compliance with the requirements and restrictions listed in the ICF and in this protocol

5.2 Exclusion Criteria

Participants are excluded from the study if any of the following criteria apply:

Medical Conditions

1. Established diagnosis of Type 1 or 2 Diabetes mellitus, or $\text{HbA1c} \geq 6.5\%$ (48 mmol/mol)
2. Autosomal dominant or autosomal recessive polycystic kidney disease
3. Lupus nephritis or ANCA-associated vasculitis or any other primary or secondary kidney disease requiring immunosuppressive therapy within 6 months prior to screening
4. History of organ transplantation

¹ capped at approximately 10% of the total population

² quantitative or semiquantitative measurement documented in the medical records

5. Acute kidney injury requiring dialysis within 6 months prior to screening
6. Uncontrolled arterial hypertension with mean sitting systolic blood pressure ≥ 160 mmHg, mean sitting diastolic blood pressure ≥ 100 mmHg at screening
7. Symptomatic heart failure with reduced ejection fraction with class 1A indication for MRAs
8. UACR > 3500 mg/g at the screening visit
Note: One re-assessment is allowed in case UACR is > 3500 mg/g in one of the three urine samples collected at the Screening Visit.
9. CV event within 3 months prior to screening (heart failure decompensation, myocardial infarction, stroke, transient ischemic attack, pulmonary embolism, elective coronary artery bypass grafting) or elective percutaneous coronary intervention within 1 months prior to screening
10. Known hypersensitivity to the study treatment (active substance or excipients)
11. Addison's disease
12. Hepatic insufficiency classified as Child-Pugh C

Prior/Concomitant Therapy

13. Concomitant therapy with eplerenone, spironolactone, esaxerenone, any renin inhibitor, sacubitril/valsartan combination, or potassium-sparing diuretic which cannot be discontinued at least 4 weeks prior to the Screening Visit
14. Concomitant therapy with both ACEI and ARBs in case one of those cannot be discontinued at least 4 weeks prior to the Screening Visit.
15. Concomitant therapy with potent CYP3A4 inhibitors or moderate and potent inducers (to be stopped at least 7 days before randomization) (see Section 10.6)

Prior/Concurrent Clinical Study Experience

16. Previous assignment to treatment during this study
17. Simultaneous participation in another interventional clinical study (e.g. Phase 1 to 3 clinical studies) or treatment with any investigational medicinal product within 30 days prior to randomization

Other Exclusions

18. Any other condition or therapy, which would make the participant unsuitable for this study and will not allow participation for the full planned study period (e.g. active malignancy or other condition limiting life expectancy to less than 12 months)
19. Any other history, condition, therapy, or uncontrolled intercurrent illness which could in the opinion of the investigator affect compliance with study requirements.
20. Pregnant or breast-feeding or intention to become pregnant during the study
21. Close affiliation with the investigational site; e.g. a close relative of the investigator, dependent person (e.g. employee or student of the investigational site).

5.3 Lifestyle Considerations

No restrictions to the participant's customary lifestyle are required.

5.4 Screen Failures

Screen failures are defined as participants who consent to participate in the clinical study but are not subsequently randomly assigned to study intervention. A minimal set of screen failure information is required to ensure transparent reporting of screen failure participants to meet the Consolidated Standards of Reporting Trials (CONSORT) publishing requirements and to respond to queries from regulatory authorities. Minimal information includes demography, screen failure details, and eligibility criteria.

Sites will also have the option of a pre-screening visit to examine UACR. This UACR value is not valid to determine eligibility, and is only to be used to assess whether the participant may progress to a formal screening visit. Participants will be required to provide informed consent for this UACR test, and very limited data will be reported. Such pre-screening failures will not be considered as part of the study population. See Section 1.3 for assessments at the pre-screening visit.

Individuals who do not meet the criteria for participation in this study (screen failure) may be re-screened once. The following conditions are pre-requisites of re-screening:

- A minimum of 1 month between the initial Screening Visit and re-screening.
- Before re-screening, new documented informed consent must be obtained
- Allocation of a new participant identification number
- All assessments for the study must be repeated

5.5 Criteria for Temporarily Delaying Randomization/ Study Intervention Administration

The following conditions may allow a participant to be randomized or started on study intervention once the conditions have resolved and the participant is otherwise eligible:

- Current febrile illness (temperature 38.0°C [100.4° F]) or other acute illness within 48 hours before study intervention administration.
- Pre-screening showing albuminuria, if not documented previously in the participant's health records. In this case, the screening visit may be delayed by at least 3 months for the inclusion criterion 2 to be fulfilled.

6. Study Intervention(s) and Concomitant Therapy

Study intervention is defined as any investigational intervention(s), marketed product(s), placebo, or medical device(s) intended to be administered to a study participant according to the study protocol.

6.1 Study Intervention(s) Administered**Table 6-1: Study intervention**

Arm Name	Finerenone	Placebo
Intervention Name	Finerenone/ BAY 94-8862	Placebo for Finerenone
Type	drug	drug
Dose	tablet	tablet
Formulation		
Unit Dose Strength(s)	10mg and 20mg	N/A
Dosage Level(s)	10mg once daily or 20mg once daily	once daily
Route of Administration	oral	oral
Use	experimental	placebo
Packaging and Labeling	Study intervention will be provided in a plastic bottle. Each bottle will be labeled as required per country requirement	Study intervention will be provided in a plastic bottle. Each bottle will be labeled as required per country requirement

Study intervention will be dispensed at the study visits summarized in SoA. Returned study intervention should not be re-dispensed to the participants.

As shown in [Table 6-2](#), the starting dose of finerenone will depend on the eGFR value at the Screening Visit.

Dosage of study intervention

At randomization, participants with an eGFR between 25 to < 60 mL/min/1.73m² at the Screening Visit (central laboratory results) will be assigned to the lower dose of finerenone (10 mg) or placebo in addition to SoC, whilst participants with an eGFR ≥ 60 mL/min/1.73m² will be assigned to the higher dose of finerenone (20 mg) or placebo in addition to SoC.

Table 6-2 Dosage of study intervention for administration

eGFR value at the Screening Visit, based on central laboratory results:	25 to < 60 mL/min/1.73m ²		≥ 60 mL/min/1.73m ²	
Participant randomized to group:	<u>Finerenone</u>	<u>Placebo</u>	<u>Finerenone</u>	<u>Placebo</u>
Receives	10 mg finerenone OD + SoC	Placebo OD + SoC	20 mg finerenone OD + SoC	Placebo OD + SoC
Study intervention intake	One tablet of study intervention once daily, preferably in the morning at approximately the same time each day. The study intervention can be taken with a glass of water, with or without food. Note: Study intervention will be administered at home, except on the days of Visit 2 and Visit 4 when the tablet will be taken at the study site.			

Table 6-2 Dosage of study intervention for administration

eGFR value at the Screening Visit, based on central laboratory results:	25 to < 60 mL/min/1.73m ²		≥ 60 mL/min/1.73m ²	
Participant randomized to group:	<u>Finerenone</u>	<u>Placebo</u>	<u>Finerenone</u>	<u>Placebo</u>
Missed intake	• If >8 hours before the next scheduled dose, the participant should take one tablet of study intervention as soon as possible. • If ≤8 hours of the next scheduled dose, the participant should wait and take the next tablet of study intervention at the usual time.			
Up-titration of dose Allowed from Visit 2 (Month 1) onwards provided that: • Potassium is ≤4.8 mmol/L ^a • eGFR decrease is less than 30% below the value measured at the last scheduled visit ^a • Must be documented in eCRF	20 mg finerenone OD, maintain SoC	Sham-titrate, maintain SoC	Not applicable	Not applicable
Down-titration of dose • Only for safety reasons • Allowed any time during the study (e.g. between scheduled visits) • Must be documented in eCRF • An unscheduled safety visit is performed within an adequate timeframe proposed by the investigator	• If at higher dose of study intervention, down-titrate to lower dose of study intervention, maintain SoC • If at lower dose of study intervention, interrupt study intervention, maintain SoC			

Abbreviations: eCRF = electronic case report form; eGFR = estimated glomerular filtration rate; OD = once daily; SoC = standard of care

^a Potassium and eGFR according to local laboratory values

Up- and down-titration of study intervention

The investigator is encouraged to up-titrate the dose of study intervention at any time once the participant has been on a stable dose for 4 weeks (± 7 days) (e.g. from Month 1 onwards for participant starting study intervention on the lower dose) either at a regular visit or an 4-week safety Visit. Up-titration can be performed provided:

- the potassium concentration (local laboratory value) is ≤ 4.8 mmol/L,

AND

- eGFR decrease (local laboratory value) is less than 30% of the value measured at the last regular visit.

Participants who started with or were up-titrated to the target dose (20 mg OD) of the study intervention but who do not tolerate this dose may be down-titrated at any point during the study, including between-scheduled visits if required for safety reasons. These participants

may be up-titrated again based on the rules provided above. If the participant is already at the lower dose, the study intervention can be interrupted at the investigator's discretion (refer to Section 6.5 for guidance). If the study intervention is interrupted for more than 7 days, the study intervention should be re-started at the lower dose (10 mg OD).

Subsequent to an up-titration or re-start of study intervention after interruption of study intervention intake for more than 7 days, the investigator should perform a 4-week safety Visit (± 7 days) after titration or restart, in order to monitor potassium levels and renal function (see Section 1.3). If a regular study visit will be scheduled to take place 4 weeks (± 7 days) after up-titration, the monitoring of potassium and renal function is assured and no 4-week safety Visit has to be performed in addition. With down-titrations, the unscheduled safety visit may need a smaller time window and should be performed at the investigator's discretion.

At any point during study intervention treatment, if blood potassium is found to be elevated, study intervention treatment should be adjusted according to guidance provided in Section 6.5.

Both up- and down-titrations will be at the investigator's discretion and will depend on the general clinical condition of the individual participant. The investigator should attempt to reach the maximum (target) dose of study intervention as long as safety is not compromised and the guidance for the up- and down titration is followed (See Section 6.5).

If up-titration will take place subsequent to a remote (+ Telehealth Video Call) visit, the participant must be instructed to continue taking the current dose until the new study intervention supply is received via DtP shipment. Therefore, actual up-titration of dose will take place several days after the completion of the Telehealth Video Call part of the visit. The start date of the new dose must be documented in the eCRF.

If laboratory results subsequent to a remote (+Telehealth Video Call) indicate the need for a down-titration, the participant must visit the site to return their current study intervention and be supplied with new intervention. Down-titration must always be performed at an on-site visit to allow participants to be supplied with the new lower dose of study intervention treatment as soon as possible.

All titrations, including the reasons for down-titration or for not dispensing the 20 mg dose, must be documented in the eCRF.

Medical Devices

No medical devices will be provided by the Sponsor in this study.

6.2 Preparation/Handling/Storage/Accountability

1. The investigator or designee must confirm study interventions have been received and, if a temperature data logger has been included, appropriate temperature conditions have been maintained during transit. Any discrepancies are to be reported and resolved before use of the study interventions
2. Only participants randomized in the study may receive study intervention and only authorized site staff (or remote pharmacy designee if applicable, where local regulations permit, for any sites utilizing remote visits) may supply or administer study intervention. All study intervention must be stored in a secure, environmentally controlled, and monitored (manual or automated) area in accordance with the labeled storage conditions with access limited to the investigator and authorized site staff.

3. The investigator or the head of the institution (where applicable) is responsible for study intervention accountability, reconciliation, and record maintenance (i.e., receipt, reconciliation, and final disposition records). Drug returns, reconciliation and destruction return information will be captured in IxRS. For sites where DtP is completed by site staff (including site pharmacy) shipping study intervention directly to individual study participants, the investigator retains responsibility for these activities. The site must receive written approval from the sponsor before engaging in direct to participant shipments that originate from the site.

For sites utilizing remote visits, where DtP distribution of study intervention occurs via a remote pharmacy designee, the investigator delegates distribution responsibility to the sponsor. In such circumstances, the investigator maintains responsibility and control for dispensing via IxRS by confirming whether or not DtP distribution of study intervention can be triggered. Where a remote pharmacy designee is used, the sponsor contracts a distribution depot with pharmacy licensing and delegates responsibility for storage and handling via DtP distribution of study intervention until handover to the participant. Participants who have a remote visit and receive study intervention via DtP will return all unused investigational product and empty bottles to the study site at their next scheduled on-site visit.

4. Further guidance and information for the final disposition of unused study interventions are provided in investigator site file.

6.3 Measures to Minimize Bias: Randomization and Blinding

All participants will be centrally assigned to randomized study intervention using IxRS. Before the study is initiated, the telephone number and call-in directions for the IxRS and/or the log in information and directions for the IxRS will be provided to each site.

Treatment allocation will be done according to a computer generated randomization list specified by the sponsor or delegate.

IxRS will be programmed with blind-breaking instructions. In case of an emergency, the investigator has the responsibility for determining if unblinding of a participant's intervention assignment is warranted. If the investigator is unavailable, and a treating physician not associated with the study requests emergency unblinding, the emergency unblinding requests are forwarded to the emergency medical advice 24 hours/7 day service. Participant safety must always be the first consideration in making such a determination. If the investigator decides that unblinding is warranted, the investigator should make every effort to contact the sponsor prior to unblinding a participant's intervention assignment unless this could delay emergency treatment of the participant. If a participant's intervention assignment is unblinded, the sponsor must be notified within 24 hours after breaking the blind. The date and reason that the blind was broken must be recorded in the source documentation and case report form, as applicable.

Sponsor pharmacovigilance staff may unblind the intervention assignment for any participant with an SAE. If the SAE requires that an expedited regulatory report be sent to one or more regulatory agencies, a copy of the report, identifying the participant's intervention assignment, may be sent to investigators in accordance with local regulations and/or sponsor policy.

6.4 Study Intervention Compliance

When participants self-administer study intervention(s) at home, compliance with study intervention will be assessed at each visit. Compliance will be assessed by direct questioning and counting of returned tablets during the site visits.

An adequate record of receipt, distribution, and return/destruction of all study treatment must be captured on the dispensing log and in the IxRS. Intervention start and stop dates, including dates for intervention delays and/or dose reductions will also be recorded.

Overall compliance with study intervention intake should be between 80% and 120% of the scheduled dose at the end of study intervention treatment.

Deviation(s) from the prescribed dosage regimen should be recorded. Study intervention tablets not returned will be considered to have been taken unless otherwise specified. At the end of the study, any remaining drugs will be collected and returned to the sponsor, destruction depot or destroyed at the site. Any discrepancies between the returned and expected returned study interventions should be explained.

6.5 Dose Modification

The dose of study intervention should be adjusted (up- or down-titration) on the basis of potassium and eGFR levels obtained from the local laboratory, except for remote visits where central lab values are used.

Monitoring of Blood Potassium and Dose Adjustment

Hyperkalemia is a frequent event in participants with CKD with decreasing eGFR, and one of the main risks identified from previous studies with MRAs including finerenone. In FIDELIO-DKD a 2-fold increase in treatment-emergent hyperkalemia was observed for participants treated with finerenone compared to those treated with placebo which was consistent throughout subgroups. However, numbers of events leading to discontinuation and hospitalization were low and no fatal events were reported. Potassium will be monitored closely during this study, with assessments of potassium concentration scheduled at visits specified in Section 1.3 SoA.

During treatment with study intervention, at on-site visits all assessments of potassium, will be performed at both local and/or central laboratories. Re-tests will be performed by local laboratory only, except for DCT visits, where central laboratory can be used. Due to sample handling and extended transportation time to the central laboratory, falsely elevated serum potassium values in centrally (but not locally) analyzed samples may be observed. Since the value of potassium in the local sample is usually not affected by this and results from the local laboratory are available earlier, local samples will be taken not only for safety purposes but also for dose adjustments.

Re-checking of potassium values must be done at the latest within 72 hours of the investigator being aware of results that need confirmation (> 5.5 mmol/L). The investigator can perform a re-test at any time if they consider it necessary to confirm the potassium concentration of the local or the central sample. For remote visits if the central laboratory serum potassium result

is > 5.5 mmol/L following blood sampling during a remote visit, as soon as the investigator is aware of the result, the participant should be contacted promptly and instructed to withhold further study intervention until the second sample can be collected. Potassium should be rechecked within 72 hours of initial potassium result awareness.

Adjustment (lowering) of participant's maximum tolerated dose of baseline ACEI/ARB medication should not be made solely for the purpose of continuing with the same study intervention dosing.

Adjustment of dose after start of study intervention intake based on blood sample potassium levels is provided in [Table 6-3](#).

Table 6-3 Potassium from blood sample: guidance for dose adjustment at Visit 2 and subsequent visits

Blood potassium (mmol/L)	Action
First sample:	
≤4.8	If on lower dose of study intervention, up-titrate to higher dose. If on higher dose of study intervention, continue on the same dose.
4.9 to 5.5	Continue on the same dose.
>5.5	Withhold study intervention and re-check potassium within 72 hours.
Second and subsequent samples:	
≤5.0	Re-start study intervention at lower dose.
>5.0	Continue to withhold study intervention; continue to monitor potassium and restart study intervention at the lower dose only if potassium is ≤5.0

NOTE: lower dose = 10 mg once daily; higher dose = 20 mg once daily.

The following aspects have also to be taken into consideration:

- Potassium should be measured 4 weeks ($\pm$ 7 days) after re-starting treatment or dose adjustment, especially after up-titration.
- If central laboratory results for potassium differ from local laboratory results to such an extent that the investigator is uncomfortable with the dose adjustment guidance shown in [Table 6-3](#), blood sampling may be repeated at the investigator's discretion.
- If the participant is already on the lower dose of study intervention but hyperkalemia recurs soon after a previous event of hyperkalemia with interruption of study intervention, and there is no explanation for the recurring hyperkalemia event other than intake of study intervention, permanent discontinuation of study intervention is recommended.
- If blood potassium is > 6.5 mmol/L, an ECG should be obtained.

6.6 Continued Access to Study Intervention after the End of the Study

No intervention is planned following the end of the study.

6.7 Treatment of Overdose

In this trial, an overdose is defined as any occasion when the participant has taken (accidentally or intentionally) any dose higher than the maximal target dose prescribed in the protocol. Overdose of study interventions should be treated as clinically indicated based on symptoms and signs. There is no specific reversal agent for finerenone and the Sponsor does not recommend specific treatment for an overdose.

In the event of an overdose, the investigator should:

- Contact the Medical Monitor.
- Evaluate the participant to determine, whether study intervention should be interrupted or whether the dose should be reduced. This may happen in consultation with the Medical Monitor.
- Closely monitor the participant for any AE/SAE and laboratory abnormalities as clinically indicated (at least 1 day).
- Document the quantity of the excess dose as well as the duration of the overdose in the eCRF.

Decisions regarding dose interruptions or modifications will be made by the investigator in consultation with the Medical Monitor based on the clinical evaluation of the participant.

Overdose per se will not be reported as an AE/SAE unless it is an intentional overdose taken with possible suicidal/self-harming intent (see Section [10.3.1](#)).

6.8 Concomitant Therapy

Relevant medications or vaccines (including over-the-counter or prescription medicines, recreational drugs, vitamins, and/or herbal supplements) or other specific categories of interest that the participant is receiving at the time of enrollment or receives during the study must be recorded along with:

- Reason for use
- Dates of administration including start and end dates
- Dosage information including dose and frequency
- Route of administration

The Medical Monitor should be contacted if there are any questions regarding concomitant or prior therapy.

All participants should be treated according to regional standards of care for CV risk factors (e.g., blood pressure, lipids, and antithrombotic treatment), and CKD complications (hyperphosphatemia, hyperparathyroidism, hyperkalaemia, acidosis and renal anaemia).

6.8.1 Standard of Care (SoC) medications

To be eligible, the participant needs to be on stable and for the participant maximum tolerated labelled daily dose of ACEI or ARB for at least 4 weeks before the screening visit

The maximum tolerated labeled dose for ACEIs or ARBs should be the highest dose within the ranges shown in [Table 6-4](#) or according to local labels, as applicable, which the participant can safely tolerate. This dose should not be below the minimum labeled dose in order to maximize therapeutic benefit of background SoC.

Table 6-4 Dose ranges of ACEIs and ARBs for adults

Angiotensin-converting enzyme inhibitors		Angiotensin receptor blockers	
Generic name	Dose Range (mg/day)*	Generic name	Dose Range (mg/day)*
Benazepril	20 – 40	Candesartan	16 – 32
Captopril	50 – 450	Eprosartan	400 – 800
Cilazapril	1 – 5	Irbesartan	150 – 300
Enalapril	10 – 40	Losartan	50 – 100
Fosinopril	20 – 80	Olmesartan	20 – 40
Lisinopril	20 – 40	Telmisartan	40 – 80
Moexipril	7.5 – 30	Valsartan	80 – 320
Perindopril	4 – 16		
Quinapril	20 – 80		
Ramipril	2.5 – 20		
Trandolapril	2 – 4		

Source: Adapted from Table 28 of reference ([Diabetes_&_CKD_Work_Group 2007](#))

If the maximum labeled dose cannot be reached, the reason(s) should be documented in the eCRF.

The dosage of SoC therapies should not be reduced in order to solely facilitate maintenance of study intervention.

6.8.2 SGLT2 inhibitors

If the participant is assessed by the investigator as having an indication for an SGLT2 inhibitor, it should be started preferably prior to inclusion into the study, and the participant should be on a stable dose for at least 1 month prior to screening.

In the exceptional case where the participant is started on an SGLT2 inhibitor after randomization, then participant must be on a stable dose of study intervention for at least 1 month before SGLT2 inhibitor initiation.

6.8.3 Potassium supplements and lowering agents

Potassium supplementation can be prescribed during the course of the study. To assure the potassium value is within the normal range, investigators are advised to monitor potassium carefully at each visit and to adapt the dose of potassium supplementation accordingly.

Potassium-lowering agents (e.g. sodium polystyrene sulfonate, calcium polystyrene sulfonate) are allowed to be started during treatment with study intervention.

Any use of potassium supplementation and potassium-lowering agents must be documented in the eCRF.

6.8.4 Low dose steroids

Immunosuppressive therapy, including therapy with high dose steroids, can be initiated during the trial if deemed necessary by the PI as long as it is not listed as a prohibited concomitant medication (see Section 6.8.5). Topical, inhaled or low dose steroids (dose equivalent of daily 5-10 mg of prednisolone) are allowed at any time if used for non-renal indication.

6.8.5 Prohibited concomitant medication

The following therapies are NOT permitted:

- Concomitant therapy with eplerenone, spironolactone, esaxerenone, or any other mineralocorticoid receptor antagonist, any renin inhibitor, or potassium-sparing diuretic which cannot be discontinued at least 4 weeks prior to the Screening Visit and during study intervention treatment
- Concomitant therapy with both an ACEI and an ARB which cannot be discontinued at least 4 weeks prior to the Screening Visit and during study intervention treatment
- Concomitant therapy with an ARNI (valsartan/sacubitril) which cannot be discontinued at least 4 weeks prior to the screening visit. ARNI can be initiated during the trial, if deemed necessary by the investigator, after discontinuation of the ACEI or ARB.
- Concomitant therapy with potent cytochrome P450 isoenzyme 3A4 (CYP3A4) inhibitors or moderate and potent inducers must be stopped at least 7 days before randomization and is not allowed to be started during study intervention treatment.

A list of the most common medication regarded as potent CYP3A4 inhibitors or moderate and potent inducers is provided in Section 10.6.

Prior medication that was discontinued to comply with the eligibility criteria of the study should also be recorded in the eCRF.

Caution: Increases in finerenone exposure in combination with moderate CYP3A4 inhibitors cannot be excluded. Examples of moderate CYP3A4 inhibitors are: amiodarone, aprepitant, bicalutamide, chloramphenicol, imatinib, mifepristone, norfloxacin, tacrolimus, verapamil, lapatinib, dasatinib, and nilotinib.

Caution should also be exercised with non-steroidal anti-inflammatory agents or trimethoprim or trimethoprim / sulfamethoxazole. Finerenone treatment may be interrupted before trimethoprim or trimethoprim / sulfamethoxazole is started.

6.8.6 Rescue Medicine

Not applicable

7. Discontinuation of Study Intervention and Participant Discontinuation/Withdrawal

Table 7-1: Discontinuation of study intervention and participant discontinuation

Study intervention discontinuation	Participant discontinuation
Must be permanently discontinued	
Pregnancy of the study participant	At their own request or at the request of their legally acceptable representative. At any time during the study and without giving reasons, a participant may decline to participate further. The participant will not suffer any disadvantage as a result
The investigator is of the opinion that continuation of treatment with study intervention is harmful to the participant's well-being	
The randomization code is broken by the investigator, or other responsible person, when knowledge of the participant's treatment is required	
Any investigational drug other than the study intervention is used.	
May be permanently discontinued	
Any suspected drug-related AE or SAE	If, in the investigator's opinion, continuation of the study would be harmful to the participant's well-being
If any exclusion criterion applies during treatment	At the specific request of the sponsor and in liaison with the investigator.
If a significant violation of the protocol occurs, as defined by the sponsor and the coordinating investigator.	

AE = Adverse event, SAE = Serious adverse event

7.1 Discontinuation of Study Intervention

In rare instances, it may be necessary for a participant to permanently discontinue study intervention. If study intervention is permanently discontinued, the participant will remain in the study and continue with the regular study visit schedule until the end of the study. Additionally, post treatment eGFR, UACR and UPCR assessments will be performed after treatment discontinuation. See the SoA for data to be collected at the time of discontinuation of study intervention and follow-up and for any further evaluations that need to be completed.

7.1.1 Temporary Discontinuation of Study Intervention

Interruption of study intervention is permitted at any time during the study for safety reasons. See Section 6.5 for guidance.

Upon temporary interruption of the study intervention due to hyperkalemia, eGFR decrease, (S)AE, OE, intolerability or any other reason, intake should be resumed as soon as medically

acceptable at the discretion of the investigator. There is no defined maximum time limit for temporary interruption. In all cases, the reason for study intervention interruption must be recorded in the eCRF and the participant's medical records.

If the study intervention is interrupted for more than 7 days, the re-start should be performed at the 10mg dose and the investigator should schedule a 4-week safety visit ($\pm$ 7 days) in order to monitor potassium levels and renal function (see [Table 6-2](#)). If a regular visit will be scheduled to take place 4 weeks $\pm$ 7 days after up-titration or re-start, the monitoring of potassium and renal function is assured and no 4-week safety visit has to be performed in addition.

7.1.2 Permanent Discontinuation of Study Intervention

- In general, participants who permanently discontinue study intervention are expected to continue in the study and to attend all protocol specified study visits, and should be encouraged to perform the required assessments described in the visit schedule. PK sampling and pregnancy testings as well as local laboratory sampling will not be continued.
- If a participant no longer on study intervention is unable to attend the clinic for a study visit, and a remote visit is not possible, a telephone consultation may be performed to determine if relevant health events / endpoints (e.g. development of CV or renal complications) have occurred.
- Ideally, a face to face visit should be performed at least once a year. Expected frequency of telephone contacts should be in line with the standard visit schedule.
- Ad hoc additional telephone contacts may also be requested (e.g. close to the end of the study). These may be made to the participant themselves or to other contact as provided by the participant, e.g. a next of kin, primary physician (or local equivalent).

7.2 Participant Discontinuation/Withdrawal from the Study

Withdrawal from treatment, and withdrawal from the study visits do not constitute withdrawal of consent. The definition of withdrawal of consent is deemed to be when the participant actively refuses any further collection of data. However, vital status should still be collected from public sources, if allowed according to local regulations. This is crucial to assure data completeness. Withdrawal of consent should only occur in rare and exceptional circumstances.

In this study, eGFR values are crucial to the analysis and must be collected until the planned end of the study, as participants will still be part of the study even if participants are no longer taking study intervention. Therefore, all efforts will be taken to motivate participants to comply with all study procedures and to continue to be followed until the end of the study. The same applies to outcome events and vital status data.

- A participant may withdraw from the study at any time at his/her own request, or may be withdrawn at any time at the discretion of the investigator for safety, behavioral, or compliance reasons. This is expected to be uncommon.

- At the time of discontinuing from the study, if possible, a PD visit should be conducted, as shown in the SoA. See SoA for data to be collected at the time of study discontinuation and follow-up and for any further evaluations that need to be completed.
- The participant will be permanently discontinued both from the study intervention and from the study at that time.
- If the participant withdraws consent for disclosure of future information, the sponsor may retain and continue to use any data collected before such a withdrawal of consent.

7.3 Lost to Follow Up

A participant will be considered lost to follow-up if he or she repeatedly fails to return for scheduled visits and is unable to be contacted by the study site.

The following actions must be taken if a participant fails to return to the clinic for a required study visit:

- The site must attempt to contact the participant and reschedule the missed visit as soon as possible and counsel the participant on the importance of maintaining the assigned visit schedule and ascertain whether or not the participant wishes to and/or should continue in the study.
- Before a participant is deemed lost to follow up, the investigator or designee must make every effort to regain contact with the participant (where possible, telephone calls and, if necessary, a certified letter to the participant's last known mailing address or local equivalent methods). These contact attempts should be documented in the participant's medical record.
- Should the participant continue to be unreachable, he/she will be considered to have withdrawn from the study.
- Site personnel, or an independent third party, will attempt to collect the vital status of the participant within legal and ethical boundaries for all participants randomized, including those who did not get study intervention. Public sources may be searched for vital status information. If vital status is determined as deceased, this will be documented and the participant will not be considered lost to follow-up. Sponsor personnel will not be involved in any attempts to collect vital status information.

Discontinuation of specific sites or of the study as a whole are handled as part of Section [10.1](#).

8. Study Assessments and Procedures

In the event of a significant trial-continuity issue (e.g. caused by a pandemic), alternate strategies for participant visits, assessments, medication distribution and monitoring may be implemented by the sponsor or the investigator, as per local health authority / ethics requirements. For example, use of remote visit activities may be expanded to include other visits than those outlined in Section [1.3](#) with prior written agreement of the sponsor.

- Study procedures and their timing are summarized in the SoA. Protocol waivers or exemptions are not allowed.
- Immediate safety concerns should be discussed with the sponsor immediately upon occurrence or awareness to determine if the participant should continue or discontinue study intervention.
- Adherence to the study design requirements, including those specified in the SoA, is essential and required for study conduct.
- All screening evaluations must be completed and reviewed to confirm that potential participants meet all eligibility criteria. The investigator will maintain a screening log to record details of all participants screened and to confirm eligibility or record reasons for screening failure, as applicable.
- Procedures conducted as part of the participant's routine clinical management (e.g., blood count) and obtained before signing of the ICF may be utilized for screening or baseline purposes provided the procedures met the protocol-specified criteria and were performed within the time frame defined in the SoA.
- Laboratory results that could unblind the study will not be reported to investigative sites or other blinded personnel until the study has been unblinded.

8.1 Efficacy Assessments

Planned time points for all efficacy assessments are provided in the SoA.

8.1.1 eGFR

The primary efficacy variable of interest is eGFR values over time, which will be collected at Month 1, every 3 months for 12 months and afterwards every 4 months until the end-of planned treatment period and an off-treatment measurement after 4-weeks post-treatment.

The laboratory parameter eGFR will be calculated by the central laboratory based on the CKD-EPI formula. The formula used for calculating eGFR (in mL/min/1.73 m²) will be the 2009 CKD-EPI equation based on creatinine and including age, sex and race ([Levey et al. 2009](#)). The primary analyses including eGFR related to the primary and key secondary efficacy endpoints will be repeated as sensitivity analyses using the updated creatinine and cystatin C-based CKD-EPI equation, and the updated CKD-EPI equation based on creatinine only without race ([Inker et al. 2021](#)). If any of the central eGFR results or cystatin C/biomarker samples are missed at a scheduled visit, sites should ask the patient to come for a retest as soon as possible, however within 30 days of the scheduled visit.

In addition, for safety reasons and titration purposes, eGFR will also be assessed locally (as specified in Section 1.3). In case of unavailability of the central eGFR measurement, computed eGFR (CKD-EPI) from local measurements may be used for the primary and secondary endpoint analyses.

8.1.2 UACR and UPCR

Early morning void urine samples will be collected at timepoints indicated in the SoA,

- on 3 consecutive days before or after the screening visit and

- on 3 consecutive days before the baseline visits and
- on 2 consecutive days before the other applicable visits described in SoA .

All samples can be collected at home.

UACR and UPCR will be assessed by Central Lab.

At screening, UACR levels for inclusion have to be confirmed by 3 samples. If one of the 3 UACR measurements is invalid but the other 2 are valid, these values can be used to assess participant's eligibility for this study.

UPCR is analyzed to support extrapolation of efficacy from the adult to future pediatric development for finerenone.

8.1.3 EuroQoL 5-dimension 5-level (EQ-5D-5L)

The EQ-5D-5L is a self-administered preference-based generic measure of health status which includes five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression ([EuroQol-GROUP 2013](#)). Participants provide a rating for each question on a five-level Likert scale: having no problems, having slight problems, having moderate problems, having severe problems and being unable to do/having extreme problems. In addition, participants are asked to self-rate their own health today on a vertical 0-100 unit visual analogue scale, with 0 corresponding to "the worst health you can imagine", and 100 corresponding to "the best health you can imagine".

EQ-5D-5L will be collected on paper questionnaire at timepoints indicated in the SoA.

The reported data will be analyzed descriptively. The utility analyses based on collected EQ-5D-5L data are planned to be conducted in separate post-hoc analyses.

8.2 Safety Assessments

Planned time points for all safety assessments are provided in the SoA.

8.2.1 Physical Examinations

A complete physical examination will include, at a minimum, assessments of the cardiovascular and respiratory systems. Height and weight will also be measured and recorded. Investigators should pay special attention to clinical signs related to previous serious illnesses.

A complete clinical examination will be conducted at timepoints indicated in the SoA.

8.2.2 Vital Signs

- Vital signs (to be taken before blood collection for laboratory tests) will consist of 1 pulse and 3 blood pressure measurements (3 consecutive blood pressure readings will be recorded at intervals of at least 1 minute). The average of the 3 blood pressure readings will be recorded.

- Blood pressure and pulse measurements will be assessed in sitting position with a completely automated device. Manual techniques will be used only if an automated device is not available.
- Blood pressure and pulse measurements should be preceded by at least 5 minutes of rest for the participant in a quiet setting without distractions (e.g., television, cell phones).

8.2.3 Electrocardiograms

- 12-lead ECG will be obtained as outlined in the SoA (see Section 1.3), in a supine position, using an ECG machine, with a preferred speed of 50 mm/s. Evaluation for safety will be performed and signed by an investigator. If there is any clinical indication, unscheduled 12-lead ECGs can be performed at any point during the study. An additional 12-lead ECG should be obtained if the potassium > 6.5 mmol/L.
- The interpretation of the tracing must be made locally by a qualified physician.
- Clinically significant abnormalities should be reported as AEs or outcome event as appropriate (e.g. new onset atrial fibrillation) when not already reported in the medical history or in one of the previous recordings and in case of worsening of previous findings.
- The date of the recording must be documented on the ECG section of the eCRF. Each ECG, paper or electronic tracing should be labeled with the study number, participant number, and date and kept in the source documents at the study site.

8.2.4 Clinical Safety Laboratory Assessments

Potassium as the key safety parameter will be collected at each visit in central and/or local laboratory. Local laboratory values will be used to guide dose titration at on site visits and central laboratory values will be used for remote visits, where applicable (See Section 6.5 for further information).

- See Section 10.2 for the list of clinical laboratory tests to be performed and the SoA for the timing and frequency.
- The investigator must review the laboratory report, document this review, and record any clinically significant changes occurring during the study as an AE. The laboratory reports must be filed with the source documents.
- Abnormal laboratory findings associated with the underlying disease are not considered clinically significant unless judged by the investigator to be more severe than expected for the participant's condition.
- All laboratory tests with values considered clinically significantly abnormal during participation in the study or within 3 days after the last dose of study intervention should be repeated until the values return to normal or baseline or are no longer considered clinically significant by the investigator or the medical monitor.

- If clinically significant values do not return to normal/baseline within a period of time judged reasonable by the investigator, the etiology should be identified, and the sponsor notified.
- All protocol-required laboratory tests, as defined in Section 10.2, must be conducted in accordance with the laboratory manual and the SoA.
- If laboratory values from non-protocol specified laboratory tests performed at the institution's local laboratory require a change in participant management or are considered clinically significant by the investigator (e.g., SAE or AE or dose modification), then the results must be recorded.

8.2.5 Pregnancy Testing

Urine or serum pregnancy test will be conducted at timepoints indicated in the SoA (Section 1.3).

8.3 Adverse Events (AEs), Serious Adverse Events (SAEs) and Other Safety Reporting

The definitions of AEs and SAEs can be found in Section 10.3.

AEs will be reported by the participant (or, when appropriate, by a caregiver, surrogate, or the participant's legally authorized representative).

The investigator and any qualified designees are responsible for detecting, documenting, and recording events that meet the definition of an AE or SAE either via on-site visit, or telephone or video-conferencing contact.

Sites utilizing remote visit activities: home nurses will not actively enquire about or collect AE/SAE information. However, if an AE/SAE is self-evident or spontaneously reported by the participant, home nurses will capture relevant details which will be made available to the investigator as soon as possible upon conclusion of the home nurse visit. In addition, in the case of suspected SAEs, the home nurse will telephone to immediately inform the investigator. If the nurse is not able to speak to the investigator or qualified designee directly, he/she is responsible for continuing to attempt this until direct contact is made and should ensure this is done on the same day that the home nurse visit took place. In all cases, the investigator is responsible for evaluating potential AEs/SAEs, ensuring their timely entry into the eCRF as well as for SAE reporting to the sponsor within the required timeframe (see Section 8.3.1). The investigator/designee is then responsible for obtaining further information if needed e.g. contacting the participant, obtaining records from other sources or deciding if an on-site follow-up visit is required.

The investigator and qualified designees remain responsible for following up AEs that are serious, considered related to the study intervention or study procedures, or that caused the participant to discontinue the study intervention (see Section 7).

The method of recording, evaluating, and assessing causality of AEs and SAEs and the procedures for completing and transmitting SAE reports are provided in Section 10.3.

8.3.1 Time Period and Frequency for Collecting AE and SAE Information

(S)AEs will be collected from the start of randomization until the last follow-up visit at the time points specified in the SoA (Section 1.3). (S)AEs which are related to protocol-required study procedures (e.g., (S)AE related to invasive study procedures) will be recorded as (S)AEs from the signing of the ICF.

Any medical occurrences/conditions that begin in the period between signing ICF and randomization, and which are not related to a protocol-required study procedure, will be recorded on the Medical History/Current Medical Conditions, not as AEs.

With the exception of hyperkalemia, which is collected as an AESI, non-serious AEs are not mandatory to be collected in the eCRF. Instead, only AEs leading to permanent discontinuation of study intervention and AEs considered related to study intervention by the investigator, or considered related to protocol required procedures will be collected. Any other AEs should only be collected if deemed relevant by the investigator.

All SAEs will be recorded and reported to the sponsor or designee immediately and under no circumstance should this exceed 24 hours of learning of the event, as indicated in Section 10.3. The investigator will submit any updated SAE data to the sponsor within 24 hours of it being available.

Investigators are not obligated to actively seek information on AEs or SAEs after conclusion of the study participation. However, if the investigator learns of any SAE, including a death, at any time after a participant has been discharged from the study, and he/she considers the event to be reasonably related to the study intervention or study participation, the investigator must promptly notify the sponsor.

8.3.2 Method of Detecting AEs and SAEs

Care will be taken not to introduce bias when detecting AEs and/or SAEs. Open-ended and non-leading verbal questioning of the participant is the preferred method to inquire about AE occurrences.

8.3.3 Follow-up of AEs and SAEs

After the initial AE/SAE report, the investigator is required to proactively follow each participant at subsequent visits/contacts. All SAEs will be followed until resolution, stabilization, the event is otherwise explained, or the participant is lost to follow-up (as defined in Section 7.3). Further information on follow-up procedures is provided in Section 10.3.

8.3.4 Regulatory Reporting Requirements for SAEs

Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of participants and the safety of a study intervention under clinical investigation are met.

The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The

sponsor will comply with country-specific regulatory requirements relating to safety reporting to the regulatory authority, IRB/ IEC, and investigators.

An investigator who receives an investigator safety report describing an SAE or other specific safety information (e.g., summary or listing of SAEs) from the sponsor will review and then file it along with the Investigator's Brochure and will notify the IRB/IEC, if appropriate according to local requirements.

Investigator safety reports must be prepared for SUSAR according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.

For region-specific requirements for the EU, refer to Section [10.7.3](#).

8.3.5 Pregnancy

- Details of all pregnancies in female participants and, if indicated, female partners of male participants will be collected after the start of study intervention and until Follow-up visit.
- If a pregnancy is reported, the investigator will record pregnancy information on the appropriate form and submit it to the sponsor within 24 hours of learning of the female participant or female partner of male participant (after obtaining the necessary signed informed consent from the female partner) pregnancy.
- For contraceptive guidance and collection of pregnancy information see Section [10.4](#).

8.3.6 Disease-Related Outcome Events Not Qualifying as AEs or SAEs

The following disease-related events (DREs) are common in participants with CKD and will not be reported as (S)AEs, but entered into the clinical trial database as OEs:

- Kidney failure defined as:
 - ESRD: Initiation of chronic dialysis (haemo- or peritoneal dialysis) for at least 30 days or renal transplantation.
 - Decrease of eGFR to less than 15 mL/min/1.73m², confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 57\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 40\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- CV death
- Non-fatal myocardial infarction
- Non-fatal stroke or transient ischemic attack (TIA)
- Hospitalization for heart failure or urgent heart failure visit
- Other CV hospitalization
- New diagnosis of atrial fibrillation or atrial flutter

- New diagnosis of heart failure
- Outpatient worsening of Heart Failure

Sustained eGFR decreases of less than 15 mL/min/1.73m, of $\geq 57\%$ from baseline, and of $\geq 40\%$ from baseline will be programmatically derived, and not entered into the eCRF by the investigator.

Because these events are typically associated with the disease under study, they will not be reported according to the standard process for expedited reporting of SAEs even though the event may meet the definition of an SAE. These events will be recorded in the eCRF. These DREs will be monitored by an unblinded independent Data Monitoring Committee, on a routine basis.

8.3.7 Adverse Events of Special Interest (AESI)

Hyperkalemia will be an AESI in this study and collected in the eCRF. Only hyperkalemia that qualifies as an SAE will be reported to PV within 24 hours of awareness.

8.4 Pharmacokinetics

For the investigation of systemic exposure to finerenone and its relationship with treatment effects, the plasma concentrations of finerenone will be determined at different time points using a sparse sampling approach in all participants. The plasma concentration versus time data collected on the visits as outlined in SoA (see Section 1.3) will be evaluated descriptively, separated by dose and visit. Plots will be prepared of all individual plasma concentrations vs. actual relative study times (time of sample collection after time of study intervention administration).

At Visits 2 and 4, a trough sample for the determination of finerenone plasma concentrations will be drawn before intake of study intervention. At this visit, study intervention will be administered at the study site by study personnel and the exact time of study intervention intake on the day before the visit and on the day of the visit and the exact sampling time will be recorded in the eCRF. The study personnel should contact the participant prior to Visits 2 and 4 to remind them not to take the study intervention as usual in the morning at home and to remind him to properly document time of intake. In case the trough sample cannot be taken at visit 2 or 4 (e.g. in case of a remote visit, or if study intervention was taken at home), this may be postponed to Visit 5.

At Visits 3 and 6 one blood sample for the determination of finerenone plasma concentrations will be drawn during the visit 1.5-10 hours after study intervention intake at home. In case the sample cannot be taken at visit 3, this may be postponed to Visit 5. The participants should be advised to take their drug as usual in the morning at home and recall the time of drug intake or note the time of drug intake on the contact card. The exact time of study intervention intake and the exact sampling times will be recorded in the eCRF.

In case an additional PK sample is collected, it will be analyzed. For DCT visits (e.g. V4) no PK sampling is required.

Pharmacokinetic and exposure-response analysis may be performed using population approaches (popPK and popPK/PD, e.g., by non-linear mixed effect modeling). Such

evaluations will be described in a separate analysis plan and will be reported separately. Such evaluations may be started prior to database lock. If this is applicable, appropriate measures will be taken to maintain blinding of the study team, e.g., data will be stored separately, and members of the study team will neither have access to the randomization list nor to individual data.

Details about the collection, processing, storage and shipment of samples will be provided separately (e.g., sample handling sheets or laboratory manual).

8.5 Genetics and/or Pharmacogenomics

Genetic as well as non-genetic analyses may be part of the biomarker investigations in this study if approved by local IECs / IRBs and competent authorities. Genetic investigations may be of any kind, including whole genome sequencing.

Participation is optional. Participants who do not wish to participate in the genetic research may still participate in the study. Details on sample handling will be provided separately. See Section 8.6. for details.

8.6 Biomarkers

In this study, genetic as well as non-genetic biomarkers will be investigated.

Timing of collection

The planned time points of sample collection are provided in the SoA in Section 1.3. If deemed necessary, the sampling time points or frequency according to the SoA may be adjusted.

Specimen Types

The following sample types will be collected for biomarker analysis:

- Plasma
- Serum
- Whole blood
- Urine

PK or safety lab samples (leftovers) may be used for biomarker investigations if deemed necessary during the study conduct. The exact specimen type for a particular biomarker (e.g. serum or EDTA plasma) will be provided in separate documents (e.g. sample handling sheets or lab manual).

Sample handling and storage

Details on the collection, processing, shipment and storage of samples will also be provided in separate documents (e.g. sample handling sheets or lab manual). Samples may be stored for a maximum of 15 years (or according to local regulations) following the end of the study at a facility selected by the Sponsor to enable further analyses.

Reporting

Biomarker investigations may be reported separately (e.g. in a biomarker evaluation report) except for biomarkers described in Section 9.4 or the SAP.

8.6.1 Genetic investigations

Genetic analyses are primarily done to address first, altered drug metabolism, in particular genetic variants that may influence finerenone exposure and second, factors predisposing to human diseases.

Genotyping solutions such as Axiom PMRA includes selected, clinically significant pathogenic variants including those implicated for actionable genetic risk. The Axiom PMRA helps to better understand the interplay between human genetics and the susceptibility to complex diseases (e.g. CKD) and to translate that understanding to potential clinical use. The Axiom PMRA includes novel risk variants that have been implicated in human diseases and conditions, including cardiovascular disease variants.

Pharmacogenetic analyses relate to gene variants such as single nucleotide polymorphisms of drug transporters and/or drug metabolizing enzymes including enzymes of the cytochrome P450 system, the Solute Carrier Family or uridine 5'-diphosphate-glucuronosyltransferase. Different variants predispose to altered metabolism of drugs and may result in differences in finerenone exposure. Specific genetic variants may arise during the clinical program.

Genetic analyses may be pivotal to understanding the pharmacology of finerenone and support its safe and efficacious use resulting in an improved benefit/risk ratio.

8.6.2 Laboratory biomarkers

Biomarkers will be evaluated in samples collected before, during and after treatment in order to determine the impact of study intervention. These markers fall into the following categories:

- A. **Biomarkers related to the mode-of-action of the study intervention and/or functional markers of the cardiovascular system:** vasoactive agents (aldosterone etc.), NT-proBNP/BNP, and Cystatin-C.
- B. **OMICS / Multiplex Analysis:** A sample will be collected for semi-targeted/untargeted omics analysis (e.g. metabolomics by mass spectrometry, OLINK Proximity Extension Assays (Explore 1536®) in order to identify 'de novo' pharmacodynamic biomarkers, biomarkers which support the understanding of the mode-of-action and/or biomarkers which may indicate disease progression.
- C. **Further biomarkers related to the mode of action or the safety of** study intervention and similar drugs may be examined. The same applies to further biomarkers deemed relevant to renal and (cardio)vascular diseases and associated health problems. These investigations may include e.g. diagnostic, safety, PD, monitoring, or potentially predictive biomarkers.

8.7 Immunogenicity Assessments

Not applicable.

8.8 Health Economics

Health Economics parameters are not evaluated in this study.

9. Statistical Considerations

9.1 Statistical Hypotheses

Testing procedure of primary and key secondary efficacy endpoints

The primary and the key secondary endpoints will be tested under a confirmatory hierarchical testing procedure. If the primary efficacy endpoint achieves the significance level of 0.05, the corresponding null hypothesis can be rejected, and the remaining secondary endpoints will be tested at a two-sided level of 0.05 according to the following hierarchy. The three hypotheses listed will be tested in the pre-specified hierarchical sequence according to their order specified below. Consequently, confirmatory testing of the first, second and third hypotheses requires rejection of each preceding null hypothesis. In this setting, each hypothesis will be assessed at a two-sided significance level of 0.05 while keeping the global type I error rate of 0.05.

1. Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or CV death
2. Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$
3. Time to the composite of HF hospitalization or CV death

9.2 Sample Size Determination

Approximately 3160 participants will be screened to achieve 1580 randomly assigned to study intervention.

The approximate power of t-tests for comparing eGFR slope parameters (mL/min/1.73m² per year) between treatment groups is based on a two-slope linear spline mixed-effects model with specified model parameters, approximately 17 months recruitment period, at least 32 months per participant follow-up, and designated sample sizes. The power estimate from the simulation is based on initial parameter values obtained from FIDELIO-DKD data, consistent with the parameters of the linear spline mixed effect model described by (Vonesh et al. 2019). A total sample size of 1500 participants (750 per treatment arm) will have > 90% power to demonstrate the superiority of finerenone to placebo with respect to total eGFR slope from baseline to month 32 at a two-sided significance level of 5%. The sample size calculation includes assuming a 5% annual event rate resulting from intercurrent events (due to kidney failure or all-cause mortality). The >90% power calculation is based on an assumption of a 0.7 mL/min/1.73m² per year total eGFR slope difference. A total sample size of 1500 will have an 80% power if we assume a 0.527 mL/min/1.73m² per year total eGFR slope difference of finerenone to placebo.

An estimated total of 1580 participants (790 per treatment arm) will be randomized, considering an approximately 5% overall discontinuation rate in the study. All participants will be randomized in a 1:1 ratio to receive treatment with either finerenone OD or placebo.

Additional participants may be enrolled in case of an exceptionally high drop-out rate due to a trial-continuity issue.

9.3 Analysis Set

Analysis sets are defined in [Table 9-1](#).

Table 9-1: Analysis sets

Participant Analysis Set	Description
Enrolled	All participants who sign the ICF
Safety analysis set (SAF)	All randomized participants who have taken at least one dose of the study intervention. Participants will be analyzed according to the intervention they received. If a participant receives both study interventions due to a bottle error, the study intervention actually received for the majority of the time in the study will be used.
Full analysis set (FAS)	All randomized participants. Participants will be analyzed according to the intervention they were randomized to.
Pharmacokinetic analysis set (PKS)	All finerenone-treated participants with at least 1 valid finerenone plasma concentration and without validity findings which would interfere with the evaluation of the PK data.

9.4 Statistical Analyses

The statistical analysis plan will be finalized prior to database lock and it will include a more technical and detailed description of the statistical analyses described in this section. This section is a summary of the planned statistical analyses of the most important endpoints, including primary and key secondary endpoints.

9.4.1 General Considerations

Statistical analysis will be performed using SAS; the version used will be specified in the SAP.

Randomization will be stratified for the use of SGLT2 inhibitor at baseline (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g).

The most important subgroups besides stratification factors are given below (with further details and subgroups provided in the SAP):

- Primary cause of CKD

- eGFR category at baseline (eGFR < 45 mL/min/1.73 m², 45 to < 60 mL/min/1.73 m², ≥ 60 mL/min/1.73 m²)
- UACR category at baseline: UACR < 300mg/g, UACR 300 to 1000mg/g, UACR > 1000 mg/g
- Baseline serum potassium (≤4.5, >4.5 mmol/L)
- Region
- Race
- Sex
- Age group
- CVD history

It is anticipated that in these proposed subgroups for analysis, differences in treatment effects may be observed according to the screening or baseline characteristics defined, due in part to the differences in the risk of clinical events expected in the different subgroups.

9.4.1.1 Planned Methods of Analysis

A log-normal distribution is assumed for serum creatinine and UACR. For all other continuous variables, a normal distribution is assumed. The distributional assumptions will be investigated. If necessary, nonparametric methods or transformation of the data will be considered.

All variables will be analyzed by treatment arm using descriptive statistical methods. The number of data available, mean, SD, minimum, median, and maximum, will be calculated for continuous data. The geometric mean and SD will be provided instead of the arithmetic mean and SD for the variables where lognormal distributions are assumed. Frequency tables will be generated for categorical data.

Baseline values will be defined as the mean central lab value from screening and randomization visit (Visits 0 and 1). If the last observation available prior to randomization is the measurement from the Screening Visit, this would be used as the baseline value. This also includes assessments from a local laboratory, in case that prior to randomization, no assessment from the central laboratory is available. Otherwise baseline values will be missing. If more than one measurement was planned for a scheduled time point, for example blood pressure measurements and heart rate, the mean value of the last set of measurements per time point prior to randomization will be used for the calculation of the baseline value.

In the event of repeated measurements for Pre-treatment (Visit 0) and Visit 1 (baseline), the closest measurement prior to randomization will be used for analysis instead of the scheduled measurements. At all visits post-randomization and unless stated otherwise, only the values at scheduled measurements will be used for analysis.

Further details on the statistical analyses will be provided in the SAP that will be approved before database release.

9.4.1.2 Demographic and Baseline Characteristics

Demographic and baseline characteristics including disease characteristics obtained before the first study intervention administration, e.g.;

- age, gender, ethnicity, race, region (North America, Europe, Asia, Latin America and others), body weight, body height, body mass index, smoking history (never, former, current smoker), alcohol consumption
- primary Cause of CKD, baseline eGFR, categories of baseline eGFR (<30 mL/min/1.73m², 30 to <45 mL/min/1.73m², 45 to <60 mL/min/1.73m², 60 to <75 mL/min/1.73m², 75 to <90 mL/min/1.73m², ≥ 90 mL/min/1.73m²) baseline UACR, categories of baseline UACR 30 to <300 mg/g, ≥ 300 mg/g), serum creatinine, serum potassium, categories of baseline serum potassium (≤ 4.5 mmol/L and > 4.5 mmol/L), vital signs parameters (i.e. systolic blood pressure, diastolic blood pressure, heart rate)

Demographic and baseline characteristics will be summarized using descriptive statistics by treatment group and all participants for full analysis set. Age for each participant will be defined as the number of years between informed consent date and birth date. For summarizing race, if multiple races are recorded in the database, then 'multiple race' will be used in the summary table. The study sample sizes will be summarized by the trial unit (site) for enrolled participants. All participants with signed ICF will be included in the enrolled participants.

9.4.1.3 Incomplete or Missing Data

It is important to minimize the amount of missing data for the analysis of the primary estimand. Certain measures will be taken to guarantee the availability of as many eGFR values as possible (e.g. DCT implementation including the option to have remote visits and access to walk-in labs, participant engagement activities including participant education on the disease and the trial, early site training on the importance of retention, option to use local lab samples in case central lab shipment is not possible).

Data handling rules for incomplete data will be described in the specified section in the SAP if needed.

9.4.2 Primary Endpoint(s)

9.4.2.1 Mean rate of change as measured by the total slope of eGFR from baseline to Month-32

The primary efficacy variable of interest is eGFR values over time collected every 3 months for 12 months and afterwards every 4 months until end-of planned treatment period. The population-level summary is the mean rate of change of eGFR as measured by total slope from baseline to month-32.

The primary analysis of the primary efficacy variable will be performed according to participants' randomized intervention and will be based on the FAS.

A descriptive summary of eGFR will be presented by treatment group and visit, including absolute and relative changes to baseline.

The rate of change of eGFR will be compared between the finerenone and the placebo group by estimating the total eGFR slope using available assessment from baseline to planned end of treatment period. We assume that changes in the mean response can be expressed in terms of a linear trend, and the treatment effect can be expressed in terms of the difference in slope between finerenone and placebo.

For analysis of the total slope, the serial change in eGFR will be modeled using a two-slope linear spline mixed-effects model in which a fixed change point will define acute and chronic eGFR slope at month-3. The two-slope linear spline mixed-effects model is based on the method described by Vonesh et al. in section 2.1 “A two-slope linear spline mixed-effects models for eGFR” (Vonesh et al. 2019). The acute and chronic phases are differentiated because finerenone showed acute effects on eGFR that differed from its long-term effects on disease progression in FIDELIO-DKD. In addition to fixed effects for the treatment (finerenone vs. placebo), time (continuous) and a treatment by time interaction, the model will include fixed effects for stratification factors used in the stratified randomization (i.e., baseline SGLT2i use (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g)). A random effect for the intercept, acute slope (baseline to month-3), and chronic slope (month-3 to planned end of treatment period) will be included to account for variation in progression rates of participants over time. Different unstructured covariance matrices will be allowed for the different treatment groups. The time variable represents the duration from baseline to each eGFR assessment during acute and chronic phases. This model will include all observed eGFR measurements except those obtained after the onset date of ESRD. The analysis will be carried out using maximum likelihood procedures. Linear contrasts will be constructed to estimate the acute, chronic, and total slope in eGFR. LS means of the acute, chronic and total eGFR slope for finerenone and placebo group with 95% confidence interval and corresponding two-sided p-value will be presented.

The time-dependent total eGFR slope until month-32 will be calculated for the two treatment groups. The difference of total eGFR slope from baseline to month-32 between the finerenone and the placebo treatment group will be calculated, and corresponding two-sided 95% confidence interval and p-value will be presented. The total slope difference of finerenone and placebo will be estimated at 32 months after baseline, considering the anticipated data availability and duration of treatment. The study will be considered successful if the p-value is statistically significant (i.e. below 0.05).

Model Specification

For reasons of clarity, only the fixed effects for treatment and time and their interaction will be included in the model specification outlined in the following. In contrast to that, the fixed effects for the stratification factors used in the stratified randomization (i.e., baseline SGLT2i use [yes/no] and screening UACR [≤ 1000 mg/g vs. > 1000 mg/g]) will not be shown in the model formula as their inclusion is straightforward.

As described in (Vonesh et al. 2019), we will formulate the two-slope linear spline mixed-effects model as follows:

$$Y_{ij} = (\beta_{0c} + \beta_{1c}t_{ij} + \beta_{2c} \max\{t_{ij} - t^*, 0\}) (1 - X_i) + (\beta_{0f} + \beta_{1f}t_{ij} + \beta_{2f} \max\{t_{ij} - t^*, 0\}) X_i + b_{0i} + b_{1i} t_{ij} + b_{2i} \max\{t_{ij} - t^*, 0\} + \epsilon_{ij}$$

where,

Y_{ij} denotes the response variable, eGFR, for the i -th participant on the j -th occasion,
 X_i be a treatment group indicator variable with $X_i = 0$ if the i -th participant is randomized to the placebo group and $X_i = 1$ if randomized to finerenone group,
 $\beta_k = (\beta_{0k} \beta_{1k} \beta_{2k})'$ denote the regression parameters for placebo ($k=c$) and finerenone ($k=f$),

ϵ_{ij} denote within-participant residuals,
 $b_i = (b_{0i} b_{1i} b_{2i})'$ denote between-participant random effects,
 t^* denotes the change point (knot at month 3),
 $\max \{ t_{ij} - t^*, 0 \}$ be the truncated line function
 where t_{ij} are the time points at which eGFR is measured for the i -th participant across v_i occasions ($j = 1, 2, \dots, v_i$).

Model assumptions:

- $(\epsilon_{ij} \mid b_i)$ are conditionally independent normally distributed random variables with mean 0 and within-participant variance σ^2
- Participant-specific random effects b_i that are independent multivariate normally distributed with mean vector 0 and 3x3 unstructured covariance matrix that varies by treatment

The interpretation of the regression parameter $\beta_k = (\beta_{0k} \beta_{1k} \beta_{2k})'$ in terms of chronic and acute slope for placebo ($k=c$) or finerenone ($k=f$), respectively, is as follows:

β_{0k} denotes the intercept for treatment group k ,
 β_{1k} is the acute slope up through but excluding time t^* (knot at month 3) for treatment group k ,
 β_{2k} is the directional change in slope starting at time t^* (knot at month 3) for treatment group k , and
 $\beta_{1k} + \beta_{2k}$ corresponds to the chronic slope for treatment group k .

Finally, the total-slope up to time t describing the rate of change in eGFR from baseline to a specific time $t > t^*$ (t = total study time, here: 32 months) is defined as

$$\{[\beta_{0k} + \beta_{1k} \cdot t + \beta_{2k} \max(t - t^*, 0)] - \beta_{0k}\} / t$$

9.4.2.2 Supportive analysis of the primary efficacy endpoint

Chronic eGFR slope, defined as the rate of eGFR change from Month 3 after randomization to the end of the planned treatment period

The eGFR slope will be determined separately, from randomization to month 3 (acute slope) and after month 3 to end of planned study treatment (chronic slope). The chronic eGFR slope will be analyzed using the same model as described for the primary analysis ([Vonesh et al. 2019](#)), with month 3 defined as the 'change point' in the model and the time variable representing the duration from baseline to each eGFR assessment during acute and chronic phases.

LS means of the chronic eGFR slope of finerenone and placebo group, as well as the treatment difference (finerenone minus placebo) with a 95% confidence interval and corresponding two-sided p-value will be presented.

eGFR change from baseline to 1 month after the end of treatment period (off-treatment)

A post-treatment eGFR measurement will be conducted for all participants 1 month after end of the treatment period.

As a supportive analysis an ANCOVA model will be fitted to the above change from baseline eGFR value from baseline to the off-treatment (1 month after last dose), including the factors treatment group, stratification factors, total time on treatment, and the baseline eGFR as a covariate. Corresponding two-sided 95% confidence intervals and p-value will be computed.

9.4.2.3 Sensitivity analysis of the primary efficacy endpoint

Sensitivity analysis related to MAR assumption

It is considered that intermittent missing eGFR values (e.g. due to one missed visit) will not lead to a biased estimation of the slope; however, for monotone missing data patterns (e.g. missingness due to lost to follow-up), several sensitivity analyses under a ‘missing not at random (MNAR)’ assumption will be considered. For this ‘control-based pattern imputation methods’ or a ‘tipping point analysis’ could be employed.

Sensitivity analysis with alternative estimands

For the ICEs of death or start of renal replacement therapy (chronic dialysis or transplant) during the trial, additional sensitivity analyses for the primary efficacy endpoint will be performed. Further details will be included in the statistical analysis plan, where the above-mentioned ICEs will be handled with different strategies.

For the ICE permanent treatment discontinuation, it is planned to investigate an estimand using a hypothetical strategy to assess the slope difference as if the participants had continued taking study medication.

Sensitivity analysis with regard to eGFR calculation

For the primary and key supportive analyses of the primary endpoint, eGFR will be calculated using the 2009 CKD-EPI formula.

To assess the potential implications of eGFR equations which remove the inclusion of race as an adjusting variable ([Inker et al. 2021](#)), two sensitivity analyses will be performed:

- eGFR calculated using the 2021 CKD-EPI formula based on creatinine only and including the demographic variables sex and age for adjustment,
- eGFR calculated using the 2021 CKD-EPI formula based on creatinine and cystatin C and including the demographic variables sex and age for adjustment.

9.4.3 Secondary Efficacy Endpoints

Secondary efficacy endpoints will be assessed by the investigator based on their clinical judgement. For each endpoint the investigator will record information specific to that type of endpoint in the eCRF. Event criteria will be provided to facilitate event evaluation and edit checks will be in place to detect potential inconsistencies.

To further explore the chronic and persistent benefit of finerenone on kidney disease progression and CV morbidity and mortality, the following secondary variables will be evaluated:

Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or CV death

The secondary efficacy variable is the time to the first occurrence of the composite endpoint of onset of kidney failure, a sustained decrease of eGFR $\geq 57\%$ from baseline, HF hospitalization or CV death.

Randomized participants without an event of the composite endpoint at the time of analysis will be censored at the date of their last visit or date of non-CV death. Participants without information about the composite endpoint after baseline will be censored at Day 1.

In order to evaluate whether finerenone is superior to placebo in prolonging the time to the first event of the composite endpoint, the corresponding null hypothesis will be tested using the log-rank test at the two-sided 5% significance level.

A Cox proportional regression model will be used in order to provide a point estimate of the hazard ratio with a corresponding two-sided 95% confidence interval. In addition, Kaplan-Meier curves will provide the cumulative proportions of events by treatment groups.

Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$

Time to the composite kidney endpoint will be analyzed analogously to the composite kidney and CV endpoint, with modifications to the censoring rules as the events will differ.

Time to the composite of HF hospitalization or CV death

Time to first event of HF hospitalization or CV death will be conducted analogously to the above secondary composite endpoints, with modifications to the censoring rules as the events will differ.

HF hospitalization events will be counted up to the day of withdrawal from the study or death. Randomized participants without a HF hospitalization event or CV death at the time of analysis will be censored at the date of their last contact up to and including the end of study visit or date.

Supportive and sensitivity analyses for the secondary endpoints are described in the SAP.

9.4.4 Secondary Safety endpoint

Number of participants with TEAEs, TESAEs and AESI:

The number and incidence of participants with TEAEs and TESAEs will be summarized by treatment group and overall using MedDRA terms grouped by Primary System Organ Class and Preferred Term. Tables and/or narratives of treatment-emergent deaths, serious and significant AEs, including early discontinuation due to AEs, will also be provided. For details on other analysis of TEAEs and TESAEs see Section [9.4.6](#).

AEs that started or worsened after the first dose of study intervention up to 3 days after any temporary or permanent interruption of study intervention will be considered as TEAEs.

9.4.5 Exploratory Endpoints

The analysis of other exploratory endpoints, including those listed below, are described in the SAP.

- Time to onset of kidney failure (ESRD or sustained eGFR decline below 15 ml/min/1.73m²)
- Time to onset of ESRD (chronic dialysis or transplant)
- Time to onset of eGFR decrease to less than 15 mL/min/1.73 m² sustained over at least 4 weeks
- Time to onset of eGFR decrease of ≥ 57 % from baseline sustained over at least 4 weeks
- Time to CV death
- Time to first hospitalization for HF or urgent HF visit
- Number of participants with UACR decrease of at least 30% from baseline at Month 6
- Change in UACR from baseline
- Time to occurrence of the composite endpoint of kidney failure or eGFR decrease of ≥ 40 % from baseline sustained over at least 4 weeks
- Time to first CV event (CV death, non-fatal MI, non-fatal stroke or transient ischemic attack, hospitalization for HF or urgent HF visit)
- Time to CV hospitalization
- Time to death from any cause (all-cause mortality)
- Time to first all-cause hospitalization
- Number of participants with new diagnosis of atrial fibrillation or atrial flutter
- Number of participants with new diagnosis of HF
- Outpatient worsening of HF (pre-existing diagnosis of HF, worsening of symptoms and signs leading into treatment augmentation)
- Change in Quality of Life (QoL) summary scores measured by the following health-related quality of life (HRQoL) questionnaire EuroQol Group 5 dimension, 5 level questionnaire (EQ-5D-5L)

9.4.6 Other Safety Analysis

All analyses on safety and tolerability data will be performed in the SAF. Details on safety analyses will be described in the SAP.

The following safety variables will be assessed during the study:

- SAEs and AEs leading to discontinuation of treatment with study intervention

- Change in serum potassium from baseline
- Number of participants with serum potassium > 5.5 mmol/L
- Number of participants with serum potassium > 6.0 mmol/L
- Number of participants with hospitalization for hyperkalemia
- Number of participants discontinuing study intervention permanently due to hyperkalemia
- Change in vital signs from baseline
- Number of participants with hospitalization for worsening of renal function
- Number of participants discontinuing study intervention permanently due to worsening of renal function
- Changes in laboratory values

9.5 Interim Analysis

No formal interim analysis is planned for this study.

10. Supporting Documentation and Operational Considerations

10.1 Appendix 1: Regulatory, Ethical, and Study Oversight Considerations

10.1.1 Regulatory and Ethical Considerations

- This study will be conducted in accordance with the protocol and with the following:
Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines
Applicable ICH Good Clinical Practice (GCP) Guidelines
Applicable laws and regulations
- The protocol, protocol amendments, ICF, Investigator Brochure, and other relevant documents (e.g., advertisements) must be submitted to an IRB/IEC by the investigator and reviewed and approved by the IRB/IEC before the study is initiated.
- Any amendments to the protocol will require IRB/IEC approval before implementation of changes made to the study design, except for changes necessary to eliminate an immediate hazard to study participants. Any substantial modification of the protocol will be submitted to the competent authorities as substantial amendments for approval, in accordance with ICH Good Clinical Practice and national and international regulations.
- Protocols and any substantial amendments to the protocol will require health authority approval prior to initiation except for changes necessary to eliminate an immediate hazard to study participants.

- The investigator will be responsible for the following:
 - Providing written summaries of the status of the study to the IRB/IEC annually or more frequently in accordance with the requirements, policies, and procedures established by the IRB/IEC
 - Notifying the IRB/IEC of SAEs or other significant safety findings as required by IRB/IEC procedures
 - Providing oversight of the conduct of the study at the site and adherence to requirements of ICH guidelines, the IRB/IEC, and all other applicable local regulations

For region-specific requirements for the EU, refer to Section [10.7.3](#).

10.1.2 Financial Disclosure

Investigators and sub-investigators will provide the sponsor with sufficient, accurate financial information as requested to allow the sponsor to submit complete and accurate financial certification or disclosure statements to the appropriate regulatory authorities. Investigators are responsible for providing information on financial interests during the course of the study and for 1 year after completion of the study.

10.1.3 Informed Consent Process

- The investigator or his/her representative will explain the nature of the study to the participants or their legally authorized representative and answer all questions regarding the study.
- Participants must be informed that their participation is voluntary. Participants or their legally authorized representative will be required to sign a statement of informed consent that meets the requirements of 21 CFR 50, local regulations, ICH guidelines, Health Insurance Portability and Accountability Act (HIPAA) requirements, where applicable, and the IRB/IEC or study center.
- The medical record must include a statement that documented informed consent was obtained before the participant was enrolled in the study and the date the written consent was obtained. The authorized person obtaining the informed consent must also sign the ICF.
- Any study specific procedure can be performed only if written informed consent form is available.
- Participants must be re-consented to the most current version of the ICF(s) during their participation in the study.
- A copy of the ICF(s) must be provided to the participants or their legally authorized representative.
- A pre-screening consent form will be signed prior to any optional pre-screening visits, which may be used at some sites to assess UACR prior to full screening.

Participants who are rescreened are required to sign a new ICF.

10.1.4 Data Protection

- Participants will be assigned a unique identifier by the sponsor. Any participant records, datasets or biological samples that are transferred to the sponsor will contain the identifier only; participant names or any information which would make the participant identifiable will not be transferred.
- The participant must be informed that his/her personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the participant who will be required to give consent for their data to be used as described in the informed consent
- The participant must be informed that his/her medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IRB/IEC members, and by inspectors from regulatory authorities.

10.1.5 Committees Structure**10.1.5.1 Steering Committee**

The Steering Committee, which consists of external experts in the area of nephrology, and cardiology, will ensure the overarching integrity of the study.

- Review feedback from the independent DMC
- Input into endpoint event assessment
- Provision of recommendations regarding amendments to the protocol
- Contribution to and oversight of publications and communication of study results
- Overseeing overall blinded event rates to ensure that they meet protocol projections.

Decisions will be made by consensus; in the absence of consensus, the co-Chairs of the steering Committee will provide their recommendation to the Sponsor for final determination.

10.1.5.2 Independent data monitoring committee

Ongoing safety monitoring during the conduct of the study will be performed by an external and independent DMC. An independent statistical analysis center will be involved in processing unblinded safety data for the DMC. Analysis periods and procedures will be defined in an operational charter (DMC charter) filed in the study file. The independent DMC will periodically review and assess all outcome events as well as efficacy and safety data from the study for imbalances in an unblinded manner. It is believed that in this way, participant safety can continue to be monitored throughout the duration of the trial, and the integrity of the study maintained. If unexpected safety issues are identified, specific amendments will be implemented based on the recommendation of the DMC. Following data review, the DMC will provide written recommendations that will be transferred to the chairmen of the Steering Committee and Bayer. DMC opinions and recommendations will be notified by Bayer as soon as possible to the competent authorities and the IECs where they qualify for expedited reporting.

10.1.6 Dissemination of Clinical Study Data

Results of clinical drug trials will be provided on the publicly funded website www.ClinicalTrials.gov and EU Clinical Trials Register in line with the applicable regulations.

In addition, Result Summaries of Bayer's sponsored clinical trials in drug development phases II, III and IV and Phase I studies in patients are provided in the Bayer Trial Finder application after marketing authorization approval in line with the position of the global pharmaceutical industry associations laid down in the "Joint Position on the Disclosure of Clinical Trial Information via Clinical Trial Registries and Databases".

Bayer commits to sharing upon request from qualified scientific and medical researchers patient-level clinical trial data, study-level clinical trial data, and protocols from clinical trials in patients for medicines and indications approved in the United States (US) and European Union (EU) on or after January 01, 2014 as necessary for conducting legitimate research.

All Bayer-sponsored clinical trials are considered for publication in the scientific literature irrespective of whether the results of the clinical trials are positive or negative.

10.1.7 Data Quality Assurance

- All participant data relating to the study will be recorded on printed or electronic CRF unless transmitted to the sponsor or designee electronically (e.g., laboratory data). The investigator is responsible for verifying that data entries are accurate and correct by physically or electronically signing the CRF.
- Guidance on completion of CRFs will be provided in the CRF Completion Guidelines.
 - Home nurses will record source data directly into a DCT study app. The clinical data recorded by home nurses is readily visible in the study app to the investigator/authorized site personnel and will be referenced when making entries into the eCRF. Source data collected in the study app can also be viewed by study monitors for ongoing source data verification purposes.
- The investigator must permit study-related monitoring, audits, IRB/IEC review, and regulatory agency inspections and provide direct access to source data documents.
- QTLs will be pre-defined in the integrated data review plan to identify systematic issues that can impact participant safety and/or reliability of study results. These pre-defined parameters will be monitored during the study and important deviations from the QTLs and remedial actions taken will be summarized in the clinical study report.
- Monitoring details describing strategy (e.g., risk-based initiatives in operations and quality such as Risk Management and Mitigation Strategies and Analytical Risk-Based Monitoring), methods, responsibilities and requirements, including handling of noncompliance issues and monitoring techniques (central, remote, or on-site monitoring) are provided in the Monitoring Plan.
- The sponsor or designee is responsible for the data management of this study including quality checking of the data.
- The sponsor assumes accountability for actions delegated to other individuals (e.g., Contract Research Organizations).

- Records and documents, including signed ICFs, pertaining to the conduct of this study must be retained by the investigator for 15 years after study completion unless local regulations or institutional policies require a longer retention period. No records may be destroyed during the retention period without the written approval of the sponsor. No records may be transferred to another location or party without written notification to the sponsor.

10.1.8 Source Documents

- Source documents provide evidence for the existence of the participant and substantiate the integrity of the data collected. Source documents are filed at the investigator's site.
- Data reported on the eCRF or entered in the eCRF that are transcribed from source documents must be consistent with the source documents or the discrepancies must be explained. The investigator may need to request previous medical records or transfer records, depending on the study. Also, current medical records must be available.
- Definition of what constitutes source data can be found in the Source Data Location List or equivalent.
- The investigator must maintain accurate documentation (source data) that supports the information entered in the CRF.
- Study monitors will perform ongoing source data verification to confirm that data entered into the CRF by authorized site personnel are accurate, complete, and verifiable from source documents; that the safety and rights of participants are being protected; and that the study is being conducted in accordance with the currently approved protocol and any other study agreements, ICH GCP, and all applicable regulatory requirements.

10.1.9 Study and Site Start and Closure

First Act of Recruitment

The study start date is the date on which the clinical study will be open for recruitment of participants.

The first act of recruitment is the first participant screened in the study and will be the study start date.

Study/Site Termination

The sponsor or designee reserves the right to close the study site or terminate the study at any time for any reason at the sole discretion of the sponsor. Study sites will be closed upon study completion. A study site is considered closed when all required documents and study supplies have been collected and a study-site closure visit has been performed.

The investigator may initiate study-site closure at any time, provided there is reasonable cause and sufficient notice is given in advance of the intended termination.

Reasons for the early closure of a study site by the sponsor or investigator may include but are not limited to:

For study termination:

- Discontinuation of further study intervention development

For site termination:

- Failure of the investigator to comply with the protocol, the requirements of the IRB/IEC or local health authorities, the sponsor's procedures, or GCP guidelines
- Inadequate or no recruitment (evaluated after a reasonable amount of time) of participants by the investigator
- Total number of participants included earlier than expected.

If the study is prematurely terminated or suspended, the sponsor shall promptly inform the investigators, the IECs/IRBs, the regulatory authorities, and any contract research organization(s) used in the study of the reason for termination or suspension, as specified by the applicable regulatory requirements. The investigator shall promptly inform the participant and should assure appropriate participant therapy and/or follow-up.

10.1.10 Publication Policy

- The results of this study may be published or presented at scientific meetings. If this is foreseen, the investigator agrees to submit all manuscripts or abstracts to the sponsor before submission. This allows the sponsor to protect proprietary information and to provide comments.
- The sponsor will comply with the requirements for publication of study results. In accordance with standard editorial and ethical practice, the sponsor will generally support publication of multicenter studies only in their entirety and not as individual site data. In this case, a coordinating investigator will be designated by mutual agreement.
- Authorship will be determined by mutual agreement and in line with International Committee of Medical Journal Editors authorship requirements.

10.2 Appendix 2: Clinical Laboratory Tests

- The tests detailed in [Table 10-1](#) will be performed by the central laboratory.
- Protocol-specific requirements for inclusion or exclusion of participants are detailed in [Section 5](#) of the protocol.
- Additional tests may be performed at any time during the study as determined necessary by the investigator or required by local regulations.

Investigators must document their review of each laboratory safety report.

Laboratory evaluations (hematology, HbA_{1c}, clinical chemistry and urinalysis parameters) are shown in [Table 10-1](#). Hematology, HbA_{1c} and (full) clinical chemistry are considered as ‘Full central laboratory’ assessments.

Laboratory evaluations will be performed according to the visit schedule (see [1.3](#)).

Table 10-1 Central laboratory parameters

Parameter	Component
Hematology	White blood cell count (WBC), red blood cell count (RBC), hemoglobin, hematocrit, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), MCH concentration (MCHC), platelets
Biomarker sample	For biomarkers blood, serum, plasma and urine samples are needed (See Section 8.6 for further information)
Genetic testing	Genetics (See Section 8.6 for further information)
Glycated hemoglobin (HbA_{1c})	HbA _{1c}
Clinical chemistry (full)	Aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (AP), gamma glutamyl transpeptidase (GGT), lactate dehydrogenase (LDH), creatine kinase (CK), cholesterol (high density lipoprotein [HDL], low density lipoprotein [LDL], total), triglycerides, creatinine, eGFR (CKD-EPI), blood urea nitrogen, uric acid, bilirubin, sodium, serum potassium, magnesium, total protein, albumin, high-sensitivity C-reactive protein (hs-CRP)
Clinical chemistry (limited)	Serum creatinine, eGFR (CKD-EPI), serum potassium, serum sodium (central lab), Serum cystatin C (biomarker lab)
Urinalysis	Urine albumin-to-creatinine ratio (UACR), Urine protein/creatinine ratio (UPCR)

10.2.1 Local laboratory

In addition to samples for the central laboratory, other blood samples will be taken for screening purposes, and also for safety from Visit 1 (Day 1; baseline) onwards (see [Section 1.3](#)) for analysis at the local laboratory. These samples may be obtained up to 72 hours before a scheduled visit. For sites utilizing remote visit activities, blood safety samples collected at remote visits will be analyzed at the central laboratory only.

The following clinical chemistry parameters must be measured for as long as participant is not prematurely discontinued from the study intervention:

- potassium from blood sample (measurement is also not needed when the participant is on temporary interruption of study intervention).
- creatinine (eGFR will be calculated and recorded in the eCRF by the investigator using the CKD-EPI formula, or the JP equation for sites located in Japan) from blood sample.

Potassium values should be recorded using a single decimal point (e.g. 4.5 mmol/L or mEq/L). In the event of hyperkalemia, see Section 6.5 for guidance on treatment.

Up-titration of the study intervention will be based on local potassium and eGFR values for on site visits. For remote visits it will be based on central laboratory potassium results and eGFR. In all cases, this must be documented in the eCRF. Down-titration of the study intervention will occur for safety reasons only (for guidance, see Section 6.5 and Table 6-3).

In women of childbearing potential, a **pregnancy test** will be performed locally, at the Screening Visit and following randomization at yearly visits as long as the participant is not permanently discontinued from the study intervention (since more than 7 days). However, pregnancy tests in participant of childbearing potential should be performed more frequently (e.g. at every visit) if this is required by national / institutional regulations. Both serological and urine tests are acceptable.

10.3 Appendix 3: AEs and SAEs: Definitions and Procedures for Recording, Evaluating, Follow-up, and Reporting

10.3.1 Definition of AE

AE Definition

- An AE is any untoward medical occurrence in a clinical study participant, associated with the use of study intervention, whether or not considered related to the study intervention.
- NOTE: An AE can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease (new or exacerbated) associated with the use of study intervention.

Events Meeting the AE Definition

- Any abnormal laboratory test results (hematology, clinical chemistry, or urinalysis) or other safety assessments (e.g., ECG, radiological scans, vital signs measurements), including those that worsen from baseline, considered clinically significant in the medical and scientific judgment of the investigator (i.e., not related to progression of underlying disease).
- Exacerbation of a chronic or intermittent pre-existing condition including either an increase in frequency and/or intensity of the condition.
- New conditions detected or diagnosed after study intervention administration even though it may have been present before the start of the study.
- Signs, symptoms, or the clinical sequelae of a suspected intervention-intervention interaction.

-
- Signs, symptoms, or the clinical sequelae of a suspected overdose of either study intervention or a concomitant medication. Overdose per se will not be reported as an AE/SAE unless it is an intentional overdose taken with possible suicidal/self-harming intent. Such overdoses should be reported regardless of sequelae.
 - “Lack of efficacy” or “failure of expected pharmacological action” per se will not be reported as an AE or SAE. Such instances will be captured in the efficacy assessments. However, the signs, symptoms, and/or clinical sequelae resulting from lack of efficacy will be reported as AE or SAE if they fulfill the definition of an AE or SAE.
-

Events NOT Meeting the AE Definition

- Any clinically significant abnormal laboratory findings or other abnormal safety assessments which are associated with the underlying disease, unless judged by the investigator to be more severe than expected for the participant’s condition.
 - The disease/disorder being studied or expected progression, signs, or symptoms of the disease/disorder being studied, unless more severe than expected for the participant’s condition.
 - Medical or surgical procedure (e.g., endoscopy, appendectomy): the condition that leads to the procedure is the AE.
 - Situations in which an untoward medical occurrence did not occur (social and/or convenience admission to a hospital).
 - Anticipated day-to-day fluctuations of pre-existing disease(s) or condition(s) present or detected at the start of the study that do not worsen.
-

10.3.2 Definition of SAE

An SAE is defined as any AE that, at any dose:

a. Results in death

b. Is life-threatening

- The term 'life-threatening' in the definition of 'serious' refers to an event in which the participant was at risk of death at the time of the event. It does not refer to an event, which hypothetically might have caused death, if it were more severe.
-

c. Requires inpatient hospitalization or prolongation of existing hospitalization

- In general, hospitalization signifies that the participant has been admitted (usually involving at least an overnight stay) at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AEs. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether "hospitalization" occurred or was necessary, the AE should be considered serious.
 - Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.
-

d. Results in persistent or significant disability/incapacity

- The term disability means a substantial disruption of a person's ability to conduct normal life functions.
 - This definition is not intended to include experiences of relatively minor medical significance such as uncomplicated headache, nausea, vomiting, diarrhea, influenza, and accidental trauma (e.g., sprained ankle) which may interfere with or prevent everyday life functions but do not constitute a substantial disruption.
-

e. Is a congenital anomaly/birth defect

f. Is a suspected transmission of any infectious agent via an authorized medicinal product

g. Other situations:

- Medical or scientific judgment should be exercised by the investigator in deciding whether SAE reporting is appropriate in other situations such as significant medical events that may jeopardize the participant or may require medical or surgical intervention to prevent one of the other outcomes listed in the above definition. These events should usually be considered serious.
 - Examples of such events include invasive or malignant cancers, intensive treatment for allergic bronchospasm, blood dyscrasias, convulsions, or development of intervention dependency or intervention abuse.
-

10.3.3 Recording and Follow-Up of AE and/or SAE**AE and SAE Recording**

- When an AE/SAE occurs, it is the responsibility of the investigator to review all documentation (e.g., hospital progress notes, laboratory reports, and diagnostics reports) related to the event.
 - The investigator will then record all relevant AE/SAE information.
-

-
- It is not acceptable for the investigator to send photocopies of the participant's medical records to Sponsor in lieu of completion of the required form.
 - There may be instances when copies of medical records for certain cases are requested by Sponsor. In this case, all participant identifiers, with the exception of the participant number, will be redacted on the copies of the medical records before submission to Sponsor.
 - The investigator will attempt to establish a diagnosis of the event based on signs, symptoms, and/or other clinical information. Whenever possible, the diagnosis (not the individual signs/symptoms) will be documented as the AE/SAE.
-

Assessment of Intensity

- The investigator will make an assessment of intensity for each AE and SAE reported during the study and assign it to 1 of the following categories:
- Mild: An event that is easily tolerated by the participant, causing minimal discomfort and not interfering with everyday activities.
- Moderate: An event that causes sufficient discomfort to interfere with normal everyday activities.
- Severe: An event that prevents normal everyday activities. An AE that is assessed as severe should not be confused with an SAE. Severe is a category utilized for rating the intensity of an event; and both AEs and SAEs can be assessed as severe.

An event is defined as 'serious' when it meets at least 1 of the predefined outcomes as described in the definition of an SAE, NOT when it is rated as severe.

Assessment of Causality

- The investigator is obligated to assess the relationship between study intervention and each occurrence of each AE/SAE.
- A “reasonable possibility” of a relationship conveys that there are facts, evidence, and/or arguments to suggest a causal relationship, rather than a relationship cannot be ruled out.
- The investigator will use clinical judgment to determine the relationship.
- Alternative causes, such as underlying disease(s), concomitant therapy, and other risk factors, as well as the temporal relationship of the event to study intervention administration will be considered and investigated.
- The investigator will also consult the Investigator’s Brochure (IB) and/or Product Information, for marketed products, in his/her assessment.
- For each AE/SAE, the investigator **must** document in the medical notes that he/she has reviewed the AE/SAE and has provided an assessment of causality.
- There may be situations in which an SAE has occurred and the investigator has minimal information to include in the initial report to Sponsor. However, **it is very important that the investigator always make an assessment of causality for every event before the initial transmission** of the SAE data to Sponsor.
- The investigator may change his/her opinion of causality in light of follow-up information and send an SAE follow-up report with the updated causality assessment.
- The causality assessment is one of the criteria used when determining regulatory reporting requirements.

Follow-up of AEs and SAEs

- The investigator is obligated to perform or arrange for the conduct of supplemental measurements and/or evaluations as medically indicated or as requested by Sponsor to elucidate the nature and/or causality of the AE or SAE as fully as possible. This may include additional laboratory tests or investigations, histopathological examinations, or consultation with other health care professionals.
 - If a participant dies during participation in the study or during a recognized follow-up period, the investigator will provide Sponsor with a copy of any post-mortem findings including histopathology.
 - New or updated information will be recorded in the originally submitted documents.
 - The investigator will submit any updated SAE data to Sponsor within 24 hours of receipt of the information.
-

10.3.4 Reporting of SAEs

SAE Reporting to Sponsor via an Electronic Data Collection Tool

- The primary mechanism for reporting an SAE to Sponsor will be the electronic data collection tool.
- If the electronic system is unavailable, then the site will use the paper SAE data transmission (see next section) to report the event within 24 hours.
- The site will enter the SAE data into the electronic system as soon as it becomes available.
- After the study is completed at a given site, the electronic data collection tool will be taken off-line to prevent the entry of new data or changes to existing data.
- If a site receives a report of a new SAE from a study participant or receives updated data on a previously reported SAE after the electronic data collection tool has been taken off-line, then the site can report this information on a paper SAE form (see next section) or to the medical monitor by telephone.
- Contacts for SAE reporting can be found in the investigator site file.

SAE Reporting to Sponsor via Paper Data Collection Tool

- Email transmission of the SAE paper data collection tool is the preferred method to transmit this information to the Sponsor.
- In rare circumstances and if email transmission is not feasible, notification by telephone is acceptable with a copy of the SAE data collection tool sent by overnight mail or courier service.
- Initial notification via telephone does not replace the need for the investigator to complete and sign the SAE data collection tool within the designated reporting time frames.
- Contacts for SAE reporting can be found in the investigator site file.

10.4 Appendix 4: Contraceptive Guidance and Collection of Pregnancy Information

Definitions:

Woman of Childbearing Potential (WCBP)

A woman is considered fertile following menarche and until becoming post-menopausal unless permanently sterile (see below).

If fertility is unclear (e.g., amenorrhea in adolescents or athletes) and a menstrual cycle cannot be confirmed before first dose of study intervention, additional evaluation should be considered.

Women in the following categories are not considered WOCBP

1. Premenarchal
2. Premenopausal female with 1 of the following:
 - Documented hysterectomy
 - Documented bilateral salpingectomy
 - Documented bilateral oophorectomy

For individuals with permanent infertility due to an alternate medical cause other than the above, (e.g., mullerian agenesis, androgen insensitivity), investigator discretion should be applied to determining study entry.

Note: Documentation can come from the site personnel's: review of the participant's medical records, medical examination, or medical history interview.

3. Postmenopausal female

- A postmenopausal state is defined as no menses for 12 months without an alternative medical cause.

A high FSH level in the postmenopausal range may be used to confirm a postmenopausal state in women not using hormonal contraception or HRT. However, in the absence of 12 months of amenorrhea, confirmation with more than one FSH measurement > 40 mIU/mL (for US only: FSH levels > 40 mIU/mL and estradiol < 20 pg/mL) is required.
- Females on HRT and whose menopausal status is in doubt will be required to use one of the non-estrogen hormonal highly effective contraception methods if they wish to continue their HRT during the study. Otherwise, they must discontinue HRT to allow confirmation of postmenopausal status before study enrollment.

Contraception Guidance for women:

According to pre-clinical and clinical data, Finerenone does not indicate teratogenicity/fetotoxicity in early pregnancy (please refer to the current version of the Investigator's Brochure for details). Based on these data, women of child-bearing potential can be included into the trial if reliable contraception is used.

Adequate contraception is defined as any combination of at least 2 effective methods of birth control. Highly effective birth control methods are hormonal contraception (progestogen only with inhibition of ovulation), combined hormonal contraception (estrogen and progestogen with inhibition of ovulation), intrauterine devices and intrauterine hormone-releasing systems. One of these methods should be combined with a supplementary barrier method (preferably male condom) due to possible drug-to drug interactions which might reduce efficacy of the hormonal contraception. Contraception should be used until 30 days after last intake of study intervention.

Male participants do not have to use condoms in the study, because there is no indication of male-mediated developmental toxicity. Therefore, female partners of male participants are also not required to use contraception.

Collection of Pregnancy Information:

Male participants with partners who become pregnant

- The investigator will attempt to collect pregnancy information on any male participant's female partner who becomes pregnant while the male participant is in this study. This applies only to male participants who receive finerenone.
- After obtaining the necessary signed informed consent from the pregnant female partner directly, the investigator will record pregnancy information on the appropriate form and submit it to the sponsor within 24 hours of learning of the partner's pregnancy. The female partner will also be followed to determine the outcome of the pregnancy. Information on the status of the mother and child will be forwarded to the sponsor. Generally, the follow-up will be no longer than 6 to 8 weeks following the estimated delivery date. Any termination of the pregnancy will be reported regardless of fetal status (presence or absence of anomalies) or indication for the procedure.

Female participants who become pregnant

- The investigator will collect pregnancy information on any female participant who becomes pregnant while participating in this study. The initial information will be recorded on the appropriate form and submitted to the sponsor within 24 hours of learning of a participant's pregnancy.
- The participant will be followed to determine the outcome of the pregnancy. The investigator will collect follow-up information on the participant and the neonate and the information will be forwarded to the sponsor. Generally, follow-up will not be required for longer than 6 to 8 weeks beyond the estimated delivery date. Any termination of pregnancy will be reported, regardless of fetal status (presence or absence of anomalies) or indication for the procedure.
 - While pregnancy itself is not considered to be an AE or SAE, any pregnancy complication or elective termination of a pregnancy for medical reasons will be reported as an AE or SAE.
 - A spontaneous abortion (occurring at <22 weeks gestational age) or still birth (occurring at >22 weeks gestational age) is always considered to be an SAE and will be reported as such.
- Any post-study pregnancy related SAE considered reasonably related to the study intervention by the investigator will be reported to the sponsor as described in Section 8.3.4. While the investigator is not obligated to actively seek this information in former study participants, he or she may learn of an SAE through spontaneous reporting.
- Any female participant who becomes pregnant while participating in the study will discontinue study intervention or be withdrawn from the study.

10.5 Appendix 5: Calculating the Child Pugh score

The severity of liver disease ([Table 10-2](#)) will determine the Child Pugh score ([Table 10-3](#)).

Table 10-2 Grading of severity of liver disease, adapted from ([Pugh et al. 1973](#))

Factor	+1	+2	+3
Bilirubin (mg/dL)	<2	2 – 3	>3
Albumin (g/dL)	>3.5	2.8 – 3.5	<2.8
International Normalized Ratio	<1.7	1.7 – 2.3	>2.3
Ascites	None	Mild	Moderate / Severe
Encephalopathy	None	Grade I - II	Grade III – IV

Table 10-3 Classification using the added score from [Table 10-2](#), adapted from ([Pugh et al. 1973](#))

Child-Pugh Class	A	B	C
Points	5 – 6	7 – 9	10 – 15

10.6 Appendix 6: Guidance on use of common CYP3A4 inhibitors or inducers

Table 10-4 lists the most common excluded medications regarded as moderate or potent CYP3A4 inducers (left column) or potent CYP3A4 inhibitors (middle column) along with examples of concomitant medications which are allowed (right column). It cannot be ruled out that medications are prescribed which are excluded comedications but which are not explicitly mentioned, e.g. because they have very recently been approved by a regulatory agency. Definitions for the categories of weak, moderate and potent inducers or inhibitors, as well as examples of drugs in these categories are given on the FDA website '<https://www.fda.gov/drugs/drug-interactions-labeling/drug-development-and-drug-interactions-table-substrates-inhibitors-and-inducers#classSub>'.

Table 10-4 Cytochrome P450: list of concomitant medication

Excluded CYP3A4 inducers	Excluded CYP3A4 inhibitors	Examples of allowed CYP3A4 inhibitors
- apalutamide - asunaprevir/beclabuvir/dacatasvir - avasimibe - bosentan - carbamazepine - cenobamate - dabrafenib - efavirenz - elagolix - enzalutamide	- adagrasib - amprenavir - atazanavir and other inhibitors of human HIV protease - boceprevir - ceritinib - clarithromycin - cobicistat - danoprevir - darunavir - dasabuvir	- amlodipine - amiodarone - aprepitant - berotralstat - bicalutamide - chloramphenicol - cilostazol - cimetidine - ciprofloxacin - conivaptan
- etravirine - fosphenytoin - hypericum perforatum / St John's wort - ivosidenib - lesinurad	- elvitegravir - ensitrelvir - fosamprenavir - grapefruit - idelalisib	- crizotinib - cyclosporine - dasatinib
- lersivirine - lorlatinib - lumacaftor - mephenytoin - metamizole - methylphenobarbital - mitotane - modafinil - nafcillin - nevirapine - oxcarbazepine - pexidartinib - phenobarbital - phenytoin - primidone - rifabutin - rifampicin	- indinavir - itraconazole - jasomycin - ketoconazole - lonafarnib - lopinavir - nefazodone - nelfinavir - ombitasvir - paritaprevir - posaconazole - ribociclib - ritonavir - saquinavir - telaprevir - telithromycin - tipranavir	- duvelisib - erythromycin - fluconazole - fluvoxamine - fosaprepitant - imatinib - isavuconazole - istradefylline - ivacaftor - letermovir - lapatinib - lomitapide - mifepristone - nilotinib - norfloxacin - ranitidine - ranolazine

▪ rifapentin	▪ troleandomycin	▪ rimegepant
▪ semagacestat		▪ tacrolimus
▪ sotorasib	tucatinib	▪ ticagrelor
▪ talviraline	voriconazole	▪ tofisopam
▪ telotristat ethyl		▪ verapamil
▪ thioridazine		▪ voxelotor
▪ troglitazone		
▪ vemurafenib		

HIV = Human immunodeficiency virus

10.7 Appendix 7: Country/Region-specific Requirements

10.7.1 China

10.7.1.1 CHN: Country-specific Changes Valid for China Only

- Genetic and non-genetic biomarker samples (except for NT-proBNP and Cystatin-C [used for calculation of eGFR]) will not be performed in China.
- For China, sample leftovers must not be repurposed after measurement.
- For China, with regard to Section 8.6.2 Laboratory biomarkers, NT-proBNP as a functional marker of the cardiovascular system, and Cystatin-C, are the only additional biomarkers that will be assessed in central laboratory.

10.7.2 Denmark

10.7.2.1 DNK-2: Country-specific Changes Valid for Denmark Only

Overview of Changes

Amendment DNK-2 (20 SEP 2021)

Overall Rationale for the Amendment:

This local protocol amendment implements country-specific modification(s) of the protocol version 2.0, dated 05 JUL 2021.

The protocol has been revised in the appropriate sections in order to meet requirements of the Danish Medicines Agency (DKMA) on reporting and assessment of SAEs and SUSARs.

Section # and Name	Description of Change	Brief Rationale
Section 1.3 Schedule of Activities, Table 1-1 Section 6.5 Dose Modification Section 10.2.1.1 Local laboratory	Local laboratory method is specified as being plasma potassium.	Change was requested by DKMA
Section 8.3.4 Regulatory Reporting Requirements for SAEs	Text was updated to include specific regulatory reporting requirements of SAEs and SUSARs in Denmark.	Change was requested by DKMA
Section 10.3.2 Definition of SAE	Example of an overnight stay for hospitalization has been deleted.	Change was requested by DKMA

Changes to the Protocol Text

Section 1.3 Schedule of Activities

Table 1-1: Schedule of Activities

Procedure	[...]	Notes
Month	[...]	
Visit number	[...]	[...]
Visit Window (days)	[...]	
[...]		
Laboratory evaluations		
Serum or urine pregnancy test	[...]	[...]
Full clinical chemistry (incl eGFR and K ⁺); central lab	[...]	
Creatinine/eGFR, K ⁺ ; Na ⁺ central lab	[...]	[...]
Creatine/eGFR, potassium (from plasma); local lab	[...]	[...]
UACR/UPCR; central lab	[...]	[...]
[...]		

Section 6.5 Dose modification

[...]

Table 6-3 Potassium from blood sample: guidance for dose adjustment at Visit 2 and subsequent visits

Blood potassium (mmol/L)	Action
First sample:	
≤4.8	If on lower dose of study intervention, up-titrate to higher dose. If on higher dose of study intervention, continue on the same dose.
4.9 to 5.5	Continue on the same dose.
>5.5	Withhold study intervention and re-check potassium within 72 hours.
Second and subsequent samples:	
≤5.0	Re-start study intervention at lower dose.
>5.0	Continue to withhold study intervention; continue to monitor potassium and restart study intervention at the lower dose only if potassium is ≤5.0

NOTE: lower dose = 10 mg once daily; higher dose = 20 mg once daily.

Local laboratory potassium should be measured from a plasma sample.

The following aspects have also to be taken into consideration:

- Potassium should be measured 4 weeks ($\pm$ 7 days) after re-starting treatment or dose adjustment, especially after up-titration.
- If central laboratory results for potassium differ from local laboratory results to such an extent that the investigator is uncomfortable with the dose adjustment guidance shown in Table 6–3, blood sampling may be repeated at the investigator’s discretion.
- If the participant is already on the lower dose of study intervention but hyperkalemia recurs soon after a previous event of hyperkalemia with interruption of study intervention, and there is no explanation for the recurring hyperkalemia event other than intake of study intervention, permanent discontinuation of study intervention is recommended.
- If local laboratory potassium in plasma is > 6.5 mmol/L, an ECG should be obtained.

Section 8.3.4 Regulatory Reporting Requirements for SAEs

Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of participants and the safety of a study intervention under clinical investigation are met.

The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The sponsor and/or the investigator will comply with country-specific regulatory requirements relating to safety reporting of all relevant events (e.g. SAEs and SUSARs) to the regulatory authority, IRB/ IEC, and investigators.

An investigator who receives an investigator safety report describing an SAE or other specific safety information (e.g., summary or listing of SAEs) from the sponsor will review and then file it along with the Investigator’s Brochure and will notify the IRB/IEC, if appropriate according to local requirements.

For all serious adverse events (SAEs), the sponsor has to carry out a separate assessment for expectedness, seriousness and causal relationship to study drug. The expectedness of AEs will

be determined by the sponsor according to the applicable reference document and according to all local regulations.

Investigator safety reports must be prepared for SUSAR according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.

Section 10.2.1.1 Local laboratory

[...]

The following clinical chemistry parameters must be measured for as long as participant is not prematurely discontinued from the study intervention:

- potassium from plasma sample.
- creatinine (eGFR will be calculated and recorded in the eCRF by the investigator using the CKD-EPI formula, or the JP equation for sites located in Japan) from blood sample.

Potassium values should be recorded using a single decimal point (e.g. 4.5 mmol/L or mEq/L). In the event of hyperkalemia, see Section 6.5 for guidance on treatment.

[...]

Section 10.3.2 Definition of SAE

[...]

c. Requires inpatient hospitalization or prolongation of existing hospitalization

- In general, hospitalization signifies that the participant has been admitted at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AEs. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether "hospitalization" occurred or was necessary, the AE should be considered serious.
- Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.

[...]

10.7.2.2 DNK-1: Country-specific Changes Valid for Denmark Only

Overview of Changes

Amendment DNK-1 (13 AUG 2021)

Overall Rationale for the Amendment:

This local protocol amendment implements country-specific modification(s) of the protocol version 2.0, dated 05 JUL 2021.

The protocol has been revised in the appropriate sections in order to meet requirements of the Danish Medicines Agency (DKMA) on reporting and assessment of SAEs and SUSARs.

Section # and Name	Description of Change	Brief Rationale
Section 8.3.4 Regulatory Reporting Requirements for SAEs	Text was updated to include specific regulatory reporting requirements of SAEs and SUSARs in Denmark.	Change was requested by DKMA
Section 10.3.2 Definition of SAE	Example of an overnight stay for hospitalization has been deleted.	Change was requested by DKMA

Changes to the Protocol Text

Section 8.3.4 Regulatory Reporting Requirements for SAEs

Prompt notification by the investigator to the sponsor of an SAE is essential so that legal obligations and ethical responsibilities towards the safety of participants and the safety of a study intervention under clinical investigation are met.

The sponsor has a legal responsibility to notify both the local regulatory authority and other regulatory agencies about the safety of a study intervention under clinical investigation. The sponsor and/or the investigator will comply with country-specific regulatory requirements relating to safety reporting of all relevant events (e.g. SAEs and SUSARs) to the regulatory authority, IRB/ IEC, and investigators.

An investigator who receives an investigator safety report describing an SAE or other specific safety information (e.g., summary or listing of SAEs) from the sponsor will review and then file it along with the Investigator's Brochure and will notify the IRB/IEC, if appropriate according to local requirements.

For all serious adverse events (SAEs), the sponsor has to carry out a separate assessment for expectedness, seriousness and causal relationship to study drug. The expectedness of AEs will be determined by the sponsor according to the applicable reference document and according to all local regulations.

Investigator safety reports must be prepared for SUSAR according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.

Section 10.3.2 Definition of SAE

[...]

c. Requires inpatient hospitalization or prolongation of existing hospitalization

- In general, hospitalization signifies that the participant has been admitted at the hospital or emergency ward for observation and/or treatment that would not have been appropriate in the physician's office or outpatient setting. Complications that occur during hospitalization are AEs. If a complication prolongs hospitalization or fulfills any other serious criteria, the event is serious. When in doubt as to whether "hospitalization" occurred or was necessary, the AE should be considered serious.
- Hospitalization for elective treatment of a pre-existing condition that did not worsen from baseline is not considered an AE.

[...]

10.7.3 European Union

10.7.3.1 EU-1: Region-specific changes valid for countries participating under EU-CTR only

Amendment EU-1 (06 AUG 2025)

Overall Rationale

This section implements region-specific modifications of the global protocol amendment 3, dated 10 APR 2025, to meet the regional requirements in the participating EU countries.

A description of changes and a brief rationale are outlined in the table below:

Section # and Name	Description of Change	Brief Rationale
8.3.4 Regulatory Reporting Requirements for SAEs	A sentence was added to clarify that the sponsor will notify SUSARs via the Eudravigilance database.	To follow the request from the competent authorities of the participating EU countries.
10.1.1 Regulatory and Ethical Considerations	The information was added that the study will be conducted in accordance with the European regulation 536/2014 (CTR Annex I, section D, no. 17, letter a).	

Changes to the Protocol Text

8.3.4 Regulatory Reporting Requirements for SAEs

[...]

Investigator safety reports must be prepared for SUSAR according to local regulatory requirements and sponsor policy and forwarded to investigators as necessary.

The sponsor will notify SUSARs via the Eudravigilance database.

10.1.1 Regulatory and Ethical Considerations

- This study will be conducted in accordance with the protocol and with the following:
Consensus ethical principles derived from international guidelines including the Declaration of Helsinki and Council for International Organizations of Medical Sciences (CIOMS) International Ethical Guidelines
Applicable ICH Good Clinical Practice (GCP) Guidelines
European Regulation No 536/2014 (CTR Annex I, section D, no. 17, letter a)
Applicable laws and regulations

[...]

10.7.4 India**10.7.4.1 IND-1: Country-specific Changes Valid for India Only****Overview of Changes****Amendment IND-1 (14 JUL 2021)****Overall Rationale for the Amendment:**

This local protocol amendment implements country-specific modification(s) of the protocol version 2.0, dated 05 JUL 2021.

The protocol has been revised in the appropriate sections in order to meet requirements of the Indian health authority regarding reporting of disease-related outcome events when these events meet the definition of an SAE.

Revision of details regarding the documentation and reporting of SAEs is clarified in the protocol. In the original protocol, the disease-related outcome events were not to be reported as (S)AEs but only entered into the clinical trial database as outcome events (OEs). In order to comply with Indian regulatory requirements, all outcome events that fulfill the definition of an SAE will be reported and documented as both SAEs and OEs in India.

Section # and Name	Description of Change	Brief Rationale
8.3.6 Disease-Related Outcome Events Not Qualifying as AEs or SAEs	Reporting of disease-related outcome events as both SAEs and OEs if they meet the definition of an SAE.	To fulfill health authority requirements on SAE reporting.

Changes to the Protocol Text

Clean amended text is provided below.

Section 8.3.6 Disease-Related Outcome Events Not Qualifying as AEs or SAEs

The following disease-related events (DREs) are common in participants with CKD and will be entered into the clinical trial database as OEs:

[...]

These events are typically associated with the disease under study and will not be reported as AEs but entered into the clinical trial database as OEs. However, when they meet the definition of an SAE, they will be reported according to the standard process for expedited reporting of SAEs, as required by the local health authority. SAE reporting is described in Section 8.3.4. In addition, these events will be documented both as Outcome Events as well as SAEs in the eCRF. These DREs will be monitored by an unblinded independent Data Monitoring Committee, on a routine basis.

However, in order to maintain the integrity of the trial, SUSARs that derive from any of these outcome events will be waived from unblinding. Blinded SUSAR reports will be sent to the Indian Authorities in an expedited manner to fulfill local regulatory requirements.

10.7.5 Japan

10.7.5.1 JPN: Country-specific Changes Valid for Japan Only

- The investigator must report immediately all non-approved medical device failures which could cause health damage, as well as any health damage that may be causally associated with a non-approved medical device failure. For this reporting, the forms provided are to be used and sent to the designated recipient.

10.7.5.2 JPN-1: Country-specific Changes Valid for Japan Only

Overview of Changes

Amendment JPN-1 (22 JUL 2021)

Overall Rationale for the Amendment:

This local protocol amendment implements country-specific modification(s) of the protocol version 2.0, dated 05 JUL 2021.

The protocol has been revised in the appropriate sections in order to meet requirements of the Japanese health authority (PMDA) regarding reporting of disease-related outcome events when these events meet the definition of SAEs.

Revision of details regarding the documentation and reporting of (S)AEs is clarified in the protocol. In protocol version 2.0, the disease-related outcomes events were not to be documented as (S)AEs but only on the Outcome Event pages of the eCRF. In order to comply with Japanese regulatory requirements, the following disease-related outcome events will also be documented as SAEs in Japan if these events fulfill the definition of SAEs.

- CV death
- Non-fatal myocardial infarction
- Non-fatal stroke or transient ischemic attack (TIA)
- Hospitalization for heart failure or urgent heart failure visit
- Other CV hospitalization
- New diagnosis of atrial fibrillation or atrial flutter
- New diagnosis of heart failure
- Outpatient worsening of heart failure

However, in order to maintain the integrity of the trial, SUSARs that derive from any of these outcome events will be waived from unblinding. Blinded SUSAR reports will be sent to the Japanese Authorities in an expedited manner to fulfill local regulatory requirements.

In addition, clarification that Japanese modification of the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation needs be used in study sites in Japan is added to the protocol. In the equation modified for Japan, the coefficient of 0.813 is to be used as multiplicative factor.

Section # and Name	Description of Change	Brief Rationale
1.1 Synopsis	The coefficient for CKD-EPI to be used in Japan added in a footnote. Reference for CKD-EPI equation was added in Section 8.1.1.	Japanese modification of the CKD-EPI with coefficient of 0.831 is to be used in Japan.
4.1 Overall Design		
5. Study Population		
8.1.1 eGFR		
10.2 Appendix 2 : Clinical Laboratory Tests		
10.2.1.1 Local laboratory		
8.3.6 Disease-Related Outcome Events Not Qualifying as AEs or SAEs	Revision that certain outcome events have to be reported as SAEs in Japan.	Required by the Japanese health authority (PMDA) that certain outcome events have to be reported also as SAEs.

Changes to the Protocol Text

Clean amended text is provided below.

Section 1.1 Synopsis

[...]

Overall Design

[...]

The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/1.73m² (CKD-EPI¹), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/1.73m² (CKD-EPI¹).

[...]

Footnote 1: In Japan, the modified coefficient of 0.813 is to be used.

Section 4.1 Overall Design

[...]

Treatment

[...]

The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/1.73m² (CKD-EPI²), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/1.73m² (CKD-EPI²).

[...]

Footnote 2: In Japan, the modified coefficient of 0.813 is to be used.

Section 5 Study Population

[...]

To ensure a pre-specified ratio for a population at risk of progressive renal function decline, the number of participants with $\text{eGFR} \geq 25$ but $< 60 \text{ mL/min/1.73 m}^2$ (CKD-EPI³) and $\text{UACR} \geq 200$ but $< 500 \text{ mg/g}$ is planned to be capped at approximately 10% of the total population.

[...]

Footnote 3: In Japan, the modified coefficient of 0.813 is to be used.

Section 8.1.1 eGFR

[...]

The laboratory parameter eGFR will be calculated by the central laboratory based on the CKD-EPI formula⁶. The formula used for calculating eGFR (in mL/min/1.73 m^2) will be the CKD-EPI equation⁶ (Horio et al. 2010, Levey et al. 2009).

In addition, for safety reasons and titration purposes, GFR will also be assessed locally (as specified in Section 1.3). In case of unavailability of the central eGFR measurement, computed eGFR (CKD-EPI⁶) from local measurements may be used for the primary and secondary endpoint analyses.

Footnote 6: In Japan, the modified coefficient of 0.813 is to be used.

Section 8.3.6 Disease-Related Outcome Events Not Qualifying as AEs or SAEs

The following disease-related events (DREs) are common in participants with CKD and will not be reported as (S)AEs, but entered into the clinical trial database as OEs:

- Kidney failure defined as:
 - ESRD: Initiation of chronic dialysis (haemo- or peritoneal dialysis) for at least 30 days or renal transplantation.
 - Decrease of eGFR to less than $15 \text{ mL/min/1.73m}^2$, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 57\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 40\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement

[...]

In Japan, in accordance with the local regulation of reporting to the authority, the following DREs will be documented both as Outcome Events as well as SAEs in the eCRF if these events fulfill the definition of SAEs:

- CV death
- Non-fatal myocardial infarction
- Non-fatal stroke or transient ischemic attack (TIA)
- Hospitalization for heart failure or urgent heart failure visit
- Other CV hospitalization

- New diagnosis of atrial fibrillation or atrial flutter
- New diagnosis of heart failure
- Outpatient worsening of heart failure

If any of the above listed outcome events fulfill the definition of SAEs as given in Section 10.3.2, the investigator must report them immediately to the sponsor and under no circumstances should this exceed 24 hours, as indicated in Section 10.3.

However, in order to maintain the integrity of the study, SUSARs that derive from any of these above listed CV events will be waived from unblinding. Blinded SUSAR reports will be sent to the Japanese Authorities to fulfill local regulatory requirement.

Section 10.2 Appendix 2 : Clinical Laboratory Tests

[...]

Table 10–1 Central laboratory parameters

Parameter	Component
Hematology	White blood cell count (WBC), red blood cell count (RBC), hemoglobin, hematocrit, mean corpuscular volume (MCV), mean corpuscular hemoglobin (MCH), MCH concentration (MCHC), platelets
Biomarker sample	For biomarkers blood, serum, plasma and urine samples are needed (See Section 8.6 for further information)
Genetic testing	Genetics (See Section 8.6 for further information)
Glycated hemoglobin HbA_{1c} (HbA_{1c})	
Clinical chemistry (full)	Aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase (AP), gamma glutamyl transpeptidase (GGT), lactate dehydrogenase (LDH), creatine kinase (CK), cholesterol (high density lipoprotein [HDL], low density lipoprotein [LDL], total), triglycerides, creatinine, eGFR (CKD-EPI ⁷), blood urea nitrogen, uric acid, bilirubin, sodium, serum potassium, magnesium, total protein, albumin, high-sensitivity C-reactive protein (hs-CRP)
Clinical chemistry (limited)	Serum creatinine, eGFR (CKD-EPI ⁷), serum potassium, serum sodium
Urinalysis	Urine albumin-to-creatinine ratio (UACR), Urine protein/creatinine ratio (UPCR)

Footnote 7: In Japan, the modified coefficient of 0.813 is to be used.

Section 10.2.1.1 Local laboratory

[...]

- creatinine (eGFR will be calculated and recorded in the eCRF by the investigator using the CKD-EPI formula, or the JP equation for sites located in Japan⁷) from blood sample.

[...]

Footnote 7: In Japan, the modified coefficient of 0.813 is to be used.

Reference List

Horio M, Imai E, Yasuda Y, Watanabe T, Matsuo S. Modification of the CKD epidemiology collaboration (CKD-EPI) equation for Japanese: accuracy and use for population estimates. *Am J Kidney Dis.* 2010 Jul;56(1):32-8.

Levey AS, Stevens LA, Schmid CH, Zhang YL, Castro AF, 3rd, Feldman HI, et al. A new equation to estimate glomerular filtration rate. *Ann Intern Med.* 2009 May 5;150(9):604-12.

10.7.6 Israel

10.7.6.1 ISR-1: Country-specific Changes Valid for Israel Only

Overview of Changes

Amendment ISR-1 (26 NOV 2021)

Overall Rationale for the Amendment:

This local protocol amendment implements a country-specific modification of the Protocol Version 2.0, dated 05 JUL 2021.

The protocol has been revised in Section 6.5 Dose Modification in order to meet requirements of the Ministry of Health (MOH) Pharmaceutical Administration of Israel on additional information regarding potassium measurements for participants in higher risk of developing hyperkalemia.

Section # and Name	Description of Change	Brief Rationale
Section 6.5 Dose Modification	A sentence was added stating that for participants in higher risk of developing hyperkalemia, additional potassium measurements may be performed at any time on unscheduled Visits.	The change was requested by MOH.

Changes to the Protocol Text

Section 6.5 Dose Modification

[...]

During treatment with study intervention, at on-site visits all assessments of potassium, including re-tests, will be performed at both local and/or central laboratories. In cases where the investigator estimates that the participant is at higher risk of developing hyperkalemia (e.g. previous hyperkalemia, high serum potassium and lower eGFR), additional potassium measurements can be performed at any time as part of unscheduled visits (see Section 10.2). Due to sample handling and extended transportation time to the central laboratory, falsely

elevated serum potassium values in centrally (but not locally) analyzed samples may be observed. Since the value of potassium in the local sample is usually not affected by this and results from the local laboratory are available earlier, local samples will be taken not only for safety purposes but also for dose adjustments.

[...]

10.8 Appendix 8: Abbreviations

ACEI(s)	Angiotensin-converting enzyme inhibitor(s)
AE(s)	Adverse event(s)
AESI(s)	Adverse event(s) of special interest
ALT	Alanine aminotransferase
ANCA	Anti-neutrophilic cytoplasmic autoantibody
ANCOVA	Analysis of covariance
AP	Alkaline phosphatase
ARB(s)	Angiotensin receptor blocker(s)
ARNI(s)	Angiotensin receptor neprilysin inhibitor(s)
ARTS	Mineralocorticoid Receptor Antagonist Tolerability Study
AST	Aspartate aminotransferase
BNP	B-type natriuretic peptide
BYOD	Bring your own device
CI	Confidence interval
CK	Creatinine kinase
CKD	Chronic kidney disease
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CONSORT	Consolidated Standards of Reporting Trials
CV	Cardiovascular
CYP3A4	Cytochrome P450 isoenzyme 3A4
DCT	Decentralized clinical trial
DKMA	Danish Medicines Agency
DMC	Data Monitoring Committee
DRE(s)	Disease-related event(s)
DtP	Direct-to-patient
ECG	Electrocardiogram
eCRF	Electronic case report form
EDTA	Ethylene diamine tetraacetic acid
eGFR	Estimated glomerular filtration rate
EMA	European Medicines Agency
ESKD	End stage kidney disease
ESRD	End stage renal disease
EQ-5D-5L	EuroQol Group 5-dimension, 5-level questionnaire
EU	European Union
EudraCT	European Union Drug Regulating Authorities Clinical Trials

FAS	Full analysis set
FDA	United States Food and Drug Administration
FSGS	Focal segmental glomerulosclerosis
FSH	Follicle-stimulating hormone
FU	Follow-up
GCP	Good Clinical Practice
GFR	Glomerular filtration rate
GGT	Gamma glutamyl transpeptidase
HbA1c	Glycated hemoglobin
HDL	High density lipoprotein
HF	Heart failure
HR	Hazard ratio
HRT	Hormone replacement therapy
hs-CRP	High-sensitivity C-reactive protein
IB	Investigator's Brochure
ICE(s)	Intercurrent event(s)
ICF	Informed consent form
ICH	International Council on Harmonisation
IgAN	Immunoglobulin A nephropathy
IEC	Independent Ethics Committee
IND	Investigatory new drug
IRB	Institutional Review Board
IxRS	Interactive voice / web response system
LDH	Lactate dehydrogenase
LDL	Low density lipoprotein
LS	Least squares
MAR	Missing at random
MCH	Mean corpuscular hemoglobin
MCHC	Mean corpuscular hemoglobin concentration
MCV	Mean corpuscular volume
MedDRA	Medical Dictionary for Regulatory Activities
MI	Myocardial infarction
MOH	Ministry of Health
MRA	Mineralocorticoid receptor antagonist
NNT	Number needed to treat
NT-proBNP	N-terminal prohormone B-type natriuretic peptide
OD	Once daily
OE	Outcome event
PD	Premature discontinuation
PK	Pharmacokinetics
PKS	Pharmacokinetic analysis set
PMRA	Precision Medicine Array
popPK	Population pharmacokinetics

QTLs	Quality tolerance limits
RBC	Red blood cell count
RRR	Relative risk reduction
SAE(s)	Serious adverse event(s)
SAF	Safety analysis set
SAP	Statistical analysis plan
SD	Standard deviation
SGLT2	Sodium-glucose co-transporter-2
SGLT2i	Sodium-glucose co-transporter-2 inhibitor
SoA	Schedule of activities
SoC	Standard of care
SUSAR(s)	Suspected unexpected serious adverse reaction(s)
T2D	Type 2 diabetes
TEAE(s)	Treatment-emergent adverse event(s)
TESAE(s)	Treatment-emergent serious adverse event(s)
TMF	Trial master file
UACR	Urinary albumin-to-creatinine ratio
UPCR	Urinary protein-to-creatinine ratio
US(A)	United States (of America)
WBC	White blood cell count
WOCBP	Women of child-bearing potential

10.9 Appendix 9: Protocol Amendment History

The Protocol Amendment Summary of Changes Table for the current amendment is located directly before the Table of Contents (TOC).

10.9.1 Amendment 1 (01 DEC 2021)

This amendment is considered to be substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the European Union.

Overall Rationale for the Amendment:

The protocol is updated to allow for applying different eGFR calculation methodologies including cystatin C based equations. This is especially important in light of the recent updates of eGFR equations ([Inker et al. 2021](#)). Applying the combined equation of cystatin and creatinine may increase precision over creatinine only.

The 2009 CKD-EPI equation will continue to be used for screening purposes and endpoint assessments, however, all analysis assessments will be repeated using the 2021 creatinine and cystatin C based equations. Keeping a variety of methodologies allows also regional differences to be addressed.

Section # and Name	Description of Change	Brief Rationale
Section 1.3 Schedule of Activities (SoA)	Informed consent may be collected prior to screening visit.	Due to the potential retests allowed for UACR, eGFR and K+ specified in section 5.1, sample handling and collection may result in a screening window >14days. Obtaining consent prior to the screening visit gives greater flexibility to the site and subject, while maintaining the standard window between the screening and baseline procedures.
Section 1.3 Schedule of Activities (SoA) Section 8.6.2 Laboratory biomarkers Section 10.2 Appendix 2: Clinical laboratory tests Section 10.2.1.1 Local laboratory Section 10.7 Appendix 7: Country specific requirements	Collection of cystatin C was added for calculation of eGFR using cystatin C.	Creatinine based equations are more commonly used in clinical practice, and will be used for eGFR calculations with the 2009 CKD-EPI equation and updated non-race adjusted 2021 CKD-EPI equation in this study. The 2009 equation will continue to be used for screening purposes and endpoint assessments of eGFR based endpoints. However, all assessments will be repeated applying the 2021 creatinine and cystatin C-based CKD-EPI equations to be able to address potential implications of applying different formulas.
Section 3 Objectives, endpoints and estimands	Outpatient worsening of HF was added to Section 3 list of the exploratory endpoints.	Correction. Outpatient worsening of HF was listed in exploratory endpoints in Section 9.4.5, but was missing from Section 3.
Section 5.2 Exclusion criteria Section 6.8.5 Prohibited concomitant medication Section 10.6 Appendix 6: Guidance on use of common CYP3A4 inhibitors or inducers	Guidance on use of common CYP3A4 inhibitors and inducers was updated.	Guidance on use of common CYP3A4 inhibitors and inducers was updated to reflect the current regulatory categorization and to include recently approved medications that are either CYP3A4 inducers or strong CYP3A4 inhibitors. Exclusion criteria 15 has been updated to reflect the information from drug-drug-interaction studies and to be consistent with the above updated guidance on use of common CYP3A4 inhibitors or inducers.
Section 6.8.5 Prohibited concomitant medication	New section heading (Section 6.8.5) was added for prohibited concomitant medications.	Clarification, a missing header was added.
Section 8.1.1 eGFR	The guidance for eGFR calculation was modified to reflect the updated eGFR formula. Missing timepoint for eGFR collection at Month 1 was added.	All efficacy analyses including eGFR will be repeated with the 2021 creatinine and cystatin C-based CKD-EPI equation, including the relative declines from baseline and $eGFR < 15 \text{ mL}/1.73\text{m}^2$.
Section 8.1.2 UACR and UPCR	Text on timepoints for urine collection for UACR and UPCR was further clarified. Screening Visit urine samples can be collected also after screening visit.	Text on timepoints for urine collection for UACR and UPCR was further clarified. In addition, to allow greater flexibility, Screening Visit urine samples can be collected also after screening visit.

Section 8.4 Pharmacokinetics	the sentence “ <i>pharmacodynamic parameters are not evaluated in this study</i> ” was deleted.	Correction, UACR is a pharmacodynamic parameter.
Section 9.4.2.3 Sensitivity analysis of the primary efficacy endpoint	Sensitivity analysis with regard to eGFR calculation were added.	All assessments will be repeated applying the 2021 creatinine and cystatin C-based CKD-EPI equations to be able to address potential implications of applying the different formulas.
Section 9.4.3 Secondary efficacy endpoints	A sentence was added stating that supportive and sensitivity analyses for the secondary endpoints are described in the SAP.	clarification
Section 9.4.5 Exploratory Endpoints	Definition of outpatient worsening of HF was added.	clarification

In addition, minor editorial and formatting revisions, including clarifications of existing text, have been made throughout the document.

10.9.2 Amendment 2 (16 AUG 2023)

This amendment is considered to be substantial based on the criteria set forth in Article 10(a) of Directive 2001/20/EC of the European Parliament and the Council of the European Union.

Overall Rationale for the Amendment:

The protocol was updated to clarify the use of immunosuppressive therapy during the trial, to update the list of common CYP3A4 inhibitors and inducers based on the most recent FDA guidance, as well as to add clarification for study specific procedures.

This amendment also included **already approved** country/region-specific changes into a unique consolidated protocol version. A description of the country-specific changes and a brief rationale is outlined in the Appendix Country/region-Specific Requirements.

Section # and Name	Description of Change	Brief Rationale
Section 1.3 Schedule of Activities (SoA) Section 8.1.1 eGFR	Update that the 4-week safety visit may be performed at home for DCT patients if necessary and samples will be sent to the central lab. Clarification was added for the case where a central eGFR measurement or cystatin C/biomarker sample is missed at a scheduled visit.	4-week safety visit can be performed as a DCT home visit for DCT patients if deemed safe by the PI per patient's preference. If eGFR results or cystatin C/biomarker sample are missing, sites should request patient to come for re-test within 30 days
Section 6.5 Dose Modification	Clarification was added to define in which laboratory “re-tests” for potassium should be performed.	Clarification.
Section 6.8.4 Low dose steroids	Description of the concomitant use of immunosuppressive therapy was clarified. Investigators can initiate these therapies during the trial if deemed necessary unless it is	To allow medically indicated use of immunosuppressive therapy during treatment unless contraindicated (Section 6.8.5)

Section # and Name	Description of Change	Brief Rationale
	among the not permitted concomitant medication.	
Section 8.2.3 Electrocardiograms	Removal of the specification that the ECG machine automatically calculates heart rate and measures PR, QRS, QT, and QTc intervals.	It is not necessary that the ECG machine automatically calculates these measures. These can be performed separately by the site.
Section 9.3 Analysis Set	Clarification of safety analysis set definition to account for participants who receive both study interventions.	If a participant receives both study interventions due to a bottle error, the study intervention actually received for the majority of the time in the study will be used.
Section 10.4 Appendix 4: Contraceptive Guidance and Collection of pregnancy information	Clarification that hormonal contraception methods are associated with inhibition of ovulation according to CTFG recommendation based upon MHRA request	Adequate contraception is defined as any combination of at least 2 effective methods of birth control. Highly effective birth control methods are hormonal contraception (progestogen only with inhibition of ovulation), combined hormonal contraception (estrogen and progestogen with inhibition of ovulation), intrauterine devices and intrauterine hormone-releasing systems.
Section 10.6 Appendix 6: Guidance on use of common CYP3A4 inhibitors or inducers	Update of common CYP3A4 inhibitors and inducers table. Text clarifies that the list is not exclusive and that some medications may not be included if they have only recently been approved.	Table was updated to display the most common allowed and excluded medications with recent updates from authorities and recently approved new medications. A statement was added to make it clear to the PIs that the list is not exclusive.
Appendix 10.7 Country/region-specific Requirements	Denmark-, India-, Japan, and Israel-specific already-approved modifications listed in the newly added Sections 10.7.2, 10.7.3, 10.7.4, and 10.7.5.	To comply with the requirement of the EU CTR 536/2014.

10.9.3 Amendment 3 (10 APR 2025)

This amendment is considered to be substantial based on the relevant criteria of the European Union clinical trial legislation.

Overall Rationale for the Amendment:

The protocol was updated to reflect a technical change to the primary endpoint analysis using a two-slope linear spline mixed effects model, as a result of a close inspection of the detailed specifications and evaluations on convergence issues. It further specifies certain aspects of planned sensitivity and supportive analyses of the primary and secondary GFR-based

endpoints, as well as the display of demographics and baseline criteria in the FAS only. This amendment also clarifies the use of certain concomitant medication during the treatment phase as well as the time points for local safety laboratory analysis. It further clarifies wording regarding the collection of vital status after the withdrawal of consent.

Section # and Name	Description of Change	Brief Rationale
1.3 Schedule of Activities 4.1 Overall Design	A footnote was added in Section 1.3 to state that Visit window of ± 7 days does not apply to EoT Visit, and to clarify that all participants can come in for their EoT visit within the month of the last randomized subject reaching M32. Wording was also modified accordingly in Section 4.1	All participants can come in for their EoT visit within the month of the last randomized subject reaching Month 32.
1.3 Shedule of activities 10.2.1 Local laboratory	Explanation was added to both Section 1.3 and 10.2.1 that potassium <u>measurement from blood sample is also not needed when the participant is on temporary interruption of study intervention.</u>	Local lab potassium and eGFR will be assessed and used for safety and titration purposes. If the participant is on a temporary interruption or on a permanent discontinuation, local potassium is not required. Local eGFR may still be used for efficacy analysis in case central eGFR is missing.
2.3.1 Risk assessment 6.8.5 Prohibited concomitant medication	ARNIs were removed from the list of prohibited concomitant medication during study intervention treatment.	ARNI use was not allowed at inclusion as it prevents use of SoC medication for CKD (ACEI or ARB); ARNIs (sacubitril/valsartan) can be started post randomization after ACEI or ARB is discontinued, without additional safety concerns.
6.8.5 Prohibited concomitant medication	It was clarified that all MRAs are prohibited during the study treatment.	All MRAs, including commercial finerenone, are prohibited concomitant medications in this study. This is mentioned as mitigation strategy for hyperkalemia in the risk assessment, but was previously not fully clarified in the prohibited concomitant medication list.

Section # and Name	Description of Change	Brief Rationale
7.2 Participant Discontinuation/Withdrawal from the study	the following change was made: The definition of withdrawal of consent is deemed to be when the participant does not agree to any kind of direct/in direct follow up and actively refuses any further collection of data. <u>However, vital status should still be collected from public sources, if allowed according to local regulations. This is crucial to assure data completeness. Therefore</u> this Withdrawal of consent should only occur in rare and exceptional circumstances.	The text was added to stress the importance of data completeness. In this context the protocol aims to give further explanation on what the consequences of withdrawal of consent are, i.e. active collection of data from a patient is not possible any longer, nor is indirect follow up through patient medical records. What is still possible is collection of data from public sources to confirm if a patient is dead or alive. This is crucial to assure data completeness.
8.1.1 eGFR	Clarification of scope of sensitivity analyses related to alternative CKD-EPI formulas for eGFR: primary analyses for eGFR-related primary and secondary efficacy endpoints instead of all eGFR-related analyses	The aim is to perform sensitivity analysis for the primary and secondary endpoints.
9.4.1.2 Demographic and Baseline Characteristics	Removal of repetitions of tables based on the safety analysis set	The safety analysis set is expected to be largely identical to the full analysis set.
9.4.2.1 Mean rate of change as measured by the total slope of eGFR from baseline to Month-32	Refinement of specification details for two-slope linear spline mixed effects model (primary analysis): omission of baseline eGFR covariate and update of covariate matrix of random effects to vary by treatment.	Omitting baseline eGFR avoids over-parameterization and convergence issues of the model due to including baseline eGFR both as the outcome and as covariate along with a treatment-specific intercept in the model. Allowing different covariance matrices by treatment removes the assumption of equal covariance structure between treatment groups and thus enables more precise estimation. These updates are not expected to impact the treatment effect estimate and overall interpretation and are primarily of technical nature.
9.4.2.2 Supportive analysis of the primary efficacy endpoint	The word "planned" is removed so that the analysis specifies that the post-treatment eGFR measurement occurs 1 month after the actual end of treatment (i.e., the actual last dose) rather than the planned end of the treatment period.	This change aligns the analysis with the schedule of events by using each participant's actual last dose date.

Section # and Name	Description of Change	Brief Rationale
9.4.2.3 Sensitivity analysis of the primary efficacy endpoint	Change in name of strategy for alternative estimand only considering eGFR measurements collected while on treatment as "hypothetical strategy"	Name of strategy was clarified because modelling of the total eGFR slope up to month 32, due to the nature of the variable, results in a prediction beyond the intercurrent event of treatment discontinuation (to month 32), leading to a hypothetical setting. Statistical methods used are unaffected by this change.

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Title Page

Protocol Title:

A randomized, double-blind, placebo-controlled, parallel-group, multicenter Phase 3 study to investigate the efficacy and safety of **F**Inerenone, in addition to standard of care, on the progression of kidney disease in patients with **Non-Diabetic Chronic Kidney Disease**

Protocol Number: 21177

Compound Number: BAY 94-8862 / Finerenone

Short Title: Efficacy and safety of finerenone in patients with non-diabetic CKD

Acronym: FIND-CKD

Sponsor Name: Bayer AG

Legal Registered Address:

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Version History

This Statistical Analysis Plan (SAP) for study BAY 94-8862 / 21177 is based on the protocol Version 2.0 dated 05JUL2021.

SAP Version	Date	Change	Rationale
1.0	17 SEP 2021	Not Applicable	Original version

1. Introduction

Diverse pathologies induce injury and inflammation resulting in a common pathophysiological mechanism in chronic kidney disease (CKD) where glomerular injury leads to proteinuria, interstitial inflammation, and, ultimately, fibrosis. Finerenone is a mineralocorticoid receptor antagonist acting by targeting inflammatory and fibrotic pathways mediated by mineralocorticoid receptor overactivation in the heart and kidney (Kolkhof and Baerfacker 1997). In the phase 3 FIDELIO-DKD study, finerenone protected patients with CKD and type 2 diabetes from kidney function decline, and reduced the risk of cardiovascular (CV) events. Efficacy was observed independent of the hyperglycemic state (Bakris et al 2020). Finerenone also reduced albuminuria in diabetic and non-diabetic patients with CKD in the ARTS ph2a study (Pitt et al 2014). Based on the available pre-clinical and clinical data, finerenone is believed to show efficacy in delaying the progression of kidney disease in diabetic and non-diabetic patients.

The statistical analysis plan (SAP) describes the final analysis of the study. No formal statistical interim analysis will be performed. Table, figure and listing specifications are contained in a separate document.

1.1 Objectives, Endpoints and Estimands

Objectives	Endpoints
Primary	
To demonstrate that finerenone in addition to standard of care (SoC), is superior to placebo in delaying the progression of kidney disease.	Mean rate of change as measured by the total slope of eGFR from baseline to Month 32.
Secondary	
- To demonstrate the beneficial effect of finerenone in addition to SoC as compared to placebo. - To assess the safety of finerenone in addition to SoC as compared to placebo	- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, heart failure (HF) hospitalization or urgent HF visit or CV death - Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$ - Time to the composite of HF hospitalization or urgent HF visit or CV death - Number of participants with treatment-emergent adverse events (TEAE), treatment-emergent serious adverse events (TESAE) and TEAEs of special interest
Exploratory	
To be supportive of the primary efficacy endpoint by further demonstrating the effect of finerenone in addition to SoC as compared to placebo.	- Chronic eGFR slope from Month 3 to end of planned treatment period - eGFR change from baseline to 1 month after end of treatment (off-treatment)

To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (supportive of the key secondary endpoints)	- Time to onset of kidney failure (ESRD or sustained eGFR decline below 15 ml/min/1.73 m²) - Time to onset of ESRD (chronic dialysis or transplant) - Time to onset of sustained eGFR decrease below 15 mL/min/1.73 m² - Time to onset of sustained eGFR decrease of $\geq 57\%$ from baseline - Time to CV death - Time to first hospitalization for HF or urgent HF visit
To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (other exploratory endpoints)	- Number of participants with UACR reduction $\geq 30\%$ from baseline at Month 6 - Change in UACR from baseline - Time to the composite of sustained eGFR decrease of $\geq 40\%$ or kidney failure - Time to the composite of sustained eGFR decrease of $\geq 40\%$ or kidney failure - Time to the first CV event (CV death, non-fatal MI, non-fatal stroke or transient ischemic attack (TIA), hospitalization for HF or urgent HF visit) - Time to CV hospitalization - Time to all-cause mortality - Time to all-cause hospitalization - Number of participants with new diagnosis of atrial fibrillation or atrial flutter - Number of participants with new diagnosis of HF
Exploratory Biomarkers	
To further investigate the study intervention and similar drugs (e.g., mode-of-action-related effects, safety) and to further investigate pathomechanisms deemed relevant to CV/CKD and associated health problems	Various biomarkers (e.g., diagnostic, safety, pharmacodynamic, monitoring, or potentially predictive biomarkers)
Exploratory: Quality of Life	
To further investigate the quality of life parameters	Change from baseline in EQ-5D-5L

Estimand

The attributes constituting the primary estimand are the following:

- Population:** The population consists of non-diabetic CKD patients, further defined by the in- and exclusion criteria.
- Treatment:** All participants will receive blinded study intervention (i.e. finerenone 10 mg/20 mg or placebo) once daily (OD) on top of individually tolerated maximum

labeled doses of either an angiotensin-converting enzyme inhibitor (ACEI) or an angiotensin-receptor blocker (ARB).

- Variable: Estimated glomerular filtration rate (eGFR) values over time collected every 1 to 4 months during the planned treatment period
- Intercurrent events (ICE): The three most important ICEs to consider are
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant, and
 - Premature treatment discontinuation/extended interruption of treatment.

The first two ICEs will be handled using a composite strategy. For all scheduled visits after occurrence of the ICE up to the planned end of treatment an unfavorable eGFR value based on lower percentiles of the distribution of eGFR within the pertaining treatment group at the respective visit will be assumed, reflecting that both ICEs are associated with an unfavorable eGFR value.

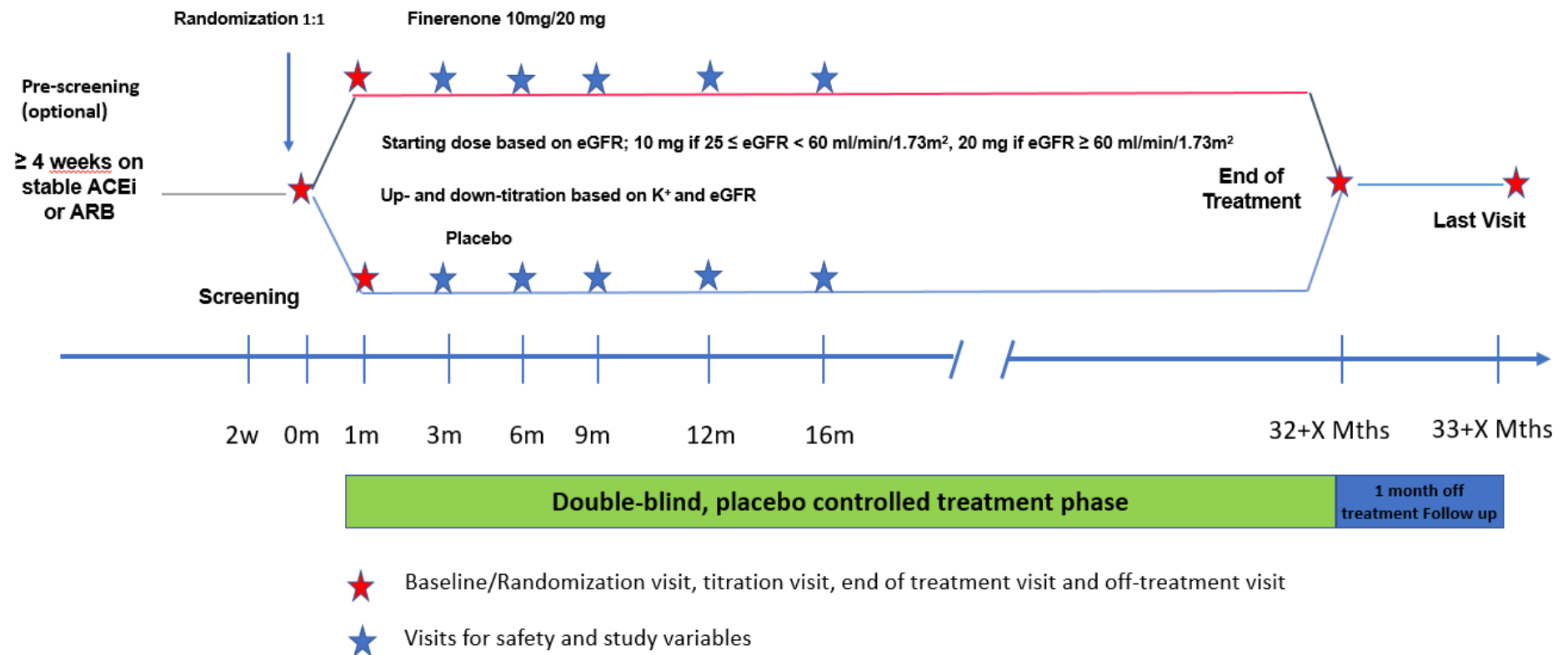
For premature treatment discontinuation/extended interruption of treatment, a treatment policy strategy will be applied. In the treatment policy strategy, participants will be followed up for their planned treatment visits after discontinuing treatment, and eGFR values after discontinuation of treatment will be included in the analysis.

- Population-level summary: Total eGFR slope from baseline to 32-months comparing finerenone treatment group with the placebo group as estimated from a random coefficient two-slope linear spline mixed-effect model.

1.2 Study Design

This is a double-blind, parallel-group intervention study in non-diabetic CKD patients with 2 treatment groups, including a placebo treatment group. The general study design as applied to this study is shown in Figure 1–1.

Figure 1–1: Study Schema



Study duration and the number of study Visits will depend on the time of enrollment of the participant

ACEi = Angiotensin-converting enzyme inhibitor, ARB = Angiotensin receptor blocker, eGFR = Estimated glomerular filtration rate, K^+ = potassium, m = Month, Mths = Months, w = Week

Pre-screening (Optional)

An assessment of Urinary Albumin Creatinine Ratio (UACR) is permitted under the Pre-screening visit and may be measured locally. This is optional and may be performed if the investigator wishes to obtain a recent UACR value to help assess if a participant is likely to be eligible. Pre-screening UACR values cannot be used in lieu of the centrally assessed urinalysis at Screening. In the exceptional case where no previous evidence of albuminuria is documented in the participant's medical records, the prescreening value can be used instead. In such situations, the screening visit can only take place 3 months after pre-screening.

Screening

Approximately 3160 participants will be screened to achieve approximately 1580 participants randomly assigned to study intervention.

Randomization

This study is a randomized, double-blind, placebo-controlled, parallel-group, multicenter study with a variable treatment duration of a minimum of 32 months, up to a maximum of approximately 49 months (final treatment duration will depend on recruitment time, the 47 months are based on an assumed recruitment time of 17 months). Participants will be randomized 1:1 to either finerenone or placebo OD in a blinded fashion on top of their SoC therapy including an ACEI or an ARB.

Randomization will be stratified by screening UACR (≤ 1000 mg/g vs. > 1000 mg/g) and baseline SGLT2 inhibitor use (yes/no) to allow equal distribution of those characteristics between the treatment arms.

Treatment

All participants will receive blinded study intervention (i.e. finerenone 10 mg/20 mg or placebo) OD on top of their individually tolerated maximum labeled doses of either an ACEI or an ARB. The starting dose will depend on the participant's eGFR level at the Screening Visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/1.73m² (CKD-EPI), or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/1.73m² (CKD-EPI). Participants who start on the lower dose of 10 mg will be uptitrated to the target dose of 20 mg from Month 1 on treatment onwards provided potassium is ≤ 4.8 mmol/L, and eGFR decrease is less than 30% below the last measured value. Down-titration or interruption of study drug is permitted at any time during the study for safety reasons and study drug has to be interrupted if potassium is > 5.5 mmol/L. Visits will take place at screening, baseline, Month 1, Month 3 and from then every 3 months up to Month 12. Thereafter, if End of Study has not been declared, participants will continue on 4-monthly visits until the End of Study.

Additional safety visits will take place 4 weeks after each up-titration or after restart of study intervention following treatment interruption for > 7 days. eGFR as the primary endpoint measure and potassium as the key safety parameter will be assessed at each visit, UACR and urinary protein/creatinine ratio (UPCR) will be collected at time points shown in the schedule of activities (SoA), see section 1.3 of the study protocol. Up- and down-titration of study drug will be based on local potassium and eGFR values.

End of treatment

The minimum planned treatment duration will be 32 months. All participants will stay in the study until the last randomized participant has reached 32 months of treatment. At this point, all participants remaining in the study and still on treatment will end treatment phase, and

move to the follow-up phase. Thus, treatment duration will be from 32 to approximately 49 months depending on the recruitment time (assumed to be approximately 17 months). Participants who have already discontinued treatment at that time, will be contacted via phone to collect outcome events and adverse events.

Follow-up

The follow-up visit will take place approximately 1 month after last administration of study intervention.

2. Statistical Hypotheses

Primary efficacy endpoint

The primary efficacy variable of interest is eGFR assessed over time. Primary endpoint is the mean rate of change of eGFR from baseline to Month 32. The rate of change of eGFR will be compared between finerenone and the placebo groups by estimating the total eGFR slope, i.e. the slope from baseline to Month 32, using available assessments from baseline to the planned end of treatment period as outlined in section 5.3.2.

In order to evaluate whether finerenone is superior to placebo in inducing a higher mean rate of change, the following null hypothesis will be tested at the two-sided significance level $\alpha = 0.05$:

$$H_0: \beta_{3,f}(32) = \beta_{3,c}(32)$$

where $\beta_{3,f}(32)$ denotes the total eGFR slope from baseline to Month 32 for finerenone and $\beta_{3,c}(32)$ denotes the corresponding total eGFR slope in the placebo group as defined in section 5.3.2. The alternative hypothesis will be:

$$H_1: \beta_{3,f}(32) \neq \beta_{3,c}(32).$$

Secondary efficacy endpoints

To evaluate whether finerenone is superior to placebo for the three key secondary time-to-event endpoints; (1) time to the composite of sustained eGFR decline of $\geq 57\%$, kidney failure, HF hospitalization or urgent HF visit or CV death, (2) time to the composite of sustained eGFR decline of $\geq 57\%$ or kidney failure, and (3) time to the composite of HF hospitalization or urgent HF visit or CV death, the following null hypotheses will be tested, where the strata are defined by the randomization stratification factors:

$$H_0: \lambda_{f,k}(t) = \lambda_{c,k}(t) \text{ for all time points } t \geq 0 \text{ and each stratum } k.$$

The alternative hypotheses will be:

$$H_1: \lambda_{f,k}(t) \neq \lambda_{c,k}(t) \text{ for at least 1 time point } t \geq 0 \text{ and at least 1 stratum } k,$$

where $\lambda_{f,k}$ denotes the hazard rate of the finerenone treatment group in stratum k and $\lambda_{c,k}$ denotes the hazard rate of the placebo treatment group in stratum k for the pertaining endpoint. In this context, the hazard rate is defined as the instantaneous risk of experiencing an event at a given time point, t , provided that no such event has yet taken place.

Multiple testing procedure of primary and key secondary efficacy endpoints

The primary and the key secondary endpoints will be tested under a confirmatory hierarchical testing procedure. If the primary efficacy endpoint achieves the significance level of 0.05, the corresponding null hypothesis can be rejected, and the three remaining key secondary

endpoints will be tested according to the pre-specified hierarchical sequence given below. Consequently, confirmatory testing of the second, third and fourth hypothesis requires rejection of each preceding null hypothesis. In this setting, each hypothesis will be assessed at a two-sided significance level of 0.05, while keeping the global type I error rate of 0.05.

The pre-defined hierarchical sequence of endpoints is given as:

1. Mean rate of change of eGFR from baseline to Month 32
2. Time to the composite of sustained eGFR decline of $\geq 57\%$, kidney failure, HF hospitalization or urgent HF visit or CV death
3. Time to the composite of sustained eGFR decline of $\geq 57\%$ or kidney failure
4. Time to the composite of HF hospitalization or urgent HF visit or CV death

3. Sample Size Determination

Approximately 3160 participants will be screened to achieve 1580 randomly assigned to study intervention.

The approximate power of t-tests for comparing eGFR slope parameters (mL/min/1.73m² per year) between treatment groups is based on a two-slope linear spline mixed-effects model specified model parameters, approximately 15 months recruitment period, at least 32 months planned time on treatment per participant, and designated sample sizes. The power estimate from the simulation is based on initial parameter values obtained from FIDELIO-DKD data, consistent with the parameters of the linear spline mixed effect model described by Vonesh et al., 2019. A total sample size of 1500 participants (750 per treatment arm) will have > 90% power to demonstrate the superiority of finerenone to placebo with respect to total eGFR slope from baseline to Month 32 at a two-sided significance level of 5%. The sample size calculation includes the assumption of 5% annual event rate resulting from ICEs (due to kidney failure or all-cause mortality). The >90% power calculation is based on an assumption of a 0.7 mL/min/1.73 m² per year total eGFR slope difference. A total sample size of 1500 will have an 80% power if we assume a 0.527 mL/min/1.73 m² per year total eGFR slope difference between finerenone and placebo.

An estimated total of 1580 participants (790 per treatment arm) will be randomized, considering an approximately 5% overall discontinuation rate in the study. All participants will be randomized in a 1:1 ratio to receive treatment with either finerenone OD or placebo.

Power calculations are presented with different scenarios in Table 3–1.

Table 3–1: Approximate power of t-tests for comparing eGFR slope parameters (mL/min/1.73m²/year) between treatment groups based on a linear spline mixed-effects model

Power calculation results for testing the following null vs alternative hypotheses at a 2-sided type I error=0.05							
H ₀ : Treatment Slope - Control Slope = 0 (no difference)							
H _A : Treatment Slope - Control Slope = DIFF (difference)							
			Total Sample Size				
Chronic Slope Treatment Difference (mL/min/1.73m ² per year)			1400	1450	1500	1550	1600
	Parameter	DIFF	Treated- Control	Treated- Control	Treated- Control	Treated- Control	Treated- Control
1.116	Chronic Slope	1.116	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Yrs)	0.33	0.376	0.385	0.397	0.408	0.418
	Total Slope (32 Months)	0.527	0.77	0.781	0.796	0.809	0.82
	Total Slope (3 Yrs)	0.592	0.864	0.873	0.884	0.894	0.903
1.224	Chronic Slope	1.224	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Yrs)	0.425	0.562	0.573	0.588	0.603	0.616
	Total Slope (32 Months)	0.625	0.893	0.9	0.91	0.919	0.926
	Total Slope (3 Yrs)	0.691	0.946	0.951	0.957	0.963	0.967
1.296	Chronic Slope	1.296	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Yrs)	0.488	0.681	0.692	0.708	0.722	0.735
	Total Slope (32 Months)	0.69	0.942	0.948	0.954	0.96	0.964
	Total Slope (3 Yrs)	0.757	0.974	0.977	0.981	0.984	0.986
1.368	Chronic Slope	1.368	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Yrs)	0.551	0.783	0.794	0.808	0.821	0.832
	Total Slope (32 Months)	0.755	0.972	0.975	0.979	0.982	0.984
	Total Slope (3 Yrs)	0.823	0.989	>0.99	>0.99	>0.99	>0.99
1.44	Chronic Slope	1.44	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Yrs)	0.614	0.863	0.872	0.884	0.894	0.903
	Total Slope (32 Months)	0.82	0.987	0.989	>0.99	>0.99	>0.99

	Total Slope (3 Yrs)	0.889	>0.99	>0.99	>0.99	>0.99	>0.99
1.512	Chronic Slope	1.512	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Yrs)	0.677	0.921	0.927	0.935	0.942	0.948
	Total Slope (32 Months)	0.886	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (3 Yrs)	0.955	>0.99	>0.99	>0.99	>0.99	>0.99

Specified model parameters: 15 months accrual period, 32 months planned time on treatment and designated sample sizes. The treated and control group acute slopes are fixed corresponding to an acute treatment effect at month-3. Further, the control group's chronic slope has been fixed. Power based on intercurrent events (kidney failure or all-cause mortality) with an overall combined rate of 5% per year.

A SAS macro program, "%GFR_Slope_Power", was provided by Vonesh et al. 2019, specifically to determine sample size/power estimates for comparing slopes between two treatment groups based on the two-slope linear spline mixed-effects model (Section 5.3.2). The power estimates are based on approximations described by Littell et al. 2006. To estimate power using the SAS macro, the user provides input values for the sample size per group and the number of measurement occasions along with hypothesized values of the parameters from the model (Section 5.3.2) and other input variables, including study accrual and follow-up periods. A table of the hypothesized parameters is included in section 6.4 (Appendix 4). The program also allows power estimates in the presence of the ICE of end-stage kidney failure, in which case power is based on an unbalanced population with missing values for the primary endpoint resulting from the ICE of kidney failure and administrative removal based on discontinuations for reasons unrelated to eGFR values, such as death.

Further participants may be enrolled in case of an exceptionally high drop-out rate due to a trial-continuity issue.

4. Analysis Sets

Analysis sets are defined in Table 4–1.

Table 4–1: Analysis sets

Participant Analysis Set	Description
Enrolled	All participants who sign the informed consent form (ICF)
Safety analysis set (SAF)	All randomized participants who have taken at least one dose of the study intervention. Participants will be analyzed according to the intervention they received.
Full analysis set (FAS)	All randomized participants. Participants will be analyzed according to the intervention they were randomized to.
Pharmacokinetic analysis set (PKS)	All finerenone-treated participants with at least 1 valid finerenone plasma concentration and without validity findings which would interfere with the evaluation of the PK data.

Final decisions regarding the assignment of participants to analysis sets will be made during the review of study data and documented in the final list of important deviations, validity findings and assignment to analysis set(s).

5. Statistical Analyses

5.1 General Considerations

The statistical evaluation will be performed by using the software SAS (release 9.4 or higher; SAS Institute Inc., Cary, NC, USA). All variables will be analyzed by descriptive statistical methods. The number of data available and missing data, mean, standard deviation (SD), minimum, quartiles, median, and maximum will be calculated for metric data. The geometric mean and SD will be provided instead of the arithmetic mean and SD for the variables where log-normal distributions are assumed. Frequency tables will be generated for categorical data.

All participants will be analyzed according to the planned study intervention in the FAS (the intent-to-treat (ITT) principle). All participants will be analyzed according to the actual study intervention in the SAF. If a participant receives both study interventions due to a bottle error, the study intervention actually received for the majority of the time in the study will be used in SAF.

The stratified analyses mentioned in this SAP will be conducted in consideration of the randomization stratification factors (1) use of SGLT2 inhibitor at baseline (yes/no) and (2) screening UACR (≤ 1000 mg/g vs. > 1000 mg/g), unless specified otherwise. For the primary analysis, participants will be assigned to the stratification categories according to the actual values as recorded on the electronic case report form (eCRF).

In case of stratification errors, the analyses of the primary endpoint and of the key secondary endpoints will also be repeated based on the stratification category used in the randomization as a sensitivity analysis.

No ICE linked to the COVID-19 Public Health Emergency is accounted for this study. However, any premature discontinuation will be summarized, and selected analyses will be repeated by confirmed COVID-19 (Yes/No) as reported by the investigator if needed.

5.1.1 Handling of Missing Data

All missing or partial data will be presented in the participant data listings as they are recorded on the eCRF.

General rules

When appropriate, the following rules will be implemented so as not to exclude participants or observations from statistical analyses due to missing or incomplete data.

Concomitant medications with missing start and stop date but flagged as being ongoing at a participant's final visit will be considered to have started prior to study medication start and ended after stopping study medication. The start and end reference period will be imputed as "before" for the medication start and as "during/after" for the medication end.

In case of (partially) missing dates for interruptions or permanent stop of study medication intake, a "worst-case" approach will be applied to impute the start and end dates of study medication intake as the minimum and maximal possible dates, i.e.:

- first month of the year, and/or first day of the month for a partially missing start date, and
- last month of the year, and/or last day of the month for a partially missing end date.

If a participant died earlier than the imputed worst study medication end date, the death date will be taken as the study medication end date. However, if these imputations lead to a temporal overlap between different exposure date records, the imputed dates will be adjusted so that no overlap exists and the time on the higher dose is maximized. The date of first exposure to treatment is not expected to be missing as the participants are instructed by the investigator to take their first dose of study drug directly at Visit 1, but in the very rare case that this date is not recorded, it will be imputed according to the rules outlined above for missing start dates, but not earlier than the randomization date.

When only partial dates are available for clinical events in the efficacy analysis, a median imputation rule will be used:

- For example, if the day is missing and the month is July, then day 16 is chosen.
- If the number of potential values is even, the lower of the 2 middle numbers is taken. For example, if the day is missing and the month is June, then day 15 is imputed. The same rule applies if the day and month are missing, e.g. if the year is 2022 and the day and month are missing, 2nd July is taken.
- In case the range of possible values is further restricted, e.g. because a participant died in the month in which the day is missing, the median in the restricted set of possible values is calculated. For example, if the clinical event occurred in June 2022 and the respective participant died on 11th June 2022, 6th June 2022 is imputed as the clinical event's date.

The same principle also applies to partially missing death dates, e.g. if only the year of death is available, but there is a last contact in the given year, then the date of death is imputed as the median of this date and the end of the given year.

In case a death date is completely missing, it will be imputed based on the last known contact when the participant was still alive and the first known contact when the participant was dead (e.g., from the participant's health status follow-up page) as the median of these two dates. As

above, if the number of potential values is even, the lower of the two middle numbers is taken.

In case both a non-fatal clinical event and death have partially missing dates, then death takes precedence and will be imputed first according to the rules outlined above. This approach also applies to non-renal and non-CV death.

However, given the importance of an accurate determination of the clinical event date in relation to the randomization date for the time to event analysis, we would expect a minimal number of such missing dates.

For the analysis of the exploratory endpoints of relative UACR reductions at Month 6, the closest post-baseline set of measurements to day 180 (+/-30 days) will be taken as Month 6. If there is no set of measurements within this time frame, the participant will be excluded from the primary analysis of UACR reduction at Month 6.

5.1.2 Data Rules

General data rules are described in this section, further data rules for specific parameters or analyses are specified in the respective subsections of section 5.

5.1.2.1 Baseline Values

For UACR, UPCR and eGFR baseline values will be defined as the mean value from the Screening Visit and Visit 1. This rule also includes assessments from a local laboratory, in case that prior to randomization, no assessment from the central laboratory is available. If the last observation available prior to randomization is the measurement from the Screening Visit, this would be used as the baseline value. Otherwise, baseline will be missing.

For all other parameters for which a baseline value needs to be defined, the baseline will be defined as the last non-missing measurement before or on the day of randomization. If the last observation available prior to randomization is the measurement from the Screening Visit, this observation would be used as the baseline value.

If more than 1 measurement was planned for a scheduled visit, for example, for blood pressure measurements, the mean value of these measurements per time point prior to randomization would be used to calculate the baseline value. This rule will not be applied for UACR and UPCR; see section 5.1.2.3 for the definition of analysis values for these parameters.

5.1.2.2 Change from Baseline

In general, change from baseline will be displayed as an absolute change from baseline defined as the difference to baseline, i.e.:

$$\text{Absolute change} = \text{Post baseline value} - \text{baseline value}$$

Some parameters will be additionally analyzed as relative change defined as

$$\text{Relative change} = 100 * [(\text{post baseline value} - \text{baseline value}) / \text{baseline value}].$$

5.1.2.3 Other Data Handling

Only the data provided by the central laboratory will be used for analysis. Values from local laboratories will not be used in the statistical analysis and will be listed only unless there are no values from the central laboratory available for baseline. An exception is made for eGFR

where local laboratory values will be used instead of missing central laboratory values at post-baseline occasions.

In case of retests for Screening visits, the closest measurement prior to randomization will be used for analysis instead of the scheduled measurements. At all visits starting at Visit 1 and if not stated otherwise, only the values at scheduled measurements will be used for analysis. In the unexpected case that more values than planned are available for a scheduled visit, only the first valid value will be used.

For those laboratory values, which are <LLOQ (Lower limit of quantification), half the value of the LLOQ will be used for analysis. Differences between 2 values <LLOQ will be assigned values of 0. Ratios between 2 values <LLOQ will be assigned a value of 1. For values which are >ULOQ (Upper limit of quantification), the ULOQ will be used for analysis. This rule does not apply to the finerenone plasma concentration data.

In case of log-normally distributed data, descriptive statistics other than minimum, maximum and median will only be calculated if at least 2/3 of the individual data were measured and were above LLOQ. In tables showing descriptive statistics, where values below LLOQ are included, these descriptive statistics will be marked.

Urinalysis is assessed 3 times per visit at screening and Visit 1 and 2 times at all post-baseline visits. According to the SoA (section 1.3 of the study protocol) UACR and UPCR have to be determined 3 times at screening and Visit 1 and 2 times at each post-baseline visit from first morning void urine samples collected on 3 or 2, respectively, consecutive days. For the analyses of UACR and UPCR, for screening and Visit 1 the measurements at one visit will be combined provided that at least 2 measurements are available. First, the coefficient of variation (CV) will be calculated for the available values as follows considering the log-normal distribution for UACR/UPCR:

$$CV = \sqrt{\exp(SD_{ln}^2) - 1}$$

where SD_{ln} is the standard deviation of the log-transformed UACR/UPCR values. If the coefficient of variation exceeds 25%, the median from the measurements will be used for the analyses. In case that only two valid measurements are available, the median will be defined as the geometric mean from these 2 values. If the coefficient of variation is 25% at the most, the geometric mean will be used. If more than 1 of the 3 scheduled assessments is missing or invalid and unscheduled assessments have been performed, then the set of unscheduled measurements closest to the planned visit will be used to determine the UACR/UPCR at the respective visit. For Visit 1, 2 out of 3 samples have to be on or before the day of randomization and have to lead to non-missing values in order for the set of samples to be valid for baseline calculation; otherwise (also including the case when there is only 1 valid sample at Visit 1), only the set of screening samples will be considered for the calculation of the baseline value.

For all post-baseline visits, in case that 2 valid measurements are available for UACR/UPCR the geometric mean of these values will be used. In case only 1 valid measurement is available, this value will be used in the statistical analyses. If the values from the scheduled assessments are missing or invalid and unscheduled assessments have been performed, then the set of unscheduled measurements closest to the planned visit will be used to determine the UACR/UPCR at the respective visit if performed within 15 days before or after the planned time point for Visit 2 (Month 1) or 30 days before or after the planned time point for all visits from Visit 3 (Month 3) onwards.

In the case of end-stage renal disease (ESRD), the UACR/UPCR and eGFR values after the onset date of ESRD will be excluded from all efficacy and safety analyses. eGFR values < 15 mL/min/1.73 m² which are measured before or on the onset date of ESRD will be included in all efficacy and safety analyses.

5.2 Participant Dispositions

The number of participants enrolled, randomized and valid for the SAF, FAS and PKS will be summarized overall and by treatment group, region, country and study site. The number of participants discontinuing each epoch, together with the primary reason for discontinuation, will be presented by the treatment group (post-randomization epochs only) and overall in separate tables.

In addition, the number of participants with important deviations and validity findings will be presented overall and by the country for each treatment group and in total. The frequencies of each important deviation and validity finding will be presented by the treatment group and in total.

5.3 Primary Endpoint Analysis

5.3.1 Definition of Endpoint

The primary efficacy variable of interest is eGFR assessed over time, collected at Month 1, then every 3 months for the first 12 months and afterwards every 4 months until the planned end of the treatment period and at an off-treatment measurement after 4 weeks post-treatment.

The laboratory parameter eGFR will be calculated by the central laboratory. The formula used for calculating eGFR (in mL/min/1.73 m²) will be the CKD-EPI equation (Levey et al. 2009).

In addition, for safety reasons, eGFR will also be assessed locally (as specified in Section 1.3 of the study protocol).

In case of unavailability of the central eGFR measurement, computed eGFR (CKD-EPI) from local measurements (if available) will be used for the analysis.

A descriptive summary of eGFR will be presented by treatment group and visit, including absolute and relative changes from baseline.

The population-level summary is the mean rate of change of eGFR as measured by total slope from baseline to Month 32.

The rate of change of eGFR will be compared between finerenone and the placebo group by estimating the total eGFR slope from baseline to Month 32 using available assessments from baseline to the planned end of treatment period. We assume the treatment effect can be expressed as difference in slope between finerenone and placebo.

For analysis of the total slope, the serial change in eGFR will be modeled using a random coefficient two-slope linear spline mixed-effects model in which a fixed change point will define acute and chronic eGFR slope at Month 3. The two-slope linear spline mixed-effects model is based on the method described by Vonesh et al. in section 2.1 of the paper "A two-slope linear spline mixed-effects models for eGFR" (Vonesh et al. 2019). The acute and chronic phases are differentiated because finerenone showed acute effects on eGFR that differed from its long-term effects on disease progression in FIDELIO-DKD. In addition to treatment specific (finerenone vs. placebo) fixed effects for the intercept, the acute slope and

the directional change in slope at the change point, time (continuous), and treatment by time interaction, the model will include fixed effects for stratification factors used in the stratified randomization (i.e., baseline SGLT2i use (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g) and for baseline eGFR (continuous). Random effects for the intercept, acute slope (baseline to Month 3), and directional change in slope at the change point (Month 3) will be included to account for variation in participants' progression rates over time. An unstructured covariance will be used to model the within-participant errors. The time variable represents the duration from baseline to each eGFR assessment during acute and chronic phases. This model will include all observed eGFR measurements during the planned treatment period.

5.3.2 Main Analytical Approach

The primary analysis of the primary efficacy variable will be performed according to participants' randomized intervention and will be based on the FAS.

The primary estimand, as defined in section 1.1 will be used for the primary analysis. The random coefficient linear spline mixed-effect model behind the population-level summary is presented in more detail.

For reasons of clarity, only the fixed effects for treatment and time and their interaction will be included in the model specification outlined in the following. In contrast to that, the fixed effects for baseline eGFR and for the stratification factors used in the stratified randomization (i.e., baseline SGLT2i use (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g)) will not be shown in the model formula as their inclusion is straightforward.

As described in Vonesh et al., 2019, we will formulate the two-slope linear spline mixed-effects model as follows:

$$Y_{ij} = (\beta_{0c} + \beta_{1c}t_{ij} + \beta_{2c} \max\{t_{ij} - t^*, 0\}) (1 - X_i) + (\beta_{0f} + \beta_{1f}t_{ij} + \beta_{2f} \max\{t_{ij} - t^*, 0\}) X_i + b_{0i} + b_{1i} t_{ij} + b_{2i} \max\{t_{ij} - t^*, 0\} + \epsilon_{ij}$$

where,

Y_{ij} denotes the response variable, eGFR, for the i -th participant on the j -th occasion,
 X_i be a treatment group indicator variable with $X_i = 0$ if the i -th participant is randomized to the placebo group and $X_i = 1$ if randomized to finerenone group,

$\beta_k = (\beta_{0k} \beta_{1k} \beta_{2k})'$ denote the regression parameters for placebo ($k=c$) and finerenone ($k=f$),

ϵ_{ij} denote within-participant residuals,

$b_i = (b_{0i} b_{1i} b_{2i})'$ denote between-participant random effects,

t^* denotes the change point (knot at month 3),

$\max\{t_{ij} - t^*, 0\}$ be the truncated line function

where t_{ij} are the time points at which eGFR is measured for the i -th participant across v_i occasions ($j = 1, 2, \dots, v_i$).

Model assumptions:

- $(\epsilon_{ij} \mid b_i)$ are conditionally independent normally distributed random variables with mean 0 and within-participant variance σ_{ij}^2
- Participant-specific random effects b_i that are independent multivariate normally distributed with mean vector 0 and 3x3 covariance matrix that reflects a common unstructured covariance matrix

The interpretation of the regression parameter $\beta_k = (\beta_{0k} \beta_{1k} \beta_{2k})'$ in terms of chronic and acute slope for placebo ($k=c$) or finerenone ($k=f$), respectively, is as follows:

β_{0k} denotes the intercept for treatment group k ,

β_{1k} is the acute slope up through but excluding time t^* (knot at month 3) for treatment group k ,

β_{2k} is the directional change in slope starting at time t^* (knot at month 3) for treatment group k , and

$\beta_{1k} + \beta_{2k}$ corresponds to the chronic slope for treatment group k .

Finally, under these assumptions the total-slope up to time t describing the rate of change in eGFR from baseline to a specific time $t > t^*$ (t = total study time, here: 32 months) is defined for treatment group k as

$$\beta_{3k}(t) = \{\beta_{1k} t + \beta_{2k} (t - t^*)\} / t = \beta_{1k} + \beta_{2k} - \beta_{2k} (t^* / t)$$

In terms of the model parameters as defined in the previous steps the hypothesis to be tested for the primary endpoint can be formulated in the way shown in Section 2.

The analysis will be carried out using maximum likelihood procedures. Linear contrasts will be constructed to estimate the acute, chronic, and total slope in eGFR. Least squares (LS) means of the acute, chronic and total eGFR slope for finerenone and placebo group with 95% confidence interval (CI) and corresponding two-sided p-value will be presented.

The time-dependent total eGFR slope until Month 32 will be calculated for the two treatment groups. The difference of total eGFR slope from baseline to Month 32 between the finerenone and the placebo treatment group will be calculated, and a corresponding two-sided 95% CI and p-value will be presented. The study will be considered successful if the p-value is statistically significant (i.e. below 0.05).

5.3.3 Sensitivity Analyses

5.3.3.1 Sensitivity Analyses Regarding the Imputation of Missing Values

We consider that intermittent missing eGFR values (e.g., due to one missed visit) will not lead to a biased estimation of the slope. In contrast to that, for monotone missing data patterns (e.g., missingness due to lost to follow-up), we will consider sensitivity analyses under a 'missing not at random' assumption. For this a "control-based pattern imputation method", a combination of imputation from retrieved drop-outs and assumed unfavorable values in case of ESRD or death as for the primary estimand, and a "tipping point analysis" will be employed. Although imputed intermittent values will not be used in the final analyses, these values will be imputed in a first step to allow further imputations for the monotone missing data.

For the control-based pattern imputation method, in the first step, intermittent missing values will be imputed using the Markov Chain Monte Carlo (MCMC) methodology, which assumes a multivariate normal distribution over all variables included in the imputation model. Variables included in the imputation model will be baseline eGFR and post-baseline eGFR values as measured over time until Month 32. This imputation will be done by the treatment group and the stratification factors. In a second step, all the monotone missing values will be multiply-imputed using the imputation model built from the placebo treatment group, i.e., assuming the missing data in the finerenone treatment group will have a profile that equals the profile of the placebo group for all time points. Both steps of the missing data imputation will be implemented using SAS procedure PROC MI with the MNAR statement.

For the combination of imputation from retrieved drop-outs with the assumption of unfavorable values for cases of death and ESKD, monotone missing data of participants with a premature discontinuation of study intervention due to other reasons than death or onset of ESKD will be imputed using the data from participants, who despite of early treatment discontinuation contribute eGFR measurements up to the planned end of study intervention (retrieved drop-outs). The imputation process is similar to the control-based pattern imputation with the imputation model in the second step now built from the retrieved drop-outs per study intervention group.

For the tipping point analysis, a delta-adjusted pattern imputation method will be applied, using the same variables for the imputation model as for the control-based pattern imputation. In a first step, intermittent missing values will again be imputed using the MCMC methodology. For the other missing values, in a second step, it will be assumed that the mean of the missing outcomes in the finerenone group deviates a certain amount, delta, to their observed counterparts in the direction of worsening the outcome, but the mean in the control group is the same between missing values and observed values. Thus, each treatment group's missing values will be imputed using the observed values in their respective treatment groups. After imputation, the imputed values in the finerenone group are shifted a certain amount in the direction of worsening the outcomes, but the placebo group's imputed values remain unchanged. Multiple shifts within a plausible range are applied. The shift that turns the conclusion from favoring the finerenone treatment group to a non-significant result will define the tipping point. The missing data imputation will be implemented using SAS procedure PROC MI with the MNAR statement together with the ADJUST option.

Technical details of the sensitivity analyses are provided in section 6.

5.3.3.2 Sensitivity Analyses Related to Stratification Factors

As already mentioned in section 5.1, in case of stratification errors, the main analysis of the primary endpoint will also be repeated with the stratification categories used in the randomization as covariates in the random coefficient two-slope linear spline mixed-effect model.

5.3.3.3 Sensitivity Analysis Related to the Statistical Model

To investigate whether a model that only specifies a single slope per study intervention leads to similar results as the two-slope model considered in the primary analysis, a standard random coefficient linear mixed-effects model for the serial change in eGFR will be considered as a sensitivity analysis.

This model includes treatment-specific fixed effects for the intercept and the (total) slope that remains unchanged over time. Further it contains fixed effects for time (continuous), a treatment by time interaction, and fixed effects for stratification factors used in the stratified randomization (i.e., baseline SGLT2i use (yes/no) and screening UACR (≤ 1000 mg/g vs. > 1000 mg/g)) and for baseline eGFR (continuous). Random effects for intercept and slope will be included to account for variation in participants' progression rates over time. This model will include all observed eGFR measurements during the 32-month treatment period.

Analyses for this model will be similar to the primary analysis for the two-slope model using maximum-likelihood methods, missing at random (MAR) assumption for intermediate missing values and imputation with unfavorable values for missing values after death or start of chronic dialysis/kidney transplantation.

5.3.3.4 Sensitivity Analysis Related to Informative Censoring

Another approach to deal with the ICEs of death and start of chronic dialysis/kidney transplant is to consider these events as nonignorable dropout and to use a shared parameter (SP) model that jointly models the trend in eGFR over time and the time to potentially nonignorable (informative) censoring events. A detailed discussion of the use of SP models in the context of the CKD setting can be found in the paper of Vonesh et al. 2019.

For this type of sensitivity analysis we will use the two-slope linear spline mixed-effects model as discussed in section 5.4.1.2 for the trend in eGFR over time and - as suggested by Vonesh et al.- a piecewise exponential survival model for the time-to-event part of the SP model.

As fitting an SP model can become problematic in case of a low number of nonignorable censoring events, this sensitivity analysis will only be performed if > 15 events are observed in total.

For details of the statistical analysis see Vonesh et al. 2019.

5.3.4 Supplementary Analyses

5.3.4.1 Supplementary Analyses with Alternative Estimands

The following four alternative estimands for the primary endpoint will be considered to evaluate different strategies for the handling of ICEs:

The first estimand is defined by the following attributes:

- Population: as defined for the primary estimand
- Treatment: as defined for the primary estimand.
- Variable: as defined for the primary estimand.
- ICEs: The same three ICEs as for the primary estimand are considered:
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant, and
 - Premature treatment discontinuation/extended interruption of treatment.

For the ICEs death and start of chronic dialysis/kidney transplantation now a “while-on-treatment”/“while alive” strategy will be applied assuming missing values fulfill the MAR assumption.

For the ICE of premature treatment discontinuation/extended interruption of treatment, the same treatment policy strategy will be applied as for the primary estimand.

- Population-level summary: Defined as for the primary estimand.

The second estimand is defined by the following attributes:

- Population: as defined for the primary estimand
- Treatment: as defined for the primary estimand.
- Variable: as defined for the primary estimand.
- ICEs: The same three ICEs as for the primary estimand are considered:
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant, and

- Premature treatment discontinuation/extended interruption of treatment.

For the ICEs death and start of chronic dialysis/kidney transplantation now a different "composite variable" strategy compared to the primary estimand will be employed where a worst-case eGFR value of 0 mL/min/1.73 m² will be assumed for all scheduled visits up to the planned end of treatment.

For the ICE of premature treatment discontinuation/extended interruption of treatment, the same treatment policy strategy will be applied as for the primary estimand.

- Population-level summary: Defined as for the primary estimand.

The third estimand is defined by the following attributes:

- Population: as defined for the primary estimand
- Treatment: as defined for the primary estimand.
- Variable: as defined for the primary estimand.
- ICEs: The same three ICEs as for the primary estimand are considered:
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant, and
 - Premature treatment discontinuation/extended interruption of treatment.

For the ICEs death and start of chronic dialysis/kidney transplantation, the same "while-on-treatment" strategy as specified for the first estimand discussed in this section will be applied.

The same "while-on-treatment" strategy will now also be applied to the ICE of premature treatment discontinuation/extended interruption of treatment to assess the slope difference while the participants were taking study medication using only the eGFR values obtained before treatment discontinuation.

- Population-level summary: as defined for the primary estimand.

The last estimand is defined by the following attributes:

- Population: as defined for the primary estimand
- Treatment: as defined for the primary estimand.
- Variable: as defined for the primary estimand.
- ICEs: The following four ICEs are considered:
 - Death,
 - Starting chronic dialysis or receiving a kidney transplant,
 - Premature treatment discontinuation/extended interruption of treatment, and
 - Start of a new SGLT2 inhibitor therapy after baseline.

The first three ICEs are handled with the same strategies as applied for the primary estimand

For the ICE of start of a new SGLT2 inhibitor therapy after baseline, a "while-on-treatment" strategy will be applied to set the eGFR values to missing that were measured after the start of the SGLT2 inhibitor.

- Population-level summary: as defined for the primary estimand.

The four alternative estimands will be evaluated analogously to the primary estimand.

5.3.4.2 Supplementary Analyses for the Primary Endpoint

Chronic eGFR slope

The eGFR slope will be determined separately from randomization to Month-3 (acute slope), i.e., the rate of eGFR decline from randomization to Month 3, and after Month-3 to end of planned study treatment (chronic slope), i.e., the rate of eGFR decline from Month-3 after randomization to the end of the treatment period.

As a supportive analysis, the chronic eGFR slope will be analyzed using the same two-slope model as described for the primary analysis (see Section 5.3.2) with Month-3 defined as the 'change point' in the model and the time variable representing the duration from baseline to each eGFR assessment during acute and chronic phases.

LS means of the chronic eGFR slope of finerenone and placebo group and the treatment difference (finerenone minus placebo) with a 95% CI and corresponding two-sided p-value, will be presented.

eGFR change from baseline to 1 month after the end of treatment period (off-treatment)

A post-treatment eGFR measurement will be conducted for all participants 1 month after end of treatment.

As a supportive analysis, an ANCOVA model will be fitted to the above change from baseline eGFR value to the off-treatment visit, including the factors treatment group, stratification factors, total time on treatment, and the baseline eGFR value as a covariate. Corresponding two-sided 95% CI and p-value will be presented.

5.4 Secondary Endpoints Analysis

The following secondary variables will be evaluated to further explore the chronic and persistent benefit of finerenone on kidney disease progression.

5.4.1 Key/Confirmatory Secondary Endpoints

The three key secondary endpoints are

- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or urgent HF visit or CV death
- Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$
- Time to the composite of HF hospitalization or urgent HF visit or CV death

Key secondary endpoints are defined as time to the first occurrence of certain outcome events (OEs) (see section 5.6.3.1). In this study, OEs will be assessed by the investigators based on their clinical judgment. For each endpoint, the investigator will record information specific to that type of endpoint in the eCRF. An exception are those OEs which are related to sustained decline of eGFR, these events will be derived programmatically, see section 5.4.1.1. Event criteria will be provided to facilitate event collection and edit checks will be in place to detect potential inconsistencies.

5.4.1.1 Definition of Endpoints

Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, heart failure (HF) hospitalization or urgent HF visit or CV death

This secondary efficacy variable is defined as the time from randomization to the first occurrence of the composite endpoint of kidney failure, a sustained decrease of $\text{eGFR} \geq 57\%$ from baseline over at least 4 weeks, HF hospitalization or urgent HF visit or CV death. Kidney failure is defined as either the occurrence of ESRD or an eGFR of less than $15 \text{ mL/min/1.73 m}^2$, confirmed by a second measurement at the earliest 4 weeks after the initial measurement. ESRD is defined as the initiation of chronic dialysis (haemo- or peritoneal-dialysis) for at least 30 days or renal transplantation, see also section 5.6.3.1.

The attributes constituting the estimand for this key secondary endpoint are the following:

- Population: The population consists of non-diabetic CKD patients, further defined by the in- and exclusion criteria.
- Treatment: All participants will receive blinded study intervention (i.e. finerenone 10mg/20mg or placebo) OD on top of an individually tolerated maximum labeled doses of either an ACEI or an ARB.
- Variable: Time to event of the first occurrence of either kidney failure, a sustained eGFR decline of $\geq 57\%$ from baseline over at least 4 weeks, HF hospitalization or urgent HF visit or CV death, whichever occurs first.
- ICEs: The two most important ICEs to consider are
 - Non-CV death
 - Premature treatment discontinuation/extended interruption of treatment.

The ICE of non-CV death will be handled using the "while alive" strategy.

For premature treatment discontinuation/extended interruption of treatment, a treatment policy strategy will be applied, which is outlined in more detail in the description of the censoring rules below.

- Population-level summary: Hazard rates of time-to-event comparing the finerenone treatment group with the placebo group using a stratified log-rank test.

The individual components “Sustained decrease of $\text{eGFR} \geq 57\%$ from baseline over at least 4 weeks” as well as “Sustained eGFR decline $< 15 \text{ mL/min/1.73 m}^2$ ” will be programmatically derived. Only in case the eGFR decrease was confirmed by at least one additional eGFR measurement taken at least 4 weeks later, it will be considered as a sustained decrease and counted for the renal endpoint. The date used for the analysis will be the date of the initial sample exceeding the threshold. In case there was no confirmatory assessment, events will only be counted for the renal endpoint when the participant died after the initial decrease or the participant went on renal replacement therapy such as dialysis or transplantation. If there is an intermediate measurement that does not confirm the initial decrease, the event will not be counted for the renal endpoint.

Outcome events for inclusion in the analysis will be counted from the day of randomization (planned at Visit 1) onwards until the planned end of treatment visit or the date of end of study notification + 4 weeks if the planned end of treatment visit has not been performed. In the event of premature discontinuation from the study with no subsequent follow-up information, events will be counted up to the day of the last visit where complete information on all components of the composite endpoint is available with an additional rule for events occurring after this visit. To account for events of kidney failure, CV death or HF hospitalization or urgent HF visit after the last eGFR is recorded at a clinic visit starting with Visit 2 (Month 1), such events will be included in the efficacy analysis if they occur in the

period up to 1 day before the next planned clinic visit, allowing a time window for late attendance of 1 month. E.g., if the next visit is planned to take place 4 months later, then this period would be a maximum of 5 months. Besides, if the participant dies due to a non-CV related cause in this period, then the date of death will be used as the censoring date. Randomized participants without an event of the composite endpoint at the time of analysis will be censored at the date of their last visit where complete information on all components of the composite endpoint is available, up to and including the planned end of treatment visit or EoS phone call for participants who have discontinued prematurely (should this visit satisfy this rule), or date of non-CV death using a time window as defined above if a subsequent clinic visit had been planned. Participants without any information about the composite endpoint after baseline will be censored at Day 1.

Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$

This secondary efficacy variable is defined as the time from randomization to the first occurrence of either kidney failure or a sustained eGFR decline of $\geq 57\%$ as defined for the first key secondary endpoint.

The attributes constituting the estimand for this key secondary endpoint are the following:

- Population: Same as in the estimand for the first key secondary endpoint.
- Treatment: Same as in the estimand for the first key secondary endpoint.
- Variable: Time to a sustained eGFR decline of $\geq 57\%$ or kidney failure.
- ICEs: The two most important ICEs to consider are
 - Death
 - Premature treatment discontinuation/extended interruption of treatment.

The ICE death will be handled using the "while alive" strategy.

For the other ICE a treatment policy strategy will be applied.

- Population-level summary: Hazard rates of time-to-event comparing finerenone treatment group with the placebo group using a stratified log-rank test.

Censoring rules for this combined renal endpoint will be similar as for the combined renal and CV endpoint. Renal events for inclusion in the analysis will be counted from the day of randomization (planned at Visit 1) onwards until the planned end of treatment visit or the date of end of study notification + 4 weeks if the planned end of treatment visit has not been performed. In the event of premature discontinuation from the study with no subsequent follow-up information, renal events will be counted up to the day of the last visit where complete information on all components of the composite endpoint is available with an additional rule for events occurring after this visit. To account for events of kidney failure, after the last eGFR is recorded at a clinic visit starting with Visit 2 (Month 1), such events will be included in the efficacy analysis if they occur in the period up to 1 day before the next planned clinic visit, allowing a time window for late attendance of 1 month. Besides, if the participant dies in this period, then the date of death will be used as the censoring date. Randomized participants without an event of the composite renal endpoint at the time of analysis will be censored at the date of their last visit where complete information on all components of the composite endpoint is available, up to and including the planned end of treatment visit or EoS phone call for participants who have discontinued prematurely (should this visit satisfy this rule), or date of death using a time window as defined above if a

subsequent clinic visit had been planned. Participants without any information about the composite renal endpoint after baseline will be censored at Day 1.

Time to the composite of HF hospitalization or urgent HF visit or CV death

This secondary efficacy variable is defined as the time from randomization to either the first HF hospitalization or urgent HF visit or CV death, whichever occurs first.

The attributes constituting the estimand for this key secondary endpoint are the following:

- Population: Same as in the estimand for the first key secondary endpoint.
- Treatment: Same as in the estimand for the first key secondary endpoint.
- Variable: Time to event of first HF hospitalization or urgent HF visit or CV death.
- ICEs: The two most important ICEs to consider are
 - Non-CV Death,
 - Premature treatment discontinuation/extended interruption of treatment.

The ICE non-CD death will be handled using the "while alive" strategy. For the other ICE, a treatment policy strategy will be applied.

- Population-level summary: Hazard rates of time-to-event comparing finerenone treatment group with the placebo group using a stratified log-rank test.

CV events will be counted up to the day of withdrawal from the study or non-CV death. Randomized participants without an event of the CV composite endpoint at the time of analysis will be censored at the date of their last contact, up to and including the planned end of treatment visit or EoS phone call for participants who have discontinued prematurely, or date of non-CV death.

5.4.1.2 Main Analytical Approach

Time to the composite of kidney failure, a sustained eGFR decline of $\geq 57\%$, HF hospitalization or urgent HF visit or CV death

In order to evaluate whether finerenone is superior to placebo in prolonging the time to the first event of the composite endpoint, the corresponding null hypothesis will be tested using a stratified log-rank test at the two-sided 5% significance level. The strata for the stratified log-rank-test will be defined by the randomization stratification factors.

A stratified Cox regression model will be used to provide point estimates of the hazard ratio with a corresponding two-sided 95% CI for each stratum. The strata for the stratified Cox model will be defined by the randomization stratification factors.

In addition, cumulative incidence plots based on the Kaplan-Meier estimator will be presented by the treatment group.

Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$

This time-to-event variable will be analyzed the same way as the time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or urgent HF visit or CV death.

Time to the composite of HF hospitalization or urgent HF visit or CV death

This time-to-event variable will be analyzed the same way as the time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, HF hospitalization or urgent HF visit or CV death.

5.4.1.3 Sensitivity Analyses

5.4.1.3.1 Sensitivity Analyses Related to the Censoring Mechanism

The censoring mechanism of participants without an event at the time of analysis is assumed to be non-informative in the analysis of the key secondary efficacy endpoints. This also implies that this analysis is valid under a MAR assumption. Sensitivity analyses will be performed, assessing the impact of a potential informative censoring of participants lost to follow-up or with withdrawn consent.

A summary of the amount of missing data by treatment arm will be given, including the number of participants who have incomplete follow-up on the respective endpoint and the ratio of the observed follow-up time to the expected follow-up time for the respective endpoint.

Robustness with respect to missing data will be investigated with a tipping point analysis on the FAS. The analysis will:

- Estimate the hazard at time of loss to follow-up, using a Weibull survival model fitted to the population of participants who prematurely discontinued study treatment and adjusted for treatment group and stratification factors.
- Inflate the hazard only in the finerenone group by a known factor, assuming non-informative censoring in the control group.
- Impute events to the end of the study 1000 times, calculate the hazard ratios with the stratified Cox model described in section 5.4.1.2 and combine them using standard multiple imputations combining rules.
- Increase the inflation factor until the upper limit of two-sided 95% CI for the hazard ratio crosses 1.0; this will be the "tipping point".

The tipping point will show how much higher the event risk after drop-out would need to be in the finerenone group so that statistical significance is lost. Since there is no particular reason to assume that the event rate should be higher, a large tipping point will indicate robustness of the study results. Technical details of the model will be provided in section 6.

5.4.1.3.2 Sensitivity Analyses Related to Stratification Factors

As already mentioned in section 5.1, in case of stratification errors the main analysis of the key secondary endpoints will also be repeated with the stratification categories used in the randomization defining the strata for the stratified log-rank test and the stratified Cox regression model.

The log-rank test, as well as the Cox regression model, will also be performed without including stratification factors.

5.4.1.4 Supplementary Analyses

5.4.1.4.1 Supplementary Analyses with Alternative Estimands

For the three key secondary time-to-event endpoints, alternative estimands applying a "while-on-treatment" strategy for ICEs will be evaluated.

The attributes of these alternative estimands are the same as for the main estimands discussed in section 5.4.1.1, with the exception of the handling of ICEs.

According to the "while-on-treatment" strategy, participants with censoring or event date more than 30 days after last study intervention in the analysis of the main estimand will now be censored at the date of last study intervention + 30 days.

5.4.1.4.2 Other Supplementary Analyses

The following other supplementary analyses of the key secondary efficacy variables will be performed:

Cumulative incidence probabilities for each treatment group based on Kaplan-Meier estimates for the time to the first endpoint event and their differences between finerenone and placebo will be derived at appropriate time points along with 95% CIs and Number Needed to Treat (NNT). The CIs will be calculated based on a normal approximation and using the Greenwood formula for the standard deviation, and the NNT will be calculated as the inverse difference of the Kaplan-Meier estimates between finerenone and placebo for the selected time points. In addition, cumulative incidence plots based on the Kaplan-Meier estimator will be presented for each treatment group. These analyses will be presented in total and for each level of the stratification factors.

For the three key secondary efficacy variables, the incidence rate per 100 patient-years will be presented by treatment group, which is defined as the number of participants with an event divided by the total sum of the times a participant is at risk for the respective event, i.e. either the time from randomization until the first event or the censoring time, multiplied by 365.25×100 .

For the composite endpoints cumulative incidence plots, log-rank tests and Cox regression for the time to first event of the individual components will also be provided as part of exploratory efficacy (see section 5.4.2).

The plausibility of the proportional hazards assumption will be assessed by visually examining the plot of the log of the negative log of Kaplan-Meier estimates of the survival function versus log time for evidence of non-parallelism and by including treatment by log-transformed time interaction into the Cox model.

The significance of the treatment by log-transformed time interaction will be tested at the 5% significance level. If the interaction is significant and there is strong evidence of non-proportionality from the plots, time-dependent hazard ratios will be estimated with the model that includes the interaction term.

5.4.2 Supportive Exploratory Endpoint(s) for the Key Secondary Endpoints

Components of the key secondary endpoints

The following components of the key secondary endpoints will be analyzed as supportive secondary endpoints:

- Time to onset of kidney failure (ESRD or sustained eGFR decline below 15 mL/min/1.73 m²)
- Time to onset of ESRD (chronic dialysis or transplant)
- Time to onset of eGFR decrease to less than 15 mL/min/1.73 m² sustained over at least 4 weeks

- Time to onset of eGFR decrease of ≥ 57 % from baseline sustained over at least 4 weeks
- Time to CV death
- Time to first hospitalization for HF or urgent HF visit

Consistent with the methods for the key secondary efficacy variables, components of the secondary time-to-event endpoints (as included in the list of other exploratory efficacy variables above) will be analyzed as exploratory time-to-event variables using the stratified log-rank test stratified by the randomization stratification factors. Point estimates of the hazard ratio and corresponding two-sided 95% CI will be calculated using a stratified Cox regression model. Cumulative incident plots will be provided for the cumulative proportions of events by treatment group.

5.5 Exploratory Endpoints Analysis

5.5.1 Other Exploratory Endpoints Specified in the Study Protocol

Apart from the components of the key secondary endpoints, other exploratory efficacy variables as specified in the study protocol are:

- Number of participants with UACR decrease of at least 30 % from baseline at Month 6
- Change in UACR from baseline
- Time to occurrence of the composite endpoint of kidney failure or an eGFR decrease of ≥ 40 % from baseline sustained over at least 4 weeks
- Time to the first CV event (CV death, MI, stroke/TIA, hospitalization for HF or urgent HF visit)
- Time to first CV hospitalization
- Time to death from any cause (all-cause mortality)
- Time to first all-cause hospitalization
- Number of participants with new diagnosis of atrial fibrillation or atrial flutter
- Number of participants with new diagnosis of HF
- Change in Quality of Life (QoL) summary scores measured by the following health-related quality of life (HRQoL) questionnaire: EuroQol Group 5 dimension, 5 level questionnaire (EQ-5D-5L)

5.5.2 Other Exploratory Endpoints Not Specified in the Study Protocol

Other exploratory endpoints not specified in the study protocol are as follows:

- Time to first occurrence of the composite endpoint of CV death, stroke/TIA or MI
- Time to onset of chronic dialysis
- Time to onset of eGFR decrease of ≥ 40 % from baseline sustained over at least 4 weeks
- Time to first stroke/TIA
- Time to first MI

- Time to first non-fatal MI, non-fatal stroke/TIA, hospitalization for HF or urgent HF visit
- Number of participants with UACR decrease of at least 30% from baseline at any time post-baseline
- Number of participants with UACR decrease of at least 40% from baseline at any time post-baseline
- Number of participants with UACR decrease of at least 50% from baseline at any time post-baseline
- Acute eGFR slope (as estimated from the two-slope model used for the primary endpoint)

5.5.3 Exploratory Time-to-Event Endpoints

The endpoint “Sustained decrease of eGFR ≥ 40 % from baseline over at least 4 weeks” will be programmatically derived as the respective components of the first and second key secondary endpoints, see section 5.4.1.1 for details of the derivation.

All exploratory time-to-event endpoints will be analyzed using the stratified log-rank test stratified by the randomization stratification factors. Point estimates of the hazard ratio and corresponding two-sided 95% CI will be calculated using a stratified Cox regression model. Cumulative incidence plots based on the Kaplan-Meier estimator will be provided for the cumulative proportions of events by treatment groups.

5.5.4 New Diagnosis of Atrial Fibrillation/Atrial Flutter and New Diagnosis of HF

Frequency tables will be generated for the number of participants with a new diagnosis of atrial fibrillation/atrial flutter and for the number of participants with a diagnosis of new HF after baseline.

The presence or absence of the event will be compared between the treatment groups using logistic regression with the factors of treatment group and randomization stratification levels. The odds ratios between the finerenone and the placebo treatment group will be calculated, and corresponding two-sided 95% CIs and p-values will be presented.

5.5.5 UACR

UACR during the study will be summarized descriptively by treatment group and visit including ratios to baseline. These analyses will be performed overall and separated by the stratification factors.

The log-transformed ratio of UACR to baseline at each visit up to the last visit with a reasonable number of participants with UACR measurements will be analyzed by a mixed model for repeated measures (MMRM) with the factors treatment group, visit, treatment-by-visit interaction, factors for the stratification levels, log-transformed baseline value as a covariate and log-transformed baseline value by visit interaction. Pairwise ratios between the finerenone and the placebo treatment group will be calculated and corresponding two-sided 95% confidence intervals will be computed.

An on-treatment analysis including measurements restricted to up to 30 days after last intake of study drug will be performed using the same model as above. The SAS procedure PROC MIXED will be used estimating covariance patterns within participants to adjust for the within participant variance. For each treatment group a separate covariance pattern will be

estimated based on an unstructured covariance. In case of convergence issues, an alternative covariance matrix will be used, with Toeplitz matrix as first choice alternative.

Frequency tables will be generated for the number of participants with a relative decrease in UACR of $\geq 30\%$, $\geq 40\%$ and $\geq 50\%$ from baseline. The analysis will be performed for each visit on which UACR is collected and for any time, post-baseline.

At Month 6, the categorical variable of a relative UACR decrease of $\geq 30\%$ from baseline will be analyzed for the presence or absence of the event using logistic regression including covariates for the treatment group and stratification levels. The odds ratio between the finerenone and the placebo treatment group will be calculated, and corresponding two-sided 95% CI and p-value will be presented.

5.5.6 Health-related Quality of Life

Health-related quality of life will be assessed using the EQ-5D-5L questionnaire. The EQ-5D-5L is a self-administered preference-based generic measure of health status, which includes five dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression (EuroQol Group 2013). Participants provide a rating for each question on a five-level Likert scale: having no problems, having slight problems, having moderate problems, having severe problems and being unable to do/having extreme problems. In addition, participants are asked to self-rate their own health today on a vertical 0-100 unit visual analogue scale (VAS), with 0 corresponding to "the worst health you can imagine", and 100 corresponding to "the best health you can imagine".

The EQ-5D-5L will be collected on paper questionnaire at time points indicated in the SoA (see section 1.3 of the study protocol).

The reported data for the five dimensions of the EQ-5D-5L will be analyzed descriptively.

Summary scores will be calculated out of the 5 dimensions according to the scoring instructions from Europe (UK as a representative country within Europe) and the US (EuroQoL Group 2013) and to the EQ-5D Value Sets (Szende et al 2007). The values and the changes from baseline of the summary scores and the EQ VAS will be summarized by treatment group and visit by means of number of observations, number of missing values, minimum, first quartile, mean, SD, median, third quartile, and maximum.

The absolute change of the summary scores and the EQ VAS scores to baseline at each visit up to the last visit with a reasonable number of participants with EQ-5D-5L measurements will be analyzed by a MMRM with the factors; treatment group, visit, treatment by visit interaction, factors for the randomization stratification levels, and baseline value as covariate and baseline by visit interaction. Pairwise differences between the finerenone and the placebo treatment group will be calculated and corresponding two-sided 95% CIs will be computed.

The SAS Procedure PROC MIXED will be used, estimating covariance patterns within participants to adjust for the within participant variance. For each treatment group a separate covariance pattern will be estimated based on an unstructured covariance. In case of convergence issues, an alternative covariance matrix will be used, with Toeplitz matrix as first choice alternative.

5.6 Safety Analyses

Extent of exposure will be analyzed in the FAS and repeated in the SAF if both populations do not coincide. All further analyses on safety and tolerability data will be performed in the SAF.

5.6.1 Extent of Exposure

The analyses described in this section will be repeated for the SAF if it differs from the FAS. All tables and figures regarding treatment duration, extent of exposure and compliance will be presented by treatment group and overall (unless otherwise stated).

Treatment duration, defined as time from start of study intervention to permanent stop of study intervention (in months) will be summarized using descriptive statistics by treatment group and overall. In addition, treatment duration will be categorized to ≤ 1 month, > 1 to 3 months, > 3 to 6 months, > 6 to 9 months, > 9 to 12 months, then 4-monthly intervals, and presented with the corresponding number and percentage of participants. Cumulative treatment duration will be categorized to at least 1 dose, at least 1 month, at least 3, 6, 9, 12 months, then further 4-monthly intervals. Cumulative treatment exposure over the study in person-years will be given. A table will be presented with the absolute and relative frequencies of participants still on study medication at each visit. Kaplan-Meier plots for “Time to end of study treatment” will be provided by treatment group.

The above analyses will be repeated for study duration, defined as time from the day of randomization to the participants’ final visit (in months).

The extent of exposure to study drug (total amount of intake in grams) and the average daily dose in mg during treatment will be summarized using descriptive statistics by treatment group.

The overall titration status, regardless of actual or sham up-titration, will be summarized with absolute and relative frequencies per treatment group, differentiated by participants starting on 10 mg or 20 mg:

- For participants starting on 10 mg: never up-titrated, up-titrated once, and up-titrated more than once.
- For participants starting on 20 mg: never down-titrated, down-titrated once, and down-titrated more than once.

The number of participants on 10 mg, 20 mg or on an interruption at Day 1, Month 1, Month 3, Month 6, Month 9, Month 12 and in 4-monthly intervals from then onwards will also be derived based on exposure information and presented by treatment group.

In addition, the number of participants with study drug down-titrated to 10 mg or temporarily interrupted as well as associated reason will be summarized with absolute and relative frequencies per treatment group.

The compliance (as percentage) will be calculated as follows:

$$100 * \text{Number of taken tablets} / \text{Number of planned tablets}$$

The number of planned tablets is calculated as follows:

$$(\text{Days from randomization to last intake of study drug} + 1) * \text{Number of planned tablets per day}.$$

All tablets, including the dummy placebo tablets, will be counted. For participants who withdraw prematurely from the study drug, compliance will be calculated up to the time of last dose. The compliance will be summarized descriptively by treatment group and overall. In addition, percent of compliance will be categorized into 3 groups, less than 80%, 80 to 120% and greater than 120%, and the categories will be summarized by treatment group and overall.

5.6.2 Adverse Events

With the exception of hyperkalemia, which is collected as an adverse event (AE) of special interest, non-serious AEs are not mandatory to be collected in the eCRF. Instead, only AEs leading to permanent discontinuation of study intervention and AEs considered related to study intervention by the investigator or considered related to protocol required procedures will be collected. Any other AEs should only be collected if deemed relevant by the investigator.

AEs will be coded using the Medical Dictionary for Regulatory Activities (MedDRA, latest version available prior to data base freeze). A listing will be provided linking the original investigator terms and the coded terms. AEs will also be presented grouped by SMQs.

AEs that started or worsened after the first dose of study drug up to 3 days after any temporary or permanent interruption of study drug will be considered as treatment-emergent AEs (TEAEs).

A worst-case approach will be applied for determining whether an AE with partially missing dates is treatment-emergent or not, i.e. if it is possible that the adverse event start date is within a period of study drug intake +3 days, then the adverse event is considered treatment emergent.

An overall summary of all reported AEs and TEAEs will be generated by treatment group. The number of participants with TEAEs, pre-randomization AEs, post-treatment AEs occurring more than 3 days after any temporary or permanent stop of study drug, treatment-emergent SAEs, treatment-emergent study drug-related AEs, treatment-emergent study drug-related SAEs, treatment-emergent SAEs and TEAEs causing permanent discontinuation of study drug, treatment-emergent non-serious AEs, non-serious AEs, TEAEs by maximum intensity, treatment-emergent SAEs by maximum intensity, drug-related TEAEs by maximum intensity, TEAEs by worst outcome and treatment-emergent SAEs by worst outcome will be summarized by treatment group using MedDRA terms grouped by Primary System Organ Class (SOC) and Preferred Term.

In case of events with different intensities within a participant, the maximum reported intensity will be used. If the intensity is missing, the event will be considered severe. If the same event is considered both unrelated and related to the study drug within a participant, it will be reported as related to study drug. If the drug relationship is missing, the event will be considered as being related to the study drug.

To comply with local regulatory requirements in Japan, cardiovascular disease-related outcome events will also be documented as (S)AEs in Japan and all serious disease-related outcome events will also be documented as SAEs in India. These will be included in the disease-related events (DRE) tables (see section 5.6.3.1), and to avoid double-counting of such events, they will not be included in the adverse event summary tables or listings. A separate listing will be generated for all AEs excluded from the AE analysis due to double

reporting in Japan and a further separate listing will be generated for all SAEs excluded from the AE analysis due to double reporting in India.

The following further safety variables related to selected AEs will be analyzed:

- Number of participants with hyperkalemia
- Number of participants with hospitalization for hyperkalemia
- Number of participants discontinuing study drug permanently due to hyperkalemia
- Number of participants with hospitalization for worsening of renal function
- Number of participants discontinuing study drug permanently due to worsening of renal function

Hyperkalemia includes AEs with MedDRA codes (preferred terms): 10020646 (Hyperkalaemia), 10005725 (Blood potassium increased).

Worsening of renal failure includes AEs with MedDRA codes (preferred terms): 10069339 (Acute kidney injury), 10001017 (Acute prerenal failure), 10005483 (Blood creatinine increased), 10018358 (Glomerular filtration rate decreased), 10059345 (Postrenal failure), 10072370 (Prerenal failure), 10038435 (Renal failure), 10062237 (Renal impairment).

These will be summarized by treatment group using frequency counts. For the number of participants with these events, the summary will be provided once for all events and once for all treatment-emergent events.

Separate tables summarizing TEAEs, study drug-related TEAEs, and SAEs that occurred in more than 5% of the participants will be provided.

Deaths, SAEs and AEs leading to permanent study drug discontinuation will be listed separately.

Cumulative incidences based on Aalen-Johansen estimates (including 95% confidence intervals) and accounting for mortality as competing risk will be provided for the first treatment-emergent hyperkalemia events (all, leading to hospitalization and leading to permanent discontinuation of study drug). For this analysis, the person-time at risk for a single participant is the number of days from first intake of study drug until the event of interest or until the minimum of (date of death, last exposure to treatment + 3 days) excluding temporary study drug interruptions (i.e. the time between start of a temporary study drug interruption + 3 days until the end of the interruption).

5.6.3 Additional Safety Assessments

5.6.3.1 Disease Related Events

The following disease-related events (DRE) are common in participants with CKD and will not be reported as (S)AEs, but entered into the clinical trial database as outcome events (OE).

- Kidney failure defined as:
 - ESRD: Initiation of chronic dialysis (haemo- or peritonealdialysis) for at least 30 days or renal transplantation.
 - Sustained decrease of eGFR to less than 15 mL/min/1.73 m², confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement

- Decrease of eGFR of $\geq 57\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- Decrease of eGFR of $\geq 40\%$ from baseline, confirmed by at least one additional standardized measurement at least 4 weeks after the initial measurement
- CV death
- Non-fatal MI
- Non-fatal stroke or TIA
- Hospitalization for HF or urgent HF visit
- Other CV hospitalization
- New diagnosis of atrial fibrillation or atrial flutter
- New diagnosis of HF
- Outpatient worsening of HF

Sustained eGFR decreases to less than 15 mL/min/1.73m², of $\geq 57\%$ from baseline, and of $\geq 40\%$ from baseline will be programmatically derived, and not entered into the eCRF by the investigator.

Because these events are typically associated with the disease under study, they will not be reported according to the standard process for expedited reporting of SAEs even though the event may meet the definition of an SAE.

DREs constitute components of the secondary and exploratory efficacy endpoints. In addition to the statistical analyses outlined in sections 5.4 and 5.5, DREs using the investigator-reported terms will be summarized by treatment group, using tables analogous to those for AEs.

An overall summary of all DREs will be generated by treatment group. The number of participants with all DREs, DREs from randomization up to 30 days after stop of study intervention, post-treatment DREs occurring more than 30 days after stop of study intervention or after the off-treatment visit, DREs by maximum intensity and DREs by worst outcome will be summarized by treatment group using MedDRA terms grouped by Primary SOC and Preferred Term.

5.6.3.2 Laboratory Parameters

The number of participants with treatment-emergent (until 3 days after any temporary or permanent interruption of study drug) abnormal laboratory values above or below the normal range will be tabulated by the laboratory parameter and treatment group. For this analysis, laboratory values from scheduled and unscheduled visits will be considered.

Summary statistics including changes to baseline will be calculated by treatment group and visit for all quantitative laboratory parameters, e.g. for hematology, clinical chemistry and quantitative parameter from urinalysis. Geometric statistics and ratios to baseline will be presented for serum creatinine, UACR and UPCR instead of arithmetic statistics with changes from baseline. For eGFR, the relative change will be displayed in addition to the absolute change from baseline.

Summary statistics for serum potassium, eGFR and serum creatinine will also be repeated by treatment group and visit separately for each level of the stratification factors.

The following special safety parameters related to laboratory measurements will be further assessed by displaying the number of participants with treatment-emergent safety events. This will also be performed by visit and by stratification factors. The summaries will be performed for the number of participants with:

- Absolute value of serum potassium >5.5 mmol/L
- Absolute value of serum potassium >6.0 mmol/L
- Relative decrease from baseline in eGFR of $\geq 30\%$, $\geq 40\%$, $\geq 50\%$ and $\geq 57\%$

The percentage of participants with the respective treatment-emergent events (non-stratified) will be compared between the finerenone and the placebo control group by applying separate explorative χ^2 tests with continuity correction. If the expected number of participants in at least 1 cell of the 2x2 contingency table is less than 5, Fisher's exact test will be applied instead of the χ^2 test. Estimates and two-sided 95% CIs will be provided for each treatment group and the treatment differences. Clopper-Pearson CIs will be calculated for each treatment group, while for treatment differences, due to the computational burden of exact unconditional confidence limits, Wald asymptotic confidence limits will be calculated.

In the same way described for treatment-emergent hyperkalemia AEs in section 5.6.2, cumulative incidences based on Aalen-Johansen estimates (including 95% confidence intervals) and accounting for mortality as competing risk will be provided for the time to first treatment-emergent serum potassium value >5.5 or 6.0 mmol/L. For this analysis, the person-time at risk for a single subject is the number of days from first intake of study drug until the event of interest or until the minimum of (date of death, last exposure to treatment +3 days) excluding temporary study drug interruptions (i.e. the time between start of a temporary study drug interruption + 3 days until the end of the interruption).

5.6.3.3 Vital Signs

The analysis of vital signs includes blood pressure and pulse measurements. Averages of non-missing values will be calculated and used for the statistical analysis. If only 1 of the planned measurements is available, this value will be used.

Vital signs values will be summarized by treatment group and visit using descriptive statistics including absolute changes from baseline. The analysis will be repeated for systolic blood pressure (SBP) stratified by baseline SBP <130 mmHg, 130 to <160 mmHg and ≥ 160 mmHg and SBP < baseline median and $\geq$ baseline median (in the SAF).

5.6.3.4 Weight and BMI

The values and the absolute changes from baseline will be summarized by treatment group and visit using descriptive statistics for weight (kg) and BMI (kg/m²).

5.7 Other Analyses

5.7.1 Other Variables and/or Parameters

5.7.1.1 Pharmacokinetics

For the investigation of systemic exposure to finerenone and its relationship with treatment effects, the plasma concentrations of finerenone will be determined at different time points using a sparse sampling approach in all participants. The plasma concentration versus time data collected on the visits as outlined in the SoA (see section 1.3 of the study protocol) will

be evaluated descriptively, separated by dose and visit. Plots will be prepared of all individual plasma concentrations vs. actual relative study times (time of sample collection after time of study intervention administration).

Pharmacokinetic and exposure-response analysis may be performed using population approaches (popPK and popPK/PD, e.g., by non-linear mixed effect modeling). Such evaluations will be described in a separate analysis plan and will be reported separately.

Pharmacodynamic parameters are not evaluated in this study.

5.7.2 Subgroup Analyses

Exploratory subgroup analysis will be performed for the primary and key secondary efficacy variables. The subgroup analyses will include analyses by randomization stratification factors. The list of key subgroups (in addition to the stratification factors) and other subgroups analyzed is specified below. Analyses will include descriptive statistics, graphical display of estimated treatment effects with 95%-CIs in a forest plot and a statistical test for interaction.

It is anticipated that in these proposed subgroups for analysis, differences in treatment effects may be observed according to the screening or baseline characteristics defined, due in part to the differences in the risk of clinical events expected in the different subgroups.

If the result for a subgroup cannot be calculated due to a small sample size or number of events, then this subgroup will be dropped.

Stratification factors:

- UACR category at screening (≤ 1000 mg/g, > 1000 mg/g)
- Baseline SGLT2 inhibitor use (yes/no)

Key subgroups as specified in the study protocol (in addition to those defined by the stratification factors):

- Primary cause of CKD (Hypertensive CKD, ischemic CKD, chronic glomerulonephritis [with IgA nephropathy analyzed as additional separate subgroup], other, unknown)
- eGFR category at baseline (eGFR < 45 , 45 to < 60 , ≥ 60 mL/min/1.73 m²)
- UACR category at baseline (< 300 mg/g, 300 to ≤ 1000 mg/g, > 1000 mg/g)
- Baseline serum potassium (≤ 4.5 , > 4.5 mmol/L)
- Region (North America, Latin America, Europe, Asia, see Appendix 6.6 for a classification of the participating countries according to geographic regions)
- Race (white, black, Asian, other)
- Sex (male, female)
- Age group (< 65 , ≥ 65 years)
- History of CV disease (absent, present)

Other subgroups:

- Systolic blood pressure at baseline ($\leq$ median and $>$ median in the FAS)
- BMI at baseline (< 30 , ≥ 30 kg/m²)
- Baseline serum potassium (by quartiles in the FAS: $\leq Q1$, $> Q1$ and $\leq Q2$, $> Q2$ and $\leq Q3$, $> Q3$)

- Baseline serum potassium (≤ 4.8 mmol/L, > 4.8 to 5.0 mmol/L, > 5.0 mmol/L)
- Baseline HbA1c ($< 5.7\%$, ≥ 5.7 to $< 6.5\%$)
- Baseline C-reactive protein (by quartiles in the FAS: $\leq Q1$, $> Q1$ and $\leq Q2$, $> Q2$ and $\leq Q3$, $> Q3$)
- SBP at baseline (< 130 mmHg, 130 to < 160 mmHg, and ≥ 160 mmHg)
- Age group (18 to 44 years, 45 to 64 years, 65 to 74 years, 75 years and over)
- Ethnicity (Hispanic or Latino, not Hispanic or Latino, not reported)
- BMI at baseline (< 20 , 20 to < 25 , 25 to < 30 , 30 to < 35 , ≥ 35 kg/m²)
- Baseline weight (< 60 , 60 to < 90 , ≥ 90 kg)
- eGFR at baseline 25 to < 45 ml/min/1.73 m² and baseline serum potassium value > 4.5 mmol/L (yes, no)
- ACEI at baseline (yes, no)
- ARB at baseline (yes, no)
- Beta-blocker at baseline (yes, no)
- Diuretic at baseline (yes, no)
- Statins at baseline (yes, no)
- Potassium supplementation at baseline (yes, no)
- Potassium lowering agents (including binders) at baseline (yes, no)
- Potency of concomitant CYP3A4 inhibitor medication at baseline (strong, unclassified, moderate, weak, none)
- Potency of concomitant CYP3A4 inducer medication at baseline (strong, unclassified, moderate, weak, none)

5.8 Interim Analyses

No formal interim analysis is planned for this study.

6. Supporting Documentation

6.1 Appendix 1: Population Characteristics

Population characteristic analyses will be performed for the FAS, if not stated otherwise.

6.1.1 Demography and Other Baseline Characteristics

Demographic and baseline characteristics, including disease characteristics obtained before the first study intervention administration includes

- age, gender, ethnicity, race, region (North America, Europe, Asia, Latin America and others), body weight (kg), body height (cm), body mass index (BMI) (kg/m²), smoking history (never, former, current smoker), alcohol consumption (abstinent, light, moderate, heavy)
- primary cause of CKD, baseline eGFR (calculated by CKD-EPI formula), categories of baseline eGFR (< 30, 30 to < 45, 45 to < 60, 60 to < 75, 75 to < 90, ≥ 90 ml/min/1.73m²), baseline UACR, categories of baseline UACR (30 to < 300, ≥ 300 mg/g), serum creatinine, serum potassium, categories of baseline serum potassium (≤ 4.5 mmol/L and > 4.5 mmol/L),
- vital signs parameters (i.e. systolic blood pressure, diastolic blood pressure, heart rate),
- additional subgroup categories as described in section 5.7.2.

Demographic and baseline characteristics will be summarized using descriptive statistics by treatment group and all participants for the FAS and SAF. Age for each participant will be defined as the number of years between informed consent date and birth date. For summarizing race, if multiple races are recorded in the database, then 'multiple race' will be used in the summary table.

6.1.2 Medical History

Medical history will be coded using the MedDRA dictionary. Medical history will be presented for each MedDRA Primary SOC and Preferred Term by treatment group and overall in a summary table. Additional medical history terms by the following Project-specific Bayer MedDRA Queries (PBMQs), Bayer MedDRA Labeling Groupings (MLGs), or selected Preferred Terms will also be presented, including the following:

- Hyperlipidemia (MLG)
- Hypertension (MLG)
- Atrial fibrillation/flutter (PBMQ)
- Ischaemic Stroke/TIA (PBMQ)
- Myocardial Infarction (Preferred Term)
- Coronary Artery Disease (Preferred Term)
- Myocardial Ischaemia and Angina pectoris (MLG)
- Peripheral Artery Disease (Preferred Term)
- Cardiac Failure (MLG)
- Coronary Artery Bypass Graft (CABG) (PBMQ)
- Percutaneous coronary intervention (PCI) (PBMQ)
- Carotid endarterectomy (Preferred Term)

The medical history tables will be repeated for SAF.

6.1.3 Concomitant Medications

Concomitant medications will be coded using the WHO Drug Dictionary (WHO-DD). The number of participants who took at least one concomitant medication, the number of participants who took at least one medication that started and ended before administration of study drug, the number of participants who took at least one concomitant medication that started after start of study drug and the number of participants who took at least one medication ongoing at baseline (i.e. starting before or on the day randomization and ending at least one day after the day of randomization) will be presented by treatment group and overall using ATC classes and subclasses. The concomitant medication tables will be repeated for SAF.

The concomitant medication tables will be repeated summarizing the number of participants with medications in the Standard or Bayer Drug Groupings of interest:

- ACEIs
- ARBs
- Beta-blockers
- Diuretics
- Loop diuretics
- Thiazide diuretics
- Potassium-sparing diuretics
- Potassium supplements
- Potassium lowering agents (including binders)
- Alpha blocking agents
- Calcium channel blockers
- Centrally acting antihypertensives
- Oral anticoagulants
- Nonsteroidal anti-inflammatory drugs (NSAIDs) (excluding acetylsalicylic acid)
- Acetylsalicylic acid and its salts
- Statins
- Erythropoietin stimulating agents
- Angiotensin receptor neprilysin inhibitors (ARNI)
- Platelet aggregation inhibitors (excluding heparin)
- Glucocorticoids
- SGLT2 inhibitors
- Immunosuppressants (excluding Glucocorticoids)

A participant will be counted only once within each ATC class / subclass or Bayer drug grouping, respectively.

The percentage of participants who are on maximum labeled dose of ACEIs and ARBs (yes / no) 4 weeks before the Screening Visit, and reasons why not on maximum labeled dose will be presented by treatment group and overall. In addition, the percentage of participants at

baseline above maximum, at maximum, between minimum and maximum, at minimum and below minimum labeled dose, using the dose ranges in Table 6-4 of the protocol, will be presented by treatment group and overall.

For potassium lowering agents, ACEIs, ARBs and diuretics, shift tables for changes of use for baseline vs. any time post-baseline will be provided.

6.2 Appendix 2: List of Abbreviations

Abbreviation	Definition
ACEI(s)	Angiotensin-converting enzyme inhibitor(s)
AE(s)	Adverse event(s)
ANCOVA	Analysis of covariance
ARB(s)	Angiotensin receptor blocker(s)
ATC	Anatomical therapeutic chemical
CI	Confidence interval
CKD	Chronic kidney disease
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
CV	Cardiovascular
CV	Coefficient of variation (in context of statistical measures)
CYP3A4	Cytochrome P450 isoenzyme 3A4
DRE	Disease related event
ECG	Electrocardiogram
eGFR	Estimated glomerular filtration rate
ESRD	End stage renal disease
EQ-5D-5L	EuroQol Group 5-dimension, 5-level questionnaire
EudraCT	European Union Drug Regulating Authorities Clinical Trials
FAS	Full analysis set
GFR	Glomerular filtration rate
HF	Heart failure
HR	Hazard ratio
ICE(s)	Intercurrent event(s)
ICF	Informed consent form
ICH	International Council on Harmonisation
IxRS	Interactive voice / web response system
LLOQ	Lower limit of quantification
LS	Least squares
MAR	Missing at random
MCMC	Markov chain Monte Carlo
MedDRA	Medical Dictionary for Regulatory Activities
MI	Myocardial infarction
MLG	(Bayer) MedDRA labeling grouping
MMRM	Mixed model for repeated measures
MNAR	Missing not at random

NNT	Number needed to treat
OD	Once daily
OE	Outcome event
PBMQ	Project specific Bayer MedDRA query
PK	Pharmacokinetics
PKS	Pharmacokinetic analysis set
popPK	Population pharmacokinetics
SAE(s)	serious adverse event(s)
SAF	Safety analysis set
SAP	Statistical analysis plan
SBP	Systolic blood pressure
SD	Standard deviation
SGLT2	Sodium-glucose co-transporter-2
SGLT2i	Sodium-glucose co-transporter-2 inhibitor
SoA	Schedule of activities
SoC	Standard of care
SOC	System organ class
SP (model)	Shared parameter model
TEAE(s)	Treatment-emergent adverse event(s)
TESAE(s)	Treatment-emergent serious adverse event(s)
TIA	Transient ischaemic attack
UACR	Urinary albumin-to-creatinine ratio
ULOQ	Upper limit of quantification
UPCR	Urinary protein-to-creatinine ratio
VAS	Visual analogue scale
WHO-DD	World Health Organization drug dictionary

6.3 Appendix 3: Changes to Protocol-planned Analyses

Not applicable.

6.4 Appendix 4: Parameters for Sample Size Estimation

The following Table 6–1 lists the initial parameters used in the SAS macro "%GFR_Slope_Power" for determination of the sample size, see section 3.

Table 6–1: Initial Parameter values for sample size estimation

Initial Parameter: The initial parameter values obtained using 36 months eGFR from FIDELIO-DKD data	Initial parameter values
knot= Assumed knot (in months) at which the acute slope transitions into a chronic slope	3
beta0= Baseline mean eGFR (intercepts: $\text{beta}_{0c} = \text{beta}_{0t} = \text{beta}_0$) for placebo Control (c) and finerenone (t) treatment groups assuming equal intercepts for the two groups based on the principals of randomization	45
beta1c= Assumed acute slope for Control group in mL/min/1.73 m ² /month	-0.2322
beta2c= Assumed delta (change) slope for Control group in mL/min/1.73 m ² /month	-0.1169
beta1t= Assumed acute slope for finerenone group in mL/min/1.73 m ² /month	-0.6630
beta2t= Assumed delta (change) slope for finerenone group in mL/min/1.73 m ² /month	0.4284
Var_e= Assumed within-subject variance component of residual errors	26.4783
Var_u0= Assumed random intercept (b_0) variance for the Control group in mL/min/1.73 m ²	134.11
Var_u1= Assumed random acute slope (b_1) variance for the Control group in mL/min/1.73 m ² /month	1.0301
Var_u2= Assumed random delta (change) slope (b_2) variance for the Control group in mL/min/1.73 m ² /month	0.9883
Cov_u0u1= Assumed covariance between b_0 and b_1 for the Control group	-0.2428
Cov_u0u2= Assumed covariance between b_0 and b_2 for the Control group	-0.2944
Cov_u1u2= Assumed covariance between b_1 and b_2 for the Control group	-0.9667
Theta= The power of mean parameter for the within subject variance	0
Kappa= The treatment-dependent constant for the variance-covariance matrix of the between-subject random effects	0
Note: the choice of Kappa=0 implies that the variance-covariance matrix of the between-subject random effects is the same for the Control and the finerenone group	
SEED = Seed used for random number generation	12345
InfCensor = Defines whether participants dropout due to ESKD Dropout = Defines a dropout rate	Yes 0.05
Months= Measurement occasions in months when eGFR is measured (study-design points)	0,3,6,9,12,16,20, 24,28,32,36
DIFF= Assumed incremental differences (in mL/min/1.73 m ² /month) between the chronic slope for the finerenone group versus the chronic slope for the Control group. Note: The DIFF values, as well as the differences in all slope parameters, are converted to units of mL/min/1.73 m ² /year in the output datasets and corresponding printout. For example, 0.10 mL/min/1.73m ² /month = 1.2 mL/min/1.73 m ² /year.	See Table 3–1 for selected values of DIFF

6.5 Appendix 5: Technical Details for Sensitivity Analyses

6.5.1 Control-based Pattern-mixture Modeling Approach

The general procedure for control-based pattern imputation as introduced by Little and Yau 1996, and later refined by Ratitch and O’Kelly 2011, is the following:

- 1) In a first step impute the intermediate, non-monotone missing data with the MCMC method and store the result of this partial imputation. In this study, the non-monotone data would represent those participants that miss some intermediate assessments but are able to come back at a later visit and complete the pertaining assessment.
- 2) Next, impute the monotone missing data with control-based pattern imputation. Here it implies that the participants who are in the finerenone group and no longer receive finerenone will assume the same conditional distribution as those in the finerenone group. We start with imputing monotone missing data at the first time-point that has some missing data (defined as time-point 1). This step is repeated for all other time points sequentially. Participants whose missing values were imputed in the last call to PROC MI will be included in the input dataset for the next call to PROC MI. Thus, data for time-point t , filled in during the last call, will be used for predictor variables in the next call to PROC MI (for time point $t + 1$). This sequential proceeding can be implemented with a single call to PROC MI to impute all time-points automatically.
- 3) Run the two-slope linear spline mixed-effects model for each of the iterations generated by PROC MI from step 2).
- 4) Combine the results from step 3) using PROC MIANALYZE.

This procedure is illustrated below by the following example code:

To use PROC MI, data needs to be in a horizontal format, i.e. there is one data row for each participant and the eGFR values for each visit are contained in a pertaining variable. Suppose the formatted data is included in an analysis dataset adegfr. The following notation will be used in this and the following sections:

“subject” represents:	unique participant identifier
“trt” represents:	study intervention (trt=1 for finerenone and trt=0 for placebo)
“strata” represents:	baseline SGLT2 inhibitor use, baseline UACR (categorized)
“visit” represents:	1, 2,...,11
“month” represents:	analysis time point by month
“month_k” represents:	max(0, month – knot) where knot denotes the change point defined in the two-slope linear spline mixed effects model
“base” represents:	eGFR value at baseline
“egfr_1”- “egfr_10” represent:	eGFR values at post-baseline visits at month 1, 3, 6, 9, 12, 16, 20, 24, 28 and 32.

Step 1.1: Check missing pattern in source data=adegfr;

```
PROC MI    data=adegfr nimpute=0;
  VAR    trt strata base egfr_1 egfr_2 ... egfr_10;
RUN;
```

If the output from PROC MI above indicates that the missing pattern is non-monotone, step 1.2 is needed.

Step 1.2: Impute only for non-monotone data first in order to obtain a monotone missing pattern

In order to obtain a monotone missing data pattern, we can first use the MCMC method in PROC MI to partially impute data as illustrated below (note the usage of impute=monotone option in combination with the MCMC statement).

```
PROC MI    data=adegfr out=data_mono nimpute=1000 seed=1234;  
    VAR   trt base strata egfr_1 egfr_2 ... egfr_10 ;  
    MCMC chain=multiple impute=monotone;  
RUN;
```

The above procedure will output the dataset data_mono, with a monotone missing data pattern for all imputations. Note that data_mono already contains nimpute iterations of the imputation process. The above Step 1 is a common step and should be done before the MNAR analysis and data_mono can be used as the partially imputed data set.

Step 2: Perform placebo based multiple imputation

Use PROC MI to impute the monotone missing data for both the experimental arm and placebo arm with MNAR statement, based on a model for placebo arm.

```
PROC MI data=data_mono nimpute=1 seed=12345 out= data_mi_pb;  
    BY _imputation_ ;  
    CLASS strata trt;  
    VAR strata base egfr_1 egfr_2 ... egfr_10;  
    MONOTONE regression (/details);  
    MNAR MODEL (egfr_1/ modelobs= (trt="0"))  
        MODEL (egfr_2/ modelobs= (trt="0"))  
        ...  
        MODEL (egfr_10/ modelobs= (trt="0"));  
RUN;
```

Step 3: Apply the two-slope linear spline mixed-effects model to each of the iterations generated by PROC MI in step 2.

For application of the two-slope linear spline mixed-effects model the dataset data_mi_pb generated in step 2 has first to be transposed to a vertical structure with one variable containing the eGFR values (say egfr) and one variable now including the value (in months) corresponding to the scheduled time point of the respective analysis visit. Let data_mi_pb_trans denote the analysis dataset adhering to the required structure then the code for the model will look like (as example only for analysis of the total slopes at month 32)

```
ODS OUTPUT Estimates=mi_est;  
PROC MIXED DATA=data_mi_pb_trans;  
    CLASS trt strata subject;  
    MODEL egfr = strata base trt trt*month trt*month_k / noint;  
    RANDOM Intercept month month_k / group=trt type=un subject=subject(trt);
```

```

ESTIMATE "Total Slope Difference (32 Mths)"    trt*month -1 1
                                                trt*month_k -0.90625 0.90625;
ESTIMATE "Treated Group Total Slope (32 Mths)" trt*month 0 1
                                                trt*month_k 0 0.90625.;
ESTIMATE "Control Group Total Slope (32 Mths)" trt*month 1 0
                                                trt*month_k 0.90625 0;

BY _imputation_;
RUN;
```

Step 4: Combine results by PROC MIANALYZE

First sort the output dataset mi_est by variable “Label”. Combination of the results is then achieved by the following call to PROC MIANALYZE.

```

PROC MIANALYZE data=mi_est;
  BY Label;
  MODELEFFECTS Estimate;
  STDERR StdErr;
  ODS OUTPUT ParameterEstimates=mi_slope;
RUN;
```

6.5.2 Pattern-mixture Modeling Approach Using Data from Retrieved Drop-outs

As eGFR values are crucial to the analysis these values must be collected until the planned end of the study, as participants will still be part of the study even if they are no longer taking study intervention. Hence it is assumed that we will have complete eGFR data up to the planned end of treatment at month 32 for some participants with an earlier discontinuation of study intervention due to other reasons than death or onset of ESKD. Such participants are denoted as retrieved drop-outs and in an alternative approach to control-based multiple imputation data from these retrieved drop-outs will be used as the basis in the imputation steps.

For participants who discontinue study intervention due to death or ESKD an imputation method will be applied as for the primary estimand setting missing eGFR values after discontinuation to an unfavorable value based on a lower percentile of the distribution of eGFR at the pertaining visit within the respective treatment group.

This procedure is illustrated below by the following example code:

Step 1: As primary step for scheduled visits after death or onset of ESKD an unfavorable value of eGFR will be assumed as described for the primary estimand.

Step 2: As in the control-based pattern mixture modelling approach in section 6.5.1 a dataset data_mono with only monotone missing data is generated in the second step.

Step 3: Perform retrieved dropout based multiple imputation

Use PROC MI to impute the monotone missing data separately for the experimental arm and the placebo arm with MNAR statement, based on a model for the retrieved dropouts within

the respective treatment arm. Let `rdrop` denote an indicator variable that flags retrieved drop-outs (`rdrop=1`), then for $x = 0, 1$ multiple imputation will be done as follows:

```
PROC MI data=data_mono (where=(trt = x)) nimpute=1 seed=12345 out= data_mi_rdx;
  BY _imputation_ ;
  CLASS strata rdrop;
  VAR strata base egfr_1 egfr_2 ... egfr_10;
  MONOTONE regression (/details);
  MNAR MODEL (egfr_1/ modelobs= (rdrop="1"))
             MODEL (egfr_2/ modelobs= (rdrop="1"))
             ...
             MODEL (egfr_10/ modelobs= (rdrop="1"));
RUN;
```

Afterwards the datasets `data_mi_rd0` and `data_mi_rd1` will be combined.

Step 4: Apply the two-slope linear spline mixed-effects model to each of the iterations generated by PROC MI in step 3.

This step is now identical to step 3 in section 6.5.1.

Step 5: For each value of delta use PROC MIANALYZE to combine the results

Again, this step is identical to step 4 in section 6.5.1.

6.5.3 Tipping Point Analysis – Primary Endpoint

We intend to perform multiple imputation analysis for a specified sequence of shift parameters, which only adjust the imputed values for observations in the finerenone group. Multiple imputation will be done using SAS PROC MI under MNAR assumption.

The delta-adjustment method will be applied sequentially where at each iteration missing values of discontinued participants are adjusted with a larger delta than at the previous iteration. This process continues until the difference between the experimental and control treatments is no longer statistically significant. This point is called a tipping point. User can specify a step by which delta parameters are increased at each iteration. Each imputed observation (missing eGFR value) will be modified according to the required delta shift. Assuming there is no penalty in eGFR in the placebo group, then for those participants with missing outcomes in the finerenone treatment arm, the eGFR values will be gradually decreased until a p-value of 0.05 or higher is produced (beyond the tipping point). The actual penalty (to 1 decimal place) that gives a p-value above and closest to 0.05 will then be identified.

This procedure is illustrated below by the following example code:

Step 1: As in the control-based pattern mixture modelling approach in section 6.5.1 a dataset `data_mono` with only monotone missing data is generated in a first step.

Step 2: Perform multiple imputation with adjustment of imputed values in the finerenone arm.

The approach in this study will follow the second variant discussed in Ratitch, O’Kelly and Tosiello 2013, where a delta adjustment is applied at each visit starting with the first visit with missing data up to the end of the observation period. In addition to using previously delta-

adjusted values as predictors it works under the assumption that discontinuation is associated with a worsening process that persists throughout the whole time period after discontinuation.

SAS code for this approach will be as follows:

```
PROC MI      data=data_mono seed=19334 nimpute=1 out=data_tpa;
  CLASS      strata trt;
  VAR        strata trt base egfr_1 egfr_2 ... egfr_10 ;
  MONOTONE    regression (/details);
  MNAR ADJUST (egfr_1 /shift=delta adjustobs=(trt='1'))
             ADJUST (egfr_2 /shift=delta adjustobs=(trt='1'))
             ...
             ADJUST (egfr_10 /shift=delta adjustobs=(trt='1'))
;
RUN;
```

A range of delta values will be considered starting with 0, with subsequent stepwise reductions of delta by 0.05.

Step 3: For each value of delta apply the two-slope linear spline mixed-effects model to each of the iterations generated by PROC MI in step 2.

For each fixed value of delta this step is identical to step 3 in section 6.5.1.

Step 4: For each value of delta use PROC MIANALYZE to combine the results

Again, for each fixed value of delta this step is identical to step 4 in section 6.5.1.

6.5.4 Tipping Point Analysis – Key Secondary Endpoints

To investigate the impact of missing data under a missing not at random (MNAR) assumption a methodology similar to pattern-mixture-models will be employed as described by Little et al, 2016. In this approach, a Weibull survival model will be fitted to all participants who prematurely discontinued study treatment in order to estimate the individual hazard rates for participants with withdrawn consent or loss to follow-up, but without a previous endpoint event. Based on these individual hazard estimates at the time of censoring (possibly penalized through inflation), event times for participants alive but censored before the global cut-off date (notification of study termination + 4 weeks) will be imputed.

Details:

Let T denote the random time-to-event variable that defines the key secondary endpoint. The SAS LIFEREG procedure will be used where the natural logarithm of T , i.e. $W = \log(T)$ is modeled which has an extreme-value distribution with survival function

$$S(w) = \exp\left(-\exp\left(\frac{w - \mu}{\sigma}\right)\right),$$

where μ is the location parameter (intercept) and $\sigma > 0$ is the scale parameter in the SAS output.

The observed hazard rates for individual participants are obtained from a Weibull model fitted to all participants who prematurely discontinued study treatment, adjusted for treatment group (coded 0=placebo, 1=finerenone) and the stratification factors. The Weibull probability density function is re-parametrized to have the following form

$$g(t|\lambda, \beta) = (\beta/\lambda) (t/\lambda)^{\beta-1} \exp(-(t/\lambda)^\beta), t > 0,$$

where $\lambda = \exp(\mu)$ is the new scale parameter $= \exp(\mu)$ and $\beta = 1/\sigma$ is the re-parametrized shape parameter, with μ and σ denoting the corresponding parameter from the extreme-value distribution used by SAS LIFEREG.

With this re-parametrization, the hazard function is given by

$$h(t|\lambda, \beta) = (\beta/\lambda) (t/\lambda)^{\beta-1}$$

and the survival function can be written as

$$S_2(t) = \exp(-(t/\lambda)^\beta).$$

For participants with consent withdrawn or who are lost to follow-up and who did not experience the key secondary efficacy endpoint, random event times will be simulated using the conditional time-to-event distribution after the last contact date (i.e. the censoring date) given that no event was observed before (see details below). If this randomly generated event time has a value less than the elapsed time between the last contact date and the global cut-off date, i.e. the date when all participants should have performed their EOS visit (notification of study termination + 4 weeks), the participant will be counted as having observed an event at the last contact date plus the random event time. Otherwise, the participant is re-adjusted to be censored at the global cut-off date.

These imputed events will be added to the observed events in the study and the key secondary efficacy variable will be re-analyzed. This process will be repeated 1000 times. For the final analysis, the hazard ratio with 95% confidence intervals will be estimated for each imputed data set and then combined with PROC MIANALYZE using standard multiple imputation combining rules.

Details on generation of the random event times:

When the survival time T follows a Weibull distribution with survival function S_2 , then the conditional distribution function of time T after the last contact date/censoring date (LC) is given for $x \geq 0$ by:

$$\begin{aligned} y &= \Pr(T \leq x + LC | T > LC) = 1 - \Pr(T > x + LC | T > LC) \\ &= 1 - \frac{\Pr(T > x + LC)}{\Pr(T > LC)} = 1 - \frac{S_2(x + LC)}{S_2(LC)} \end{aligned}$$

With the shape parameter β and the scale parameter λ from the underlying Weibull model, this results in:

$$\begin{aligned} y &= 1 - \exp\left(-\left(\frac{x + LC}{\lambda}\right)^\beta\right) / \exp\left(-\left(\frac{LC}{\lambda}\right)^\beta\right) \\ &\equiv \log(1 - y) = -\left(\frac{x + LC}{\lambda}\right)^\beta + \left(\frac{LC}{\lambda}\right)^\beta \\ &\equiv x = \lambda \left(\left(\frac{LC}{\lambda}\right)^\beta - \log(1 - y) \right)^{1/\beta} - LC \end{aligned}$$

If y is replaced with a realization of a random variable with uniform distribution on the interval $[0, 1]$, a realization of the conditional distribution of T under the condition $T > LC$ can be generated using the inverse transformation technique via

$$\lambda \left(\left(\frac{LC}{\lambda} \right)^\beta - \log(1 - rand('uniform')) \right)^{1/\beta} - LC.$$

In addition, the estimated hazard rates for the finerenone group will be inflated by a known factor k , leading to individual hazard rates

$$k \times (\beta/\lambda) (t/\lambda)^{\beta-1}$$

with new scale parameter $s = \lambda/(k^{1/\beta})$. At each rate of inflation 1000 simulations will be performed. Inflation will be done until the upper limit of the confidence interval of the combined estimate of the hazard ratio crosses 1.

6.6 Appendix 6: Region Grouping

For the subgroup analyses by region the participating countries will be pooled according to the following rule

- Europe: Australia*, Bulgaria, Czechia, Denmark, Greece, Israel*, Italy, Spain, Russia
- Asia: China, India, Japan, Malaysia, South Korea
- North America: Canada, United States of America
- Latin America: Argentina, Mexico

*Although not geographically located in Europe, Australia and Israel have been included in this pooled region.

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Title Page**Protocol Title:**

A randomized, double-blind, placebo-controlled, parallel-group, multicenter Phase 3 study to investigate the efficacy and safety of **F**inerenone, in addition to standard of care, on the progression of kidney disease in patients with **Non-Diabetic Chronic Kidney Disease**

Protocol Number: 21177

Compound Number: BAY 94-8862 / Finerenone

Short Title: Efficacy and safety of finerenone in patients with non-diabetic CKD

Acronym: FIND-CKD

Sponsor Name: Bayer AG

Legal Registered Address:

Non-US: Bayer AG, 51368 Leverkusen, Germany

US territory: Bayer HealthCare Pharmaceuticals Inc., 100 Bayer Boulevard, P.O. Box 915, Whippany NJ 07981-0915, USA

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Version History

This statistical analysis plan (SAP) for study BAY 94-8862 / 21177 is based on protocol Version 5.0, dated 10APR2025.

Version	Date	Change	Rationale
1.0	17SEP2021	Not applicable	Original version
2.0	23MAY2025	Formatting adjustments, language improvements, non-substantive corrections, and other editorial clarifications have been made throughout the SAP, including: - The exploratory endpoint, "Time to the composite of sustained eGFR decrease of $\geq 40\%$ or kidney failure", has been removed due to duplication. - Clarifies that pregnancy test outcomes, cystatin C, and NT-proBNP collected in the study will be summarised by visit. - Added boxplots to quantitative laboratory summaries to visualize data distributions alongside tables. - Safety eGFR summaries will use the 2009 Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation (race-adjusted), aligning with the primary efficacy variable. - Vital sign summaries now include boxplots to visualize distribution shifts across visits.	These editorial corrections improve readability, consistency, and alignment with study terminology without altering the analytic content or intent of the SAP.
2.0	23MAY2025	Clarified the classification of missing data as "expected" for instances where data collection was planned but did not occur, making it eligible for imputation. "Non-expected" missing data arising from unscheduled assessments is not considered missing and, therefore, not imputed.	Imputation only applies to data expected per the study protocol.
2.0	23MAY2025	The way post-ESKD/death eGFR values are handled has been revised: Instead of inserting a single "unfavorable" value derived from the lower percentiles of each treatment group distribution, the updated SAP now specifies a multiple imputation procedure in which, for every scheduled visit after ESKD or death, values are drawn (with replacement) from a pre-defined pool of clearly unfavorable eGFR measurements (≤ 15 mL/min/1.73 m ²) obtained from the FIDELIO-DKD, FIGARO-DKD, and FIND-CKD studies.	Lower percentile values within a treatment group can still be clinically acceptable; a donor pool restricted to eGFR ≤ 15 mL/min/1.73 m ² guarantees that every imputed value is unequivocally unfavorable. Implementing the draw via multiple imputation (rather than single substitution) reproduces appropriate uncertainty into the final variance estimates.

Version	Date	Change	Rationale
2.0	23MAY2025	The control-based pattern-mixture multiple imputation method has been removed.	Copying the placebo trajectory after treatment discontinuation is implausible for finerenone because of its mode of action.
2.0	23MAY2025	A new multiple imputation analysis conditioning on visit-specific treatment discontinuation flags (Guizzaro, et al. 2021) has been added, replacing the earlier "retrieved dropout" approach.	Guizzaro et al. (2021) showed that including a compliance/on-treatment indicator in the imputation model preserves unbiasedness and improves efficiency when estimating a treatment policy estimand in trials with incomplete follow-up.
2.0	23MAY2025	The "worst-case" eGFR = 0 imputation has been relabeled as a sensitivity analysis of the primary eGFR endpoint (instead of a separate supplemental estimand). After death or initiation of dialysis/kidney transplant, every remaining scheduled eGFR is set to 0 mL/min/1.73 m ² up to the planned end of treatment.	Positions the extreme worst-case scenario as a robustness check on the primary estimand, making a direct comparison with the primary analysis possible while preventing the addition of alternative estimands.
2.0	23MAY2025	A hypothetical estimand analysis was added that censors eGFR data at ESKD or death and adjusts for informative censoring (e.g., via a joint longitudinal-survival model) instead of imputing post-event values.	Targets the estimand "what the eGFR trajectory would have been had ESKD or death not occurred", by assuming those participants would have followed the same pattern as event-free participants and correcting for any bias from informative dropout.
2.0	23MAY2025	The definition of the heart failure (HF) endpoint in the SAP has been updated to exclude urgent HF visits, aligning it with the protocol.	This adjustment provides consistency between the SAP and the protocol.
2.0	23MAY2025	Outpatient worsening of HF has been included as an exploratory endpoint, correcting an oversight in the previous version of the SAP.	This inclusion aligns the SAP with the protocol.
2.0	23MAY2025	The SAP now includes a new definition for "extended treatment interruption", which is defined as a continuous period during which a participant does not receive study medication for more than 240 consecutive days.	Sets the exact threshold already intended in the ICE definition.
2.0	23MAY2025	The section addressing potential additional participant enrollment due to high dropout rates has been removed from the SAP as recruitment for the trial has been closed.	The contingency plan for addressing dropout rates is no longer relevant.
2.0	23MAY2025	The sensitivity analysis that uses stratification factors from randomization when there were errors in their calculation has been removed.	Given the large sample size, this additional analysis is deemed unnecessary and removed for simplicity.

Version	Date	Change	Rationale
2.0	23MAY2025	The SAP originally included a sensitivity analysis to refit the Cox proportional hazards model for secondary efficacy endpoint analyses without the randomization stratification factors. This sensitivity analysis has been removed.	Omitting the randomization stratification factors from a statistical model can compromise both type I and type II errors by ignoring the allocation scheme on which the trial was designed. In line with this, EMA and FDA guidance both recommend retaining the stratification factors used at randomization in prespecified efficacy analyses.
2.0	23MAY2025	No further analyses will be conducted based on the confirmed COVID-19 status of participants.	COVID-19 is no longer considered a public health emergency; therefore, additional analyses by COVID-19 status are no longer planned.
2.0	23MAY2025	It is now clarified that since only the year of birth is collected, the date of birth will be imputed as 1 January.	This rule was added to ensure that a complete date of birth is available for calculations in the analysis (e.g., summarizing age and computing eGFR), which requires full date information.
2.0	23MAY2025	New guidance has been added on how to handle missing data for the severity and intervention relationship of adverse events (AEs) and outcome events (OEs).	This imputation rule addresses the potential underreporting of severe or intervention-related events.
2.0	23MAY2025	Clarification on the method for calculating baseline values for urinary albumin-to-creatinine ratio (UACR) and urinary protein/creatinine ratio (UPCR) using the geometric mean and for eGFR using the arithmetic mean has been added.	The geometric mean presents a more representative measure of the central tendency for the skewed UACR and UPCR distributions.
2.0	23MAY2025	Definitions for study day (relative to randomization) and study month (calculated as study day ÷ 30) were added, with baseline eGFR set to Month 0 for longitudinal modeling.	Clarification of time-variable calculations.
2.0	23MAY2025	A new section on visit mapping has been added, specifying that by-visit summaries will analyze data according to their nominal schedule and that the end-of-treatment visits coinciding with Month 32 will be remapped to Month 32.	The guidelines align directly with the study's structured design.
2.0	23MAY2025	The new SAP text provides additional detail on handling eGFR data from local laboratories when central laboratory data are not available due to specific discontinuation scenarios. Local laboratory eGFR values will be used only in the analyses for the primary endpoint and will not be incorporated into the sustained-decline time-to-event analysis.	This clarification ensures the inclusion of the most complete eGFR dataset possible.

Version	Date	Change	Rationale
2.0	23MAY2025	The previous rule regarding unscheduled UACR/UPCR measurements was removed as it was redundant. This has been replaced with a clarification that focuses solely on re-testing during screening. If a re-test is necessary because initial UACR values at screening disqualified the participant from enrollment, these re-test values will be used.	The triple measurement requirement applies only to the screening and randomization visits, where 3 measurements are specifically mandated.
2.0	23MAY2025	The update introduces "pooled regions" and consolidates data presentation by treatment group and geographical categorizations, omitting the breakdown by study site.	This update streamlines data summaries by prioritizing broader regions.
2.0	23MAY2025	A summary is now planned to summarize randomized participant counts by categorizing them into those who completed treatment and those who did not, with further subdivision to indicate if participants completed the follow-up period. The SAP also now clarifies that participants' overall study completion status will be summarized.	Outline participant progression throughout the study, specifying completion rates for the treatment phase, the follow-up phase, and overall study completion.
2.0	23MAY2025	Clarified that the planned treatment period runs from randomization through the scheduled end-of-treatment visit, or, if that visit is not completed, extends up to 4 weeks after the end-of-study (EoS) notification.	Adds an explicit 4-week window for participants who miss the end-of-treatment visit.
2.0	23MAY2025	The SAP now includes descriptions of two alternative eGFR calculation methods as specified in the protocol: one using the CKD-EPI equation with the race factor omitted and another based on serum creatinine, serum cystatin C, age, and gender. The SAP now specifies which efficacy analyses will be performed using the two alternative eGFR formulas.	The change aligns the SAP with the protocol.
2.0	23MAY2025	The general overview of the primary analysis model has been refined to align with the eGFR model in the analysis section. Time has been removed as the main effect, and interactions between time and fixed effects have been highlighted.	The update was made to ensure the consistency of the primary analysis model formulation across the SAP.
2.0	23MAY2025	The covariance matrix of the primary analysis mixed model now varies between treatments, replacing the previously assumed common covariance matrix across all treatments.	This modification provides a more customized approach to modeling, considering potential differences in variance structures among treatment groups.

Version	Date	Change	Rationale
2.0	23MAY2025	The unscheduled eGFR sample collected about 1 month after a participant first meets an event criterion (either a $\geq 57\%$ decline from baseline or an absolute eGFR < 15 mL/min/1.73 m ²) will be removed from the mixed model analysis of eGFR.	The confirmation draw is taken only for participants who have already met an event criterion, which would add extra post-event data for that subgroup and introduce informative sampling bias into the mixed model trajectories and related estimates.
2.0	23MAY2025	The notation for the within-participant error variance of eGFR, σ_e^2 , has been corrected to remove incorrect indices previously suggested to vary by subject and treatment.	This update clarifies that the error variance σ_e^2 is identical for all subjects and treatment groups, aligning the SAP's notation with the model's specifications.
2.0	23MAY2025	The primary analysis model has been updated to utilize the containment method for calculating the denominator degrees of freedom.	Specifying the containment method clarifies the approach used to calculate the degrees of freedom in the statistical analysis.
2.0	23MAY2025	Added a step-wise fallback plan for convergence problems: if the primary mixed model fails, (i) fit a model with a pooled within-participant variance-covariance matrix across treatment arms; if still problematic, (ii) move to a heterogeneous compound-symmetry structure; if needed, (iii) reduce to simple compound symmetry and sequentially drop time-by-stratification interactions, then stratification main effects.	The tiered simplification sequence guards against non-convergence by gradually easing model complexity, letting the analysis proceed while keeping the most flexible structure the data can reliably support.
2.0	23MAY2025	A trajectory plot illustrating the primary analysis was added.	The figure provides a visual summary of the results from the primary analysis.
2.0	23MAY2025	A sensitivity analysis of eGFR employing a mixed model for repeated measures (MMRM) has been introduced to examine the robustness of the primary analysis linearity assumption.	The MMRM analysis addresses potential concerns about the linearity assumption inherent in the primary analysis' two-slope model. Given the extensive duration and multiple visit assessments in the study, there is a possibility of non-linearity in the eGFR decline over time.
2.0	23MAY2025	For the sodium-glucose co-transporter 2 (SGLT2) inhibitor use sensitivity analysis, the population is now restricted to participants who are SGLT2 inhibitor-naïve at baseline, and all eGFR values after the first SGLT2 inhibitor use are set to missing, regardless of whether the SGLT2 inhibitor is later discontinued.	This estimand is formulated to evaluate eGFR in participants who are SGLT2 inhibitor-naïve at baseline, under the hypothetical scenario that those who later start an SGLT2 inhibitor would have followed the same trajectory had they remained untreated.
2.0	23MAY2025	Total-slope analyses were added for 36, 40, 44, and 48 months using the same method as the 32-month primary analysis.	Assesses whether the eGFR slope treatment effect holds over longer follow-up periods.

Version	Date	Change	Rationale
2.0	23MAY2025	The SAP now specifies that the Efron method will be used to handle ties in time-to-event data for survival endpoints. It also clarifies using SAS procedures: PROC LIFETEST for conducting stratified log-rank tests and PROC PHREG for calculating hazard ratios.	These methods and procedures are specified for clarity in the statistical analysis framework of the time-to-event endpoints.
2.0	23MAY2025	Time-to-event efficacy endpoints will now be analyzed with a competing-risk approach that treats death as a competing event, replacing the previously planned Kaplan-Meier method.	Censoring deaths in a Kaplan-Meier curve can lead to an overestimation of the endpoint's incidence while treating deaths as a competing risk provides cumulative incidence estimates that more accurately reflect patients' real-world experiences.
2.0	23MAY2025	The SAP now includes a method for calculating confidence intervals (CIs) for the incidence rates per 100 patient years of the time-to-event efficacy outcomes.	The update provides the approach that will be used to estimate the precision of the incidence rates of the efficacy endpoint.
2.0	23MAY2025	Added urgent HF visits to the renal-cardiovascular (CV) composite endpoint (kidney failure, sustained eGFR decline, HF hospitalization, CV death) as an exploratory endpoint.	This endpoint reflects current practice, where clinically significant HF deteriorations are managed in urgent-care settings without an overnight stay.
2.0	23MAY2025	A new exploratory endpoint has been added on the time to onset of eGFR decrease of $\geq 50\%$ from baseline sustained over at least 4 weeks.	Introduces an exploratory 50% threshold that captures additional meaningful eGFR declines, adding broader context alongside the primary 57% criterion.
2.0	23MAY2025	The only exploratory endpoints retained (not in the protocol) are: - Time to kidney failure, sustained $\geq 57\%$ eGFR decline (≥ 4 weeks), HF hospitalization, urgent HF visit, or CV death. - Time to kidney failure or sustained $\geq 50\%$ eGFR decline (≥ 4 weeks).	Keeps focus on the two most meaningful exploratory efficacy outcomes.
2.0	23MAY2025	The MMRM analysis of UACR will be repeated using data collected up to 30 days after the last study drug dose.	Gives an on-treatment estimate of UACR change by excluding off-treatment measurements.
2.0	23MAY2025	A win-odds analysis has been added to describe a hierarchical composite endpoint (HCE) analysis.	The win-odds analysis considers the clinical importance of different renal events in the trial to assess treatment effects across multiple dimensions of clinical outcomes.
2.0	23MAY2025	Implemented a mean cumulative function analysis to monitor all recurrent hyperkalemia episodes resulting in hospitalization or permanent drug discontinuation.	Unlike a first-event approach, which overlooks repeat occurrences, the mean cumulative function captures the cumulative burden of hyperkalemia over time.
2.0	23MAY2025	A count of all worsening renal function AEs will now be presented alongside the existing counts for hospitalization and permanent discontinuation cases.	Including milder events provides a fuller picture of renal safety.

Version	Date	Change	Rationale
2.0	23MAY2025	The safety outputs will no longer summarize disease-related events; they remain covered under the efficacy analyses.	Avoids duplicate reporting and keeps safety and efficacy results separated.
2.0	23MAY2025	The PK section now specifies the exact descriptive statistics to be reported and states that values below the lower limit of quantification (LLOQ) will be imputed as $\frac{1}{2}$ LLOQ before analysis.	Explicitly defining the summary statistics and the LLOQ imputation rule ensures transparent reporting of the PK data.
2.0	23MAY2025	The subgroup analysis section was revised to provide detailed modeling information, while several low-priority subgroups were removed.	More detailed technical information clarifies the planned analyses while limiting non-essential subgroups, which reduces multiplicity and keeps the analysis set manageable.
2.0	23MAY2025	Population characteristics summaries will no longer be produced for the safety analysis set (SAF).	Aligns with protocol Version 5 and avoids duplicating tables when the SAF and full analysis set (FAS) overlap.
2.0	23MAY2025	The list of concomitant medications of interest has been updated.	Keeps the focus on drugs most relevant to study outcomes.
2.0	23MAY2025	Added shift tables summarizing SGLT2 inhibitor use from baseline status to any post-baseline initiation.	Tracks treatment-emergent SGLT2 inhibitor use to assess its potential impact on study outcomes.
2.0	23MAY2025	Prior and concomitant procedures will be tabulated and summarised.	Clarifies how already-collected procedure data will be summarised.
2.0	23MAY2025	Details on the classification of AEs linked to intervention down-titration or interruption have been added. AEs will be categorized as either "hyperkalemia" or "other than hyperkalemia" to refine the assessment of dosage adjustment reasons.	This classification allows for a more granular analysis of AEs associated with intervention changes.
2.0	23MAY2025	The calculation method for the number of tablets taken has been specified.	Specifying the calculation method for tablet counts clarifies compliance measurement.
3.0	11FEB2026	Added explicit clarification that any other intercurrent events (e.g., initiation of prohibited or rescue medication) are handled using the treatment policy strategy; no changes to the estimand or analysis methods.	Agency feedback requesting explicit specification that rescue medications are handled under the treatment policy.
3.0	11FEB2026	Added a date imputation rule specifying that when an event start date is completely missing, the study intervention start date will be used as the imputed event start date.	Prevent data loss by including events with missing start dates in analyses.
3.0	11FEB2026	Added an imputation rule that if baseline height is missing, the first available post-baseline height will be used to impute baseline height; baseline body mass index (BMI) will be calculated using this imputed baseline height.	Preserve baseline BMI data for analyses when baseline height is missing.

Version	Date	Change	Rationale
3.0	11FEB2026	Added a by-visit imputation/remapping rule for summaries and mixed model for repeated measures (MMRM) analyses (use closest qualifying alternative within visit windows; remap coincident end-of-treatment to the scheduled visit; summarize follow-up separately; restrict imputation to the planned treatment period).	End-of-treatment, premature discontinuation, and follow-up visits (for prematurely discontinued participants) have flexible timing and may overlap with scheduled assessments; remapping prevents duplication and avoids data loss when overlaps occur.
3.0	11FEB2026	Added a pairing rule for 2021 CKD-EPI creatinine-cystatin C eGFR: use the creatinine measurement's date/visit as reference; pair with the closest cystatin C within ± 14 days (scheduled or unscheduled).	Creatinine and cystatin C are occasionally measured on different days; the ± 14 -day pairing prevents loss of usable data.
3.0	11FEB2026	Added an analysis-level rule to compute UACR/UPCR when the lab reports the ratio as missing due to component censoring: impute components as $\frac{1}{2}$ LLOQ (if below LLOQ) or upper limit of quantification (ULOQ) (if above ULOQ), then calculate the ratio.	Labs may suppress ratio values when component results are censored at the quantification limits (LLOQ/ULOQ); imputing components prevents data loss.
3.0	11FEB2026	Added rule that if only one valid UACR or UPCR measurement is available at screening, that single result will be used.	Ensure the UACR-based stratification factor is available at screening and avoid missingness for randomization/analysis.
3.0	11FEB2026	Added calculation of the race-adjusted 2012 CKD-EPI creatinine-cystatin C eGFR for comparison and specified Japanese calibration factors (0.813 for creatinine-only; 0.908 for creatinine+cystatin C).	Provide a benchmark against the established race-adjusted equation and ensure standardized Japanese calibration based on published sources.
3.0	11FEB2026	Added specification that treatment-specific slope estimates are obtained by evaluating stratification fixed effects and their time interactions at the population-average distribution of randomization strata, using weights proportional to stratum sample sizes in the analysis population.	Ensure marginal (population-average) treatment-slope estimates are comparable across arms and not driven by stratum imbalances.
3.0	11FEB2026	Restricted the donor pool for unfavorable eGFR imputation to the current study (last pre-ESKD eGFR per participant), replacing the prior use of FIDELIO-DKD, FIGARO-DKD, and FIND-CKD.	Agency feedback to avoid heterogeneity from diabetic kidney disease trials; using only current non-diabetic (CKD) data aligns donor values with the study population.
3.0	11FEB2026	Defined the on-treatment period for Sensitivity Analysis 1 as ending at the earlier of the last dose date or the start of an extended treatment interruption.	Clarify the exposure window used in imputation.
3.0	11FEB2026	Specified that MMRM model-based means and between-treatment contrasts will use observed margins, averaging over the observed distribution of stratification factors at each visit.	Produce population-average estimates that reflect the actual strata mix and reduce sensitivity to stratum imbalances.

Version	Date	Change	Rationale
3.0	11FEB2026	Added additional MMRM fallback measures: remove stratification-by-visit interaction terms that cause nonconvergence, and exclude visits with insufficient sample size from the analysis.	Improve model convergence and avoid unstable estimates when data are sparse at certain visits.
3.0	11FEB2026	Added explicit shared-parameter (SP) model specifications (fixed Month 3 knot; piecewise exponential baseline hazard with quantile-based cutpoints, max 6-month intervals and ≤ 9 intervals; constant longitudinal residual variance; Cholesky covariance; estimation via <code>PROC NLMIXED</code> with Laplace and informed starts), and refined censoring to the earliest of planned end-of-treatment or last study contact.	Improve clarity, reproducibility, and consistency of terminology and censoring rules.
3.0	11FEB2026	Expanded the existing analysis of covariance (ANCOVA)-based 1-month off-treatment eGFR analysis to repeat using additional CKD-EPI equations (CKD-EPI 2021 creatinine race-unadjusted; CKD-EPI 2021 creatinine-cystatin C race-unadjusted; CKD-EPI 2012 creatinine-cystatin C race-adjusted).	Added as sensitivity analyses across CKD-EPI equations (race-adjusted vs unadjusted; creatinine-only vs creatinine+cystatin C).
3.0	11FEB2026	Removed the "starting from Visit 2 (Month 1)" qualifier so kidney failure and CV events occurring after the last recorded eGFR at any clinic visit are included in the efficacy analyses.	Avoids artificial timing restrictions and ensures full capture of post-visit events potentially attributable to the study drug.
3.0	11FEB2026	Added UPCR as an exploratory endpoint.	Captures proteinuria as a marker of renal injury.
3.0	11FEB2026	Updated the missing data tipping point sensitivity analysis of the time-to-event endpoints to (1) estimate the post-discontinuation hazard using only patients who discontinued treatment prior to the event of interest, using their follow-up from treatment discontinuation onwards, and (2) perform proper multiple imputation by incorporating uncertainty in the estimated hazard rates (via repeated parameter draws) when imputing post-dropout event times.	This update bases the sensitivity analysis on empirically observed post-discontinuation risk among patients who discontinued treatment before the event of interest, rather than on stronger extrapolation assumptions.
3.0	11FEB2026	Removed cumulative incidence function (Aalen-Johansen) and mean cumulative function (recurrent event) analyses for hyperkalemia, and removed cumulative incidence function analyses for time to first treatment-emergent serum potassium exceedances (5.5/6.0 mmol/L).	Descriptive incidence (proportion of participants with treatment-emergent hyperkalemia) is sufficient; the cumulative incidence function and the mean cumulative function do not provide additional insight.
3.0	11FEB2026	Removed selected subgroups from the efficacy subgroup analyses.	These subgroups are redundant with other planned subgroup categories and are not expected to provide material additional insight.

Version	Date	Change	Rationale
3.0	11FEB2026	Added clarification on how the subgroup-by-treatment interaction p -value will be calculated under multiple imputation for the acute, delta, chronic, and total slope endpoints.	Improves transparency and reproducibility by explicitly describing the multiple imputation-based inference approach for interaction testing.
3.0	11FEB2026	Removed cumulative incidence analyses for worsening renal function.	Cumulative incidence functions for worsening renal function do not provide additional insight.
3.0	11FEB2026	Removed the planned "repeat" win-odds analysis that replaced the 7 th endpoint with a dichotomous indicator of ≥ 1 mL/min/1.73 m ² /year eGFR-slope improvement vs placebo.	Dichotomizing the slope at ≥ 1 mL/min/1.73 m ² /year discards information, may reduce statistical power, and complicates interpretation versus the continuous slope component.
3.0	11FEB2026	UPCR has been excluded and will not be included in the planned analyses.	UPCR is deemed non-essential, and removing it reduces analytical complexity.

1. Introduction

Diverse pathologies induce injury and inflammation, resulting in a common pathophysiological mechanism in chronic kidney disease (CKD) where glomerular injury leads to proteinuria, interstitial inflammation, and, ultimately, fibrosis. Finerenone is a mineralocorticoid receptor antagonist acting by targeting inflammatory and fibrotic pathways mediated by mineralocorticoid receptor overactivation in the heart and kidney (Kolkhof and Bäracker 2017). In the Phase 3 FIDELIO-DKD study, finerenone protected patients with CKD and type 2 diabetes from kidney function decline and reduced the risk of cardiovascular (CV) events. Efficacy was observed independent of the hyperglycemic state (Bakris, et al. 2020). Finerenone also reduced albuminuria in diabetic and non-diabetic patients with CKD in the ARTS Phase 2a study (Pitt, et al. 2014). Based on the available pre-clinical and clinical data, finerenone is believed to show efficacy in delaying the progression of kidney disease in diabetic and non-diabetic patients.

The statistical analysis plan (SAP) describes the final analysis of the study. No formal statistical interim analysis will be performed. Table, listing, and figure (TLF) specifications are contained in a separate document.

1.1 Objectives, Endpoints and Estimands

Objectives	Endpoints
Primary	
To demonstrate that finerenone in addition to SoC, is superior to placebo in delaying the progression of kidney disease	Mean rate of change as measured by the total slope of eGFR from baseline to Month 32
Secondary	
- To demonstrate the beneficial effect of finerenone in addition to SoC as compared to placebo - To assess the safety of finerenone in addition to SoC as compared to placebo	- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, or CV death - Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$ - Time to the composite of hospitalization for HF, or CV death - Number of participants with TEAEs, TESAEs and AESIs
Exploratory	
To be supportive of the primary efficacy endpoint by further demonstrating the effect of finerenone in addition to SoC as compared to placebo	- Chronic eGFR slope from Month 3 to end of planned treatment period - eGFR change from baseline to 1 month after end of treatment (off-treatment)
To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (supportive of the key secondary endpoints)	- Time to onset of kidney failure (ESKD or sustained eGFR decline below 15 mL/min/1.73 m²) - Time to onset of ESKD (chronic dialysis or transplant) - Time to onset of sustained eGFR decrease below 15 mL/min/1.73 m² - Time to onset of sustained eGFR decrease of $\geq 57\%$ from baseline - Time to CV death - Time to first hospitalization for HF or urgent HF visit

Objectives	Endpoints
To further demonstrate the effect of finerenone in addition to SoC as compared to placebo (other exploratory endpoints)	- Number of participants with UACR reduction $\geq 30\%$ from baseline at Month 6 - Change in UACR from baseline - Time to the composite of sustained eGFR decrease of $\geq 40\%$ or kidney failure - Time to first CV event (CV death, non-fatal MI, non-fatal stroke or TIA, hospitalization for HF, or urgent HF visit) - Time to CV hospitalization - Time to all-cause mortality - Time to all-cause hospitalization - Number of participants with new diagnosis of atrial fibrillation or atrial flutter - Number of participants with new diagnosis of HF - Outpatient worsening of HF - Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, urgent HF visit, or CV death[†] - Time to the composite of sustained eGFR decrease of $\geq 50\%$ or kidney failure[†]
Exploratory Biomarkers	
To further investigate the study intervention and similar drugs (e.g., mode-of-action-related effects, safety) and to further investigate pathomechanisms deemed relevant to CV/CKD and associated health problems	Various biomarkers (e.g., diagnostic, safety, pharmacodynamic, monitoring, or potentially predictive biomarkers)
Exploratory: Quality of Life	
To further investigate the quality of life parameters	Change from baseline in EQ-5D-5L

[†]Not explicitly specified as an exploratory endpoint in the protocol; included as an exploratory analysis. AESI = Adverse event of special interest, CKD = Chronic kidney disease, CV = Cardiovascular, eGFR = Estimated glomerular filtration rate, EQ-5D-5L = EuroQol 5 Dimensions, 5 Levels, ESKD = End-stage kidney disease, HF = Heart failure, MI = Myocardial infarction, SoC = Standard of care, TIA = Transient ischemic attack, TEAE = Treatment-emergent adverse event, TESAE = Treatment-emergent serious adverse event, UACR = Urinary albumin-to-creatinine ratio

Estimand

The attributes constituting the primary estimand are the following:

- Population:** The population consists of non-diabetic CKD patients, further defined by the in- and exclusion criteria.
- Treatment:** All participants will receive blinded study intervention (i.e., finerenone 10 mg/20 mg or placebo) once daily (OD) on top of individually tolerated maximum labeled doses of either an angiotensin-converting enzyme inhibitor (ACEI) or an angiotensin-receptor blocker (ARB).
- Variable:** Estimated glomerular filtration rate (eGFR) values over time collected every 1 to 4 months during the planned treatment period
- Intercurrent events (ICEs):** The 3 most important ICEs to consider are
 - death,

- starting chronic dialysis or receiving a kidney transplant, and
- premature treatment discontinuation/extended interruption of treatment.

The first 2 ICEs (ESKD and death) will be handled using a *composite* strategy. For all scheduled visits after the occurrence of the ICE up to the planned end of treatment, unfavorable eGFR values will be used (see Section 5.3.2).

An extended treatment interruption is defined as a continuous period during which a participant does not receive the study medication for more than 240 consecutive days.

For premature treatment discontinuation/extended interruption of treatment, a *treatment policy* strategy will be applied: participants will continue to be followed up for all planned visits after they discontinue study treatment or have an extended interruption, and any eGFR values collected during these periods will be included in the analysis.

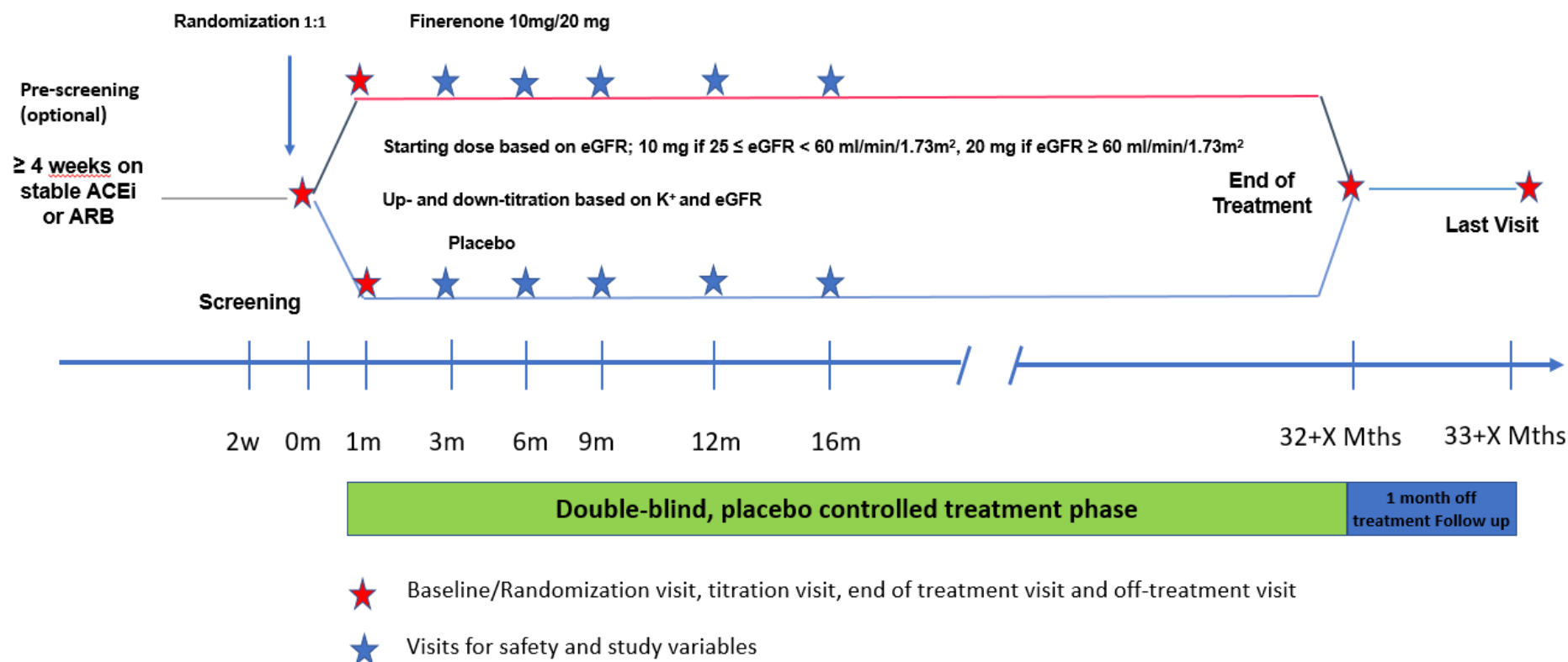
- **Population-level summary:** Total eGFR slope from baseline to 32 months comparing the finerenone treatment group with the placebo group as estimated from a random coefficient 2-slope linear spline mixed-effects model.

Any other conceivable intercurrent events not explicitly mentioned above (e.g., the initiation of prohibited or rescue medication) are handled according to the treatment policy.

1.2 Study Design

This is a double-blind, parallel-group intervention study in non-diabetic CKD patients with 2 treatment groups, including a placebo treatment group. The general study design of this study is shown in Figure 1–1.

Figure 1–1: Study Schema



Study duration and the number of study visits will depend on the enrollment time of the participant.

ACEi = Angiotensin-converting enzyme inhibitor, ARB = Angiotensin receptor blocker, eGFR = Estimated glomerular filtration rate, K^+ = Potassium, m = Month, Mths = Months, w = Week

Pre-screening (optional)

An assessment of a urinary albumin-to-creatinine ratio (UACR) is permitted during the pre-screening visit and may be measured locally. This is optional and may be performed if the investigator wishes to obtain a recent UACR value to help assess if a participant is likely to be eligible. Pre-screening UACR values cannot be used in lieu of the centrally assessed urinalysis at screening. In the exceptional case where no previous evidence of albuminuria is documented in the participant's medical records, the pre-screening value can be used instead. In such situations, the screening visit can only take place 3 months after pre-screening.

Screening

Approximately 3160 participants will be screened to achieve approximately 1580 participants randomly assigned to study intervention.

Randomization

This study is a randomized, double-blind, placebo-controlled, parallel-group, multicenter study with a variable treatment duration of a minimum of 32 months up to a maximum of approximately 49 months (final treatment duration will depend on recruitment time; the 49 months are based on an assumed recruitment time of 17 months). Participants will be randomized 1:1 to either finerenone or placebo OD in a blinded fashion on top of their standard of care (SoC) therapy, including an ACEI or an ARB.

Randomization will be stratified by screening UACR (≤ 1000 mg/g vs > 1000 mg/g) and baseline sodium-glucose co-transporter 2 (SGLT2) inhibitor use (yes, no) to allow equal distribution of those characteristics between the treatment arms.

Treatment

All participants will receive blinded study intervention (i.e., finerenone 10 mg/20 mg or placebo) OD on top of their individually tolerated maximum labeled dose of either an ACEI or an ARB. The starting dose will depend on the participant's eGFR level at the screening visit: a lower dose of 10 mg OD if eGFR is between 25 to < 60 mL/min/ 1.73 m² (2009 Chronic Kidney Disease Epidemiology Collaboration [CKD-EPI]) or the higher (target) dose of 20 mg OD if eGFR is ≥ 60 mL/min/ 1.73 m² (CKD-EPI 2009). Participants who start on the lower dose of 10 mg will be up-titrated to the target dose of 20 mg from Month 1 on treatment onwards, provided potassium is ≤ 4.8 mmol/L and the decrease in eGFR is less than 30% below the last measured value. Down-titration or interruption of the intervention is permitted at any time during the study for safety reasons, and the intervention has to be interrupted if potassium is > 5.5 mmol/L. Visits will take place at screening, baseline, Month 1, Month 3, and then every 3 months up to Month 12. Thereafter, if the end-of-study (EoS) has not been declared, participants will continue on 4-monthly visits until the EoS.

Additional safety visits will take place 4 weeks after each up-titration or after restart of the study intervention following treatment interruption for > 7 days. eGFR as the primary endpoint measure and potassium as the key safety parameter will be assessed at each visit; UACR and urinary protein/creatinine ratio (UPCR) will be collected at time points shown in the schedule of activities (SoA), see Section 1.3 of the study protocol. Up- and down-titration of the intervention will be based on local potassium and eGFR values.

End of treatment

The minimum planned treatment duration will be 32 months. All participants will stay in the study until the last randomized participant has reached 32 months of treatment. At this point,

all participants remaining in the study and still on treatment will end the treatment phase and move to the follow-up phase. Thus, the treatment duration will be from 32 to approximately 49 months, depending on the recruitment time (assumed to be approximately 17 months). Participants who have already discontinued treatment at that time will be contacted via phone to collect outcome events (OEs) and adverse events (AEs).

Follow-up

The follow-up visit will take place approximately 1 month after the last administration of the study intervention.

2. Statistical Hypotheses

Primary efficacy endpoint

The primary efficacy variable of interest is eGFR, assessed over time. The primary endpoint is the mean rate of change of eGFR from baseline to Month 32. The rate of change of eGFR will be compared between finerenone and the placebo groups by estimating the total eGFR slope, i.e., the slope from baseline to Month 32, using available assessments from baseline to the end of the planned treatment period as outlined in Section 5.3.2.

In order to evaluate whether finerenone is superior to placebo in inducing a higher mean rate of change, the following null hypothesis will be tested at the 2-sided significance level $\alpha = 0.05$:

$$H_0: \beta_{3,f}(32) = \beta_{3,c}(32),$$

where $\beta_{3,f}(32)$ denotes the total eGFR slope from baseline to Month 32 for finerenone and $\beta_{3,c}(32)$ the total eGFR slope in the placebo group as defined in Section 5.3.2. The alternative hypothesis is:

$$H_1: \beta_{3,f}(32) \neq \beta_{3,c}(32).$$

Secondary efficacy endpoints

To evaluate whether finerenone is superior to placebo for the 3 key secondary time-to-event endpoints: (i) time to the composite of sustained eGFR decline of $\geq 57\%$, kidney failure, hospitalization for heart failure (HF), or CV death; (ii) time to the composite of sustained eGFR decline of $\geq 57\%$ or kidney failure; and (iii) time to the composite of hospitalization for HF, or CV death, the following null hypotheses will be tested, where the strata are defined by the randomization stratification factors:

$$H_0: \lambda_{f,k}(t) = \lambda_{c,k}(t) \text{ for all time points } t \geq 0 \text{ each stratum } k.$$

The alternative hypotheses is:

$$H_1: \lambda_{f,k}(t) \neq \lambda_{c,k}(t) \text{ for at least 1 time point } t \geq 0 \text{ and at least 1 stratum } k,$$

where $\lambda_{f,k}$ denotes the hazard rate of the finerenone treatment group in stratum k and $\lambda_{c,k}$ denotes the hazard rate of the placebo treatment group in stratum k for the pertaining endpoint. In this context, the hazard rate is defined as the instantaneous risk of experiencing an event at a given time point, t , provided that no such event has yet occurred.

Multiple testing procedure of primary and key secondary efficacy endpoints

The primary and the key secondary endpoints will be tested under a confirmatory hierarchical testing procedure. If the primary efficacy endpoint achieves the significance level of 0.05, the corresponding null hypothesis can be rejected, and the 3 remaining key secondary endpoints

will be tested according to the pre-specified hierarchical sequence given below. Consequently, confirmatory testing of the second, third, and fourth hypotheses requires rejecting each preceding null hypothesis. Each hypothesis will be assessed at a 2-sided significance level of 0.05 in this setting while maintaining the global type I error rate at 0.05.

The pre-defined hierarchical sequence of endpoints is given as:

1. Mean rate of change of eGFR from baseline to Month 32
2. Time to the composite endpoint of sustained eGFR decline of $\geq 57\%$, kidney failure, hospitalization for HF, or CV death
3. Time to the composite endpoint of sustained eGFR decline of $\geq 57\%$ or kidney failure
4. Time to the composite endpoint of hospitalization for HF, or CV death

The definition of kidney failure is presented in Section [5.4.1.1](#).

3. Sample Size Determination

Approximately 3160 participants will be screened to achieve 1580 randomly assigned to study intervention.

The approximate power of *t*-tests for comparing eGFR slope parameters (mL/min/1.73 m²/year) between treatment groups is based on a 2-slope linear spline mixed-effects model specified model parameters, approximately 17 months recruitment period, at least 32 months planned time on treatment per participant, and designated sample sizes. The power estimate from the simulation is based on initial parameter values obtained from FIDELIO-DKD data, consistent with the parameters of the linear spline mixed-effects model described by Vonesh et al. (2019). A total sample size of 1500 participants (750 per treatment arm) will have > 90% power to demonstrate the superiority of finerenone to placebo with respect to the total eGFR slope from baseline to Month 32 at a 2-sided significance level of 5%. The sample size calculation includes the assumption of a 5% annual event rate resulting from ICEs (due to kidney failure or all-cause mortality). The > 90% power calculation is based on an assumption of a 0.7 mL/min/1.73 m²/year total eGFR slope difference. A total sample size of 1500 will have an 80% power when assuming a 0.527 mL/min/1.73 m²/year total eGFR slope difference between finerenone and placebo.

An estimated total of 1580 participants (790 per treatment arm) will be randomized, considering an approximate 5% overall discontinuation rate in the study. All participants will be randomized in a 1:1 ratio to receive treatment with either finerenone OD or placebo.

Power calculations are presented with different scenarios in [Table 3–1](#).

Table 3–1: Approximate Power of t-tests for Comparing eGFR Slope Parameters (mL/min/1.73 m²/Year) Between Treatment Groups Based on 2-Slope Linear Spline Mixed-Effects Model

Power calculation results for testing the following null vs alternative hypotheses at a 2-sided type I error = 0.05							
H ₀ : Treatment Slope – Control Slope = 0 (no difference)							
H ₁ : Treatment Slope – Control Slope = DIFF (difference)							
Total Sample Size							
Chronic Slope Treatment Difference (mL/min/1.73 m ² /Year)	Parameter	DIFF	1400	1450	1500	1550	1600
			Treated– Control	Treated– Control	Treated– Control	Treated– Control	Treated– Control
1.116	Chronic Slope	1.116	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Years)	0.33	0.376	0.385	0.397	0.408	0.418
	Total Slope (32 Months)	0.527	0.77	0.781	0.796	0.809	0.82
	Total Slope (3 Years)	0.592	0.864	0.873	0.884	0.894	0.903
1.224	Chronic Slope	1.224	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Years)	0.425	0.562	0.573	0.588	0.603	0.616
	Total Slope (32 Months)	0.625	0.893	0.9	0.91	0.919	0.926
	Total Slope (3 Years)	0.691	0.946	0.951	0.957	0.963	0.967
1.296	Chronic Slope	1.296	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Years)	0.488	0.681	0.692	0.708	0.722	0.735
	Total Slope (32 Months)	0.69	0.942	0.948	0.954	0.96	0.964
	Total Slope (3 Years)	0.757	0.974	0.977	0.981	0.984	0.986
1.368	Chronic Slope	1.368	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Years)	0.551	0.783	0.794	0.808	0.821	0.832
	Total Slope (32 Months)	0.755	0.972	0.975	0.979	0.982	0.984
	Total Slope (3 Years)	0.823	0.989	>0.99	>0.99	>0.99	>0.99
1.44	Chronic Slope	1.44	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Years)	0.614	0.863	0.872	0.884	0.894	0.903
	Total Slope (32 Months)	0.82	0.987	0.989	>0.99	>0.99	>0.99
	Total Slope (3 Years)	0.889	>0.99	>0.99	>0.99	>0.99	>0.99

Table 3–1: Approximate Power of t-tests for Comparing eGFR Slope Parameters (mL/min/1.73 m²/Year) Between Treatment Groups Based on 2-Slope Linear Spline Mixed-Effects Model

Power calculation results for testing the following null vs alternative hypotheses at a 2-sided type I error = 0.05							
H ₀ : Treatment Slope – Control Slope = 0 (no difference)							
H ₁ : Treatment Slope – Control Slope = DIFF (difference)							
Total Sample Size							
Chronic Slope Treatment Difference (mL/min/1.73 m ² /Year)	Parameter	DIFF	1400	1450	1500	1550	1600
			Treated– Control	Treated– Control	Treated– Control	Treated– Control	Treated– Control
1.512	Chronic Slope	1.512	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (2 Years)	0.677	0.921	0.927	0.935	0.942	0.948
	Total Slope (32 Months)	0.886	>0.99	>0.99	>0.99	>0.99	>0.99
	Total Slope (3 Years)	0.955	>0.99	>0.99	>0.99	>0.99	>0.99

Specified model parameters: 15 months accrual period, 32 months planned time on treatment, and designated sample sizes. The treated and control group acute slopes are fixed, corresponding to an acute treatment effect at Month 3. Further, the control group's chronic slope has been fixed.

The power is based on intercurrent events (kidney failure or all-cause mortality), with an overall combined rate of 5% per year.

A SAS macro program, "%GFR_Slope_Power", was provided by Vonesh et al. (2019) specifically to determine sample size/power estimates for comparing slopes between 2 treatment groups based on the 2-slope linear spline mixed-effects model (Section 5.3.2). The power estimates are based on approximations described by Littell et al. (2006). To estimate power using the SAS macro, the user provides input values for the sample size per group and the number of measurement occasions, along with hypothesized values of the parameters from the model (Section 5.3.2) and other input variables, including study accrual and follow-up periods. A table of the hypothesized parameters is included in Section 6.3. The program also allows power estimates in the presence of the ICE of end-stage kidney disease (ESKD), in which case power is based on an unbalanced population with missing values for the primary endpoint resulting from ESKD the ICE kidney failure and from administrative removal due to discontinuations for reasons unrelated to eGFR values, such as death.

4. Analysis Sets

The analysis sets are defined in Table 4–1.

Table 4–1: Analysis Sets

Participant Analysis Set	Description
Enrolled	All participants who sign the informed consent form (ICF)
Safety analysis set (SAF)	All randomized participants who have taken at least 1 dose of the study intervention. Participants will be analyzed according to the intervention they received.
FAS	All randomized participants. Participants will be analyzed according to the intervention they were randomized to.
Pharmacokinetic analysis set (PKS)	All finerenone-treated participants with at least 1 valid finerenone plasma concentration and without validity findings that would interfere with the evaluation of the pharmacokinetic (PK) data.

Final decisions regarding the assignment of participants to analysis sets will be made during the review of study data and documented in the final list of important deviations, validity findings, and assignment to analysis sets.

5. Statistical Analyses

5.1 General Considerations

The statistical analyses will be performed using the software SAS (release 9.4 or higher; SAS Institute Inc., Cary, NC, USA). All variables will be analyzed using descriptive statistical methods. The number of data available and missing data, mean, standard deviation (SD), minimum, quartiles, median, and maximum will be calculated for metric data. The geometric mean and geometric SD will be provided instead of the arithmetic mean and SD for the variables where log-normal distributions are assumed. Frequency tables will be generated for categorical data.

All participants will be analyzed according to the planned study intervention for the FAS (the intent-to-treat [ITT] principle) and the actual study intervention for the SAF. If a participant receives both study interventions due to a bottle error, the study intervention actually received for the majority of the time in the study will be used for the SAF.

The stratified analyses mentioned in this SAP will be conducted in consideration of the randomization stratification factors (i) use of SGLT2 inhibitor at baseline (yes, no) and (ii) screening UACR (≤ 1000 mg/g vs > 1000 mg/g), unless specified otherwise. For the primary

analysis, participants will be assigned to the stratification categories according to the actual data as recorded on the electronic case report form (eCRF).

No ICE linked to the coronavirus disease 19 (COVID-19) Public Health Emergency is accounted for in this study.

5.1.1 Handling of Missing Data

All missing or partial data will be presented in the participant data listings as they are recorded on the eCRF.

When appropriate, the following rules will be implemented so as not to exclude participants or observations from statistical analyses due to missing or incomplete data.

Only the year is captured for the date of birth; the date of birth will be imputed to 1st January of the recorded year.

Concomitant medications with missing start and stop dates but flagged as being ongoing at a participant's final visit will be considered to have started prior to the study medication start and ended after stopping the study medication. The start and end reference periods will be imputed as "before" for the medication start and as "during/after" for the medication end.

In case of (partially) missing dates for interruptions or permanent stop of study medication intake, a "worst-case" approach will be applied to impute the start and end dates of study medication intake as the minimum and maximal possible dates, i.e.:

- first month of the year and/or first day of the month for a partially missing start date, and
- last month of the year and/or last day of the month for a partially missing end date.

If a participant died earlier than the imputed study medication end date, the death date will be taken as the study medication end date. If imputation leads to a temporal overlap between different exposure date records, the imputed dates will be adjusted so that no overlap exists and the time on the higher dose is maximized. The date of first exposure to treatment is not expected to be missing as the participants are instructed by the investigator to take their first dose of the intervention directly at Visit 1, but in the very rare case that this date is not recorded, it will be imputed according to the rules outlined above for missing start dates, but not earlier than the randomization date.

When only partial dates are available for clinical events in the efficacy analysis, a median imputation rule will be used:

- For example, if the day is missing and the month is July, then day 16 is chosen.
- If the number of potential values is even, the lower of the 2 middle numbers is taken. For example, if the day is missing and the month is June, then day 15 is imputed. The same rule applies if the day and month are missing, e.g., if the year is 2022 and the day and month are missing, 2nd July is taken.
- In case the range of possible values is further restricted, e.g., because a participant died in the month in which the day is missing, the median in the restricted set of possible values is calculated. For example, if the clinical event occurred in June 2022 and the respective participant died on 11th June 2022, 6th June 2022 is imputed as the clinical event's date.

The same principle above also applies to partially missing death dates, e.g., if only the year of death is available, but there is a last contact in the given year, then the date of death is imputed as the median of this date and the end of the given year.

In case a death date is completely missing, it will be imputed based on the last known contact when the participant was still alive and the first known contact when the participant died (e.g., from the participant's health status follow-up page) as the median of these 2 dates. As mentioned above, if the number of potential values is even, the lower of the 2 middle numbers is taken.

In case where both a non-fatal clinical event and death have partially missing dates, death takes precedence and will be imputed first according to the rules outlined above. This approach also applies to non-renal and non-CV death. However, given the importance of an accurate determination of the clinical event date in relation to the randomization date for the time-to-event analysis, a minimal number of such missing dates would be expected.

In case an event start date is completely missing, the start date of the intervention will be used as the imputed date.

Missing severity and intervention relationship of AEs and OEs will be handled in the following way:

- If the intensity is missing, the event will be considered severe.
- If the intervention relationship is missing, the event will be considered related to the intervention.

5.1.2 Data Rules

General data rules are described in this section, further data rules for specific parameters or analyses are specified in the respective subsections of Section 5.

5.1.2.1 Baseline Values

For UACR and eGFR, baseline values will be defined as the mean value from the screening visit and Visit 1. The geometric mean will be calculated for UACR, and the arithmetic mean will be calculated for eGFR. This rule also includes assessments from a local laboratory for eGFR in case, prior to randomization, no assessment from the central laboratory is available. If the last observation available prior to randomization is the measurement from the screening visit, this will be used as the baseline value. Otherwise, the baseline will be considered missing.

For all other parameters for which a baseline value needs to be defined, the baseline will be defined as the last non-missing measurement before or on the day of randomization. If the last observation available prior to randomization is the measurement from the screening visit, this observation will be used as the baseline value. An exception will be made for glycated hemoglobin (HbA1c), where, in case no value is measured before or on the day of randomization, values obtained up to 3 months (90 days) after randomization will qualify as baseline.

For participants with a missing baseline height measurement, the first available post-baseline height will be used for imputation. The baseline body mass index (BMI) will then be calculated using this imputed baseline height.

If more than 1 measurement was planned for a scheduled visit, such as blood pressure measurements, the mean value of these measurements per time point prior to randomization

will be used to calculate the baseline value. This rule will not be applied for UACR; see Section 5.1.2.4 for the definition of analysis values for these parameters.

Study day is defined as the actual number of calendar days between a visit's assessment (collection) date and the date of randomization, with one additional day added for assessments (collections) occurring on or after randomization. For longitudinal analyses, the study month is derived as the study day divided by 30. The baseline eGFR value is referenced as Day 0 (Month 0).

5.1.2.2 Change from Baseline

In general, the change from baseline will be presented as the absolute change, defined as the difference from the baseline value:

$$\text{Absolute change} = \text{Post-baseline value} - \text{Baseline value}.$$

Additionally, some parameters will be analyzed in terms of relative change, calculated as

$$\text{Relative change} = 100 \times [(\text{Post-baseline value} - \text{Baseline value}) / \text{Baseline value}].$$

5.1.2.3 Visit Mapping

In this study, there is a distinction between scheduled visits and permanently off-treatment scheduled visits. For non-safety data, by-visit summaries will be generated separately for the on-treatment and off-treatment phases, and a combined summary will be generated to integrate data from both phases. Only the pooled (on- and off-treatment combined) summaries will be presented for safety analyses.

For by-visit summaries, including mixed model for repeated measures (MMRM) analyses, if a scheduled assessment is missing or invalid, the value will be imputed from the closest qualifying alternative within the applicable visit window. Qualifying alternatives include premature treatment discontinuation, follow-up, end-of-treatment, or unscheduled assessments (excluding unscheduled visits to confirm sustained eGFR decline). For Visit 2 (Month 1), the visit window is ± 15 days; for Visit 3 (Month 3) and all subsequent visits, the visit window is ± 30 days. As a consequence, if a participant's sponsor-declared end-of-treatment visit coincides with a scheduled visit (for example, Month 32), the end-of-treatment assessment will be remapped to that scheduled visit. Follow-up visits will also be summarized separately, even if they are used to impute a scheduled visit for participants who discontinue treatment prematurely. For analyses restricted to the planned treatment period, only imputations based on collections occurring within that period will be included.

Data from the restart, up-titration, and safety visits will not be included in by-visit summaries.

5.1.2.4 Other Data Handling

Only the data provided by the central laboratory will be used for analysis. Values from local laboratories will not be used in the statistical analysis and will be listed only. Exceptions are made for eGFR: local laboratory values will be used for baseline measurements if central laboratory values are missing for baseline and screening (see Section 5.1.2.1) and in cases where a participant prematurely discontinues the study intervention. In the latter case, the local eGFR values obtained during the planned treatment period, after the last administration of the study intervention, and the last available central value during the planned treatment period will be used in the analysis. For this purpose, eGFR values will be recalculated from local serum creatinine values. Intermediate missing central eGFR values will not be substituted by local values.

For the calculation of eGFR based on the 2021 CKD-EPI creatinine-cystatin C equation, the date and visit label of the creatinine measurement will be used as the reference. Each creatinine value will be paired with the closest available cystatin C measurement within a window of ± 14 days, regardless of whether the cystatin C sample was collected at a scheduled or unscheduled visit. If 2 cystatin C samples are equidistant from the creatinine date, the earlier one will be selected. In this way, the derived creatinine-cystatin C eGFR inherits both the collection date and visit designation of the corresponding creatinine measurement.

In the case of retests for screening visits, the closest measurement prior to randomization will be used for analysis instead of the scheduled measurements. At all visits starting at Visit 1, and if not stated otherwise, only data from scheduled measurements will be used for analysis. In the unexpected case that more values than planned are available for a scheduled visit, only the first valid measurement will be considered.

For laboratory values below the lower limit of quantification (LLOQ), half the LLOQ will be used for analysis. When comparing 2 values that are both less than LLOQ, differences will be assigned a value of 0, while ratios will be assigned a value of 1. For values exceeding the upper limit of quantification (ULOQ), the ULOQ itself will be used for analysis. However, these rules do not apply to finerenone plasma concentration data.

For UACR, if the laboratory reports one or both components as below LLOQ or above ULOQ, the laboratory may report the overall ratio as missing. In such cases, UACR will be calculated at the analysis level using the imputed component values ($\frac{1}{2}$ LLOQ if below LLOQ, ULOQ if above ULOQ).

For data expected to follow a lognormal distribution, descriptive statistics other than minimum, maximum, and median will only be calculated if at least $\frac{2}{3}$ of the individual data points are measured and exceed the LLOQ. In tables presenting descriptive statistics that include values below the LLOQ, these statistics will be marked to indicate this inclusion.

Urinalysis is conducted 3 times per visit at screening and Visit 1, and twice at all post-baseline visits. As specified in Section 1.3 of the study protocol, UACR determinations must be made from first-morning void urine samples collected over 3 consecutive days at screening and Visit 1 and over 2 consecutive days at each post-baseline visit. For the analysis of UACR at screening and Visit 1, results from a single visit will be combined if at least 2 measurements are available. The coefficient of variation for these measurements will be calculated using the following formula, which accounts for the log-normal distribution of UACR values:

$$CV = \sqrt{\exp(SD_{\ln}^2) - 1},$$

where $SD_{\ln}$ is the SD of the log-transformed UACR values. If the coefficient of variation for UACR measurements exceeds 25%, the median value will be used for analysis. When only 2 valid measurements are available, the median will be calculated as the geometric mean of these values. Conversely, if the coefficient of variation is 25% or less, the geometric mean of all measurements will be used. For retests conducted during screening due to initial UACR values that do not qualify a participant for enrollment, the data from the retest will be considered valid if at least 2 measurements are available. If only 1 valid UACR measurement is available at screening, this single result will be used. For Visit 1, to constitute a valid set for baseline calculations, at least 2 out of 3 samples must be collected on or before the day of randomization and must yield non-missing values. If these conditions are not met, or if only 1

valid sample is available from Visit 1, then only the set of samples from screening will be used to calculate the baseline value. For all post-baseline visits, in case 2 valid measurements are available for UACR, the geometric mean of these values will be used. In case only 1 valid measurement is available, this value will be used in the statistical analyses.

In the case of ESKD, the UACR and eGFR values after the onset date of ESKD will be excluded from all efficacy and safety analyses.

5.2 Participant Disposition

The number of participants enrolled, randomized, and valid for the SAF, FAS, and PKS will be summarized overall and by treatment group, pooled region (see Section 6.6), country/region, and trial unit (site). The number of participants discontinuing each epoch and the primary reason for discontinuation will be presented by treatment group (post-randomization epochs only) and presented in separate tables. The disposition tables will summarize randomized participant counts, categorizing them into those who completed treatment and those who did not. Each category will be further subdivided to indicate whether participants completed the follow-up period. Participants' study completion status will also be summarized. A participant will be considered as having completed the study if there was contact with the participant after the EoS notification or if the participant died. In addition, the number of participants with important deviations and validity findings will be presented overall and by treatment group.

5.3 Primary Endpoint Analysis

5.3.1 Definition of Endpoint

The primary efficacy variable of interest is eGFR assessed over time, collected at Month 1, then every 3 months for the first 12 months, and afterward every 4 months until the end of the planned treatment period. A participant's planned treatment period refers to the time from randomization up to the end-of-treatment visit or up to 4 weeks after the EoS notification if the planned end-of-treatment visit is not completed.

The laboratory parameter eGFR will be calculated by the central laboratory. The equation used for calculating eGFR (in mL/min/1.73 m²) will be the 2009 CKD-EPI equation (Levey, et al. 2009) based on serum creatinine, age, gender, and race.

To account for recent developments in the estimation of GFR excluding race, which is considered a social but not biological measure, 2 alternative race-unadjusted equations for eGFR will be applied:

- equation based on serum creatinine, age, and gender;
- equation based on serum creatinine, serum cystatin C, age, and gender (Inker, et al. 2021).

In addition, the race-adjusted 2012 CKD-EPI creatinine-cystatin C equation (Inker, et al. 2012), which incorporates serum creatinine, serum cystatin C, age, gender, and race, will be calculated for comparison.

For Japanese patients, creatinine-only CKD-EPI equations (2009 and 2021) will be calibrated by multiplying the results by a correction factor of 0.813 (Horio, Imai, et al. 2010), and CKD-EPI equations based on creatinine plus cystatin C (2012 and 2021) will be modified by the factor 0.908 (Horio, Imai, et al. 2013).

Key efficacy analyses that include eGFR will be repeated as sensitivity analyses using the updated creatinine and cystatin C-based CKD-EPI equations and the updated CKD-EPI equation based on creatinine only without race.

If the date of birth is partially missing and subsequently imputed, eGFR values will be derived based on the imputed date of birth (see Section 5.1.1). Summary statistics for eGFR will be provided by visit.

In addition, for safety reasons and titration, eGFR will also be assessed locally, as specified in Section 1.3 of the study protocol.

In case the central eGFR measurement is not available, computed eGFR (CKD-EPI) from local measurements (if available) will be used for the analysis in those cases described in Section 5.1.2.4.

A descriptive summary of central eGFR will be presented by treatment group and visit, including absolute and relative changes from baseline.

The population-level summary is the mean rate of change of eGFR as measured by the total slope from baseline to Month 32.

The rate of change of eGFR will be compared between the finerenone and placebo groups by estimating the total eGFR slope from baseline to Month 32 using eGFR values aligned with the specific rules defined by the relevant estimand. It is assumed that the treatment effect can be expressed as the difference in the slope between finerenone and placebo.

For analysis of the total slope, the serial change in eGFR will be modeled using a random coefficient 2-slope linear spline mixed-effects model in which a fixed change point will distinguish between the acute and chronic eGFR slopes at Month 3. The 2-slope linear spline mixed-effects model will be based on the methods described by Vonesh et al. (2019) in Section 2.1 of the paper "*A two-slope linear spline mixed-effects models for eGFR*". The acute and chronic phases are differentiated because finerenone showed acute effects on eGFR that differed from its long-term effects on disease progression in FIDELIO-DKD. In addition to treatment-specific fixed effects (finerenone vs placebo) for the intercept, time from randomization, and time difference relative to the change point, the model will include fixed effects for stratification factors used in the stratified randomization (i.e., baseline SGLT2 inhibitor use (yes, no) and screening UACR (≤ 1000 mg/g vs > 1000 mg/g)). The model will also include interactions between these fixed effects and the time components. Random effects for the intercept, acute slope (baseline to Month 3), and directional change in slope at the change point (Month 3) will be included to account for variation in participants' progression rates over time. The unstructured covariance matrix of the random effects will be allowed to vary freely between treatments. In the random coefficient 2-slope linear spline mixed-effects model, time for each eGFR assessment during both the acute and chronic phases is measured relative to randomization, with the baseline measurement designated as Day 0. This model will include all observed eGFR measurements in accordance with the specified estimand. Measurements obtained at unscheduled confirmatory visits for previously observed declines in eGFR will not be included. The analysis will include only participants' eGFR data, whether observed or imputed, up to and including the end of their planned treatment period.

5.3.2 Main Analytical Approach

The primary analysis of the primary efficacy variable will be performed according to the participants' randomized intervention and will be based on the FAS.

The primary estimand, as defined in Section 1.1, will be used for the primary analysis. The random coefficient 2-slope linear spline mixed-effects model behind the population-level summary is presented in this section in more detail.

For clarity, the model specification outlined below includes only the treatment-specific intercepts and time variables. In contrast to that, the fixed effects for the stratification factors used in the stratified randomization (i.e., baseline SGLT2 inhibitor use (yes, no) and screening UACR (≤ 1000 mg/g vs > 1000 mg/g)) and their interactions with time are not shown in the model formula as their inclusion is straightforward.

As described in Vonesh et al. (2019), the 2-slope linear spline mixed-effects model will be formulated as follows:

$$Y_{ij} = (\beta_{0c} + \beta_{1c}t_{ij} + \beta_{2c}\max\{t_{ij} - t^*, 0\})(1 - X_i) \\ + (\beta_{0f} + \beta_{1f}t_{ij} + \beta_{2f}\max\{t_{ij} - t^*, 0\})X_i \\ + b_{0i} + b_{1i}t_{ij} + b_{2i}\max\{t_{ij} - t^*, 0\} + \epsilon_{ij},$$

where

Y_{ij} denotes the response variable, eGFR, for the i^{th} participant on the j^{th} occasion,
 X_i is a treatment group indicator variable with $X_i = 0$ if the i^{th} participant is randomized to the placebo group and $X_i = 1$ if randomized to the finerenone group,

$\boldsymbol{\beta}_k = (\beta_{0k}, \beta_{1k}, \beta_{2k})'$ denotes the vector of regression parameters for placebo ($k = c$) and finerenone ($k = f$),

ϵ_{ij} denote within-participant residuals,

$\mathbf{b}_i = (b_{0i}, b_{1i}, b_{2i})'$ denotes the vector of between-participant random effects,

t^* denotes the change point (knot at Month 3),

$\max\{t_{ij} - t^*, 0\}$ is the truncated line function, and

t_{ij} are the time points at which eGFR is measured for the i^{th} participant across v_i occasions ($j = 1, 2, \dots, v_i$).

Model assumptions:

- The errors ($\epsilon_{ij}|\mathbf{b}_i$) are conditionally independent normally distributed random variables with mean 0 and within-participant variance σ_e^2
- The participant-specific random effects $\mathbf{b}_i$ are independent multivariate normally distributed with mean vector $\mathbf{0}$ and a 3×3 covariance matrix $\boldsymbol{\Omega}_k$, where $\boldsymbol{\Omega}_k$ denotes an unstructured covariance matrix that varies by treatment k .

The interpretations of the regression parameters $\boldsymbol{\beta}_k = (\beta_{0k}, \beta_{1k}, \beta_{2k})'$ in terms of chronic and acute slope for placebo ($k = c$) or finerenone ($k = f$) are as follows:

β_{0k} denotes the intercept for treatment group k ,

β_{1k} is the acute slope up to before (excluding) time t^* (the knot at Month 3) for treatment group k ,

β_{2k} is the directional change in slope starting at time t^* (the knot at Month 3) for treatment group k , and

$\beta_{1k} + \beta_{2k}$ corresponds to the chronic slope for treatment group k .

Finally, under these assumptions, the total slope up to time t describing the rate of change in eGFR from baseline up to a specific time $t > t^*$ is defined for treatment group k as

$$\beta_{3k}(t) = \frac{\{\beta_{1k}t + \beta_{2k}(t - t^*)\}}{t} = \beta_{1k} + \beta_{2k} - \beta_{2k}\left(\frac{t^*}{t}\right).$$

Specifically, this total slope calculation will be performed for up to $t = 32$ months.

The model parameters, as defined in the previous steps, can be used to formulate the hypothesis to be tested for the primary endpoint, as shown in Section 2.

The analysis will be carried out using maximum likelihood procedures. The model will be fitted using the SAS procedure MIXED, with the *containment method* applied to calculate the denominator degrees of freedom for tests of the fixed effects. Linear contrasts will be constructed to estimate the acute, chronic, and total eGFR slopes. Estimates of these slopes for both the finerenone and placebo groups will be presented along with 95% confidence intervals (CIs). For treatment-specific slope estimates, the fixed-effect terms for the stratification factors and their interactions with time will be evaluated at a population-average distribution of the randomization strata, using weights proportional to the number of participants in each stratum in the analysis population.

All eGFR values recorded after the occurrence of ESKD will be set to missing. A set of multiply-imputed "unfavorable-value" datasets will then be created; in each copy, every scheduled visit after ESKD or death (and falling within the participant's planned end-of-treatment period) will be imputed by drawing, with replacement, from the empirical donor pool of clinically unfavorable eGFR measurements. Each completed dataset will be analyzed with the 2-slope linear-spline mixed-effects model described above, and the resulting parameter estimates and standard errors will be combined across imputations according to Rubin's rules using PROC MIANALYZE to obtain the final pooled treatment effects and precision measures. Implementation steps for the unfavorable-value resampling imputation under the composite strategy are provided in Section 6.4.

In the event of convergence issues with the initial model, a simpler approach will be employed where a common within-participant variance-covariance matrix will be estimated using pooled data from both finerenone and placebo groups (hence, $\boldsymbol{\Omega}_k = \boldsymbol{\Omega}$). The model will be adjusted to a heterogeneous compound symmetry structure if convergence issues still persist. Should further simplification be necessary, the model will be refined to a compound symmetry structure, which assumes uniform variances and covariances across all random effects. All stratification interaction terms with time will be excluded from the model if convergence issues continue. If convergence issues still persist, the main effects for stratification factors will also be removed.

The time-dependent total eGFR slope until Month 32 will be calculated for the 2 treatment groups. The difference in total eGFR slope from baseline to Month 32 between the finerenone and the placebo treatment group will be calculated, and a corresponding 2-sided 95% CI and p -value will be presented. The study will be considered successful if it demonstrates that the eGFR slope for finerenone is significantly smaller than that for placebo, achieving statistical significance at an alpha level of 5%.

A plot will be generated from the 2-slope model to visually assess the treatment effect of finerenone compared to placebo. This plot will use only the model's fixed effects, with all random effects set to 0, and will present model-based mean profiles averaged across the randomization strata, weighted by the number of participants in each stratum.

5.3.3 Sensitivity Analyses

5.3.3.1 Sensitivity Analyses: Missing Data Imputation

Missing data is categorized based on whether it was expected to be collected according to the SoA for the purpose of a given estimand. Missing data classified as "expected" refers to instances where data collection did not occur as planned, such as broken samples or missed visits. This data is considered for imputation. Conversely, "non-expected" missing data includes instances where data is not collected during unscheduled assessments; such data is not considered missing and thus does not need to be imputed. Only data expected to be collected according to the study's protocol and required for a given estimand is considered when applying imputation techniques.

It is considered that intermittent missing eGFR values (e.g., due to a single missed visit) will not lead to a biased slope estimation. In contrast, for monotone missing data patterns (i.e., missingness due to loss of eGFR follow-up), sensitivity analyses under different assumptions for the missingness mechanism will be performed. For this, a multiple imputation that conditions on visit-specific on-treatment indicators and a tipping-point analysis will be employed. In both sensitivity analyses, the same imputation approach as for the primary estimand will be used in case of ESKD or death.

5.3.3.1.1 Sensitivity Analysis 1: Multiple Imputation with Treatment Discontinuation Indicators

For participants who permanently discontinue study intervention for reasons other than death or ESKD, monotone missing eGFR values will be imputed using a multiple imputation regression model that conditions on visit-specific treatment discontinuation indicators, as proposed by Guizzaro et al. (2021). For this analysis, a participant is considered to be on treatment through the earliest of the last dose date or the start date of an extended treatment interruption. Intermittent gaps will first be imputed using Markov chain Monte Carlo (MCMC) multiple imputation draws in SAS PROC MI using the MCMC

IMPUTE = MONOTONE option, assuming a multivariate normal distribution for baseline and post-baseline eGFR, thereby producing a monotone dataset. The MCMC model will include the fully observed treatment group and randomization stratification factors as predictors, so these variables inform the draws while not themselves being imputed. The remaining monotone missing values will then be imputed with the MONOTONE REG method, each visit being modeled as a function of treatment group, randomization stratification factors, baseline eGFR, the observed values of earlier visits, and the relevant treatment discontinuation flags; an indicator will only be included when at least 10 post-discontinuation observations are available at that visit. One regression draw will be created for each initial MCMC dataset, yielding m fully imputed datasets that will be analyzed separately and combined with PROC MIANALYZE according to Rubin's rules.

5.3.3.1.2 Sensitivity Analysis 2: Tipping Point Analysis

A delta-adjusted pattern imputation method will be utilized for the tipping point analysis. This method will employ the same variables used in the multiple imputation. In the first step, intermittent missing values will be imputed using the MCMC methodology. In the subsequent step, it will be assumed that the average of the missing outcomes in the finerenone group deviates by a specified magnitude, referred to as delta, from the values predicted by the multiple imputation model described above, indicating a worsening in outcomes. In contrast, the mean in the control group will be assumed to be consistent between missing and observed

data. Therefore, missing values for each treatment group will be imputed based on observed values within their respective groups based on the multiple imputation model described above. Post-imputation, imputed values in the finerenone group will be adjusted by a specified amount to reflect a worsening of outcomes, whereas the placebo group's imputed values will remain unchanged. A series of shifts within a plausible range will be applied. The shift that changes the conclusion from favoring the finerenone group to yielding a statistically non-significant result will define the tipping point. This approach to handling missing data will be implemented in SAS using the PROC MI procedure with the MNAR statement and the ADJUST option.

Technical details of the multiple imputation sensitivity analyses are provided in Section 6.5.2.

5.3.3.2 Sensitivity Analysis 3: Statistical Model

In order to determine if a model specifying a single slope per study intervention yields results similar to those of the 2-slope model used in the primary analysis, a standard random coefficient linear mixed-effects model will be employed as a sensitivity analysis.

This model will include treatment-specific fixed effects for the intercept and the (total) slope, which remain unchanged over time. Further, it includes fixed effects for time (continuous), the treatment-by-time interaction, fixed effects for randomization stratification factors (i.e., baseline SGLT2 inhibitor use [yes, no] and screening UACR [≤ 1000 mg/g vs > 1000 mg/g]), along with their respective interactions with time. Random effects for the intercept and slope will be included to account for variations in participants' progression rates over time.

Analyses based on the linear model will be similar to the primary analysis based on the 2-slope model, which employs maximum likelihood methods under the assumption of missing-at-random (MAR) for intermediate missing values and multiple imputation incorporating unfavorable values for missing data following death or the initiation of chronic dialysis/kidney transplantation.

5.3.3.3 Sensitivity Analysis 4: Linearity Assumption

As the single- and 2-slope linear spline mixed-effects models rely on the linearity assumption for the eGFR decline over time, an MMRM for the primary estimand will also be applied to provide a visual inspection of the appropriateness of the linearity and change point assumptions. Thus, this analysis is not meant to reproduce the findings regarding the significance of the treatment effect in the primary analysis. It should be noted that such an MMRM analysis requires estimating additional model parameters and may require a larger sample size to achieve the same power.

The change from baseline in eGFR at each nominal visit will be analyzed using an MMRM with fixed effects for treatment, visit (categorical), treatment-by-visit interaction, stratification levels, and their respective interactions with visit. Pairwise differences between the finerenone and placebo treatment groups at each visit will be calculated with corresponding 2-sided 95% CIs. Model-based means and between-treatment contrasts will be computed using observed margins, such that estimates at each visit are averaged over the observed distribution of stratification factors in the analysis data. For comparability with the total slope from the 2-slope mixed model, the treatment difference in the mean change from baseline at the nominal visits from the MMRM will be annualized.

For the MMRM analysis, the SAS procedure MIXED will be employed to estimate covariance patterns within participants and adjust for the within-participant variance. A separate covariance pattern will be estimated for each treatment group based on an unstructured

covariance matrix. In the event of convergence issues, the analysis will first estimate a common covariance matrix for both the finerenone and placebo groups to simplify the covariance structure. If further simplification is needed, a heterogeneous banded Toeplitz matrix, $TOEPH(r)$, will be assumed, where r represents the total number of visits applicable to the study. The simplification process will begin by reducing the band width from $r - 1$ and continue decrementing until convergence is achieved. If the inclusion of stratification-by-visit interaction terms results in model nonconvergence, the responsible interaction term(s) will be removed to simplify the fixed-effects structure and aid convergence. If the sample size at a given visit is too small to support stable estimation of the model (e.g., sparse data within one or both treatment groups), that visit will be excluded from the MMRM analysis.

If the results of the MMRM indicate that the change point at 3 months is questionable, an additional analysis will be performed to estimate the change point based on the data.

5.3.3.4 Sensitivity Analysis 5: Worst-Case Imputation

An alternative "composite variable" strategy will be employed as an additional sensitivity analysis for the ICEs of death and the initiation of chronic dialysis or kidney transplantation. This approach will assume a worst-case eGFR value of 0 mL/min/1.73 m² for all scheduled visits leading up to the planned end of treatment, in contrast with the approach of imputation based on unfavorable eGFR values.

5.3.4 Supplementary Analyses

5.3.4.1 Supplementary Analyses with Alternative Estimands

The following 3 alternative estimands for the primary endpoint will be considered to evaluate different strategies for the handling of ICEs:

The first estimand is defined according to the following attributes:

- **Population:** as defined for the primary estimand
- **Treatment:** as defined for the primary estimand
- **Variable:** as defined for the primary estimand
- **ICEs:** The same 3 ICEs as for the primary estimand are considered:
 - Death
 - Starting chronic dialysis or receiving a kidney transplant
 - Premature treatment discontinuation/extended interruption of treatment

For the ICEs of death and the start of chronic dialysis/kidney transplantation, a *hypothetical* strategy will be applied. eGFR data collected up to the time point of these ICEs will be included in the analysis, and missing post-ICE values will be addressed using a shared-parameter (SP) model that allows the ICE process to be associated with the underlying eGFR trajectory via shared random effects. This estimand addresses the scenario where participants do not die or experience ESKD and would have similar outcomes to those without the ICEs.

For the ICE of premature treatment discontinuation/extended interruption of treatment, the same *treatment policy* strategy will be applied as for the primary estimand.

- **Population-level summary:** as defined for the primary estimand

To apply the *hypothetical* strategy for the ICEs of death and the start of chronic dialysis/kidney transplant, an SP model will be used. This model jointly analyzes the trend in

eGFR over time and the time to the first occurrence of death or the start of chronic dialysis/kidney transplantation, up to the planned end-of-treatment visit or up to 4 weeks after the EoS notification if the planned end-of-treatment visit is not completed. An overview of the application of SP models in the context of CKD is provided by Vonesh et al. (2019). For this type of analysis, the 2-slope linear spline mixed-effects model discussed in Section 5.4.1.2 (with a fixed knot at Month 3) will be used to analyze the trend in eGFR over time. As suggested by Vonesh et al. (2019), a piecewise exponential survival model will be used for the time-to-event component of the SP model. The baseline hazard will be modeled as piecewise constant over intervals, with cutpoints chosen using event-time quantiles to target approximately equal numbers of observed events per interval, subject to a maximum interval width of 6 months and an upper bound of 9 on the number of intervals. The same covariates, interactions, and random effects used for the primary analysis will be applied to the longitudinal part of the SP model. The longitudinal residual variance will be assumed constant over time. The piecewise exponential survival model will include the following parameters:

- the treatment covariate,
- a random intercept, and
- random acute and chronic slope effects (with the random chronic slope defined as the sum of the random acute and post-knot slope effects).

The random effects covariance matrix will be parameterized using a Cholesky factorization to ensure positive definiteness during estimation.

The joint model will be fitted using maximum likelihood (PROC NLMIXED, Laplace approximation), with starting values obtained from PROC MIXED and a standalone piecewise exponential survival fit.

In the event of convergence issues, the same simplifications described for the primary analysis will be applied to the longitudinal component. If this model still does not converge, a simpler approach will be adopted, relying solely on a time-to-event model with the treatment covariate, thus assuming no association between the longitudinal and survival components. Participants without an ICE will be right-censored at the earliest of the planned end-of-treatment visit or their last study contact (up until the completed end-of-treatment visit, or, if the end-of-treatment visit is missed, up to 4 weeks after the EoS notification).

The second estimand is defined by the following attributes:

- **Population:** as defined for the primary estimand
- **Treatment:** as defined for the primary estimand
- **Variable:** as defined for the primary estimand
- **ICEs:** The same 3 ICEs as for the primary estimand are considered:
 - Death
 - Starting chronic dialysis or receiving a kidney transplant
 - Premature treatment discontinuation/extended interruption of treatment

For the ICEs of death and the start of chronic dialysis/kidney transplantation, the same *hypothetical* strategy as specified for the first supplementary estimand discussed in this section will be applied.

The same *hypothetical* strategy will be applied to the ICE of premature treatment discontinuation/extended interruption of treatment to assess the slope difference while

the participants were taking study medication using only the eGFR values obtained before treatment discontinuation/extended interruption of treatment.

- **Population-level summary:** as defined for the primary estimand

This alternative estimand will be assessed in a manner similar to that of the primary estimand. All data collected after the occurrence of any ICEs will be discarded, and missing data will be handled implicitly using the 2-slope linear spline mixed-effects model.

This estimand addresses the scenario where participants do not die, experience ESKD, permanently discontinue or have an extended interruption of the study intervention and would have similar outcomes to those without the ICEs.

The third estimand is defined by the following attributes:

- **Population:** The population consists of non-diabetic CKD patients (as defined for the primary estimand) who are naïve to SGLT2 inhibitor therapy at baseline.
- **Treatment:** as defined for the primary estimand
- **Variable:** as defined for the primary estimand
- **ICEs:** The following 4 ICEs are considered:
 - Death
 - Starting chronic dialysis or receiving a kidney transplant
 - Premature treatment discontinuation/extended interruption of treatment
 - Start of a new SGLT2 inhibitor therapy after baseline

The first 3 ICEs are handled using the same strategies as those applied to the primary estimand.

For the ICE of start of a new SGLT2 inhibitor therapy after baseline, a *hypothetical* strategy will be applied to set the eGFR values to missing that were measured after the start of the SGLT2 inhibitor with no imputation. eGFR values up to the end of the planned treatment will be set to missing, irrespective of the potential discontinuation of the SGLT2 inhibitor therapy.

- **Population-level summary:** as defined for the primary estimand

This alternative estimand will be evaluated analogously to the primary estimand.

This estimand reflects a scenario in which participants who are SGLT2-naïve and initiate SGLT2 inhibitor therapy after baseline experience outcomes comparable to those who do not initiate such therapy.

5.3.4.2 Supportive Analyses for Primary Endpoint

Chronic eGFR slope from Month 3 to end of planned treatment period

The eGFR slope will be calculated separately for 2 periods: from randomization to Month 3 (acute slope), representing the rate of eGFR decline during the initial phase, and from Month 3 to the end of the planned study treatment (chronic slope), representing the rate of decline from Month 3 until the end of the planned treatment period.

The chronic eGFR slope will be analyzed using the same 2-slope model employed in the primary analysis (see Section 5.3.2), with Month 3 serving as the "change point" and the *time* variable representing the duration from baseline to each eGFR assessment during the acute and chronic phases.

Fixed effects estimates for the chronic eGFR slope of both the finerenone and placebo groups, including the treatment difference (finerenone minus placebo), will be presented. These estimates will be accompanied by 95% CIs and corresponding 2-sided *p*-values.

Total slope after 32 months

The total slope calculation will be performed for up to 36, 40, 44, and 48 months, analogous to the total slope calculation of up to 32 months in the primary analysis. 95% CIs and corresponding 2-sided *p*-values for the difference of the total slope parameters between finerenone and placebo will be reported.

eGFR change from baseline to 1 month after end of treatment (off-treatment)

A post-treatment eGFR measurement will be collected for all participants 1 month after the end of the treatment period. As a supportive analysis, an analysis of covariance (ANCOVA) will be used to assess the change from baseline in eGFR at the 1-month post-treatment visit. This model will include treatment group, stratification factors, total time on treatment, and the baseline eGFR value as covariates. The estimate for the mean change from baseline will be presented along with the corresponding 2-sided 95% CIs and *p*-values. The supportive ANCOVA analysis of the change from baseline in eGFR at the 1-month post-treatment visit will be performed using eGFR computed by CKD-EPI 2009 creatinine (race-adjusted), CKD-EPI 2021 creatinine (race-unadjusted), CKD-EPI 2021 creatinine-cystatin C (race-unadjusted), and CKD-EPI 2012 creatinine-cystatin C (race-adjusted).

5.4 Secondary Endpoints Analysis

The following secondary variables will be evaluated to further explore the chronic and persistent benefit of finerenone on kidney disease progression.

Key secondary analyses involving eGFR will be repeated as sensitivity analyses using (i) the 2021 CKD-EPI creatinine equation (race-unadjusted), (ii) the 2021 CKD-EPI creatinine-cystatin C equation (race-unadjusted), and (iii) the 2012 CKD-EPI creatinine-cystatin C equation (race-adjusted).

The secondary efficacy analyses will be conducted on the FAS.

5.4.1 Key/Confirmatory Secondary Endpoints

The 3 key secondary endpoints are as follows:

- Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, or CV death
- Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$
- Time to the composite of hospitalization for HF, or CV death

The key secondary endpoints are defined as the time to the first occurrence of certain OEs. In this study, OEs will be assessed by the investigators based on their clinical judgment. For each endpoint, the investigator will record information specific to that type of endpoint in the eCRF. An exception applies to OEs related to sustained declines in eGFR; these events will be derived programmatically, as detailed in Section 5.4.1.1. Event criteria will be provided to facilitate event collection, and edit checks will be put in place to detect potential inconsistencies.

5.4.1.1 Definition of Endpoints

Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, or CV death

This secondary efficacy variable is defined as the time from randomization to the first occurrence of the composite endpoint of kidney failure, a sustained decrease of eGFR $\geq 57\%$ from baseline over at least 4 weeks, hospitalization for HF, or CV death. Kidney failure is defined as either the occurrence of ESKD or an eGFR of $< 15 \text{ mL/min/1.73 m}^2$, confirmed by a second measurement at the earliest 4 weeks after the initial measurement. ESKD is defined as the initiation of chronic dialysis (haemo- or peritoneal dialysis) for at least 30 days or renal transplantation.

A hospitalization for HF is defined as a stay in a hospital or hospital equivalent due to HF that has a duration of at least 24 hours.

The attributes constituting the estimand for this key secondary endpoint are as follows:

- **Population:** The population consists of non-diabetic CKD patients, further defined by the in- and exclusion criteria.
- **Treatment:** All participants will receive blinded study intervention (i.e., finerenone 10 mg/20 mg or placebo) OD on top of an individually tolerated maximum labeled dose of either an ACEI or an ARB.
- **Variable:** Time to the event of the first occurrence of either kidney failure, a sustained eGFR decline of $\geq 57\%$ from baseline over at least 4 weeks, hospitalization for HF, or CV death, whichever occurs first.
- **ICEs:** The 2 most important ICEs to consider are
 - Non-CV death
 - Premature treatment discontinuation/extended interruption of treatment

The ICE of non-CV death will be handled using the "while alive" strategy.

For premature treatment discontinuation/extended interruption of treatment, a *treatment policy* strategy will be applied, as outlined in more detail in the description of the censoring rules below.

- **Population-level summary:** Hazard rates of time-to-event comparing the finerenone treatment group with the placebo group using a stratified log-rank test.

The individual components "*Sustained decrease of eGFR $\geq 57\%$ from baseline over at least 4 weeks*" as well as "*Sustained eGFR decline $< 15 \text{ mL/min/1.73 m}^2$* " will be programmatically derived. Only if the eGFR decrease was confirmed by at least 1 additional eGFR measurement taken at least 4 weeks later will it be considered a sustained decrease and counted for the renal endpoint. The date used for the analysis will be the date of the initial sample that meets the threshold. In case there was no confirmatory assessment, events will only be counted for the renal endpoint when the participant died after the initial decrease or the participant went on renal replacement therapy such as dialysis or transplantation. If an intermediate measurement does not confirm the initial decrease, the event will not be counted for the renal endpoint. Only central laboratory results for eGFR will be used to determine sustained declines; local laboratory results will not be considered.

OEs will be included from the day of randomization, scheduled at Visit 1, extending to the planned end-of-treatment visit or up to 4 weeks after the EoS notification if the planned end-

of-treatment visit is not completed. For participants who prematurely discontinue the study without further follow-up, events will be counted up to the last visit, where complete information on all components of the composite endpoint was recorded. To account for events of kidney failure, CV death, or hospitalization for HF, any occurrences after the last recorded eGFR at a clinic visit will be included in the efficacy analysis. These events must occur up to 1 day before the next scheduled clinic visit, with an allowance for a 1-month late attendance window. For instance, if the next scheduled visit is 4 months later, the allowable period extends to a maximum of 5 months. Should a participant die from a non-CV-related cause within this time frame, their date of death will serve as the censoring date. Randomized participants who do not experience an event of the composite endpoint by the time of analysis will be censored on the date of their last visit, where complete data on all components of the composite endpoint are available. This includes data up to and including the planned end-of-treatment visit or the EoS phone call, applicable for participants who have discontinued the study prematurely, provided this visit meets the censoring criteria. If a participant experiences a non-CV death within the defined time window and a subsequent clinic visit was scheduled, the date of death will be used as the censoring date. Participants lacking any post-baseline information on the composite endpoint will be censored on Day 1.

Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$

This secondary efficacy variable is defined as the time from randomization to the first occurrence of either kidney failure or a sustained eGFR decline of $\geq 57\%$, as defined for the first key secondary endpoint.

The attributes constituting the estimand for this key secondary endpoint are the following:

- **Population:** Same as the estimand for the first key secondary endpoint.
- **Treatment:** Same as the estimand for the first key secondary endpoint.
- **Variable:** Time to a sustained eGFR decline of $\geq 57\%$ or kidney failure.
- **ICEs:** The 2 most important ICEs to consider are
 - Death
 - Premature treatment discontinuation/extended interruption of treatment

The ICE of death will be handled using the "*while alive*" strategy. For the other ICE, a *treatment policy* strategy will be applied.

- **Population-level summary:** Hazard rates of time-to-event comparing the finerenone treatment group with the placebo group using a stratified log-rank test.

The censoring rules for this combined renal endpoint will be similar to those for the combined renal and CV endpoint. Renal events will be included from the day of randomization, scheduled at Visit 1, extending to the planned end-of-treatment visit or up to 4 weeks after the EoS notification if the planned end-of-treatment visit is not completed. For participants who prematurely discontinue the study without further follow-up, renal events will be counted up to the last visit, where complete information on all components of the composite endpoint was recorded. To account for kidney failure events, any occurrences after the last recorded eGFR at a clinic visit will be included in the efficacy analysis. These events must occur up to 1 day before the next scheduled clinic visit, with an allowance for a 1-month late attendance window. If the participant dies within this period, then the date of death will be used as the censoring date. Randomized participants who do not experience an event of the composite renal endpoint by the time of analysis will be censored on the date of their last visit, where complete data on all components of the composite endpoint are available. This includes data

up to and including the planned end-of-treatment visit or the EoS phone call, applicable for participants who have discontinued the study prematurely, provided this visit meets the censoring criteria. If a participant dies within the defined time window and a subsequent clinic visit is scheduled, the date of death will be used as the censoring date. Participants lacking any post-baseline information on the composite renal endpoint will be censored on Day 1.

Time to the composite of hospitalization for HF or CV death

This secondary efficacy variable is defined as the time from randomization to either the first hospitalization for HF, or CV death, whichever occurs first.

The attributes constituting the estimand for this key secondary endpoint are the following:

- **Population:** Same as the estimand for the first key secondary endpoint.
- **Treatment:** Same as the estimand for the first key secondary endpoint.
- **Variable:** Time to the event of the first hospitalization for HF or CV death.
- **ICEs:** The 2 most important ICEs to consider are
 - Non-CV death
 - Premature treatment discontinuation/extended interruption of treatment

The ICE of non-CV death will be handled using the "*while alive*" strategy. For the other ICE, a *treatment policy* strategy will be applied.

- **Population-level summary:** Hazard rates of time-to-event comparing the finerenone treatment group with the placebo group using a stratified log-rank test.

CV events will be counted up to the day of withdrawal from the study or non-CV death. Randomized participants without an event of the CV composite endpoint at the time of analysis will be censored at the date of their last contact, up to and including the planned end-of-treatment visit or EoS phone call for participants who have discontinued prematurely, or date of non-CV death.

5.4.1.2 Main Analytical Approach

Time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, or CV death

In order to evaluate whether finerenone is superior to placebo in prolonging the time to the first event of the composite endpoint, the corresponding null hypothesis will be tested using a stratified log-rank test at the 2-sided 5% significance level to compare the survival curves between treatment groups. The SAS procedure LIFETEST will be utilized to conduct the stratified log-rank test.

A stratified Cox regression model will be used to provide point estimates of the hazard ratio with a corresponding 2-sided 95% CI. For the estimation of hazard ratios, the SAS procedure PHREG will be employed. The Efron method will be used to handle ties in the event times.

The strata for the stratified log-rank test and the stratified Cox regression model will be defined by the randomization stratification factors.

If convergence issues are encountered, stratification will be removed for the affected analysis.

In addition, cumulative incidence plots accounting for competing risks (non-CV death or all-cause death, as appropriate) will be presented by treatment group.

Time to the composite of kidney failure or sustained eGFR decline of $\geq 57\%$

This time-to-event variable will be analyzed the same way as the time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, or CV death.

Time to the composite of hospitalization for HF or CV death

This time-to-event variable will be analyzed the same way as the time to the composite of kidney failure, sustained eGFR decline of $\geq 57\%$, hospitalization for HF, or CV death.

5.4.1.3 Sensitivity Analysis Related to Censoring Mechanism

The censoring mechanism of participants without an event at the time of analysis is assumed to be non-informative in the analysis of the key secondary efficacy endpoints. This implies that this analysis is valid under the MAR assumption. Sensitivity analyses will be performed, assessing the impact of potentially informative censoring due to participants lost to follow-up or who have withdrawn consent.

A summary of the missing data will be provided by treatment arm, including the number of participants with incomplete follow-up for each endpoint and the ratio of the observed follow-up time to the expected follow-up time over all participants.

Robustness with respect to missing data will be investigated with a tipping point analysis. The analysis will:

- Estimate the hazards and subject-specific survival function within each treatment arm, using a piecewise exponential model fitted to the population of participants who prematurely discontinued study treatment prior to the occurrence of the event of interest (i.e., who, after treatment discontinuation, either had the event or were followed until the end of the study without having the event), adjusted for the stratification factors.
- Inflate the hazard only in the finerenone group by a predetermined factor, assuming non-informative censoring in the control group.
- Impute events to the end of the study M times by taking M draws from the Bayesian posterior distribution of model parameters, and for each draw, simulating times to event by inverting the piecewise exponential cumulative distribution function.
- For each imputation iteration, calculate the hazard ratios using the stratified Cox model described in Section 5.4.1.2, and combine these estimates using Rubin's rule and log-transformation of the hazard ratio.
- Incrementally increase the inflation factor until the upper limit of the 2-sided 95% CI for the hazard ratio crosses 1.0, identifying the "tipping point".

The tipping point analysis will help determine how much higher the event risk after dropout would need to be in the finerenone group for the study's statistical significance to be lost. Since there is no inherent reason to believe that the event rate should be disproportionately higher post-dropout, a large tipping point would indicate the robustness of the study results. Further technical details of the models used are provided in Section 6.5.3.

5.4.1.4 Supplementary Analyses**5.4.1.4.1 Supplementary Analyses with Alternative Estimands**

For the 3 key secondary time-to-event endpoints, alternative estimands employing a "*while-on-treatment*" strategy for handling ICEs will be assessed. The attributes of these alternative estimands align with those discussed for the main estimands in Section 5.4.1.1, except for the

specific approach to handling ICEs. Under the "*while-on-treatment*" strategy, participants whose censoring or event dates occur more than 30 days after the earliest of (i) the last receipt of study intervention or (ii) the start of an extended treatment interruption, as analyzed in the main estimand, will now be censored at 30 days after that earliest date.

5.4.1.4.2 Other Supplementary Analyses

Additional supplementary analyses will be conducted for the key secondary efficacy variables as follows:

Cumulative incidence probabilities for each treatment group, accounting for competing risks using the Aalen-Johansen estimator, will be derived for the time to the first endpoint event and their differences between finerenone and placebo at designated time points along with 95% CIs and the number needed to treat (NNT). The CIs will be calculated using the Greenwood formula for the standard error (Collett 2023). The NNT will be calculated as the inverse difference of the cumulative incidence estimates between finerenone and placebo for the selected time points. In addition, cumulative incidence plots accounting for competing risks will be presented for each treatment group. These analyses will be presented in total and for each level of the stratification factors.

For the 3 key secondary efficacy variables, the incidence rate per 100 patient years will be presented by treatment group. This rate is calculated as the number of participants experiencing an event divided by the total sum of the time each participant is at risk for the event. The time at risk in days for each participant is defined as the period from randomization until the occurrence of the first event or the point of censoring, multiplied by 365.25 to annualize it and scaled to 100 patient years. CIs for the incidence rates per 100 patient years will be calculated using the chi-squared distribution method. The degrees of freedom for this calculation are derived from twice the number of participants experiencing an event. The lower and upper limits of the CIs are determined by dividing the 2.5th and 97.5th percentiles, respectively, of the chi-squared distribution by twice the total sum of times at risk, adjusted for 100 patient years (Nelson 1982).

For the composite endpoints, cumulative incidence plots, log-rank tests, and Cox regression analysis for the time to the first event of the individual components will also be provided as part of exploratory efficacy analyses (see Section 5.4.2).

The plausibility of the proportional hazards assumption will be assessed by visually examining the plot of the log of the negative log of Kaplan-Meier survival estimates vs log time, looking for evidence of non-parallelism (i.e., a test for proportional hazards). Additionally, a treatment-by-log-time interaction will be included in the Cox regression model. The significance of this interaction will be tested at the 5% significance level to assess proportionality statistically. If the interaction term is statistically significant, indicating a violation of the proportional hazards assumption, and the plots strongly suggest non-proportionality, time-dependent hazard ratios will be estimated using a model incorporating the interaction term.

5.4.2 Supportive Exploratory Endpoints for Key Secondary Endpoints

Components of the key secondary endpoints

The following components of the key secondary endpoints will be analyzed as supportive secondary endpoints:

- Time to onset of kidney failure (ESKD or sustained eGFR decline below 15 mL/min/1.73 m²)

- Time to onset of ESKD (chronic dialysis or renal transplantation)
- Time to onset of eGFR decrease to $< 15 \text{ mL/min/1.73 m}^2$ sustained over at least 4 weeks
- Time to onset of eGFR decrease of $\geq 57\%$ from baseline sustained over at least 4 weeks
- Time to CV death
- Time to first hospitalization for HF or urgent HF visit

An urgent HF visit is defined as a presentation at an acute care facility with a duration of less than 24 hours if at least 1 of the following measures is required:

- Intravenous diuretic or vasoactive agent
- Mechanical intervention
- Implantation of a cardiac resynchronization pacemaker
- Implantation of an implantable cardiac defibrillator device

Consistent with the methods for the key secondary efficacy variables, components of the secondary time-to-event endpoints (as included in the list of other exploratory efficacy variables above) will be analyzed as exploratory time-to-event variables using the stratified log-rank test stratified by the randomization stratification factors. Point estimates of the hazard ratio and corresponding 2-sided 95% CI will be calculated using a stratified Cox regression model. Cumulative incident plots will be provided for the cumulative proportions of events by treatment group.

5.5 Exploratory Endpoints Analysis

Censoring rules for time-to-event exploratory endpoints will align with those established for the secondary endpoints.

The exploratory efficacy analyses will be conducted on the FAS.

5.5.1 Other Exploratory Endpoints Specified in Study Protocol

Apart from the components of the key secondary endpoints, other exploratory efficacy variables, as specified in the study protocol, are as follows:

- Number of participants with UACR decrease of at least 30% from baseline at Month 6
- Change in UACR from baseline
- Time to occurrence of the composite endpoint of kidney failure or an eGFR decrease of $\geq 40\%$ from baseline sustained over at least 4 weeks
- Time to first CV event (CV death, non-fatal myocardial infarction [MI], non-fatal stroke/transient ischemic attack [TIA], hospitalization for HF, or urgent HF visit)
- Time to first CV hospitalization
- Time to death from any cause (all-cause mortality)
- Time to first all-cause hospitalization
- Number of participants with a new diagnosis of atrial fibrillation or atrial flutter
- Number of participants with a new diagnosis of HF
- Number of participants with outpatient worsening of HF (pre-existing diagnosis of HF, worsening of symptoms, and signs leading into treatment augmentation)

- Change in quality of life summary scores measured by the following health-related quality of life (HRQoL) questionnaire: EuroQol 5 Dimensions, 5 Levels questionnaire (EQ-5D-5L)

5.5.2 Other Exploratory Endpoints Not Specified in Study Protocol

Other exploratory endpoints not specified in the study protocol are as follows:

- Time to occurrence of the composite endpoint of kidney failure, eGFR decrease of $\geq 57\%$ from baseline sustained over at least 4 weeks, hospitalization for HF, urgent HF visit, or CV death
- Time to occurrence of the composite endpoint of kidney failure or an eGFR decrease of $\geq 50\%$ from baseline sustained over at least 4 weeks

5.5.3 Exploratory Time-to-Event Endpoints

The endpoint involving the sustained decrease of eGFR $\geq 50\%$ from baseline over at least 4 weeks will be programmatically derived, similar to those described in Section 5.4.1.1.

All exploratory time-to-event endpoints will be analyzed using the stratified log-rank test stratified by the randomization stratification factors. Point estimates of the hazard ratio and corresponding 2-sided 95% CI will be calculated using a stratified Cox regression model. Cumulative incidence plots accounting for competing risks (non-CV death or all-cause death, as appropriate) will be provided for the cumulative proportions of events by treatment groups.

5.5.4 New Diagnosis of Atrial Fibrillation/Atrial Flutter and New Diagnosis of Heart Failure

Frequency tables will be generated for the number of participants with a new diagnosis of atrial fibrillation/atrial flutter and the number of participants with a diagnosis of new HF after baseline.

The presence or absence of the event will be compared between the treatment groups using logistic regression incorporating factors of treatment group and randomization stratification levels. Odds ratios comparing the incidence in the finerenone vs the placebo group will be calculated, along with corresponding 2-sided 95% CIs and *p*-values.

5.5.5 Outpatient Worsening of Heart Failure

For participants with a prior medical history of HF, frequency tables will be generated to display the number of participants experiencing an outpatient worsening of HF. *Outpatient worsening* of HF is defined as any worsening of HF that does not fulfill the criteria of hospitalization but merely requires an augmentation of the treatment for HF.

The presence or absence of the event will be compared between the treatment groups using logistic regression incorporating factors of treatment group and randomization stratification levels. Odds ratios comparing the incidence in the finerenone vs the placebo group will be calculated, along with corresponding 2-sided 95% CIs and *p*-values.

5.5.6 Urinary Albumin-to-Creatinine Ratio

The UACR will be summarized descriptively by treatment group and visit throughout the study, including ratios relative to baseline values. These summaries will be provided for both the overall study and stratified according to the randomization stratification factors.

The change from baseline in log-transformed UACR at each nominal visit during the planned treatment period, with a sufficient number of participants who have UACR measurements,

will be analyzed using an MMRM. Visits with insufficient numbers of participants with non-missing post-baseline and baseline UACR data (overall or within treatment group) will be excluded from the analysis to ensure stable model estimation. For each participant and visit, the response will be defined as the natural logarithm of UACR at that visit minus the natural logarithm of the corresponding baseline value. The model will include fixed effects for treatment, visit (categorical), treatment-by-visit interaction, baseline SGLT2 inhibitor use, and the interaction between SGLT2 inhibitor use and visit, with the log-transformed baseline UACR value as a covariate, and an interaction term between the log-transformed baseline value and visit. From this model, exponentiation of the model-based means will provide geometric mean ratios to baseline within each treatment group at each visit, and exponentiation of the between-treatment differences will provide pairwise geometric mean ratios of these ratios to baseline between the finerenone and placebo treatment groups, along with 2-sided 95% CIs. Model-based means and between-treatment contrasts will be computed using observed margins, such that estimates at each visit are averaged over the observed distribution of stratification factors in the analysis data, and with continuous baseline covariates evaluated at their observed mean values. The SAS procedure MIXED will be employed to estimate covariance patterns within participants and to adjust for the within-participant variance. A separate covariance pattern will be estimated for each treatment group based on an unstructured covariance matrix. An alternative covariance matrix will be used in cases of convergence issues, following the same strategy for addressing convergence issues as outlined in Section 5.3.3.3 for the MMRM analysis. If model nonconvergence occurs due to the inclusion of visit interaction terms, the responsible interaction term(s) will be removed to simplify the fixed-effects structure and aid convergence. This may include removal of the stratification factor-by-visit interaction term(s) and/or the interaction between the log-transformed baseline value and visit, as needed.

The presence or absence of a relative decrease in UACR of $\geq 30\%$ from baseline at Month 6 will be analyzed using logistic regression, with covariates for treatment group and randomization stratification levels. Odds ratios comparing the incidence in the finerenone vs the placebo group will be calculated, along with corresponding 2-sided 95% CIs and *p*-values.

The analyses described above will be conducted based on all participants' UACR measurements, irrespective of whether they prematurely stopped study intervention in accordance with the treatment policy strategy. In addition to those analyses, an on-treatment analysis of UACR will be conducted, including measurements restricted to up to 30 days after the earliest of (i) the last intake of the intervention or (ii) the start of an extended treatment interruption, using the same MMRM approach described above.

5.5.7 Health-Related Quality of Life

Health-related quality of life will be assessed using the EQ-5D-5L questionnaire. The EQ-5D-5L is a self-administered preference-based generic measure of health status, which includes 5 dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression (Oemar, et al. 2013). Participants provide a rating for each question on a 5-level Likert scale: having no problems, having slight problems, having moderate problems, having severe problems, and being unable to do/having extreme problems. In addition, participants are asked to self-rate their own health today on a vertical 0-100 unit visual analogue scale (VAS), with 0 corresponding to "the worst health you can imagine", and 100 corresponding to "the best health you can imagine".

The EQ-5D-5L will be collected on paper questionnaires at time points indicated in the SoA (see Section 1.3 of the study protocol).

The reported data for the 5 dimensions of the EQ-5D-5L will be analyzed descriptively.

Summary scores will be calculated out of the 5 dimensions according to the scoring instructions from Europe (the UK as a representative country within Europe) and the US (Oemar, et al. 2013) and to the EQ-5D Value Sets (Szende, Oppe and Devlin 2007). The values and changes from baseline of the summary scores and the EQ VAS will be summarized by treatment group and visit. These summaries will include the number of observations and missing values and descriptive statistics such as the minimum, first quartile (Q1), median, third quartile (Q3), maximum, mean, and SD.

The absolute change from baseline in the summary scores and EQ VAS scores at each visit up to the last visit, where a reasonable number of participants have EQ-5D-5L measurements, will be analyzed using an MMRM. Visits with insufficient numbers of participants with non-missing post-baseline and baseline EQ-5D-5L measurements (overall or within treatment group) will be excluded from the analysis to ensure stable model estimation. The model will include fixed effects for the treatment group, visit, treatment-by-visit interaction, and randomization stratification levels. Additionally, the baseline value of each score will be included as a covariate, along with an interaction term between the baseline value and visit. Pairwise differences between the finerenone and placebo treatment groups at each visit will be calculated, with corresponding 2-sided 95% CIs. Model-based means and between-treatment contrasts will be computed using observed margins, and continuous baseline covariates will be evaluated at their observed mean values. The SAS procedure MIXED will be employed to estimate covariance patterns within participants and to adjust for the within-participant variance. A separate covariance pattern will be estimated for each treatment group based on an unstructured covariance matrix. An alternative covariance matrix will be utilized in case of convergence issues following the same strategy for addressing convergence issues as outlined for the MMRM analysis in Section 5.3.3.3. If the inclusion of stratification-by-visit interaction term(s) results in model nonconvergence, the responsible interaction term(s) will be removed to simplify the fixed-effects structure and aid convergence.

5.5.8 Win-Odds Analysis

The established composite renal endpoint in clinical trials combines clinical events with sustained large changes in eGFR but does not weigh the relative clinical importance of the endpoint components. By contrast, a hierarchical composite endpoint (HCE) accounts for the clinical importance of the endpoint components. In order to support the analysis of the primary endpoint and key secondary renal endpoints, a renal HCE by Heerspink et al. (2023) will be considered, including 7 components which are ranked in order of highest to lowest effect as:

1. All-cause mortality
2. ESKD
3. Sustained eGFR < 15 mL/min/1.73 m² for at least 4 weeks
4. Sustained eGFR decline $\geq$ 57% from baseline for at least 28 days
5. Sustained eGFR decline $\geq$ 50% from baseline for at least 28 days
6. Sustained eGFR decline $\geq$ 40% from baseline for at least 28 days
7. Total eGFR slope at 32 months

In this, patient-level total slopes will be derived by fitting a linear model to each patient's eGFR values from Month 3 up until the end of the treatment period. The difference from baseline to the predicted value at Month 32, divided by 32 months and multiplied by 12, will then be the patient's annualized total slope estimate at 32 months.

The HCE is analyzed using the win-odds method, which involves pairwise comparisons between every participant in the finerenone and placebo groups. In each comparison, only the most severe event among the 7 predefined endpoints is considered for each participant. The participant experiencing the more severe event is considered to "lose" against the participant with a less severe or no event. If both participants experience their most severe event of equal severity, the timing of the event determines the "loser". Thus, the participant whose event occurred earlier is deemed to have lost. If neither participant experiences any qualifying event, the total eGFR slope magnitude is used to decide the outcome. A tie results when both participants have experienced the most severe outcome of the same severity at the same time and no further comparisons are made after a tie is established. Once all comparisons are made, the win-odds statistic is calculated by taking the total number of wins for finerenone, adding half the number of ties, and dividing this sum by the total number of losses plus half the number of ties. The win-odds statistic quantifies the overall efficacy of finerenone relative to placebo based on the ranked clinical importance of the events.

In order to account for variations in the participants' follow-up times, only events occurring within 32 months post-randomization will be considered in the pairwise comparisons.

The win-odds ratio for finerenone vs placebo and its corresponding 95% CI and *p*-value will be presented. Additionally, supporting tables will detail the contribution of each of the 7 components of the HCE to the overall analysis. The number of wins, losses, and ties for each treatment group will also be presented.

5.6 Safety Analyses

The SAF will be used for all analyses related to safety and tolerability.

5.6.1 Adverse Events

With the exception of hyperkalemia, which is collected as an AE of special interest (AESI), non-serious AEs are not mandatory to be collected in the eCRF. Instead, only AEs leading to permanent discontinuation of study intervention and AEs considered related to study intervention by the investigator or related to protocol-required procedures will be collected. Any other AEs should only be collected if the investigator deems them relevant.

AEs will be coded using the latest version of the Medical Dictionary for Regulatory Activities (MedDRA) version up to before the database freeze. A listing will be provided that links each AE reported by the investigator to its corresponding coded term in MedDRA.

AEs that started or worsened after the first dose of the intervention up to 3 days after any temporary or permanent interruption will be classified as treatment-emergent AEs (TEAEs). A worst-case approach will be used to determine if an AE with partially missing dates falls within the treatment-emergent window. Specifically, if it is possible that the AE start date is within the period of intervention intake plus 3 days, the AE will be considered treatment-emergent.

An overall summary of all reported AEs and TEAEs will be generated for each treatment group, including the number and percentage of participants with at least 1 event for the following categories:

- TEAEs
- Post-treatment AEs occurring more than 3 days after the temporary or permanent discontinuation of treatment
- Treatment-emergent serious AEs (TESAEs)

- Treatment-emergent intervention-related AEs
- Treatment-emergent intervention-related serious AEs (SAEs)
- TEAEs leading to permanent discontinuation of intervention
- Treatment-emergent intervention-related AEs leading to permanent discontinuation of intervention
- TESAEs leading to permanent discontinuation of intervention
- Treatment-emergent non-serious AEs
- Non-serious AEs
- SAEs
- SAEs with fatal outcome
- TEAEs by maximum intensity
- TESAEs by maximum intensity
- Intervention-related TEAEs by maximum intensity
- TEAEs by worst outcome
- TESAEs by worst outcome

All AEs will be summarized using MedDRA terms grouped by Primary System Organ Class (SOC) and Preferred Term.

In the case of events with different intensities within a participant, the maximum reported intensity will be used. If the intensity is missing, the event will be considered severe. If the same event is considered both unrelated and related to the intervention within a participant, it will be reported as related to the intervention. If the intervention relationship is missing, the event will be considered as being related to the intervention.

In order to comply with local regulatory requirements in Japan, cardiovascular disease-related OEs will also be documented as AEs or SAEs in Japan, and all serious disease-related OEs will also be documented as SAEs in India. Disease-related events will not be included in the AE summary tables or listings to avoid double-counting of such events. A separate listing will be generated for all AEs excluded from the AE analysis due to double reporting in Japan, and a separate listing will be generated for all SAEs excluded from the AE analysis due to double reporting in India.

The cumulative incidence estimates for TEAEs leading to permanent discontinuation of the intervention will be presented. The cumulative incidence probabilities will be derived using the Aalen-Johansen estimator, with the discontinuation of the study intervention due to reasons other than AEs being treated as competing events. Participants who complete treatment as planned will be censored.

The following additional safety variables related to selected TEAEs will be summarized by treatment group using frequency counts:

- Number of participants with hyperkalemia
- Number of participants with hospitalization for hyperkalemia
- Number of participants discontinuing intervention permanently due to hyperkalemia
- Number of participants with worsening renal function
- Number of participants with hospitalization for worsening renal function

- Number of participants discontinuing intervention permanently due to worsening of renal function

Hyperkalemia includes AEs with MedDRA codes (Preferred Terms): 10020646 (Hyperkalaemia), 10005725 (Blood potassium increased).

Worsening of renal function includes AEs with MedDRA codes (Preferred Terms): 10069339 (Acute kidney injury), 10005483 (Blood creatinine increased), 10018358 (Glomerular filtration rate decreased), 10059345 (Postrenal failure), 10072370 (Prerenal failure), 10038435 (Renal failure), 10062237 (Renal impairment).

For hyperkalemia, the number of participants who develop an event will also be summarized by the intervention action taken and the recorded event outcome.

All deaths, SAEs, and AEs leading to permanent intervention discontinuation will be listed separately.

5.6.2 Additional Safety Assessments

5.6.2.1 Laboratory Parameters

The number of participants with treatment-emergent abnormal laboratory values (until 3 days after any temporary or permanent interruption of the intervention) will be tabulated by laboratory parameter and treatment group. These abnormalities include values both above and below the normal range and will be assessed using data from both scheduled and unscheduled visits.

The laboratory parameters scheduled for collection at both baseline and post-baseline visits will be summarized for each visit. Summary statistics will include both the observed values at each visit and evaluations of changes from baseline. The same approach will apply to cystatin C. Results will also be presented graphically as plots of the mean, with ± 1 SD error bars.

Pregnancy test results for female participants will be summarized by visit.

For serum creatinine and UACR, geometric means and ratios to baseline will be presented instead of arithmetic means. For eGFR based on the 2009 CKD-EPI equation (Levey, et al. 2009), both relative and absolute changes from baseline will be summarized.

Furthermore, summary statistics for serum potassium, eGFR, and serum creatinine will be conducted by treatment group and visit for each level of the stratification factors.

The assessment of special safety parameters related to laboratory measurements will include the number of participants with treatment-emergent safety events. These assessments will also be presented per visit, as well as stratification factors. Summaries will specifically quantify participants exhibiting:

- Serum potassium levels exceeding 5.5 mmol/L and 6.0 mmol/L.
- Relative decreases from baseline in eGFR of $\geq 30\%$, $\geq 40\%$, $\geq 50\%$, and $\geq 57\%$.

The percentage of participants experiencing treatment-emergent events will be compared between the finerenone and placebo control groups using exploratory χ^2 tests with a continuity correction. If the expected number of participants in any cell of the 2×2 contingency table is less than 5, Fisher's exact test will be used instead. Estimates and 2-sided 95% CIs will be provided for each treatment group and the treatment differences. Clopper-Pearson CIs will be calculated for each treatment group. Clopper-Pearson CIs will be calculated for individual treatment groups, while Wald asymptotic CIs will be calculated for treatment differences due to the computational burden of exact unconditional CIs.

5.6.2.2 Vital Signs

Vital signs, taken before blood collection for laboratory tests, will include 1 pulse and 3 blood pressure measurements. The 3 consecutive blood pressure readings will be taken at intervals of at least 1 min. The average of these 3 readings will be recorded in the eCRF by the investigator.

Vital signs data will be summarized by treatment group and visit using descriptive statistics, including absolute changes from baseline. Results will also be presented graphically as plots of the mean with accompanying ± 1 SD error bars. The analysis of systolic blood pressure (SBP) will be repeated and stratified by baseline SBP categories: < 130 mmHg, 130 to < 160 mmHg, and ≥ 160 mmHg. Additionally, SBP will be stratified by $<$ baseline median and $\geq$ baseline median of the SAF.

5.6.2.3 Weight and Body Mass Index

The values and absolute changes from baseline for weight (kg) and BMI (kg/m^2) will be summarized by treatment group and visit. Descriptive statistics will be used to present these data, which will also be displayed graphically as mean plots with ± 1 SD error bars.

5.7 Other Analyses

5.7.1 Other Variables and Parameters

Pharmacokinetics

To investigate systemic exposure to finerenone and its relationship with treatment effects, plasma concentrations of finerenone will be determined at various time points using a sparse sampling approach across all participants. As outlined in the SoA (refer to Section 1.3 of the study protocol), plasma concentration vs time data will be descriptively evaluated by dose and visit. All pharmacokinetic analyses will be conducted using the PKS.

Descriptive summaries will include the following statistics: the number of participants with values above the LLOQ, geometric mean, geometric SD, geometric coefficient of variation (%), arithmetic mean, arithmetic SD, arithmetic coefficient of variation (%), median, and the minimum and maximum values. For the calculation of statistics, values below the LLOQ will be substituted by half the LLOQ.

Plots of individual plasma concentrations vs actual relative study times (time post-study intervention administration) will be prepared.

Pharmacokinetic and exposure-response analyses may be conducted using population approaches (population PK [popPK] and popPK/pharmacodynamics), potentially through non-linear mixed-effects modeling. These evaluations will be detailed in a separate analysis plan and reported separately.

Pharmacodynamic parameters

Pharmacodynamic parameters will not be evaluated in this study.

5.7.2 Subgroup Analyses

Exploratory subgroup analyses will be conducted for the safety and primary, key secondary, and selected exploratory efficacy variables, focusing on variations by randomization stratification factors:

- For the primary efficacy endpoint, the subgroup analysis will be conducted based on the 2-slope linear spline mixed-effects model as specified for the primary analysis.
- For the key secondary endpoints, the subgroup analysis will include hazard ratios from the Cox model for time-to-event analyses.
- Selected exploratory efficacy variables related to UACR will be analyzed for subgroups. The analysis will cover (i) the presence or absence of a relative decrease in UACR of $\geq 30\%$ from baseline at Month 6, based on logistic regression analysis, and (ii) UACR over time, analyzed using an MMRM. These analyses will follow the approach described in Section 5.5.6.
- Number and percentage of participants with TEAEs, TESAEs, treatment-emergent hyperkalemia events, and worsening of renal function

These analyses will incorporate descriptive statistics of eGFR and UACR, over time, graphical displays of estimated treatment effects with 95% CIs in forest plots, and statistical tests for heterogeneity of treatment effects across subgroup levels. For the primary efficacy endpoint (2-slope linear spline mixed-effects model), subgroup-specific treatment effects for the acute, delta, chronic, and total slopes at Month 32 will be estimated using linear combinations of the fitted model coefficients. Heterogeneity of treatment effects across subgroup levels for each slope estimand will be assessed using Wald tests of appropriate linear contrasts (difference-in-differences) comparing treatment differences between subgroup levels, with joint multi-degree-of-freedom tests used for subgroups with more than 2 levels; 2-sided p -values will be reported. For the primary efficacy endpoint, where multiple imputation is used, the relevant model coefficient estimates and their associated variance-covariance matrices will be obtained within each imputed dataset and pooled across imputations using Rubin's rules implemented in PROC MIANALYZE; Wald tests of the required linear contrasts will be performed using TEST statements. For subgroup analyses of other endpoints (e.g., Cox models for time-to-event endpoints, logistic regression for the Month 6 UACR responder analysis, and MMRM analyses for UACR over time), heterogeneity across subgroup levels will be assessed using the relevant interaction terms (or equivalent Wald tests) from the corresponding model, and 2-sided p -values will be reported. For MMRM analyses of UACR over time, the model will include the subgroup factor and its interactions with treatment and visit, including the treatment-by-visit-by-subgroup interaction. Heterogeneity of treatment effects across subgroup levels will be assessed using Wald tests of the treatment-by-subgroup interaction (overall heterogeneity) and the treatment-by-visit-by-subgroup interaction (time-varying heterogeneity); 2-sided p -values will be reported. If a subgroup model does not converge after applying prespecified model simplifications (e.g., visit trimming and covariance structure simplification), it may be fitted separately within each subgroup level, and no p -values for heterogeneity will be reported.

The incidence of safety events will be summarized graphically by subgroup using dot plots.

It is anticipated that in these proposed subgroups for analysis, differences in treatment effects may be observed according to the screening or baseline characteristics, due in part to the differences in the risk of clinical events expected in the different subgroups.

The key subgroups, in addition to the stratification factors and additional subgroups for analysis, are detailed in the following subsections.

All eGFR-based subgroups will be defined using values calculated with the 2009 CKD-EPI formula.

5.7.2.1 Efficacy Analyses

Stratification factors:

- UACR at screening (≤ 1000 , > 1000 mg/g)
- Baseline SGLT2 inhibitor use (yes, no)

Key subgroups as specified in the study protocol (in addition to those defined by the stratification factors):

- Primary cause of CKD (hypertensive/ischemic nephropathy, chronic glomerulonephritis, other/unknown)
- Detailed CKD etiology (hypertensive/ischemic nephropathy, immunoglobulin A [IgA] nephropathy, focal segmental glomerulosclerosis [FSGS; primary or secondary], membranous nephropathy, mesangial proliferative glomerulonephritis [MesPGN], other chronic glomerulonephritis, CKD other/unknown)
- Baseline eGFR (< 45 , 45 to < 60 , ≥ 60 mL/min/1.73 m²)
- Baseline UACR (< 300 , 300 to ≤ 1000 , > 1000 mg/g)
- Baseline serum potassium (≤ 4.5 , > 4.5 mmol/L)
- Region (North America, Latin America, Europe, Asia; see Section 6.6 for a classification of the participating countries according to geographic regions)
- Ethnicity (Hispanic or Latino, not Hispanic or Latino, not reported)
- Race (White, Black, Asian, other)
- Gender (male, female)
- Age group (< 65 , ≥ 65 years)
- History of atherosclerotic CV disease (absent, present)

Chronic glomerulonephritis includes IgA nephropathy, FSGS, membranous nephropathy, MesPGN, and other chronic glomerulonephritis.

5.7.2.2 Safety Analyses

The same subgroups defined for efficacy will apply to the safety analyses with the following additional subgroups:

- ACEI at baseline (yes, no)
- ARB at baseline (yes, no)
- Baseline BMI (< 30 , ≥ 30 kg/m²)
- Baseline SBP (< 130 , 130 to < 160 , ≥ 160 mmHg)
- Baseline UACR (30 to < 300 , ≥ 300 mg/g)
- Age group (< 65 , 65 to < 75 , 75 to < 85 , ≥ 85 years)
- Baseline weight (< 60 , 60 to < 90 , ≥ 90 kg)

- eGFR at baseline 25 to < 45 mL/min/1.73 m² and baseline serum potassium > 4.5 mmol/L (yes, no)
- eGFR at baseline < 25, 25 to < 45, 45 to < 60, ≥ 60 mL/min/1.73 m²
- eGFR at baseline < 30, 30 to < 45, 45 to < 60, 60 to < 75, 75 to < 90, ≥ 90 mL/min/1.73 m²
- Baseline serum potassium < 4.8, 4.8 to ≤ 5.0, > 5.0 mmol/L
- Beta-blocker at baseline (yes, no)
- Diuretics (including loop diuretics, thiazide diuretics, potassium-sparing diuretics and vasopressin antagonists) at baseline (yes, no)
- Potassium supplementation at baseline (yes, no)
- Potassium lowering agents (including binders) at baseline (yes, no)
- Oral anticoagulants at baseline (yes, no)
- Platelet aggregation inhibitors (excluding heparin) at baseline (yes, no)
- Immunosuppressants (excluding glucocorticoids and anti-inflammatory drugs) at baseline (yes, no)
- Potency of concomitant cytochrome P450 isoenzyme 3A4 (CYP3A4) inducer medication at baseline (strong, unclassified, moderate, weak, none)
- Potency of concomitant CYP3A4 inhibitor medication at baseline (strong, unclassified, moderate, weak, none)
- History of hyperkalemia

Focal segmental glomerulosclerosis includes primary and secondary focal segmental glomerulosclerosis.

MesPGN is defined as the occurrence of one of the following: glomerulonephritis membranoproliferative, glomerulonephritis proliferative, and mesangioproliferative glomerulonephritis.

A *history* of atherosclerotic cardiovascular disease is defined as a history of any of the following: peripheral artery disease, myocardial infarction, stroke, or coronary artery stenosis.

In case a participant used more than 1 concomitant CYP3A4 inhibitor/inducer medication at baseline, the highest potency will be used for classification (in the order strong, unclassified, moderate, and weak).

5.8 Interim Analyses

No formal interim analysis is planned for this study.

6. Supporting Documentation

6.1 Appendix 1: Population Characteristics

Population characteristic analyses will be conducted for the FAS.

6.1.1 Demography and Other Baseline Characteristics

Demographic and baseline characteristics, including disease characteristics obtained before the first study intervention, are as follows:

- Individual stratification factors: UACR at screening (≤ 1000 mg/g [low UACR], > 1000 mg/g [high UACR]) and baseline SGLT2 inhibitor use (SGLT2 inhibitor use at baseline [SGLT2+], no SGLT2 inhibitor use at baseline [SGLT2-]).
- Combined stratification factors (low UACR + SGLT2+, low UACR + SGLT2-, high UACR + SGLT2+, high UACR + SGLT2-).
- Demographics: Age, gender, ethnicity, race (including Asian subcategories), region (North America, Europe, Asia, Latin America, and others), body weight (kg), body height (cm), and BMI (kg/m²)
- Lifestyle factors: Smoking history (never, former, current smoker), and alcohol consumption (abstinent, light, moderate, heavy)
- Disease and health parameters: Primary cause of CKD, baseline eGFR (calculated by CKD-EPI equation), categories of baseline eGFR (< 30 , 30 to < 45 , 45 to < 60 , 60 to < 75 , 75 to < 90 , ≥ 90 mL/min/1.73 m²), baseline UACR, categories of baseline UACR (30 to < 300 , ≥ 300 mg/g), serum creatinine, serum potassium, and categories of baseline serum potassium (≤ 4.5 mmol/L and > 4.5 mmol/L)
- Vital signs at baseline: SBP, diastolic blood pressure (DBP), and heart rate
- Additional subgroups: As described in Section 5.7.2

Each participant's age will be defined as the number of years from the date of informed consent to their birth date. In cases where multiple races are recorded for a participant, the category "multiple race" will be used in the summary tables.

Demographic and baseline characteristics will be summarized using descriptive statistics by treatment group. In addition to this, demographic and baseline characteristics will be summarized separately for participants who completed the treatment phase and who did not.

6.1.2 Medical History

Medical history will be coded using the MedDRA dictionary. Medical history will be presented for each MedDRA Primary SOC and Preferred Term by treatment group and overall. Additional medical history terms by the following Project-specific Bayer MedDRA Queries (PBMQs), Bayer MedDRA Labeling Groupings (BMLGs), or selected MedDRA Preferred Terms will also be presented, including the following:

- Hyperlipidemia (BMLG)
- Hypertension (BMLG)
- Atrial fibrillation/flutter (PBMQ)
- Ischaemic stroke/TIA (PBMQ)
- Myocardial infarction (Preferred Term)
- Coronary artery disease (PBMQ)
- Myocardial ischaemia and angina pectoris (BMLG)
- Peripheral arterial occlusive disease (Preferred Term)
- Cardiac failure (BMLG)
- Coronary artery bypass graft (CABG) (PBMQ)

- Percutaneous coronary intervention (PCI) (PBMQ)
- Carotid endarterectomy (Preferred Term)

CKD medical history summaries will be presented by etiology.

6.1.3 Concomitant Medications

The concomitant medication usage will be summarized as follows:

- Number of participants who took at least 1 medication before the start of the study intervention regardless of when they ended
- Number of participants who took at least 1 concomitant medication during the study period
- Number of participants who took at least 1 medication that started and ended before starting the intervention
- Number of participants who initiated at least 1 concomitant medication after starting the intervention
- Number of participants with ongoing medications at baseline (starting before or on the day of randomization and continuing at least 1 day after randomization)

These data will be categorized by Anatomical Therapeutic Chemical (ATC) classes and subclasses and presented by treatment group and overall. They will include the number of participants with medications in the following Standardized and Bayer Drug Groupings of interest:

- ACEIs
- ARBs
- Beta-blockers
- Diuretics (excluding loop diuretics, thiazide diuretics, potassium-sparing diuretics, and vasopressin antagonists)
- Loop diuretics
- Thiazide diuretics
- Potassium-sparing diuretics
- Potassium supplements
- Potassium-lowering agents (including binders)
- Alpha blocking agents
- Calcium channel blockers
- Other antihypertensives, excluding renin inhibitors
- Oral anticoagulants
- Nonsteroidal anti-inflammatory drugs (NSAIDs) (excluding acetylsalicylic acid [ASA] and its salts)
- ASA and its salts
- Statins
- Erythropoietin stimulating agents
- Angiotensin receptor neprilysin inhibitors (ARNIs)
- Platelet aggregation inhibitors (excluding heparin and ASA)

- Platelet aggregation inhibitors (excluding heparin)
- Heparins
- Trimethoprim and derivatives
- Glucocorticoids
- SGLT2 inhibitors
- Immunosuppressants (excluding glucocorticoids and anti-inflammatory drugs)
- CYP3A4 inhibitors (strong, unclassified, moderate, weak)
- CYP3A4 inducers (strong, unclassified, moderate, weak)
- Glucagon-like peptide-1 (GLP-1) agonists
- Insulins
- Dipeptidyl peptidase-4 (DPP-4) inhibitors
- Other antidiabetics
- Anti-inflammatory drugs (excluding NSAIDs and ASA)
- Antiarrhythmics (excluding cardiac glycosides, calcium channel blockers, and beta blockers, but including sotalol)
- Cardiac glycosides
- Renin inhibitors
- Vasopressin antagonists
- Thrombolytic drugs
- COVID-19 therapies

Participants will be counted only once within each ATC class/subclass or Bayer Drug Grouping.

The percentage of participants on the maximum labeled dose of ACEIs and ARBs 4 weeks before the screening visit will be reported. This report will include reasons for not being on the maximum labeled dose, presented by treatment group and overall. Additionally, at baseline, the percentage of participants above the maximum labeled dose, at the maximum labeled dose, between the minimum and maximum labeled doses, at the minimum labeled dose, or below the minimum labeled dose, as specified in the dose ranges outlined in Table 6-4 of the protocol, will be presented for each treatment group and overall.

For potassium-lowering agents, ACEIs, ARBs, SGLT2 inhibitors, and diuretics, shift tables will be provided to show changes in usage from baseline to any time post-baseline.

6.1.4 Prior and Concomitant Procedures

Prior and concomitant procedures will be coded using MedDRA. The number of participants who underwent at least 1 prior procedure (performed before the day of randomization) and the number of participants who underwent at least 1 concomitant procedure (performed on or after the day of randomization) will be presented by MedDRA Primary SOC and Preferred Term for each treatment group and overall in summary tables.

6.1.5 Extent of Exposure

Unless otherwise specified, all statistical outputs related to treatment or study duration, extent of exposure, and compliance will be presented by treatment group and overall.

Treatment duration, defined as the time from the start of the study intervention to the last study drug intake (in months), will be summarized using descriptive statistics by treatment group and overall. Additionally, treatment duration will be categorized into ≤ 1 month, > 1 to 3 months, > 3 to 6 months, > 6 to 9 months, > 9 to 12 months, and subsequent 4-month intervals. These categories will be presented along with the corresponding number and percentage of participants. Cumulative treatment duration will be categorized into at least 1 month, at least 3, 6, 9, 12 months, and 4-month intervals. The total cumulative treatment exposure over the study, expressed in person-years, will also be provided. The tables will show the absolute and relative frequencies of participants remaining on study medication at Months 1, 3, 6, 9, 12, and subsequently at 4-month intervals. Additionally, Kaplan-Meier plots for the time up to the last study drug intake will be provided for each treatment group.

The above analyses will be repeated for the study duration, which is defined as the time from the day of randomization to the participants' final visit (in months).

The extent of exposure to the intervention (total amount of intake in grams) and the average daily dose in mg during treatment will be summarized using descriptive statistics by treatment group.

The overall titration status will be summarized for each treatment group, with absolute and relative frequencies presented for participants who started on either 10 mg or 20 mg, categorized as follows:

- For participants starting on 10 mg, categories will include never up-titrated, up-titrated once, and up-titrated more than once.
- For participants starting on 20 mg, categories will include never down-titrated, down-titrated once, and down-titrated more than once.

Additionally, the distribution of participants on 10 mg or 20 mg or experiencing a medication interruption will be tracked and reported at key time points: Day 1, Month 1, Month 3, Month 6, Month 9, Month 12, and after that, in 4-month intervals. This information will be derived from exposure data and presented by treatment group.

In addition, the number of participants whose intervention dosage was down-titrated to 10 mg or temporarily interrupted will be summarized, along with the reasons for these changes, presenting both absolute and relative frequencies for each treatment group. If down-titration or interruption is due to an AE, it will be further categorized into "AE: hyperkalemia" and "AE: non-hyperkalemia".

Compliance, expressed as a percentage, will be calculated using the formula:

$$100 \times \text{Number of taken tablets} / \text{Number of planned tablets}$$

The number of tablets taken, not directly captured in the eCRF, will be calculated as follows:

$$\text{Number tablets dispensed} - \text{Number tablets returned} - \text{Number tablets lost}$$

The number of tablets planned is calculated from the total duration of the intervention administration:

$$(\text{Days from randomization to the last intake of intervention} + 1) \times \text{Number of planned tablets per day.}$$

Compliance will be calculated using all tablets, including placebo tablets. For participants who withdraw the intervention prematurely, compliance will be calculated up to the time of their last dose. Compliance rates will be summarized descriptively for each treatment group

and the overall study population. In addition, the percentage of compliance will be categorized into 3 groups: < 80%, 80% to $\leq$ 120%, and > 120%. These categories will be summarized both by treatment group and overall.

6.2 Appendix 2: Changes to Protocol-Planned Analyses

Not applicable.

6.3 Appendix 3: Parameters for Sample Size Estimation

Table 6–1 presents the initial parameters used in the SAS macro "%GFR_Slope_Power" for determination of the sample size in Section 3.

Table 6–1: Initial Parameter Values for Sample Size Estimation

Initial Parameter: The initial parameter values obtained using 36 months eGFR from FIDELIO-DKD data	Initial parameter values
Knot = Assumed knot (in months) at which the acute slope transitions into a chronic slope	3
beta0 = Baseline mean eGFR (intercepts: $\beta_{0c} = \beta_{0t} = \beta_{00}$) for placebo control (c) and finerenone (t) treatment groups assuming equal intercepts for the 2 groups based on the principals of randomization	45
beta1c = Assumed acute slope for control group in mL/min/1.73 m ² /month	-0.2322
beta2c = Assumed delta (change) slope for control group in mL/min/1.73 m ² /month	-0.1169
beta1t = Assumed acute slope for finerenone group in mL/min/1.73 m ² /month	-0.6630
beta2t = Assumed delta (change) slope for finerenone group in mL/min/1.73 m ² /month	0.4284
Var_e = Assumed within-subject variance component of residual errors	26.4783
Var_u0 = Assumed random intercept (b_0) variance for the control group in mL/min/1.73 m ²	134.11
Var_u1 = Assumed random acute slope (b_1) variance for the control group in mL/min/1.73 m ² /month	1.0301
Var_u2 = Assumed random delta (change) slope (b_2) variance for the control group in mL/min/1.73 m ² /month	0.9883
Cov_u0u1 = Assumed covariance between b_0 and b_1 for the control group	-0.2428
Cov_u0u2 = Assumed covariance between b_0 and b_2 for the control group	-0.2944
Cov_u1u2 = Assumed covariance between b_1 and b_2 for the control group	-0.9667
Theta = The power of mean parameter for the within subject-variance	0
Kappa = The treatment-dependent constant for the variance-covariance matrix of the between-subject random effects Note: the choice of Kappa = 0 implies that the variance-covariance matrix of the between-subject random effects is the same for the control and the finerenone group	0
SEED = Seed used for random number generation	12345
InfCensor = Defines whether participants drop out due to ESKD Dropout = Defines a dropout rate	Yes 0.05
Months = Measurement occasions in months when eGFR is measured (study-design points)	0, 3, 6, 9, 12, 16, 20, 24, 28, 32, 36

Initial Parameter: The initial parameter values obtained using 36 months eGFR from FIDELIO-DKD data	Initial parameter values
DIFF = Assumed incremental differences (in mL/min/1.73 m ² /month) between the chronic slope for the finerenone group vs the chronic slope for the control group. Note: The DIFF values and the differences in all slope parameters are converted to units of mL/min/1.73 m ² /year in the output datasets and corresponding printouts. For example, 0.10 mL/min/1.73 m ² /month = 1.2 mL/min/1.73 m ² /year.	See Table 3–1 for selected values of DIFF

6.4 Appendix 4: Composite Strategy Donor Pool Imputation

The composite strategy resampling imputation procedure will be implemented using the following prespecified steps:

- For each participant, identify the earliest occurrence of death or ESKD.
- For eGFR assessments expected under the primary estimand within the planned treatment period and those that occur after this time point, treat the analysis value as missing and impute it using multiple imputation.
- Imputed values are generated by resampling with replacement from an empirical donor pool of clinically unfavorable eGFR values observed in this study, defined using eGFR derived from the 2009 CKD-EPI creatinine equation only.
 - Identify potential donors as participants with ESKD in the FAS.
 - For each potential donor, derive the last available eGFR measurement prior to the onset of ESKD (based on either central or local laboratory).
 - If the value above is 15 mL/min/1.73 m² or less, it will be included in the donor pool; otherwise, it will be removed.
 - For each missing assessment due to ESKD and death, draw with replacement from this donor pool.
- m completed datasets will be generated, analyzed using the prespecified primary model, and combined using Rubin's rules.

6.5 Appendix 5: Technical Details for Sensitivity Analyses

6.5.1 Multiple Imputation with Treatment Discontinuation Indicators

Under the treatment policy strategy, eGFR values are intended to be collected for every participant throughout the planned treatment period, even after the permanent discontinuation of the study intervention. Values after loss of eGFR follow-up will be imputed with a multiple imputation procedure that conditions on visit-specific treatment discontinuation indicators.

For the scheduled visits occurring after the ICEs of ESKD and death, the same multiple imputation approach under the composite strategy will be employed as for the main analytical approach.

As we expect to capture ESKD and death status even for participants who discontinue eGFR follow-up, all missing values after the loss of eGFR follow-up are first imputed under the treatment policy approach before overwriting values after ESKD and death with the composite strategy imputation approach.

This procedure is illustrated below by the following example code:

In order to use PROC MI, data needs to be in a horizontal format, i.e., there is 1 data row for each participant, and the eGFR values for each visit are contained in a pertaining variable.

Suppose the formatted data is included in an analysis dataset `adegfr`. The following notation will be used in this and the following sections:

Variable	Description
<code>subject</code>	Represents unique participant identifier
<code>trt</code>	Represents study intervention (<code>trt=1</code> for finerenone and <code>trt=0</code> for placebo)
<code>strata</code>	Represents baseline SGLT2 inhibitor use, baseline UACR (categorized)
<code>visit</code>	Represents Visits 1, 2, ..., 15
<code>month</code>	Represents analysis time point by month
<code>month_k</code>	Represents $\max(0, \text{month} - \text{knot})$ where "knot" denotes the change point defined in the 2-slope linear spline mixed-effects model
<code>base</code>	Represents eGFR value at baseline
<code>egfr_1 – egfr_15</code>	Represents eGFR values at post-baseline visits at Months 1, 3, 6, 9, 12, 16, 20, 24, 28, 32, 36, 40, 44, and 48
<code>disc_flag_1 – disc_flag_15</code>	<code>disc_flag_1 – disc_flag_15</code> are binary variables coded 1 = participant permanently discontinued study treatment on or before the corresponding scheduled visit (Months 1, 3, 6, 9, 12, 16, 20, 24, 28, 32, 36, 40, 44, and 48) and 0 = still on treatment at that visit (i.e., no permanent discontinuation or extended interruption before or at that visit).

Step 1: Create a monotone missing data pattern (MCMC).

For every participant, all eGFR values recorded after ESKD are first set to missing.

After setting all eGFR values recorded after ESKD to missing, the data may still contain intermittent gaps. The following PROC MI call performs an MCMC imputation with the option `IMPUTE=MONOTONE`:

```
proc mi data=adegfr out=data_mono nimpute=<m> seed=21177;
  var <xxx_trt_dummies> <xxx_strata_dummies> base egfr_1-
  egfr_15;
  mcmc chain=multiple impute=monotone;
run;
```

`data_mono` now contains m partially imputed, fully-monotone copies.

Step 2: Sequential regression imputation with treatment discontinuation flags

The remaining monotone missing values are imputed with PROC MI / MONOTONE REG. Each visit-specific model contains `trt`, `strata`, `base`, all earlier eGFR values, and every `disc_flag_k` for which ≥ 10 post-treatment discontinuation observations exist; flags with fewer than 10 such observations are omitted to avoid sparse cells.

Example code (replace angle-bracket placeholders with the retained flags):

```
proc mi data=data_mono out=mi_full nimpute=1 seed=21177;
  by _imputation_;
  class trt strata <retained disc_flag_k's>;
  var trt strata base <retained disc_flag_k's> egfr_1-egfr_15;
  monotone reg(egfr_1=trt strata base <flag_1>);
  monotone reg(egfr_2=trt strata base egfr_1 <flags_1-2>);
  ...
  monotone reg(egfr_15=trt strata base egfr_1-egfr_14
    <flags_1-15>);
run;
```

mi_full now contains m fully imputed datasets. As 1 monotone-regression draw is added within each of the m MCMC datasets, the final analysis uses m complete imputations in total.

All imputed eGFR values less than 0 will be set to 0.

Step 3: Composite strategy fill for ESKD/death

All eGFR values imputed in the previous step but recorded after ESKD or death are discarded. Those cells are then replaced in each of the multiply imputed datasets, using the composite strategy procedure described in Section 5.3.2.

Step 4: Analyze each imputed copy and pool the results.

Reshape every imputed dataset from wide to long so that there is exactly 1 record per participant visit. Remove any measurements taken after the planned end-of-treatment date that are not meant to enter the analysis. Add any unscheduled, premature treatment discontinuation, follow-up, or end-of-treatment measurements that occur on or before the participant's planned end-of-treatment date.

Then fit the 2-slope linear-spline mixed-effects model to each imputed copy; an example is shown below.

```
ODS OUTPUT Estimates=mi_est;
PROC MIXED DATA=mi_long;
  BY _imputation_;
  CLASS trt (ref='0') strata subject;
  MODEL egfr=strata trt trt*month trt*month_k strata*month
    strata*month_k / ddfm=contain noint;
  RANDOM Intercept month month_k / group=trt type=un
    subject=subject(trt);
  ESTIMATE "Total Slope Difference (32 Mths) (Treated -
    Placebo)" trt*month 1 -1 trt*month_k -0.90625 0.90625;
  ESTIMATE "Treated Group Total Slope (32 Mths)" trt*month 1 0
    trt*month_k 0.90625 0;
  ESTIMATE "Control Group Total Slope (32 Mths)" trt*month 1 0
    trt*month_k 0 0.90625;
RUN;
```

Step 5: Use PROC MIANALYZE to combine the m sets of estimates.

First, sort the output dataset `mi_est` by the variable "Label". Then, combine the results by calling PROC MIANALYZE as follows:

```
PROC MIANALYZE data=mi_est;  
  BY Label;  
  MODELEFFECTS Estimate;  
  STDERR StdErr;  
  ODS OUTPUT ParameterEstimates=mi_slope;  
RUN;
```

`mi_slope` provides the pooled point estimates, standard errors, and 95% CIs for all predefined contrasts under the 2-slope linear-spline mixed-effects model.

6.5.2 Tipping Point Analysis – Primary Endpoint

A multiple imputation analysis will be performed for a specified sequence of shift parameters, which only adjusts the imputed values for observations in the finerenone group. Multiple imputation will be done using SAS PROC MI under the MNAR assumption.

The delta-adjustment method will be applied sequentially, with missing eGFR values occurring after study discontinuation in the finerenone arm adjusted with a larger delta at each iteration. This process continues until the difference between the experimental and control treatments is no longer statistically significant. This point is called a tipping point. The user can specify a step by which delta parameters are increased at each iteration. Each imputed observation (missing eGFR value) will be modified according to the required delta shift. Assuming there is no penalty in eGFR in the placebo group, then for those participants with missing outcomes in the finerenone treatment arm, the eGFR values will be gradually decreased until a p -value of 0.05 or higher is produced (beyond the tipping point). The actual penalty (to 1 decimal place) that gives a p -value above and closest to 0.05 will then be identified.

This procedure is illustrated below by the following example code:

Step 1: As in Step 1 of Section 6.5.1, a dataset `data_mono` with only monotone missing data is generated in the first step. `data_mono` now contains m partially imputed copies.

Step 2: Perform multiple imputation with adjustment of imputed values in the finerenone arm.

The approach in this study will follow the second variant discussed in Ratitch, O'Kelly, and Tosiello (2013), where a delta adjustment is applied at each visit, starting with the first visit with missing data up to the end of the observation period. In addition to using previously delta-adjusted values as predictors, it works under the assumption that study discontinuation is associated with a worsening process that persists throughout the whole time period after study discontinuation.

The call below takes `data_mono` (wide format) from Step 1 as input and, using the MNAR ADJUST statement, creates 1 delta-adjusted wide dataset (`data_tpa`) for each of the m imputations, applying the shift only to the finerenone arm (`trt=1`). Should stratification factors trigger convergence issues, they will be omitted from the imputation models.

```

PROC MI data=data_mono seed=21177 nimpute=1 out=data_tpa;
  BY _imputation_;
  CLASS strata trt;
  VAR strata trt base egfr_1 egfr_2 ... egfr_15;
  MONOTONE regression (/ details);
  MNAR
  ADJUST(egfr_1 / shift=<delta> adjustobs=(trt="1"))
  ADJUST(egfr_2 / shift=<delta> adjustobs=(trt="1"))
  ...
  ADJUST(egfr_15 / shift=<delta> adjustobs=(trt="1"));
RUN;

```

The delta adjustment is applied only to eGFR values that are missing after study discontinuation in the finerenone arm. All other missing values (including those occurring while participants remain in the study or in the placebo arm) retain their original MAR-based imputations.

A range of delta values will be considered starting with 0, with subsequent stepwise reductions of delta.

All imputed eGFR values less than 0 will be set to 0.

Step 3: As described in the main analytical approach, values for scheduled visits after death or onset of ESKD will be imputed (overwritten) under the composite strategy.

Step 4: For each value of delta, apply the 2-slope linear spline mixed-effects model to each data_tpa replicate generated by PROC MI in Step 3.

For each fixed value of the delta, this step follows the same data handling, mixed-model fitting and pooling described in Section 6.5.1 (Step 4).

Step 5: Use PROC MIANALYZE to combine the results for each delta value.

6.5.3 Tipping Point Analysis – Key Secondary Endpoints

In order to investigate the impact of missing data under an MNAR assumption, methodology based on the general framework of pattern-mixture models combined with a delta adjustment approach ("tipping point analysis") will be employed as described by Lipkovich et al. (2016). In this approach, a piecewise exponential proportional hazards model will be fitted to all participants who prematurely discontinued study treatment prior to the occurrence of the key secondary endpoint. This imputation model will be used to impute event times for participants alive but censored before the global cutoff date (notification of study termination + 4 weeks), possibly penalized via an inflation factor.

Details:

With the piecewise exponential proportional hazards model for K intervals, the hazard function within each treatment arm is modeled as $h_0(t)\exp(\mathbf{x}_i^T\boldsymbol{\beta})$, where $\mathbf{x}_i$ is a baseline covariate vector, the baseline hazard is given as

$$h_0(t) = \lambda_1 I(0 < t \leq \tau_1) + \lambda_2 I(\tau_1 < t \leq \tau_2) + \cdots + \lambda_K I(\tau_{K-1} < t \leq \tau_K),$$

$0 = \tau_0 < \tau_1 < \cdots < \tau_K = \infty$, and $I(\cdot)$ is an indicator function.

Let $\lambda_j(\mathbf{x}_i) = \delta \lambda_j \exp(\mathbf{x}_i^T \boldsymbol{\beta})$, $j = 1, \dots, K$, where $\delta > 1$ is an inflation factor and $\boldsymbol{\beta}$ is the vector of model parameters. Then the following subject-specific survival function can be obtained (with $K = 3$):

$$S(t | \mathbf{x}_i) = \begin{cases} e^{-t\lambda_1(\mathbf{x}_i)}, & t \leq \tau_1 \\ e^{-\tau_1\lambda_1(\mathbf{x}_i)} e^{-(t-\tau_1)\lambda_2(\mathbf{x}_i)}, & \tau_1 < t \leq \tau_2 \\ e^{-\tau_1\lambda_1(\mathbf{x}_i)} e^{-(\tau_2-\tau_1)\lambda_2(\mathbf{x}_i)} e^{-(t-\tau_2)\lambda_3(\mathbf{x}_i)}, & t > \tau_2 \end{cases}$$

Random event times will then be imputed for all patients lost to follow-up before experiencing the key secondary efficacy endpoint event as:

$$t_i^* = \begin{cases} -\frac{\log(1-u_i)}{\lambda_1^*(\mathbf{x}_i)}, & u_i < 1 - e^{-\tau_1\lambda_1^*(\mathbf{x}_i)} \\ \tau_1 - \frac{\log(1-u_i) + \tau_1\lambda_1^*(\mathbf{x}_i)}{\lambda_2^*(\mathbf{x}_i)}, & 1 - e^{-\tau_1\lambda_1^*(\mathbf{x}_i)} \leq u_i < 1 - e^{-\tau_1\lambda_1^*(\mathbf{x}_i)} e^{-(\tau_2-\tau_1)\lambda_2^*(\mathbf{x}_i)} \\ \tau_2 - \frac{\log(1-u_i) + \tau_1\lambda_1^*(\mathbf{x}_i) + (\tau_2-\tau_1)\lambda_2^*(\mathbf{x}_i)}{\lambda_3^*(\mathbf{x}_i)}, & 1 - e^{-\tau_1\lambda_1^*(\mathbf{x}_i)} e^{-(\tau_2-\tau_1)\lambda_2^*(\mathbf{x}_i)} \leq u_i < 1 \end{cases}$$

where $(\lambda_1^*, \lambda_2^*, \dots, \lambda_K^*, \boldsymbol{\beta}^*)$ are sampled from a Bayesian posterior distribution and u_i is generated from a uniform distribution $U(1 - \hat{S}(c_i | \mathbf{x}_i, \boldsymbol{\beta}), 1)$, i.e., based on the estimated survival function of patient i at the censoring time c_i . The M samples of the Bayesian joint posterior distribution of parameters $(\lambda_1^*, \lambda_2^*, \dots, \lambda_K^*, \boldsymbol{\beta}^*)$ from the piecewise exponential survival model will be generated using SAS PROC PHREG procedure based on a gamma prior:

```
PROC PHREG data=<data>;
  BY trt;
  MODEL time * cnsr(1) = [covariates];
  BAYES seed=21177 piecewise=hazard (ninterval=3 prior=gamma)
  nbi=<> thinning=<> nmc=<>;
  ODS OUTPUT PosteriorSample = PSamp;
/* nbi: number of burn-in iterations */
/* thinning: controls thinning of the Markov chain */
/* nmc: number of iterations after the burn-in */
```

If the randomly generated event time is after the global cutoff date, the participant's imputed data are re-adjusted to be censored at the global cutoff date.

These imputed events will be added to the observed events in the study, and the key secondary efficacy variable will be re-analyzed. This process will be repeated M times for each value of the inflation factor δ . For the final analysis, the hazard ratio with 95% CIs will be estimated for each imputed dataset and then combined with PROC MIANALYZE using Rubin's rule and log-transformation of the hazard ratio (see Moscovici & Ratitch (2017)).

Inflation of the hazard in the finerenone group will be done until the upper limit of the 95% CI of the combined estimate of the hazard ratio crosses 1.

6.6 Appendix 6: Region Grouping

For the subgroup analyses by region, the participating countries will be pooled according to the following rule:

- North America: United States of America
- Latin America: Argentina, Mexico
- Europe: Australia*, Belgium, Bulgaria, Czechia, Denmark, Greece, Hungary, Israel*, Italy, Portugal, Spain, Russia, United Kingdom
- Asia: China, Hong Kong, India, Japan, Malaysia, South Korea, Taiwan, Singapore

*Although not geographically located in Europe, Australia and Israel have been included in this pooled region.

6.7 Appendix 7: List of Abbreviations

Abbreviation	Definition
ACEI	Angiotensin-converting enzyme inhibitor
AE	Adverse event
AESI	Adverse event of special interest
ANCOVA	Analysis of covariance
ARB	Angiotensin receptor blocker
ARTS	Mineralocorticoid Receptor Antagonist Tolerability Study
ASA	Acetylsalicylic acid
ATC	Anatomical Therapeutic Chemical
BMI	Body mass index
BMLG	Bayer MedDRA Labeling Grouping
CABG	Coronary artery bypass graft
CI	Confidence interval
CKD	Chronic kidney disease
CKD-EPI	Chronic Kidney Disease Epidemiology Collaboration
COVID-19	Coronavirus disease 19
CV	Cardiovascular
CYP3A4	Cytochrome P450 isoenzyme 3A4
DBP	Diastolic blood pressure
DPP-4	Dipeptidyl peptidase-4
eCRF	Electronic case report form
eGFR	Estimated glomerular filtration rate
EoS	End-of-study
EQ-5D-5L	EuroQol 5 Dimensions, 5 Levels
ESKD	End-stage kidney disease
EudraCT	European Union Drug Regulating Authorities Clinical Trials
FAS	Full analysis set
FIDELIO-DKD	F inerenone in reducing kiDnEy faiLure and dIsease prOgression in D iabetic K idney D isease (Study 16244)
FIGARO-DKD	F inerenone in reducin G cArdiovascular moRtality and mOrbidity in D iabetic K idney D isease (Study 17530)
FIND-CKD	F inerenone efficacy on progression of kidney disease in Non-Diabetic Chronic Kidney Disease (Study 21177)
FSGS	Focal segmental glomerulosclerosis
GLP-1	Glucagon-like peptide-1
HbA1c	Glycated hemoglobin
HCE	Hierarchical composite endpoint
HF	Heart failure
HR	Hazard ratio
HRQoL	Health-related quality of life

Abbreviation	Definition
ICE	Intercurrent event
ICF	Informed consent form
ICH	International Council on Harmonisation
IgA	Immunoglobulin A
ITT	Intent-to-treat
LLOQ	Lower limit of quantification
MAR	Missing-at-random
MCMC	Markov Chain Monte Carlo
MedDRA	Medical Dictionary for Regulatory Activities
MesPGN	Mesangial proliferative glomerulonephritis
MI	Myocardial infarction
MMRM	Mixed model for repeated measures
MNAR	Missing-not-at-random
NNT	Number needed to treat
NSAID	Nonsteroidal anti-inflammatory drug
OD	Once daily
OE	Outcome event
PBMQ	Project-specific Bayer MedDRA Query
PCI	Percutaneous coronary intervention
PK	Pharmacokinetic(s)
PKS	Pharmacokinetic analysis set
popPK	Population pharmacokinetics
Q1	First quartile
Q2	Second quartile
Q3	Third quartile
SAE	Serious adverse event
SAF	Safety analysis set
SAP	Statistical analysis plan
SBP	Systolic blood pressure
SD	Standard deviation
SGLT2	Sodium-glucose co-transporter 2
SoA	Schedule of activities
SoC	Standard of care
SOC	System Organ Class
SP	Shared-parameter
TEAE	Treatment-emergent adverse event
TESAE	Treatment-emergent serious adverse event
TLF	Table, listing, and figure
TIA	Transient ischaemic attack
UACR	Urinary albumin-to-creatinine ratio
ULOQ	Upper limit of quantification
UPCR	Urinary protein-to-creatinine ratio
US	United States
USA	United States of America
VAS	Visual analogue scale
WHO-DD	World Health Organization Drug Dictionary

7. References

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